

CHAPTER ONE

BASIC SCIENCES for GYNAECOLOGY

Bharat Bassaw

VULVA

The vulva (or pudendum) applies to the female external genitalia. The labia majora are the prominent hair-bearing folds extending backwards from the mons pubis to meet in the midline of the perineum posteriorly. The labia majora are the equivalent to the scrotum in the male. The labia minora lie medial to the labia majora. The labia minora consist of soft skin without hair. Anteriorly, the labia minora enclose the clitoris.

The perineum consists of two parts, a larger posterior, anal region and a smaller urogenital region which contains the external genitalia. The perineal body comprises the bulbospongiosus, superficial and deep transverse perineii, external anal sphincter, and pubovaginalis of the levator ani. The perineal body is a midline fibromuscular mass between the anal canal and the vagina. This body provides support to the vagina and by extension to the uterus. Injury to the perineal body during childbirth weakens the pelvic floor and contributes to prolapse.

The main blood supply and nerve supply to the perineum are the internal pudendal vessels and pudendal nerve (anterior divisions S2, 3, and 4 nerves). The pudendal nerve is medial to the blood vessels at the ischial spines. This is important to remember when performing a pudendal block.

ANUS

The anal canal is about 4cm long and it is the distal extremity of the gastrointestinal tract. It is a tubular structure consisting of circular muscle fibres.

The external anal sphincter comprises deep, superficial and subcutaneous parts of the skeletal muscle. The puborectalis muscle fuses with the sphincter at the anorectal ring. The anococcygeal ligament connects the anus and to the coccygx.

The internal anal sphincter is a thickened downward continuation of the inner circular smooth muscle of the rectum. Damage to the external and internal anal sphincter during childbirth is considered a third degree perineal tear whereas injury to the anal mucosa constitutes a fourth degree laceration.

For further details on the anatomy of the rectum, anus and bladder refer to the Last's Anatomy.

CLITORIS

The clitoris is homologous to the penis in the male. It is formed by the corpora cavernosa without any corpus spongiosus. Blood supply is via the dorsal artery of the clitoris.

PELVIC JOINTS, LIGAMENTS

1. Sacroiliac joint is a synovial joint between the auricular surfaces of the ilium and the sacrum.
2. Pubic symphysis is about 4cm long behind the mons pubis. This is a secondary cartilaginous joint.
3. Sacrococcygeal joint is a secondary cartilaginous joint.
4. Hip joint is a ball and socket type.
5. Sacrospinous ligament lies on the pelvic part of the sacrotuberous ligament. It is attached to the side of the lower part of the sacrum and the upper part of the coccyx and inserts in the ischial spine.
6. Sacrotuberous ligament is a flat band joining the posterior border of the ilium and the ischial spines to the transverse tubercles of the sacrum and upper coccyx.

Branches of the internal iliac artery

Anterior division

1. Superior vesical
2. Uterine
3. Inferior vesical
4. Middle rectal
5. Vaginal
6. Obturator
7. Inferior gluteal
8. Internal pudendal

Posterior division

1. Iliolumbar
2. Superior gluteal
3. Lateral sacral

Nerve roots

1. Obturator: anterior divisions L2, 3, 4
2. Pudendal: anterior divisions S2, 3, 4
3. Femoral: posterior divisions L2, 3, 4
4. Pelvic splanchnic (nervi erigentes): anterior roots S2, 3, 4
5. Iliohypogastric and Ilioinguinal: L1
6. Genitofemoral: L1, 2

7. Sacral plexus: L4, 5 and S 1, 2, 3, 4
8. Dorsal nerve of the clitoris: terminal branch of the pudendal nerve
9. Perineal nerve: terminal branch of the pudendal nerve
10. Nerve to the obturator internus: L5, S1, 2
11. Superior gluteal: L4, 5, S1
12. Inferior gluteal: L5, S1, 2

Embryology

1. Ovary: paramesonephric ridge of intermediate cell mass
2. Remnants of the mesonephric tubules and Wolffian ducts:
 - Cyst of Morgagni
 - Epoophoron
 - Paraoophoron
 - Cyst of Gartner's duct
3. Gubernaculum gives rise to the ovarian and round ligaments
4. Uterus, Fallopian tubes, upper vagina arise from the paramesonephric (or Mullerian) ducts
5. Lower vagina, bladder and urethra develop from the urogenital sinus
6. Labia minora: urogenital folds
7. Labia majora: labioscrotal swellings
8. Clitoris: genital tubercle
9. Vestibule: urogenital groove

Gametogenesis

Both the male and female germ cells undergo a number of changes in preparation for possible fertilization.

1. To reduce the number of chromosomes to half that of a normal somatic cell (46 to 23). This is accomplished by 2 divisions.
2. To alter the shape of the germ cells.

Human somatic cells contain 23 pairs of chromosomes (diploid) whereas the germ cell (oocyte or spermatozoon) has 23 chromosomes (haploid).

Steps in gametogenesis:

1. Replication of the DNA in the primitive germ cell (primary oocyte and primary spermatocyte). Each germ cell has twice the amount of DNA and each of the 46 chromosomes is double structure.
2. First meiotic division is extremely long. Pairing of homologous chromosomes occur. The homologous pair consists of 4 chromatids. Separation of the homologous pair occurs after the exchange or crossing over of blocks of genes. The result of this first meiosis is each daughter cell contains one member of each chromosome pair and thus each has 23

double-stranded chromosomes, and the amount of DNA is equal to that of a normal somatic cell ($2n$).

3. Second meiotic division begins immediately after the end of the first meiotic division. The 23 double-structured chromosomes divide at the centromere thus each daughter cell has 23 single chromosomes and half of the DNA (n) in a mature somatic cell.

Summary

The primary oocyte gives rise to 4 daughter cells, each with 22 plus 1-X chromosome. Only one of these daughter cells develops into a mature oocyte (ovum) ready for fertilization. The other 3 daughter cells become polar bodies which contain little cytoplasm.

The primary spermatocyte gives rise to 2 daughter cells with 22 plus 1-X chromosome and 22 plus 1-Y chromosome.

If fertilization occurs between the ovum and a spermatozoon which has the Y chromosome, a male fetus develops while fertilization with a spermatozoon with the X chromosome will result in a female fetus.

Non-dysjunction results in one daughter cell with 24 chromosomes and the other cell with 22 chromosomes. If fertilization takes place with a cell with 24 chromosomes and one with 23 chromosomes, the result is an offspring with 47 chromosomes in each cell i.e. Trisomy.

Non-dysjunction is thought to occur in female germ cells and not in the male germ cells. Women of advanced maternal age (>35) are at high risk for trisomic babies due to non-dysjunction.

Fetal circulation

1. Before birth, oxygenated blood from the placenta returns to the fetus via umbilical vein (usually one vein).
2. The major part of the oxygenated blood bypasses the liver and enters the inferior vena cava via the ductus venosus. This blood is then short-circuited from the right atrium through the foramen ovale into the left atrium.
3. A small amount of blood mainly from the lower limbs and the liver also enters the right atrium and it mixes with the venous return from the head and neck in the superior vena cava. This deoxygenated blood is then directed to the right ventricle. From there the blood goes towards the fetal lungs in the pulmonary trunk. Since the resistance in the fetal lungs is high in fetal life, the majority of the blood in the pulmonary trunk is channelled into the descending aorta via the ductus arteriosus where it mixes with blood from the proximal aorta.
4. Blood from the fetal lungs is returned to the left atrium via the pulmonary veins. This blood is then ejected into the left ventricle from where it enters the aorta.

5. Blood is returned via the 2 umbilical arteries which originate from the descending aorta to the placenta. Here the exchange of oxygen and carbon dioxide occurs by diffusion. The maternal and fetal blood does not usually mix.

Changes at birth

Fetal circulation

1. Fetal lungs become aerated suddenly with the first cry.
2. Ductus arteriosus closes by muscular contraction of its wall, thereby the amount of blood flowing from the heart to the lungs increases significantly.
3. Pressure in the right atrium decreases and the septum primum is then opposed to the septum secundum. This results in closure of the foramen ovale.
4. Closure of the umbilical arteries occurs a few minutes after birth. The umbilical arteries give rise to the lateral umbilical ligaments (distal) and the superior vesical arteries (proximal).
5. The umbilical vein and ductus venosus close shortly after the umbilical arteries. The umbilical vein develops into the ligamentum teres while the ductus venosus forms the ligamentum venosum.
6. The complete closure of the atrial walls takes about one year. Sometimes it does not happen and an atrial septal defect results.
7. The ductus arteriosum leads to the ligamentum arteriosum.

Menstrual cycle

Ovarian cycle

At the start of each ovarian cycle, several primordial follicles enlarge with the formation of a cavity around the ovum, antrum formation (Figure 1). At about day 6, one of these follicles becomes dominant (Graffian follicle), and the others undergo atresia. Granulosa cells and theca interna cells under the influence of follicle stimulating hormone (FSH) released from the anterior pituitary gland secrete oestrogen. This represents a negative feedback mechanism. At the start of the cycle the level of oestrogen is low and because of a negative feedback on the hypothalamus produces gonadotrophin-releasing hormone (GnRH) which stimulates the basophils in the pituitary to secrete FSH.

At about day 12, the level of oestrogen is high (FSH is now low) and via a positive feedback a luteinizing hormone (LH) surge from the anterior pituitary occurs. This LH surge is responsible for ovulation (release of the oocyte from the Graffian follicle). The fimbriae of the Fallopian tube collect the oocyte.

The remaining follicle becomes a corpus luteum by proliferation of the theca and the accumulation of lipid. The corpus luteum is a major source of progesterone and oestrogen. If fertilization occurs the corpus luteum survives in order to continue to produce progesterone to maintain the pregnancy until the placenta is formed. When there is no fertilization, the corpus luteum undergoes atresia and becomes corpus albicans. This results in a sudden withdrawal of progesterone and this is believed to play a major role in the initiation of the next menstruation. In fetal life there are over 7 million primordial follicles (and oogonia). At birth about 2 million ova survive. About half of these start the first meiotic division which ends just before ovulation. Immediately, the second meiotic division commences and this is only completed if the sperm penetrates the oocyte. In the normal reproductive life there are only about 500 ova, the others degenerate.

After ovulation, FSH and LH levels are low. Ovulation can be confirmed with an elevated day-21 serum progesterone ($>10\text{ng/ml}$). If fertilization occurs, the trophoblast (mainly fetal in origin) produces human chorionic gonadotrophin (HCG). The presence of HCG in serum or urine is used to confirm a pregnancy. HCG rises exponentially in early pregnancy, doubling approximately every 36-48 hours until it reaches a peak at about 10-12 weeks' gestation.

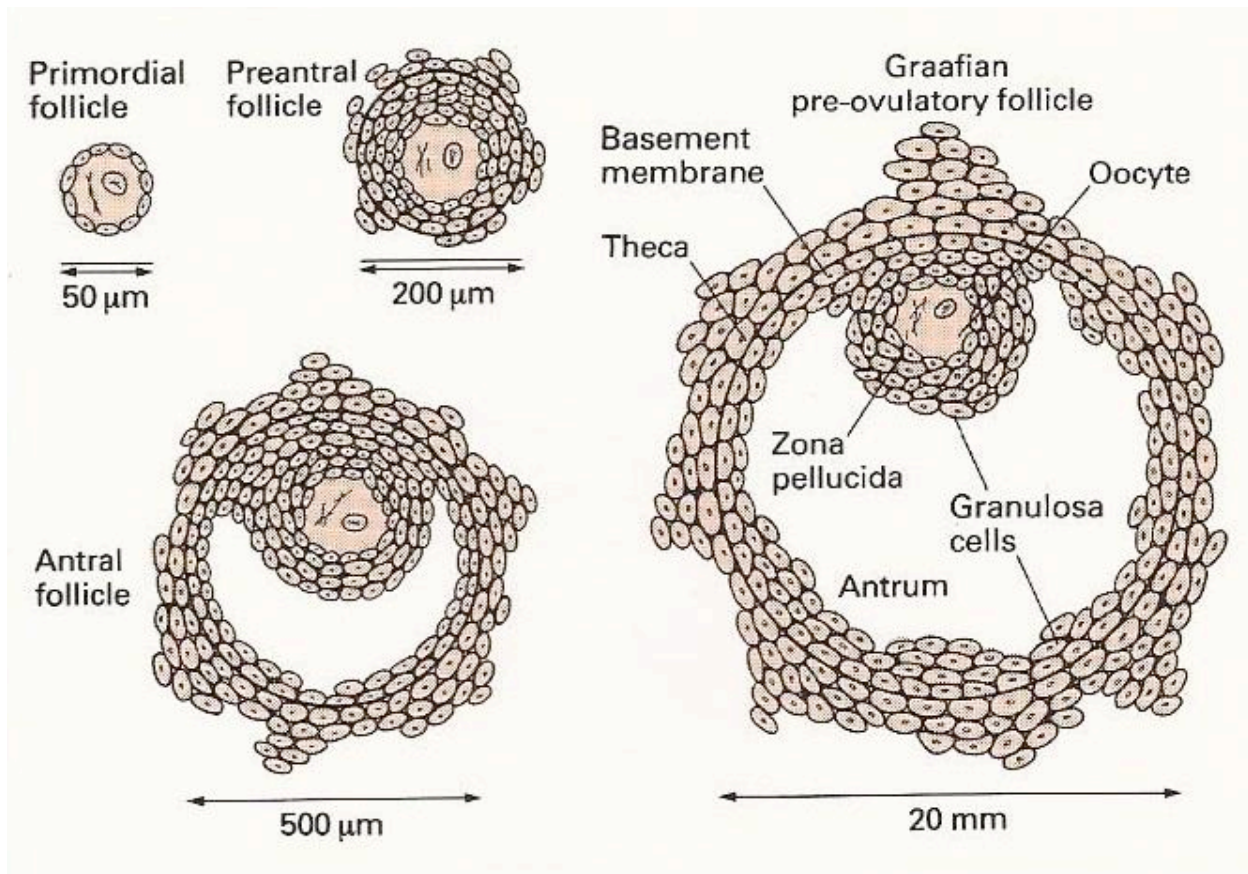


Figure 1. Development of the ovarian follicle

Uterine cycle

At the end of menstruation (usually day 5), only the deep layers of the endometrium remain. The endometrium proliferates under the influence of oestrogen (from the theca interna and the granulosa cells). The endometrium thickens and the uterine glands become tall and straight. This first half of the uterine cycle is referred as the proliferative phase.

After ovulation with the formation of the corpus luteum and hence, the secretion of progesterone, the endometrium undergoes secretory changes. The glands become coiled and tortuous and they produce glycogen and lipid. The endometrium also becomes highly vascularised and oedematous (Figure 2).

If the corpus luteum dies with subsequent cessation of progesterone production the endometrium becomes thinner and foci of necrosis appear which then coalesce. In addition, spasm and degeneration of the spiral arteries take place leading to spotty haemorrhages which become confluent and produce the menstrual flow. Vasospasm is as a result of prostaglandins which exist in secretory endometrium and in the menstrual flow.

Normal menses lasts between 3-7 days and the normal cycle is between 21-35 days. Menstrual flow is predominantly arterial (only 25% is venous). It also contains tissue debris, prostaglandins, and large amount of fibrinolysin (which lyses clots). Normal menses should not clot. Heavy menses (menorrhagia) is defined as a mean menstrual blood loss in excess of 80ml. In the first half of the cycle, the vaginal epithelium comprises cornified epithelial cells (which are large) due to high levels of oestrogen. In contrast, during the second half of the cycle (secretory or progestational phase), the vagina has a predominance of mainly navicular or boat-shaped cells which is due to progesterone.

Cervical mucus just before ovulation is thin and watery. The mucus at this time undergoes ferning if left to dry on a slide. The mucus can be stretched (it is elastic) to 8-12mm in length. This is referred to as spinnbarkeit. If ovulation occurs, there is no spinnbarkeit. Progesterone causes the mucus to become thick, tenacious and cellular.

Inhibin A is low in the first half of the cycle and is elevated in the second half. The opposite effect is seen with Inhibin B i.e. high in the first half and low in the second half of the cycle. Just after ovulation, basal body temperature rises and this persists in the second half as a result of the thermogenic property of progesterone. In the past recording of basal body temperature (BBT) charting was performed to monitor for ovulation (presence of a biphasic chart). This is now superseded by day-21 progesterone.

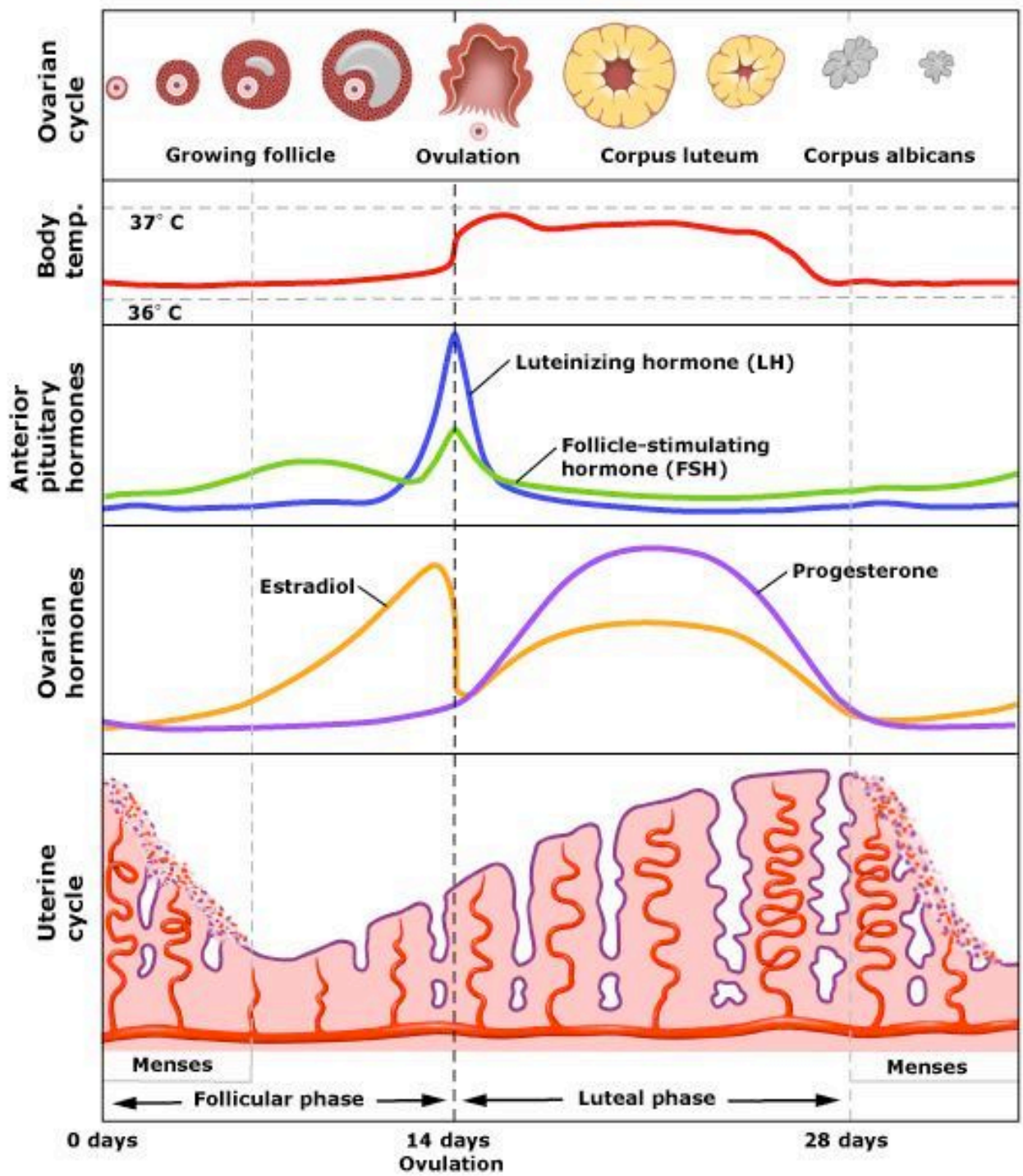


Figure 2. Endometrial and hormonal changes during the menstrual cycle

CHAPTER TWO

GYNAECOLOGICAL HISTORY AND EXAMINATION

Hugh H. Wynter

It will be presumed in this Chapter that in evaluating a woman's complaints, a general medical history, a careful systematic enquiry and a general physical examination are conducted. Only the approach to the gynaecological patient is presented here. It should be emphasized that a good doctor regards each patient as a whole person, and as Hofmeister (1971) expressed it, *"at the moment you assume the responsibility of examining a woman, you become her primary Physician, the source of her information and comfort: your obligation is to give the best, most complete care of your patient."*

HISTORY

Age and parity

The patient's age should be recorded since the significance attached to a woman's symptoms is age-related. For example, irregular uterine bleeding during adolescence is almost certainly of endocrine origin, while over the age of 40 years, it may suggest a cervical carcinoma. Parity refers to the number of births after 28 weeks gestation (in many developed countries, 24 weeks is considered the cut-off for viability). We still use 28 weeks in the Caribbean for viability.

The number of abortions or ectopic pregnancies is affixed to the parity. For example, a woman who has had 2 miscarriages, 1 ectopic pregnancy and delivered 2 babies after 28 weeks of gestation is regarded as a para 2⁺³.

Marital status

In the West Indies, it is not uncommon for a woman to have a partner or two, without being married. The duration of marriage or the period of time she has been sexually active with her common-law partner, should be recorded. The frequency of coitus and type of sexual experience should be sought. As we shall see, this information is useful in the infertile patient.

Presenting complaints

This is best recorded in the patient's own words. The onset of her symptoms is best related to an event i.e. what was she doing when her symptoms first appeared? The student should guide the history-taking. For example, if she complains of "heavy periods", she should be asked questions relating to this symptom, such as duration of bleeding, flooding and number of pads used, before acquiring a history of other complaints. It is worthwhile to remember that much

information is obtained by simply listening to the patient and if you listen carefully, invariably, she gives you the diagnosis!

Some common complaints are:

1. Abnormal uterine bleeding. You should decide whether it is menorrhagia, intermenstrual bleeding or post-coital bleeding.
2. Abdominal pain. Was it of sudden or gradual onset? What is its location? Is it constant or intermittent? Does it radiate? Is there associated vomiting or any urinary complaint?
3. Feeling of a lump. What is its site? Is it growing? Is there any associated discomfort?
4. Vaginal discharge. What are its colour, nature, consistency and odour? Is it associated with scratching (pruritus vulvae)?
5. Urinary retention and incontinence. Is dysuria, urgency, strangury or haematuria present?
6. Constipation, diarrhoea and rectal bleeding.
7. Protrusion of the "front passage". Patients with genital prolapse complain of something coming down the front passage. There may also be urinary symptoms.
8. Low backache.
9. Infertility.

Menstrual History

The following points should be noted:

- a. Age at menarche, length and regularity of the cycle, duration of the menstrual flow, the date of the last menstrual period and its duration. This is recorded as, for example, 11, 5/28-30 meaning menarche at age 11, with a flow of 5 days and a cycle of 28-30 days. She is also asked whether or not there are any clots and flooding in the menstrual flow.
- b. Dysmenorrhoea: onset and duration; primary or secondary; congestive or spasmodic.
- c. Dyspareunia, whether superficial or deep. The former implies a lesion at the introitus e.g. Bartholin's cyst or an abscess; the latter suggests a pelvic condition such as endometriosis or pelvic inflammatory disease.
- d. Discharge: amount, colour, odour, consistency and associated pruritus.
- e. Intermenstrual bleeding: amount and time of cycle.
- f. Post-coital bleeding: amount.
- g. Contraceptive or other hormonal agents.
- h. Pap smear.

Obstetric History

The outcome of each pregnancy should be recorded in chronological order stating the mode and time of delivery, the duration labour (prolonged or precipitate), birthweight, and any postpartum complications.

Miscarriages should be listed; were they spontaneous or induced, did they occur in the first or

second trimester, were there any associated complications, was curettage performed? The patient may not easily divulge a previous termination of pregnancy. A good rapport should be established and important data such as post-abortion sepsis should be obtained.

Medical and Surgical History

A brief medical and surgical history should be elicited.

It should be remembered that drugs such as Phenothiazines may cause amenorrhoea; that abnormal bleeding in a diabetic may signify an endometrial carcinoma; and menstrual disorders and hirsutism may occur in women with Cushing's syndrome.

Any previous pelvic operation is noted e.g. myomectomy, appendicectomy, ovarian cystectomy, cervical biopsy, tubal repair or dilatation and curettage.

Sexual History

A great deal of tact is required when obtaining a sexual history and this should be done in privacy and with a professional attitude.

EXAMINATION

General examination

The examination of the pelvic organs is next performed. The examination should be conducted in a cubicle that is aesthetically pleasant, comfortable, clean and private, and preferably in the presence of a nurse or nursing assistant. A general physical and breast examination is done with the patient in the supine position with both arms outstretched at her sides and the pubic area covered.

Abdominal examination

A careful abdominal examination is then performed. Inspection includes hair distribution, irregularity of contour, presence of scars and striations, and discoloration of the umbilicus. The abdomen is first gently palpated and then more deeply for tenderness, guarding, rebound tenderness and rigidity. A painful area is best left until last for deep palpation. If the hand can get above a mass but not below it, it is presumed to be arising out of the pelvis. Percussion of the abdomen is performed to distinguish between ascites and it is extremely important to palpate the inguinal areas for lymphadenopathy as this may suggest a vulval lesion, inflammatory or neoplastic.

Pelvic examination

1. Inspection of the external genitalia.
2. Speculum examination: inspection of the cervix and obtaining a smear for cytological diagnosis.
3. Inspection of the vagina.
4. Palpation of the internal genitalia (Figure 1).
5. Recto-vaginal examination.



Figure 1. Female pelvic organs

The patient should be in the dorsal position, with her thighs abducted, flexed at the hips and flexed at the knee joints with both heels together in the midline. The majority of hospitals in the West Indies are not equipped with couches for the lithotomy position.

1. The external genitalia comprise the mons pubis, labia majora and minora, clitoris, introitus and the hymenal or vaginal opening. Inspection of the external genitalia is performed for hair distribution, scratch marks, discharge or presence of vulval lesions such as warts or ulcers.

The urethral meatus is inspected and the lower vaginal walls and perineal body are surveyed for laxity. The labia are then separated and the patient is asked to bear down or strain in order to elicit stress incontinence (escape of urine from the urethral meatus) or descent of the vaginal walls and cervix.

2. Speculum examination is now performed using a bivalve Cusco's speculum, moistened with water but without a lubricant jelly in order to visualize the vagina and cervix. The

cervix is scrutinized for discharge, erosion, cervicitis or a polyp protruding through the external os. A Pap smear is then performed using a wooden scraper (Ayre's spatula) and a brush for the endocervix, care being taken to ascertain that the transformation zone is adequately sampled. The goal here is to scrape the cervix of cells that have exfoliated or are loosely held on the surface and to transfer them to a glass slide. The specimen is spread on the slide and the smear is fixed by placing it in a container with 95% Ethyl alcohol or by spraying with a fixative. Any contact bleeding is noted. The shape of the external os is noted; it is slit-like in multigravidae and circular or pinpoint in nulliparae.

There are three main methods of collecting exfoliated cells:

- a) scraping of the portio vaginalis gently with a wooden spatula;
- b) aspiration of the posterior fornix pool with a glass pipette; and
- c) rotation of a cytobrush in the cervical canal.

The bivalve or Cusco's speculum is preferable to the Sims' speculum for visualizing the cervix. The Sims' speculum is used to demonstrate vaginal wall laxity or a fistula (Figure 2).



Cusco's speculum



Sims' speculum

Figure 2. Speculum

- 3. The vaginal walls are inspected as the speculum is withdrawn, for discharge, colour and rugosity. A swab of the discharge or a smear of exfoliated cells for cornification index is taken, if indicated. If vaginal wall prolapse is present, the patient is then put in the left lateral position and examined with a Sims' speculum, which affords a clearer view of a cystocele and a rectocele.
- 4. The examination continues with a digital palpation. The index and middle fingers of the right hand are lubricated with K-Y jelly or Hibitane cream, and introduced into the

introitus.

a) The right Bartholin's gland is palpated between the fingers and the thumb. To palpate the left gland, the hand is pronated in an anti-clockwise movement. Normally, these glands are not palpable.

b) The urethra is then "milked" to express discharge from the urethra or Skene's glands.

c) The cervix is felt for consistency, regularity and mobility. Cervical excitation pain is elicited by moving the cervix to one side or the other. The cervix normally lies centrally in the vault of the vagina, but in the presence of a posteriorly-situated fibroid or a markedly retroverted uterus, it is displaced behind the symphysis pubis. If the patient has cervical excitation tenderness, then there is almost certainly a pelvic peritonism or peritonitis, due to either to a ruptured ectopic pregnancy or to pelvic inflammatory disease.

d) A bimanual examination is carefully done, defining the size of the uterus, its mobility, consistency and position, whether anteverted, upright or retroverted.

If the cervix moves with a pelvic mass, the latter is almost certainly of uterine origin.

The adnexae (Fallopian tubes and ovaries) are then palpated, for masses or tenderness. This is done by pushing the adnexae down to the pelvis with the abdominal hand, which creates counter-pressure against the vaginal fingers in each lateral fornix.

Normal-sized ovaries (3 cm in diameter) are only just palpable in a slim patient. The width of the lateral fornices should be noted; the fornix may be narrowed in women with cervical carcinoma beyond Stage 1.

Palpating through the posterior fornix may reveal a mass in the pouch of Douglas. The mass could be an ovarian cyst, or if fluctuant, a pelvic abscess, haematoma or haemoperitoneum.

e) A rectal or recto-vaginal examination may then be performed but is not part of the routine. However, it would be considered in suspected cases of endometriosis and in those with carcinoma of the cervix.

Terminology

Gynaecological problems may occur at the two extremes of, as well as during, the years of reproduction. It would be useful to define some of the terms employed to describe events or conditions, during these three phases of a woman's life-span.

At the commencement of reproductive life:

Puberty: The period between the appearance of secondary sex characteristics and completion of somatic growth.

Adolescence: The period beginning with the appearance of secondary sex characteristics and termination with the cessation of somatic growth.

Pubarche: The beginning of growth of pubic hair.

Menarche: The beginning of menstruation.

Thelarche: The beginning of development of the breasts at puberty.

During reproductive life:

Oligomenorrhoea: Abnormally scanty or infrequent menstruation greater than 40 days apart.

Amenorrhoea: Absence of menstruation.

Menorrhagia: Uterine bleeding occurring cyclically at regular intervals, excessive in amount and/or in duration (mean menstrual blood loss > 80ml).

Metrorrhagia: Uterine bleeding at anytime between the menstrual periods which are essentially normal in frequency and flow.

Meno-metrorrhagia: Excessive uterine bleeding, irregular in frequency and duration of episodes.

At the end of reproductive life:

Climacteric: The syndrome of endocrine, somatic and psychic changes occurring at the end of the reproductive period.

Menopause: The cessation of menstruation. Problems arising in the patient more than six months after this event are referred to as being postmenopausal.

KEY POINTS

- 1. It is advisable that the learner see as many patients as possible**
- 2. There is an art in eliciting a proper history from the patient or her relative.**
- 3. After obtaining the presenting complaints, consider what are the likely causes and ask some leading questions.**
- 4. For example, if the patient complains of “heavy periods” then quiz her on the number of pads, towels, tampons, etc and the symptoms of anaemia. Enquire about dysmenorrhoea, dyspareunia, discharge and pressure symptoms.**
- 5. A history of predisposing or precipitating factors will help significantly in making the diagnosis.**
- 6. Practise examination of the abdomen and pelvic organs.**
- 7. Always obtain the patient’s permission, and if a pelvic examination is planned, ensure that a chaperone is present**

CHAPTER THREE

REPRODUCTIVE ENDOCRINOLOGY

Catherine Minto-Bain

Amenorrhoea

Amenorrhoea, or the absence of menstruation, may be physiological (pregnancy, breast feeding, pre-puberty and post-menopausal) or pathological. It is defined as either primary, when menarche has never occurred or secondary. Of these two types of amenorrhoea, the secondary form is seen frequently, and is a common reason for visits to the primary care physician or gynaecologist. Secondary amenorrhoea is defined as the cessation of menstruation for at least three cycles or a length of time of six months or more.

The average age of menarche fell slightly throughout the 20th century and in most countries the average age at menarche is now 12-13 years. When menstruation has not started by the age of 16 years in a teenager with normal secondary sexual characteristics and growth parameters, investigations for primary amenorrhoea should be undertaken. Most clinicians will also investigate primary amenorrhoea at the age of 14 years where there is an absence of secondary sexual characteristics.

Primary amenorrhoea

Causes of primary amenorrhoea can be classified according to the site of dysfunction (Table 1). A review of the physiology of the menstrual cycle helps to understand how these conditions lead to amenorrhoea. A delicate feedback loop between the central nervous system and the gonads, called the hypothalamic-pituitary-ovarian (HPO) axis, controls menstruation, female hormone production and fertility. The hypothalamus produces Gonadotrophin releasing hormone (GnRH) which travels to the anterior pituitary in a pulsatile manner to trigger release of the gonadotrophin hormones FSH and LH. The gonadotrophin hormones have their action at the ovary and promote development and maturation of oocytes in follicles and hormone production. Release of various hormones and factors from the ovary completes the feedback loop to the pituitary and hypothalamus. As oocyte maturation progresses the level of oestrogens increases, and at a critical level triggers a change from negative to positive feedback in the CNS. With this positive feedback a sudden rise in LH occurs leading to ovulation 36-48 hours later. After ovulation the ruptured follicle changes to become a corpus luteum and progesterone levels rise promoting pro-pregnancy changes in the endometrium. Without a pregnancy these progesterone levels will then fall and trigger shedding of the endometrium (menstruation) and a beginning to the entire cycle again.

Table 1. Causes of Primary Amenorrhoea

Hypothalamic & Pituitary	Functional hypothalamic hypogonadism e.g. weight loss, excessive exercise, eating disorders Congenital GnRH deficiency Hyperprolactinaemia Tumours (e.g. craniopharyngioma, hamartoma) Irradiation
Gonadal	Turner's syndrome (45 X0) Gonadal dysgenesis (e.g. XY gonadal dysgenesis) Intersex conditions (e.g. androgen insensitivity syndrome) Genetic mutations of FSH and FSH receptor Ovarian agenesis Chemotherapy Irradiation
Uterine	Müllerian agenesis (Uterine and Utero-vaginal agenesis or MRKH syndrome) Cervical agenesis Vaginal septum Imperforate hymen
Pubertal dysfunction	Chronic disease Endocrine disorders Constitutional pubertal delay
Other	Pregnancy Severe PCOS Congenital adrenal hyperplasia Idiopathic

Hypothalamic & Pituitary Dysfunction

Any CNS disorder or pathology that disrupts hypothalamic function may lead to primary amenorrhoea. Functional hypothalamic hypogonadism is a relatively common cause of primary amenorrhea (up to 20% of cases). GnRH release is disrupted by stress, emotion and chronic disease states. Menarche requires a minimum of 17% total body fat (Frisch's theory) for hypothalamic maturity and therefore excessive exercise or eating disorders will often present with primary amenorrhoea. Clinical signs of low BMI, lanugo-type body hair and dry skin are highly suggestive of anorexia nervosa, and should lead to appropriate referral.

Other causes of hypothalamic hypogonadism are genetic disorders, e.g. GnRH deficiency (idiopathic hypothalamic hypogonadism or when associated with anosmia - Kallmann

syndrome), X-linked adrenal hypoplasia, GnRH receptor defects, FSH receptor defects, Prader-Willi (associated with obesity and mental retardation) and other rare disorders.

Hyperprolactinaemia is an unusual cause of primary amenorrhoea and can be caused by pituitary adenomas. Symptoms of severe or persistent headaches or symptoms suggesting other central nervous system pathology should lead to investigations for very rare tumours or other space-occupying lesions with either MRI or CT scanning of the brain.

Gonadal dysfunction

These are common causes of primary amenorrhoea and will be found in up to 50% of cases. Complete absence of the ovaries (karyotype 46XX) is rare and the aetiology is unknown. Polycystic ovary syndrome is the commonest cause of secondary amenorrhoea but may rarely lead to primary amenorrhoea.

The commonest form of primary ovarian failure is Turner's syndrome, caused by genes missing from the short arm of one X chromosome. Most cases have a karyotype 45X with a missing X chromosome. The loss of this genetic material leads to an accelerated rate of atresia of ovarian follicles leading to primary ovarian failure early in life. The ovaries become small and fibrous and are described as "streak gonads". Associated clinical signs are short stature, webbed neck, low hair-line, barrel-shaped chest and signs of hypo-oestrogenism. The incidence of Turner's syndrome is around 1 in 2000 live female births. Investigations will show hypergonadotrophic hypogonadism (high LH and FSH with low oestradiol levels). Imaging of the pelvic area will reveal a normal but small uterus. Due to the small size of the streak gonads they will be difficult to easily visualise with ultrasound.

Karyotype analysis is essential in management as there are rarer forms of gonadal dysgenesis (gonads that are both abnormal structurally and functionally), such as mosaic forms of Turner's syndrome (45X0/46XX and 45X0/46XY) and the intersex condition XY gonadal dysgenesis (Swyer's syndrome). Whenever there is a presence of Y chromosome material the streak gonads have a high chance of tumour formation, primarily gonadoblastomas, and therefore laparoscopic gonadectomy will then be required.

Intersex conditions are rare and may present with a female external appearance and primary amenorrhoea (Table 2). An intersex person is someone born with a mix or blend of the physically defining features, i.e. karyotype, gonadal structure, internal genitalia and external genitalia, usually associated with standard males and females. The term intersex therefore covers a diverse range of conditions encompassing individuals with standard male or female genitalia, who may have a variety of internal genital organs and karyotypes, and also those with ambiguous external genitalia. There are currently over fifteen recognised intersex conditions. Referral to an expert paediatric endocrinologist, gynaecologist or urologist is essential for accurate diagnosis, psychological assistance and management.

Table 2. Intersex

INTERSEX CONDITION	INTERNAL GENITAL ORGANS	EXTERNAL GENITALIA APPEARANCE	KARYOTYPE
17-hydroxylase deficiency	Testes	Female	46XY
Complete androgen insensitivity syndrome	Testes	Female	46XY
Partial androgen insensitivity syndrome (PAIS)	Testes	Variable from: normal female, clitoral enlargement, ambiguous to normal male appearance.	46XY
Congenital Lipoid Adrenal Hypoplasia	Testes	Female	46XY
Denys-Drash syndrome	Streak or dysgenetic testes	Female to ambiguous	46XY
Frasier syndrome	Streak	Female to ambiguous	46XY
Gonadal agenesis	Absent	Female	46XY
Leydig cell hypoplasia	Testes with absent/ diminished leydig cells	Variable from: normal female, clitoral enlargement, ambiguous to male genitalia with micropenis.	46XY
Mosaic Turner's syndrome	Bilateral streak gonads	Female	45X/46XY
Smith-Lemli-Opitz syndrome	Testes	Female to ambiguous	46XY
Campomelic Dysplasia with sex reversal	Testes	Female to ambiguous	46XY
Swyer's syndrome	Bilateral streak gonads	Female	46XY
5 α -Reductase-2 deficiency	Testes	Variable from: clitoral enlargement, ambiguous to male with hypospadias and	46XY

		micropenis.	
17 β -hydroxysteroid dehydrogenase deficiency	Testes	Variable from: clitoral enlargement, ambiguous to male with hypospadias and micropenis.	46XY
True hermaphroditism	Variable-consist of ovarian and testicular tissue, either separately or combined in ovotestes.	Variable from: normal female, clitoral enlargement, ambiguous to normal male appearance	71% XX 20% XX/XY 7% XY

Uterine Abnormalities

Abnormal embryonic development of the Müllerian system that develops into the uterus, cervix and vagina, is responsible for up to 20% of cases of primary amenorrhoea.

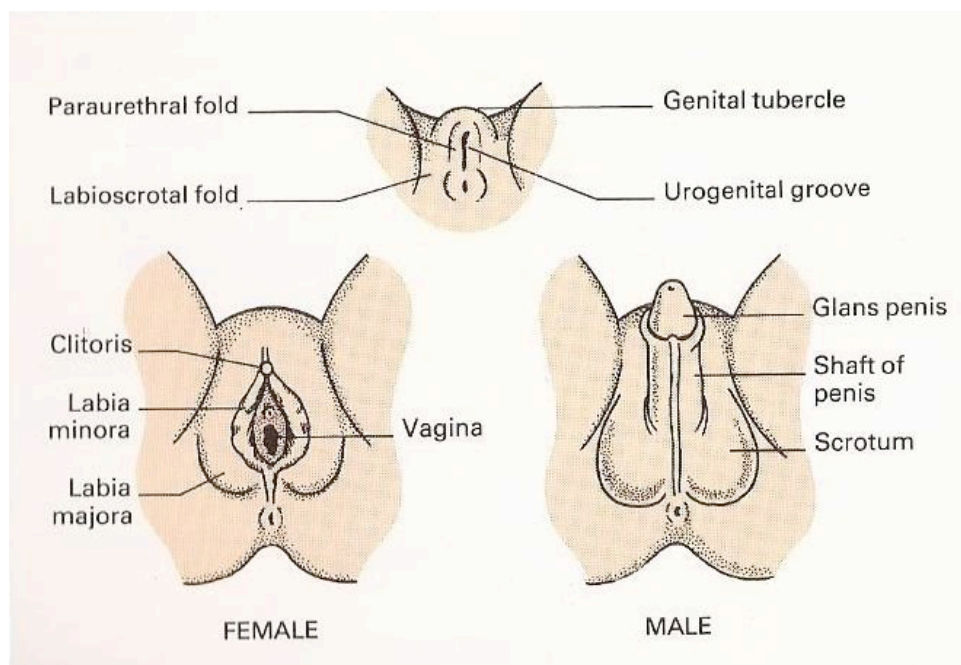


Figure 1. Embryology of external genitalia

Mayer-Rokitansky-Kuster-Hauser syndrome (MRKH syndrome) is vaginal agenesis with variable or absent cervical and uterine development. Often there are small uterine buds bilaterally without functional endometrial tissue. Ovarian development is unaffected and sexual secondary characteristics are normal. Renal imaging is required as there is a 40-50% incidence of abnormal renal development. Creation of a neovagina is achieved with dilatation treatment

or surgical reconstruction. Fertility treatment is possible through ovarian hyperstimulation, egg collection and the use of a surrogate.

Rarer structural causes often present earlier such as imperforate hymen. However, imperforate hymen or isolated cervical and/or vaginal agenesis with normal uterine development may present with primary amenorrhoea and cyclical abdominal pain or “cryptomenorrhoea”. In extreme cases the menstrual blood will accumulate in the uterus or vagina to such an extent that a pelvic mass will be found (haematocolpos or haemometra). In these situations severe endometriosis is often found during the required surgical treatment. Figure 1 demonstrates the embryological origin of the external genitalia in the male and female.

A detailed history and physical examination is important to exclude some causes of primary amenorrhoea and direct further investigations appropriately. Body mass index should be calculated using the appropriate adolescent ranges. Inspection of both pubertal status (Tanner staging) and the external genitalia is important but must be done sensitively. Speculum and vaginal examination should only be undertaken in teenagers who have been sexually active. Absence of pubic and axillary hair with otherwise normal secondary sexual characteristics is pathognomonic of complete androgen insensitivity syndrome. Virilising signs of the genitalia require urgent investigation with karyotype and hormonal evaluation for intersex conditions, adrenal or gonadal tumours and late-onset congenital adrenal hyperplasia (CAH). Figure 2 shows a haematocolpos from an imperforate hymen in a young girl with primary amenorrhoea.

Initial investigations are listed in Table 3.

A sensitive enquiry about sexual activity, without the parents present, is essential and will direct possible investigation and treatment.

Table 3. Initial investigations for amenorrhoea

Blood	FSH, LH and Oestradiol
	Prolactin and Thyroid function
Urine	Pregnancy test
Pelvic ultrasound	Trans-abdominal ultrasound is the appropriate route.
	When regular sexual activity is known trans-vaginal ultrasound is preferable.
CNS imaging	MRI of the brain where high prolactin levels are found or symptoms suggest tumours.

When physical examination is suggestive of chronic disease and the initial investigations are normal, then referral to a paediatrician is appropriate for further investigation and treatment. Abnormal gonadotrophins, virilising signs, absent vagina or abnormal secondary sexual characteristics will require further investigation with karyotype and MRI of the pelvis to define Müllerian and gonadal structures.

Treatment options are individual to the causes of primary amenorrhoea. Where chronic disease or hypothalamic causes are found resolution of these will usually allow menarche. Hyperprolactinaemia invariably responds well to bromocriptine or other prolactin lowering drugs and surgery is rarely required. In Turner's syndrome expert paediatric endocrine advice is required to maximise growth potential before commencing combined hormone replacement therapy. Once oestrogens are commenced epiphyseal closure will occur within two years.

Careful discussion of the underlying issues with both the adolescent and parents is essential. In cases of intersex disorders psychological therapy is the mainstay of management and surgical and medical management need expert direction. As with children born through adoption and sperm donation, it is essential to aim for full disclosure of the true situation guided by expert advice.



Figure 2. Haematocolpos from an imperforate hymen

KEY POINTS

- A careful and sensitive approach is essential when managing adolescents with primary amenorrhoea.
- Sexual activity is always a possibility and must be enquired about.
- Management may involve restoring menstruation or providing hormone replacement therapy.
- Prevention of disease must be considered, with karyotype evaluation for Y chromosome material in gonadal dysgenesis, and upper urinary tract imaging for MRKH syndrome.
- Future reproductive and sexual health must be considered and discussed.

Secondary amenorrhoea

As with primary amenorrhoea it is useful to classify the disorders by the level of dysfunction (Table 4). Pregnancy, breast feeding and hysterectomy are easily differentiated causes. In Trinidad and Tobago the next commonest cause is polycystic ovary syndrome (PCOS) which is discussed later in this chapter, and accounts for up to 50% of cases of secondary amenorrhoea.

Table 4. Causes of secondary amenorrhoea

Hypothalamic & Pituitary	Functional hypothalamic hypogonadism e.g. weight loss, excessive exercise, eating disorders, and poor nutrition. Psychiatric illness and stress Hyperprolactinaemia Sheehan's syndrome (ischaemic necrosis of the pituitary) Tumours (e.g. craniopharyngioma, hamartoma) Irradiation
Gonadal	Premature ovarian failure PCOS Chemotherapy Irradiation Virilising ovarian tumours (e.g. arrhenoblastoma)
Uterine	Missed miscarriage

	Hysterectomy and endometrial ablation
	Uterine artery embolisation
	Asherman's syndrome (intra-uterine scarring, often post D&C)
	Endometrial tuberculosis
Other	Pregnancy
	Pseudocyesis (phantom pregnancy)
	Congenital adrenal hyperplasia
	Galactosaemia (inborn error of galactose metabolism)
	Chronic illness (diabetes, thyroid disease, Cushing's syndrome, Addison's disease etc)
	Drugs (e.g. Depo-Provera, phenothiazines and metoclopramide)

As with primary amenorrhoea, hypothalamic dysfunction is a common cause of secondary amenorrhoea. Severe emotional stress is a common reason for disruption of the menstrual cycle. Disrupted oestrogen and leptin production are thought to unbalance the feedback loop in the hypothalamic-pituitary axis leading to reduced GnRH secretion, lack of the required FSH and LH pulsatile release and ultimately anovulation and amenorrhoea.

Up to 30% of secondary amenorrhoea will be due to hyperprolactinaemia, which is described later in this chapter. Other pituitary causes such as Sheehan's syndrome after severe hypovolaemia, are rare.

Premature ovarian failure, due to a lack of follicles and oocytes, is usually defined as a menopause before the age of either 35 years or 40 years. The aetiology may be genetic, drug-related, induced by chemotherapy or radiotherapy, after surgery or autoimmune. The diagnosis is confirmed by very high FSH and LH levels and low oestradiol levels. Anti-Müllerian hormone (AMH) levels will be in the undetectable range and pelvic ultrasound imaging will usually show a small uterus and small volume ovaries. Fertility is possible with donor egg IVF. Hormone replacement therapy is important to consider as young women often have severe symptoms due to oestrogen withdrawal. A bone density scan and treatment to prevent rapid onset of osteopenia and osteoporosis are important.

History taking and investigations for secondary amenorrhoea are essentially the same as those for primary amenorrhoea (Table 3). In the history, symptoms of thyroid disease, galactorrhoea and hypo-oestrogenism should be asked about. At examination signs of hypo-oestrogenism (vaginal atrophy, dry skin, breast atrophy), hirsutism, acanthosis nigricans and androgen excess (hirsutism, excess body hair, acne etc) should be looked for. Speculum, bimanual vaginal examination and transvaginal pelvic ultrasound are undertaken unless the patient has never been sexually active. A pregnancy test must always be performed. Where androgen excess is found, serum testosterone, androstenedione, DHEA, DHEAS, 17-hydroxy progesterone (17-OHP) and sex hormone binding globulin (SHBG) should be assessed.

In suspected Asherman's syndrome hysteroscopy with adhesiolysis and subsequent oestrogen therapy may occasionally restore menstruation. Other treatments depend on the patient's

circumstances and wishes. Where fertility is desired medications such as clomiphene citrate or gonadotrophins can be used to restore ovulation. When fertility is not desired then the endometrial lining should have a breakdown bleed at least every three months to prevent hyperplasia and an increased risk of future endometrial cancer. This can be achieved with either the combined oral contraceptive or cyclical progestogens. Women with premature ovarian failure do not require cyclical bleeds for endometrial protection.

KEY POINTS

- **Pregnancy testing should always be preformed**
- **Management is dependent on the patient's requirements – fertility or restoration of cyclical bleeding.**
- **Regular cyclical bleeding is required to prevent future endometrial pathology, except in premature ovarian failure.**
- **Long-term hypo-oestrogenic states will cause osteopenia and osteoporosis and suitable prevention is required.**

Hyperprolactinaemia

Hyperprolactinaemia is raised serum prolactin levels, and accounts for up to 30% of cases of secondary amenorrhoea. Two thirds of cases will have associated galactorrhoea (milk production from the breasts).

Secretion of the peptide hormone prolactin from the anterior pituitary is under tonic inhibition by dopamine from the hypothalamus. Physiologically raised levels occur in pregnancy (to promote breast growth) and during lactation. Pathologically raised levels occur due to reduction in this inhibition or increased production (Table 5). The high prolactin levels inhibit GnRH release from the hypothalamus leading to low gonadotrophin output and subsequent impaired follicular development, hypo-oestrogenism, amenorrhoea and infertility.

Table 5. Causes of hyperprolactinaemia

Reduced inhibition	Drugs 1. Phenothiazines 2. Haloperidol 3. Anti-psychotic medication 4. Domperidone 5. Methyl Dopa 6. Reserpine 7. Metoclopramide 8. Cimetidine Pituitary stalk compression (e.g. tumours causing acromegaly) Division of pituitary stalk
Increased production	Prolactinoma or pituitary adenoma 1. 90% are microadenomas - diameter <10mm 2. 10% are macroadenomas – diameter >10mm
Other	Severe stress Hypothyroidism (TSH stimulates prolactin release) Chronic renal failure Sarcoidosis Idiopathic

A pituitary adenoma or prolactinoma is a benign growth in the pituitary gland. Local pressure in macroadenomas may cause symptoms like visual disturbances or headache. Additionally symptoms of hypo-oestrogenism such as loss of libido and vaginal dryness can occur. When the condition is chronic the hypo-oestrogenism will lead to loss of bone mineral density. The cause of prolactinomas is unclear. Researchers have looked for gene mutations that are involved in development of tumours in other parts of the body and so far none have been identified. Oestrogens do promote the growth of prolactin-producing cells, but a link to prolactinomas has not been found. Both men and women may get prolactinomas but diagnosis is usually earlier in women, especially pre-menopausal women, because symptoms are more obvious. Prolactinomas that cause symptoms occur in about 14 people in every 100,000 each year.

Table 6. Investigation of hyperprolactinaemia

Serum	Serum prolactin, thyroid function, renal function
MRI scan (head)	When there is no relevant drug history and the prolactin level is above a certain level (e.g.100ng/ml)
Urine	Pregnancy test

Treatment of pituitary adenomas centres on drug treatments to reduce the production of prolactin, and occasionally surgery in the case of macroprolactinomas which are causing local pressure effects.

Bromocriptine, a dopamine agonist, inhibits prolactin production and is the first line treatment. Drug treatment is started at between 1.25-2.5mg daily and is increased at 4 weekly intervals in response to prolactin levels. As prolactin levels slowly reduce, the levels of LH and FSH start to rise again, and ovulation resumes. Side effects are common, including headache and effects on the nervous system leading to anxiety and restlessness. Lack of compliance is therefore common. Cabergoline works in a similar way to bromocriptine and is the second-line therapy for patients who have intolerable side effects with bromocriptine. Side effects may be similar to Bromocriptine – although often much less marked – there may also be leg cramps. Where fertility is not an issue and there is a microprolactinoma, the combined oral contraceptive pill can be used to re-establish menstruation and protects against the risk of osteoporosis. There is, however, a theoretical risk that this may cause any prolactinoma to increase in size.

Limited removal of part of the pituitary gland is sometimes necessary when symptoms do not respond to drug therapy, side effects of drug treatment are too great, or prolactin levels fail to fall. Surgery may also be needed with macroprolactinomas when drug therapy does not reduce symptoms or adenoma size.

KEY POINTS

- **Galactorrhoea is not always present in cases of hyperprolactinaemia.**
- **A detailed drug history is vital to exclude drug-induced causes.**
- **A head MRI scan is essential when prolactin levels exceed 100ng/ml.**
- **Surgery is rarely required for pituitary adenomas.**
- **Pituitary adenomas are benign tumours.**

Polycystic Ovarian Syndrome

Polycystic ovarian syndrome (PCOS) has a wide range of symptoms which vary from person to person (Table 7). These include irregular periods, weight gain, infertility and excess hair on the body. PCOS is usually associated with the ultrasound finding of polycystic ovaries, in which the ovaries are found to be enlarged and contain multiple (twelve or more) cysts 2-9mm in diameter (Figure 3). Recently a consensus agreement in Rotterdam defined PCOS as having two out of three of the following criteria; oligo or anovulation, excess androgen activity and polycystic ovaries on ultrasound.

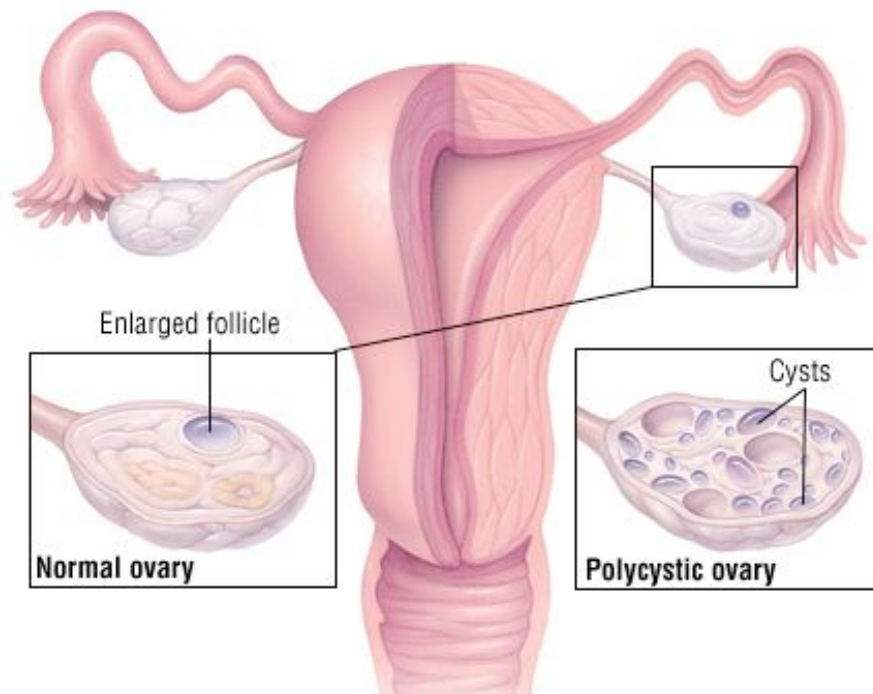


Figure 3. Polycystic ovary

Polycystic ovary syndrome (PCOS) was originally described in 1935 by Stein & Leventhal with three features; irregular periods, excessive body and facial hair and a high body mass index. People were found to have multiple cysts within their ovaries, hence the name. We now know that the condition has a range of symptoms and these vary from person to person. It is found in between 7-10% of women of childbearing age. A number of other conditions are associated with PCOS. There is an increased risk of type 2 diabetes, dyslipidaemia and hypertension. Recent studies have also shown that there may also be a small risk of gynaecological cancers in people with PCOS.

Table 7. Features of PCOS

Irregular menstrual cycles or amenorrhoea
Infertility (anovulation)
Hirsutism and increased body hair
Acne
Obesity
Obstructive sleep apnoea
Alopecia
Psychological aspects
Increased miscarriage risk

The exact cause of polycystic ovarian syndrome remains unclear but there may be a genetic component and hormonal changes are responsible for some of the symptoms. Women who have PCOS have higher levels of androgens leading to acne, excessive hair on the face, chest, abdomen, the tops of the thighs and male-pattern baldness. Problems with irregular periods are caused by a failure to ovulate properly each month and the ovaries produce a large number of cysts and rarely proceed to actual ovulation. This leads to chronically high levels of unopposed oestrogen, thought to cause the increased risk of endometrial cancer in PCOS. Studies have shown that even those people who have regular periods fail to ovulate on many occasions.

Obesity is related to an insulin resistance, causing weight gain and a tendency to lay down fat around the abdomen. Insulin levels in the body are actually chronically raised and are responsible for changes in the lipid (fat) profile with raised cholesterol levels and particularly raised levels of LDL cholesterol. Unfortunately, the cycle of insulin resistance becomes a vicious cycle as weight gain causes worsening of insulin resistance and this promotes further weight gain. Eventually, in some cases, the pancreas can no longer keep up with the demands made upon it for insulin production and diabetes is diagnosed. The lifetime risk of NIDDM in PCOS is estimated as between three to seven times that of the general population. In real terms this translates to 40% of PCOS women having type-2 diabetes by the age of forty.

Table 8 – PCOS history, examination and investigation

History	Irregular periods Secondary amenorrhoea Primary or secondary infertility Weight gain Depression Family history of diabetes and metabolic syndrome
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Examination	Acne Hirsutism Increased body hair Acanthosis nigricans Raised BMI
Investigations	FSH, LH and oestradiol Prolactin, testosterone & sex hormone binding globulin (SHBG) Thyroid, renal and liver function Complete blood count, glucose tolerance test and lipid profile Pelvic ultrasound
Differential Diagnosis	Cushing's syndrome (high cortisol levels) Thyroid disease Hyperprolactinaemia (commoner in PCOS)

Treatments are chosen according to the patient's concerns. One problem for many younger women with polycystic ovarian syndrome (PCOS) is the excess hair. This is particularly distressing when it occurs in the beard and moustache areas. Bleaching, shaving, laser photothermolysis and electrolysis provide the answer for many people but for others advice and reassurance is all that is needed. For a detailed review of treatments for hirsutism see the later section on hirsutism.

Weight loss is very important for long-term health and for general feelings of well-being. Often with dietary advice and a modest degree of exercise results can be impressive. With only a loss of 5-10% of body weight, this is sometimes enough to break the cycle of insulin resistance and restore ovulation and fertility. When diet and exercise fail to reduce BMI, then surgical approaches such as gastric banding or stapling should be considered.

When fertility is not required, the combined oral contraceptive pill, particularly those containing Cyproterone acetate or Drospirenone, is ideal to restore a regular cyclical bleed and may also help to reduce hirsutism and acne. Alternatively cyclical progestagens can be used. The insulin sensitising drug Metformin is effective in achieving ovulation and reduces fasting insulin levels, but in meta-analyses seems to have little effect on reducing BMI or weight. Side effects (in about 10% of cases) of Metformin limit its use and include gastrointestinal upset, flatulence, diarrhoea and bloating.

Fertility drugs are those to promote ovulation. Clomiphene citrate is the first line treatment, and is given in doses of between 50-150mg a day. Clomiphene is an anti-oestrogen and has its effect by blocking oestrogen feedback in the CNS and thereby increasing endogenous FSH release. Doses higher than 150mg/day are likely to be less effective than lower doses due to the anti-implantation effect on the endometrium at higher doses. The drug is taken from day two to day six or from day five to day 9 of the menstrual cycle and usually follicle tracking ultrasound scans are used to monitor response. There is a small risk of multiple births and the

drug shouldn't be used for more than 6-12 cycles.

When pregnancy is not achieved with Clomiphene citrate, the second-line treatment is now aromatase inhibitors (e.g. letrozole 2.5-5mg daily for five days). Sometimes tamoxifen is also tried. When these fail to produce ovulatory cycles or pregnancy then the next step is either surgery or exogenous Gonadotrophins. Gonadotrophin medications are synthetic or human-derived forms of FSH and LH that are given by injection. They carry risks of over-stimulating the ovaries (ovarian hyperstimulation syndrome) and of multiple pregnancy.

Surgery can be an effective form of treatment for PCOS, but only for women with good egg reserve who require fertility. It inevitably destroys oocytes, but with the benefit of improved ovarian function, in a manner that is poorly understood. Often previously clomiphene-resistant patients will spontaneously resume ovulation after treatment or become responsive to clomiphene citrate. It is always performed via the laparoscopic approach and point diathermy is applied in a limited and controlled manner to the ovaries, with cooling irrigation immediately to limit reduction in egg reserve. Excessive diathermy can lead to premature ovarian failure and its use should be limited to once or twice in a woman's lifetime. Surgery is an inappropriate method of controlling the other symptoms of PCOS.

In pregnancy women with PCOS are at increased risk for gestational diabetes, miscarriage and possibly pre-eclampsia. Research and future directions for PCOS lie in understanding better the aetiology of PCOS, particularly elucidating early phenotypes, the intra-uterine environment and possible endocrine disrupters. Future treatments may be other insulin sensitisers such as pioglitazone and D-chiro-inositol.

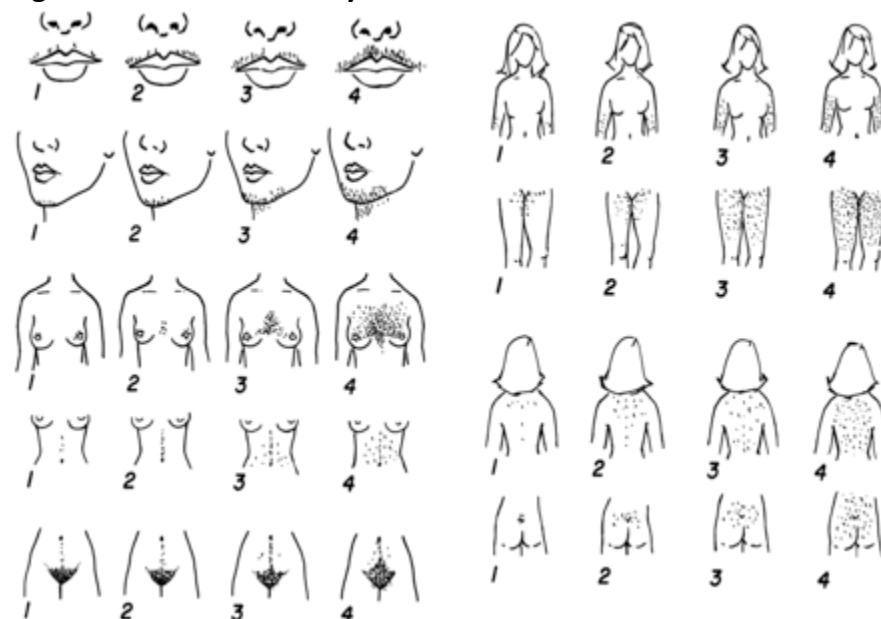
KEY POINTS

- **PCOS is the commonest female endocrine disorder.**
- **It is a heterogeneous disorder with a wide variety of symptoms.**
- **Management of PCOS is based on symptoms and the fertility desires of the patient.**
- **For women with obesity, weight loss is critical in management.**
- **Screening for hyperlipidaemia and NIDDM should commence at 35 years or earlier where indicated.**
- **Diet and exercise are critical to preventing NIDDM.**
- **Medical and surgical treatment can control symptoms, but no cure exists.**
- **In pregnancy, women with PCOS should be screened for gestational diabetes.**

Hirsutism

Hirsutism is excessive growth of hair in females that has a male type of distribution. The severity varies and is classified by the Ferriman-Galway score (Figure 4). In many ethnic groups body hair is naturally increased, however when the hair is present in excessive amounts or becomes coarse and thick, this is hirsutism.

Figure 4. Ferriman-Galway score for hirsutism



Hirsutism is a feature of increased androgen production or reduced serum androgen binding leading to increased free serum testosterone. Associated signs and symptoms can be acne & oily skin, and in severe cases male pattern baldness, deepening of the voice and clitoromegaly. The commonest cause of hirsutism is PCOS (see earlier). Drug therapy (e.g. danazol, androgens, phenytoin, diazoxide and acetazolamide) and genetic predisposition are the second commonest. Endocrine disorders such as congenital adrenal hyperplasia (CAH), adrenal tumours, arrhenoblastoma, Cushing's syndrome and acromegaly will be the cause on rarer occasions and very rarely an intersex condition will leading to virilising signs in a female (e.g. 5-alpha-reductase deficiency, true hermaphroditism and 17-ketosteroid deficiency)

Androgens are either gonadal or adrenal in origin. Adrenal androgen production is controlled by ACTH. Ovarian androgen production is mediated by LH and to a lesser extent FSH and is produced in the theca and interstitial cells (Figure 5). Testosterone circulates in the serum predominantly bound to sex hormone binding globulin (SHBG) and the only bio available testosterone is the unbound portion. At the target tissue, e.g. the hair follicle, testosterone is converted to the more potent Dihydrotestosterone (DHT) by the enzyme 5-alpha-reductase.

Figure 5. Adrenal and ovarian steroidogenesis

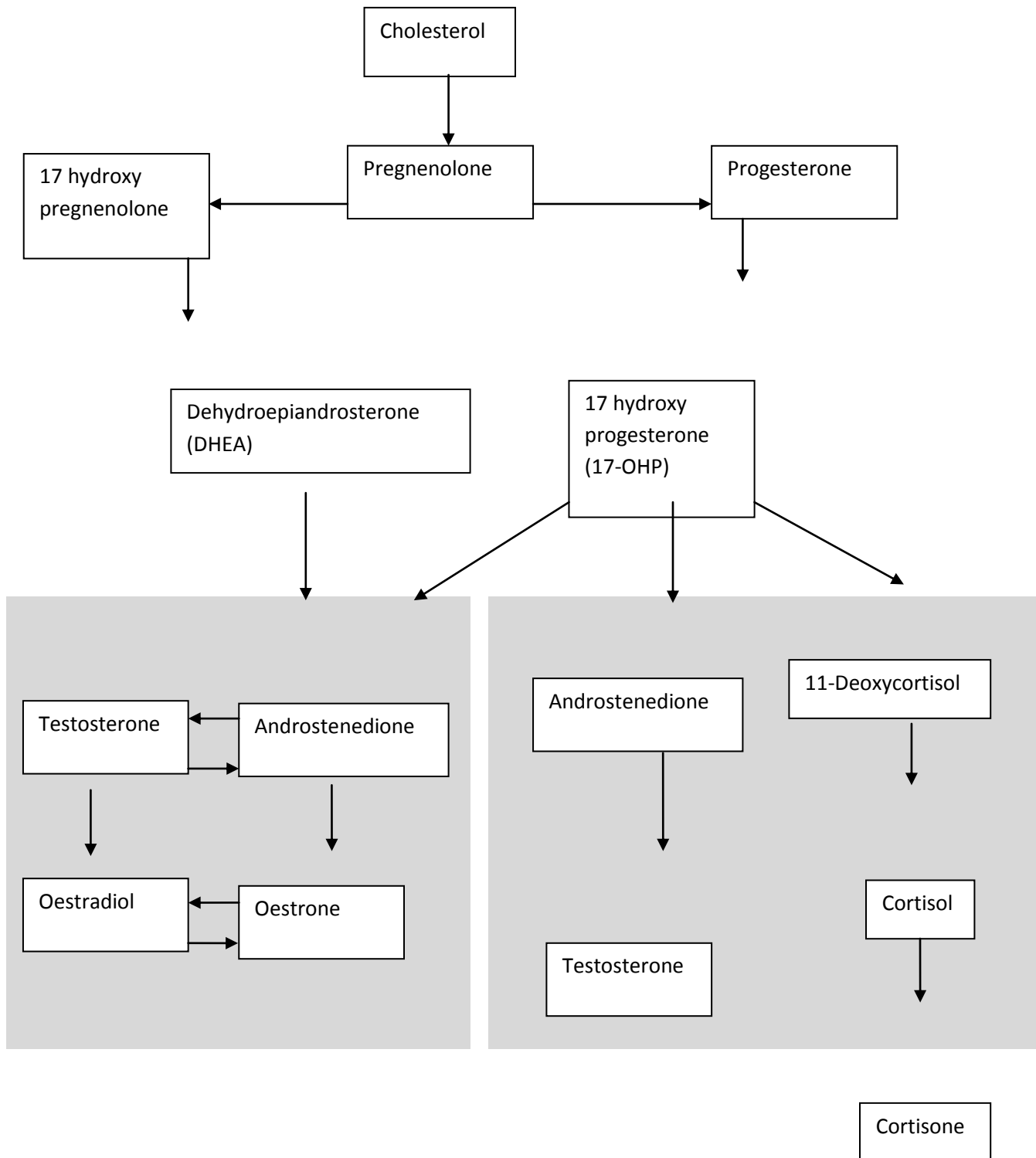


Table 9. Investigations for hirsutism

Investigations should be directed by the severity of hirsutism and associated symptoms	
Serum	Free testosterone Sex hormone binding globulin (SHBG) Ratio of LH:FSH Prolactin DHEA, DHEAS and androstenedione 17-hydroxy-progesterone (high in CAH) – when testosterone levels are very high
Ovary	Trans-vaginal pelvic ultrasound – to diagnose PCOS or ovarian tumours Pelvic MRI – when pelvic tumours are suspected
Adrenal	CT scan of abdomen – suspected CAH or adrenal tumour
Urine	24 hour collection for cortisol levels – when Cushing’s syndrome is suspected
Other	Dexamethasone suppression test – when CAH is suspected

Treatments for hirsutism will be directed by the cause. Tumours will require surgical excision. Referral to an endocrinologist for treatment of endocrinopathies is recommended. At times the hirsutism will be persistent and additional treatments may be necessary. Anti-androgens are a good first line therapy, particularly when combined with oestrogen in the combined oral contraceptive – e.g. Cyproterone acetate and Drospirenone. Cyproterone acetate (12.5-100 mg daily) or Spironolactone 100-300 mg daily can be very effective. More potent anti-androgens such as Flutamide and Finasteride must be used with caution and appropriate monitoring. Pregnancy is absolutely contraindicated and good contraception is essential. The topical cream Eflornithine (Vaniqa) is now available. All these treatments will take 3-6 months before a response is seen as the hair follicles respond to the change in androgen levels. Once treatment is ceased the hirsutism will reoccur.

More immediate treatments are laser photothermolysis, electrolysis and shaving. The main drawback with these options is the potential scarring to the skin that can occur. When PCOS is the cause of hirsutism insulin sensitisers such as Metformin may reduce hirsutism for some women.

KEY POINTS

- 1. Hirsutism is caused by excess androgen production or increased serum androgen bioavailability.**
- 2. The commonest cause of hirsutism is PCOS.**
- 3. Cosmetic and medical treatments are often used in combination.**
- 4. When fertility is desired only cosmetic treatments can be used.**

CHAPTER FOUR

MENORRHAGIA

Robert W. Shaw

Introduction

Excessive menstrual loss has an enormous impact on many women's lives and is one of the main reasons for which women seek medical advice. The condition causes significant burden to the woman, her family and society in general and its treatment accounts for a significant proportion of health care costs in Gynaecology. In the Caribbean, as in many other developing countries especially Africa, uterine fibroids are a leading cause of heavy menstrual losses and surgery is commonly resorted in order to control the bleeding. In many other patients, heavy menstrual bleeding is found in the absence of pathology, and there are many other medical and less invasive surgical approaches now available to treat these patients.

Definition

Menorrhagia is defined as heavy, cyclical menstrual bleeding taking place over several consecutive cycles. The menstrual cycle is judged as normal by the duration, the amount and the periodicity of the menstrual flow. Usually the menses lasts for 5-6 days with an occurrence of every 28 days. There are wide variations of this at different times of a woman's reproductive life. The average blood loss is about 50 ml (range 20-80 ml). About 75 per cent of menstrual blood loss occurs during the first two days of bleeding. Women who lose more than 80 ml per menstrual period for six months or more are likely to develop iron-deficiency anaemia, thus the upper limit of normal menstruation is taken as 80 ml per menses.

Measurement of menstrual blood loss

The perception of menstrual blood loss is highly subjective and it has been found that almost 50 per cent of women complaining of excessively heavy periods actually have losses within the normal limits. The number of pads soaked or tampons used is an unreliable indicator of the actual amount of bleeding. While blood loss can be measured accurately by extracting blood from used sanitary protection, using the alkaline haematin technique, this is tedious and not used in routine practice. An accurate measurement of blood loss can be obtained utilising a Pictorial Blood Assessment Chart (PBAC) which has a strong correlation with the measured menstrual loss. This is simple to use and patients are asked to quantify the amount of blood loss on each item of sanitary protection as minimal, mild or soaked and record this on a simple chart. A PBAC score of well in excess of 100 is equivalent to a blood loss exceeding 80 ml per cycle.

Physiology of menstruation

Menstrual blood loss does not usually clot and is characteristically of a dark-reddish brown colour similar to that of venous blood. However, when menstrual bleeding is profuse, the discharge is often bright red and there can be clot formation in the vagina. Clots are aggregates of red blood cells, mucoid substances, muco-proteins and glycogen. At the onset of menstruation with the withdrawal of oestrogen and progesterone support from the waning corpus luteum, the sub-epithelial capillaries in the upper two thirds of the endometrium functionalis/superficial layer, or the stratum compactum and stratum spongiosum are shed. Menstruation is preceded by the regression of the endometrium and an intense vasospasm of the spiral arterioles which supply the endometrial surface, which produce ischaemia and tissue damage and death of the overlying endometrium. Figure 1 demonstrates blood supply to the endometrium.

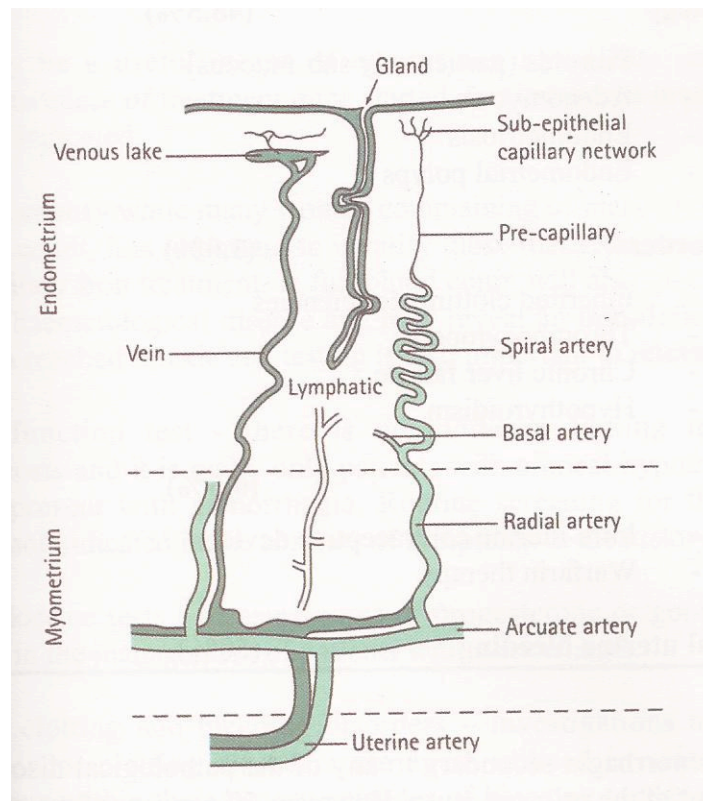


Figure 1. Blood supply

This vasoconstriction is induced by the local production of prostaglandins within the myometrium from precursors of arachidonic acid released from the degenerating endometrial surface cells. It is the maintenance of this vasoconstriction of the smooth muscle surrounding the basal and spiral arterioles which limits the vast majority of blood loss from the uterine wall. Within the first 36 hours of the onset of menstruation, capillary haemostasis from platelet aggregates and fibrin plugs are important but beyond the first two days it is vasoconstriction

which is of most relevance until the endometrium begins to regenerate from the basal layers of the endometrium (the stratum basale).

This regeneration of the endometrium is stimulated by various growth factors including vascular endothelial growth factor (VEGF), a primary initiator of blood vessel development. The speed of regrowth and regeneration of the endometrium is the key controller of the number of days of menstrual blood loss and where this is delayed prolonged menstrual bleeding may occur.

It can therefore, be seen that impaired vasoconstriction, prolonged vasodilatation of the spiral arterioles and delayed regeneration of the endometrium are all factors contributing to increased heavy menstrual bleeding.

Causes of Menorrhagia

Menorrhagia may be caused by a number of factors as indicated in Table 1. These include:

1. Specific pelvic pathology
2. Systemic disorders
3. Iatrogenic causes

Table 1. Causes of Menorrhagia

1. Pelvic pathology (48.5%)
Fibroids
Adenomyosis
Endometrial polyp
Endometrial hyperplasia/carcinoma
Endometriosis
2. Systemic disorders (1.0%)
Inherited clotting deficiencies
Thrombocytopenia
Chronic liver failure
Hypothyroidism
3. Abnormal uterine bleeding (50%)
4. Iatrogenic (0.5%)
Intra-uterine contraceptive devices
Warfarin therapy

Management of menorrhagia secondary to any of the pathological disorders should be directed at treatment of the primary cause. However, 50 per cent of women complaining of menorrhagia will have idiopathic menorrhagia or abnormal uterine bleeding. Most women with

idiopathic menorrhagia have regular ovulatory cycles and the excessive bleeding is attributed to a disorder of the local control of menstruation. Although heavy menses can be to anovulatory cycles, this tends to occur at the extremes of reproductive life such as puberty and the perimenopause and does not usually affect women of childbearing age unless they have an underlying disorder such as polycystic ovary syndrome.

History and Examination

In the history a patient complaining of menorrhagia, it is important to demonstrate the duration of the complaint, cyclicity of the menstrual cycle and to exclude other abnormal bleeding episodes (intermenstrual bleeding, post-coital bleeding). The key to making the diagnosis of menorrhagia is the verification that the bleeding is regular. Within the history, evidence of overt thyroid disease or clotting disorders can be excluded and it is important to discuss the woman's contraceptive needs or intention of having children in the future in order to determine appropriate treatment options.

An abdominal and pelvic examination is required to ensure there is no obvious pelvic pathology, particularly fibroids, before treatment is initiated. If a cervical smear is due, it would be good practice to take one at the same time. A speculum examination will also exclude local cervical lesions or a prolapsing submucosal fibroid or polyp.

Investigations

The PBAC (Figure 2) may be a useful means of quantifying blood loss and particularly of evaluating effectiveness of treatment once started, but important investigations include the following as indicated:

1. Full blood count: while many women complaining of menorrhagia have a normal full blood count, it is important to identify those that are already anaemic to give supplementary iron treatment. A full blood count will also give extra information about the haematological disease and may reveal an iron-deficiency state before anaemia is reached. Sick cell testing is also important in the Caribbean.
2. Thyroid function test: there is no evidence linking menorrhagia with thyrotoxicosis and it is really only patients with clinical hypothyroidism who are likely to present with heavy menses. Routine screening is therefore not indicated in the absence of symptoms in the history.
3. Other endocrine tests: measurements of progesterone or gonadotrophins are of little help in the management of menorrhagia.
4. Tests for clotting and bleeding disorders: investigations to exclude platelet dysfunction or clotting disorders are again utilised in selected patients who usually have within their history episodes of bleeding tendencies (post-deliveries, following surgical procedures, dental extraction etc.) or excessive bruising.

5. Endometrial biopsy: at one time a diagnostic curettage was undertaken on the vast majority of women with menorrhagia, this has been shown to be of little help in the management or detection of the likely pathological causes, polyps or sub-mucous fibroids. Endometrial sampling is indicated if there are other abnormal bleeding patterns in which case endometrial pathology must be excluded before medical treatment is commenced.
6. Pelvic ultrasound: this can be of great benefit in diagnosing endometrial polyp and fibroids as well as confirming the origin of pelvic masses if found clinically.
7. Hysteroscopy and directed biopsy: hysteroscopy is a useful investigative procedure in patients in whom a sub-mucous fibroid or polyp is thought to be present and indeed treatment may be applied at the same time. It is not necessary to undertake hysteroscopy in all patients with menorrhagia but perhaps in selective patients where medical treatment has already failed.

Pictorial Bleeding Assessment
Calendar

NAME:
DAY START:

SCORE:

	DAY							
	1	2	3	4	5	6	7	8
TOWEL								
								
								
								
CLOTS/ FLOODING								
TAMPON	1	2	3	4	5	6	7	8
								
								
								
CLOTS/ FLOODING								

Figure 2. Pictorial blood assessment chart (PBAC)

Initial Treatment

Drug treatment is the preferred initial option for women in their childbearing years as it is reversible and preserves fertility. It is important that only drugs known to be effective in reducing menstrual blood are used.

There are a number of treatment options as outlined in Table 2.

Table 2. Factors influencing choices of drug therapy for menorrhagia

First Line Treatment			
Drug	Reduction in MBL (%)	Side- effects	Additional Use
Luteal Progestogens	10-15	Nausea, vomiting, bloatedness, weight gain, skin rashes	Irregular cycle
Mefenamic acid	20-40	Nausea, vomiting, diarrhoea, gastric irritations, rashes	Dysmenorrhoea, menstrual migraine
Tranexamic acid	45-50	Nausea, vomiting, tinnitus, dizziness	-
Oral contraceptive pill	50	Nausea, weight gain, thrombosis	Dysmenorrhoea, irregular cycle
Second Line Treatment			
Danazol	60-80	Acne, hirsutism, voice changes, muscle cramps,	Dysmenorrhoea, pre-endometrial ablation
GnRH analogues	90	Climacteric changes, loss of libido	Pre-endometrial ablation

Mefenamic acid

This is a prostaglandin synthetase inhibitor and is a popular first choice of treatment especially when patients also have dysmenorrhoea. It is usually given at a dose of 500 mg three times a day for the duration of the menses and can be continued over long periods of time. One must keep in mind the risk of ulceration in the gastro-intestinal tract with this drug.

Tranexamic acid

This drug competitively inhibits the activation of plasminogen to plasmin and counteracts the high fibrinolytic activity in the endometrium that may be one of the causes of menorrhagia. Doses usually used are 1 g three times daily at the onset of the menses for the heaviest flow

days. Tranexamic acid is contra-indicated in women with a history of thrombo-embolic disorders. However, it does not seem to cause thrombosis per se from large studies of its use.

Combined oral contraceptive pill (COC)

The pill can be useful in reducing menstrual bleeding although their mode of action is unclear. It is probably via the induction of endometrial atrophy. Use of COCs has the advantage that they also provide effective contraception and help in reducing period pain by producing anovulatory cycles. Although the pill is generally well tolerated with minimal side-effects, patients must be screened for contra-indications to pill usage. Increasing age, increased weight and smoking are important factors to evaluate in planning long-term use of these agents in patients with menorrhagia.

Oral Progestogens

Although oral progestogens are commonly prescribed for patients with menorrhagia, their efficacy is questionable, especially when used in the short duration. There is a limited role for oral progestogens in patients with anovulatory cycles but other their use for regulating menses, delaying or bringing forward the start of menses, they are not recommended as a first line treatment in heavy menstrual bleeding.

Long-acting Progestogens

Medroxyprogesterone acetate is available as a depot injection for contraception. Dosage of 150 mg administered every three months initially cause irregular spotting and bleeding but then in many patients can induce amenorrhoea. For this reason long-acting progestogens have been tried as treatment for menorrhagia in patients who also require contraception but more detailed studies are required.

Progestogen Releasing Intra-uterine Device (Mirena)

It has been observed that the Mirena intra-uterine system, which releases 20 micrograms of Levonorgestrel daily for up to 5 years, reduces menstrual blood loss by up to 86 per cent by three months and up to 97 per cent by 12 months. One disadvantage is the high incidence of breakthrough bleeding in the first 6 months. Its' possible modes of action are reduction in endometrial prostaglandin synthesis, the induction of endometrial atrophy and decreased angiogenesis. These devices are now widely used as the first line treatment of menorrhagia in the absence of intra-uterine pathology.

Ethamsylate

Ethamsylate is a non-hormonal agent that is thought to act by increasing capillary wall resistance and platelet adhesiveness in the presence of a vascular lesion. It is used for the

duration of a heavy period and has minimal side-effects. Evidence from various trials has been inconclusive but overall it is not particularly effective in reducing menses.

What if the initial treatment fails?

Several studies that have looked at the effectiveness of drug therapy have found that if appropriate drug therapy fails to reduce menstrual blood loss to an acceptable level, intra-uterine pathology such as the presence of sub-mucous fibroids are more likely to be present. If pathology has been excluded, second line drug therapy may be used as interim treatments but their usage is usually limited because of side-effects, costs and availability of the drugs.

Danazol

Danazol is an isoxazol derivative of 17-alpha-ethinyl testosterone. It inhibits ovulation, causes low circulating oestrogen levels and endometrial atrophy. It has mild androgenic properties. A dose of 200 mg daily reduces the menstrual blood loss by about 60 per cent but its incidence of side-effects, weight gain, oily skin and muscle cramps often inhibits its long-term use and it is imperative that barrier contraception is used during Danazol administration in non-sterilised women.

GnRH analogues

This group of drug act on GnRH receptors of gonadotrophs in the anterior pituitary leading to down-regulation, desensitisation and suppression of LH and FSH release. This leads to suppression of ovarian function and in the vast majority of patients, low oestrogen and amenorrhoea result. They are highly effective in reducing menstrual blood loss. They are also useful as pre-treatment methods prior to endometrial ablation since the endometrium is thinned given 4-8 weeks pre-surgery. They are also used before myomectomy in order to reduce intra-operative blood loss. Our study revealed that this benefit is only with small fibroids (<4cm in diameter). They create a pseudo-menopause and hence the common side-effects are those of the climacteric. Of importance, they are associated with bone loss which can lead to osteoporosis and pathological fractures later in life. To offset these problems, add-back therapy with Tibolone is recommended.

Endometrial Ablation Techniques

Transcervical resection of the endometrium and electrocoagulation destruction (rollerball) have been utilised for many years in the treatment of menorrhagia. The techniques involve complex hysteroscopic skills and there can be complications related to the distending medium when excessive absorption occurs with the potential to produce hyponatraemia. These initial methods have been replaced by the new generation of treatments which include microwave

endometrial ablation, radio-frequency wave and thermal balloon destruction. These second generation techniques are being more widely used since they reduce the risks of perforation of the uterus, non-use of distension fluids and potentially can be done under local anaesthesia. It would be expected that some 40 per cent of patients would become amenorrhoeic following such techniques, a further 50 per cent experience a reduced menstrual blood loss while there may be a failure in 10 per cent or so of patients with no response.

Results of outcome of treatment are improved by pre-thinning the endometrium with agents such as Danazol or GnRH analogues or performing the procedures immediately at the end of the menses.

In about 20 per cent of cases of menorrhagia with no obvious cause, medical treatment fails and these patients may need a hysterectomy. Other indications for hysterectomy include symptomatic fibroids, adenomyosis and endometriosis. In young women and in those who do not wish to have a hysterectomy, a myomectomy is an appropriate surgical method for dealing with fibroids. A 3-month course of a GnRH analogue will lead to shrinkage of the fibroid and decreased blood supply, and this will facilitate an endoscopic myomectomy and less intra-operative blood loss. We have shown that the above is true only with few fibroids and small ones.

If the menorrhagia is due to a genital tract malignancy, then accurate staging and definitive treatment are required. For example, endometrial carcinoma Stage 1 is treated with a pan-hysterectomy.

KEY POINTS

1. Menorrhagia presents a major challenge to Gynaecologists.
2. **Patients' history of menorrhagia is often unreliable**
3. There is a wide variety of medical treatments available.
4. The new generation of endometrial ablation techniques and the Mirena intra-uterine system mean that many patients can now be offered effective treatment which avoids the need for hysterectomy.
5. If the patient has an obvious organic pathology, then the appropriate surgical intervention is required.
6. For example, in a 45-year-old woman whose parity is 5 with fibroids causing menorrhagia, then it is justifiable to offer her a hysterectomy.

CHAPTER FIVE

PELVIC PAIN and DYSMENORRHOEA

Bharat Bassaw

Pain may be produced by a variety of physiological and pathological processes. Pelvic pain can result from conditions involving organs both within, and outside the pelvis.

In Britain, pelvic pain is the most common complaint among women seeking gynaecological consultation. A survey in the United Kingdom also found that the primary indication for diagnostic laparoscopy was chronic pelvic pain. Another study revealed that the causes of chronic pelvic pain included pelvic inflammatory disease (20%), endometriosis (12%) and others (3%), and of note in 65% of cases, no abnormalities were detected. A similar finding was observed in terms of acute pelvic pain where a normal pelvis was seen with laparoscopy in 44% of cases.

Chronic pelvic pain refers to pain which is present, either intermittently or continuously, for at least six months and which is so severe that it affects the woman's quality of life and analgesia is required.

Acute pain reflects recent tissue damage and resolves as the tissues heal. In chronic pain, additional factors are involved and pain may persist long after the original insult.

Pelvic and lower abdominal nerve supply

The autonomic nerves supply the abdomino-pelvic organs. Damage or irritation to any of these nerves can stimulate pelvic pain. Pain impulses from the uterus, medial aspect of the Fallopian tubes, broad ligament and the upper vagina are conducted along afferent fibres to the paracervical plexus and then to the hypogastric plexus to enter the thoracolumbar sympathetic chain. These afferents then enter the spinal cord at T10-L1. The nerves fibres from the ovaries, lateral aspect of the Fallopian tubes and peritoneum travel to the main sympathetic chain and then to the spinal cord at T9-10. The parasympathetic fibres travel along the pudendal nerves and enter the spinal cord at S2-4. Pain originating from any of these pelvic organs is also experienced in areas of the skin supplied by the somatic afferent fibres. This is called referred pain.

History

A detailed history is the single most important aspect in making a diagnosis in medicine, and this is certainly the case with pelvic pain. While eliciting the history, observe the patient's general condition and demeanour. Some of the details of the pain should include the time of onset, site, nature, radiation, aggravating and relieving factors and any relationship with sexual intercourse or with the menses.

Pain of sudden onset tends to suggest an acute event such as torsion of an ovarian cyst or rupture of an ectopic pregnancy whereas pain caused by infection or inflammation e.g. appendicitis or pelvic inflammatory disease, tends to be slower to arrive, dull in nature and more poorly localized (Table 1). Accident to an ovarian cyst especially torsion may lead to nausea and vomiting.

Colicky pain suggests muscle contraction or spasm such as incomplete miscarriage, ureteric calculus or intestinal obstruction. Constant pain occurs with an ischaemic event or an inflammatory problem. A large fibroid can cause a dragging sensation in the lower abdomen and back pain. It is common for fibroids to lead to abnormal uterine bleeding especially heavy menses with the passage of clots and associated flooding especially with submucosal ones. Depending on the quantity of blood loss there may also be symptoms of anaemia. Pressure symptoms on the adjacent viscera e.g. urinary bladder may lead to urinary frequency, urge, urge incontinence, nocturia (with anterior fibroids), and urinary retention, hesitancy, poor stream, and strangury and associated features of a cystitis (low posterior fibroids). Constipation may result from pressure with posterior fibroids on the rectum. Spasmodic dysmenorrhoea may occur if the fibroids are intramural and if there are clots with the menses.

Radiation of the pain to the medial and anterior aspects of the ipsilateral thigh suggests an ovarian aetiology with irritation of the obturator nerve (L2-4). This nerve runs in close relationship with the ovary and thus an ovarian cyst may lead to both lower abdominal pain and also pain in the thigh. Irritation of the phrenic nerve (C2-5) by blood (e.g. ruptured ectopic pregnancy) or contents of a perforated viscus can lead to shoulder tip pain (Kehr's sign).

Pelvic inflammatory disease apart from causing pelvic pain usually bilaterally, also presents with a history of an abnormal vaginal discharge, dysuria, menstrual problems such as intermenstrual bleeding or heavy periods, post-coital bleeding and deep dyspareunia, and occasionally right hypochondriac pain referred to as Fitz-Hugh-Curtis syndrome seen with Chlamydial infection in the Fallopian tubes.

Pain relieved by lying still and exacerbated by movement including deep breathing points to peritonism or peritonitis. This is seen with an ectopic pregnancy when ruptured. Deep dyspareunia may indicate peritoneal inflammation. This is a common feature of both pelvic inflammatory disease and endometriosis (Figure 1).

Pelvic pain which varies markedly over the menstrual cycle is likely to be attributable to a hormonally derived condition such as endometriosis.

Appendicitis which is common in young people is usually associated with anorexia, nausea, vomiting and change in bowel movements. The pain usually begins periumbilically and moves to right lower quadrant especially over Mc Burney's point. Note this may not be the case in pregnancy since the appendix moves upwards and thus the pain may be referred to the lumbar region.

Ureteric calculus is commonly associated severe colicky pain that radiates from loin to groin, haematuria, vomiting and a possible past or family history of stones. Rectal carcinoma seen in more elderly people frequently causes blood and mucus in the stools.

Abdominal pain is the commonest symptom of an ectopic pregnancy. The classical triad also includes a period of amenorrhoea and vaginal bleeding. As opposed to an incomplete miscarriage, the pain starts before the onset of the bleeding in an ectopic pregnancy. Furthermore, the bleeding is much less with an ectopic pregnancy.

Ovulation pain (Mittelschmerz) occurs at the midcycle with sudden onset, and the sharp lower pain is followed by a dull ache in the pelvis for several hours. A young girl with an imperforate hymen or a transverse vaginal septum may present with an abdominal mass, pain, and urinary retention as well as primary amenorrhoea. Adenomyosis may cause chronic pelvic pain as well as menorrhagia and dysmenorrhoea.

Ovarian hyperstimulation syndrome (OHSS) characterized by enlargement of the ovaries, ascites and sometimes, pleural and pericardial effusion, acid-base and electrolyte imbalance is a complication of assisted reproduction. These ovarian cysts are fragile and thus easy to rupture. Severe abdominal pain would result. The usual clinical picture is one of acute pelvic pain occurring 5 to 10 days after the HCG injection. There may be a history of nausea, vomiting and diarrhoea. Pelvic examination should not be done as this can cause rupture of the enlarged ovaries.

In a postmenopausal woman, acute pelvic pain can result from torsion of an ovarian cyst, pyometra which is commonly associated with endometrial cancer, or urinary retention from utero-vaginal prolapse. Diverticulitis which occurs in the elderly usually leads to more diffuse abdominal pain. Irritable bowel is a non-gynaecological condition seen with colicky abdominal pain and frequent loose stools.

Women with have pain due to pelvic congestion are always in the reproductive age with a mean age of 32 years. Typically, the pain is dull and aching in nature similar to dysmenorrhoea, and is interspersed with backache, urinary symptoms and headache. Congestive dysmenorrhoea is common. There may also be a history of deep dyspareunia and post-coital ache.

Pelvic pain following a hysterectomy with ovarian conservation can result from residual ovarian syndrome. It is characterized by pelvic pain, deep dyspareunia and a fixed tender ovary felt at the angle of the vaginal vault. Dense adhesions from previous surgery can lead to chronic pelvic pain. The patient is also at risk for intestinal obstruction.

It is always necessary to enquire about the last normal period and to obtain a sexual history in women of the reproductive age. It is recognized that many patients who present with pelvic pain do have a psycho-sexual problem which may have been due to sexual assault in their early years.

Table 1. Important causes of acute pelvic pain

1. Gynaecological causes
 - A. Uterine
 - Miscarriage
 - Fibroids
 - Adenomyosis
 - Endometrial polyp
 - Endometritis
 - Cancer
 - B. Ovarian
 - Cyst (accident)
 - Endometrioma
 - Sepsis
 - Cancer
 - Hyperstimulation
 - C. Tubal
 - Salpingitis
 - Ectopic pregnancy
2. Non-Gynaecological
 - A. Gastrointestinal
 - Appendicitis
 - Hepatitis
 - Gall bladder disease
 - Obstruction
 - Irritable bowel syndrome
 - Inflammatory bowel disease

Malignancy

B. Urinary tract

Cystitis

Interstitial cystitis

Calculi

Pyelonephritis

C. Retroperitoneal structures

Musculoskeletal pain

Psoas tumour

Tumour

Abscess

Adhesions

D. Psychological and social (sexual)

Examination

The first observation to make is whether the patient looks ill or not. If she is in shock, she will be pale and her extremities cold and clammy. Other important signs include tachycardia and hypotension. Resuscitation and call for help are required immediately. With peritonism, the patient lies still and there is shallow breathing. The patient who has colicky pain from spasm with a hollow organ e.g. ureteric colic will be restless. Pyrexia points to an infectious cause.

Figure 1. Endometriosis

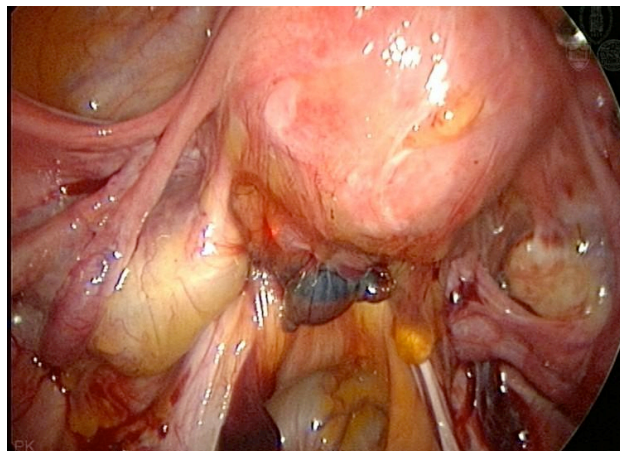


Table 2. Causes of chronic pelvic pain

Endometriosis
PID
Chronic pelvic congestion
Adhesions
Ovarian cyst
Residual ovarian syndrome
Fibroids
Utero-vaginal prolapse
Cervical stenosis
Malignancies of the genital tract
Hernia
Musculo-skeletal problems
Osteoarthritis
Inflammatory bowel disease
Diverticulitis
Urinary calculus
Psychogenic
Psycho-sexual problems
Idiopathic

Abdominal distension may be due to a mass, ascites or distended bowel. The latter may be associated with an ileus caused by intra-peritoneal blood or pus, or due to bowel obstruction. Note the presence of any surgical scars and hence, the possibility of adhesions which may be the cause of the pain. Classically, maximum tenderness occurs over Mc Burney's point with appendicitis whereas pelvic inflammatory disease causes lower abdominal tenderness.

Suprapubic tenderness is a sign of cystitis. Rebound tenderness and tachycardia without pyrexia should cue one to the possibility of a ruptured ectopic pregnancy. If a mass is present, always check for ascites and organomegaly. A mass should be defined in terms of size, site, surface, consistency, mobility, symmetry, tenderness, and whether the margins are well-defined or not. Also check for dullness with percussion, transillumination, and if you can get above and not below the mass which indicates that the mass is arising from the pelvis.

A large ovarian mass becomes central but an endometrioma is usually confined to the iliac fossa. Uterine fibroids are usually associated with an asymmetrically enlarged uterus whereas adenomyosis cause the uterus to be symmetrically enlarged. With an enlarged uterus, there may be compression of the ureters where they enter the pelvis over the sacro-iliac joints with

the development of proximal hydronephrosis and hydronephrosis. Rarely, pelvic inflammatory disease when chronic can lead to a mass in the lower abdomen.

A bimanual pelvic examination will determine if the mass is arising from the uterus or not. The uterine mass will move when the cervix is moved or vice versa. This is not to be expected with an ovarian mass except if the mass is adherent to the uterus. Cervical excitation pain indicates the presence of peritonitis or peritonism. This may occur with an ectopic pregnancy, pelvic inflammatory disease or endometriosis.

A speculum examination may reveal an abnormal discharge, the string of an intra-uterine contraceptive device or a fungating cauliflower mass typically found with cervical carcinoma. Products of conception are seen with an incomplete miscarriage. Infrequently, endometrial deposits may be visualized in the posterior fornix or on the cervix. The uterus will be enlarged with fibroids or adenomyosis. Fixed retroversion of the uterus is a characteristic feature of endometriosis. Adnexal tenderness and/or masses are detected with ectopic pregnancy, endometriosis, ovarian cyst or pelvic inflammatory disease.

Adequate time should be allowed for the initial assessment of women with chronic pelvic pain. Many women want an explanation for the pain even if no definite treatment is possible.

Specific investigations

Suitable samples to screen for infection especially *Chlamydia trachomatis* and gonorrhea should be taken if there is any suspicion of PID. All sexually active women with pelvic pain should be offered screening for sexually transmitted infections (STIs).

Ultrasound

Ultrasound is an ideal non-invasive method for scrutiny of the pelvis particularly for the identification and assessment of adnexal masses. It can detect pelvic masses such as ovarian cysts and fibroids as well as it can confirm an intra-uterine pregnancy. Together with Beta HCG measurements, ultrasound especially transvaginally is a useful tool in the diagnosis of an unruptured ectopic pregnancy. This investigation is also the best method to localize a lost intra-uterine contraceptive device. Magnetic resonance imaging (MRI) is superior for the diagnosis of adenomyosis. While MRI lacks sensitivity in the detection of endometrial deposits, it may be useful in the assessment of palpable nodules in the pelvis and for rectovaginal disease.

Laparoscopy

This is the most popular investigation to study chronic pelvic pain. Laparoscopy remains the cornerstone in diagnosing endometriosis as well as in its staging. Conservative surgical treatment may be performed simultaneously. If the diagnosis of pelvic inflammatory disease is in doubt then laparoscopy is indicated. Other conditions which can be diagnosed with

endoscopic procedure include ectopic pregnancy, ovarian cyst, polycystic ovaries and adhesions.

In the past diagnostic laparoscopy was regarded as the 'gold standard' in the diagnosis of chronic pelvic pain. It may be better as a second-line investigation if other therapeutic interventions fail.

Estimated risk of death with laparoscopy is approximately 1 in 10,000, and the risk of injury to bowel, bladder or blood vessel is 2.4 per 1,000, of whom two-thirds will require laparotomy.

Other investigations

Depending on the likely cause, abdominal X-rays including IVP, urine for microscopy and culture, swabs, pelvic venography (with pelvic congestion), MRI or CT scans may be necessary. Other investigations may include cystoscopy, barium meal or enema, sigmoidoscopy, or colonoscopy. Many patients with chronic pelvic pain have psychological or sexual problems, and hence psychological evaluation may be required.

In women over 50 who have vague gastrointestinal complaints such as bloating, dyspepsia or abdominal discomfort should have CA 125 as part of work-up for ovarian cancer.

Treatment

The treatment of pelvic pain depends on the cause. The management of non-gynaecological conditions would not be discussed here. Pelvic inflammatory disease is treated with broad-spectrum antibiotics, and surgery is confined to severe cases e.g. tubo-ovarian abscess. The male partner must also be treated. Endometriosis is treated medically, surgically, or both. If the main complaint is pain, then the treatment is usually medical with analgesia, hormonal manipulation, GnRH analogues, Mirena, or Danazol (or aromatase inhibitors) while surgery either via laparoscopy or open laparotomy is indicated with infertility. Symptomatic fibroids are usually treated surgically either with myomectomy in younger women and hysterectomy in the older ones. Newer forms of treatment include medical methods with modulation of receptors by GnRH analogues and progestogen, uterine artery embolization and MRI focused ultrasound.

An accident to an ovarian cyst requires surgical intervention. A ruptured ectopic pregnancy needs resuscitation and immediate surgery. Pelvic congestion syndrome is treated medically with Medroxy progesterone acetate 30-50 mg/day for six months. Further details concerning the management of pelvic pain from gynaecological causes are found elsewhere in this text.

Other treatment options include amitriptyline or gabapentin for neuropathic pain, and dietary modification and mebeverine hydrochloride for irritable bowel syndrome. Laparoscopic uterosacral nerve ablation (LUNA) is ineffective in the management of chronic pelvic pain.

DYSMENORRHOEA

The term dysmenorrhoea, of Greek derivation, means “difficult menstrual flow”, and is used to refer to painful menstruation. It is a symptom-complex and not a disease and is one of the most frequent gynaecological complaints among young girls.

Dysmenorrhoea has been aptly called “the neglected syndrome”. For a long time, it was regarded as a time of illness and its cause was often ascribed to a woman’s rejection of her “feminine role”. Today, however, there is a changing attitude towards menstruation. Attention is now focused on the limitation of physical activity, the significant impact on work and life-style and the resulting socio-economic implications of dysmenorrhoea. Efforts are being made to understand the mechanism of pain production and to provide relief for the vast number of young women who suffer from this disorder.

Classification

Dysmenorrhoea is best divided into two categories: primary and secondary.

In **primary dysmenorrhoea**, there is no detectable pelvic pathology and there is no physical explanation for the symptoms.

Secondary dysmenorrhoea constitutes painful menstrual cramps due to an underlying pathological condition in the pelvis, such as uterine fibroids and endometriosis.

Spasmodic dysmenorrhoea is associated with intermittent, cramping abdominal pain especially on the second and third days of the menses. It is due to increased myometrial contractility.

Congestive dysmenorrhoea is typically a dull lower ache in the lower abdomen, thighs and back. It is thought to be due to pelvic venous congestion.

PRIMARY DYSMENORRHOEA

This is most frequent in virgins, less frequent in married women, and is practically never a problem in parous women, who have delivered vaginally. It arises a few months after the menarche, at a time when the menstrual cycles become ovulatory; the cycles within the first few months of the menarche are anovulatory and they tend to be painless.

The dominant feature of dysmenorrhoea is the “menstrual cramp”. Classically, the pain is colicky in type, tending to occur in waves which are superimposed on a constant ache in the lower abdominal region. The pain is situated in the suprapubic region and although it starts soon after the onset of the menstrual flow, it is most severe on the second or third day, after which it gradually wanes.

Some patients also complain of low backache, leg cramps, a feeling of nausea, vomiting, diarrhoea and fatigue. A few experience headaches, tingling of the breasts and emotional reactions such as depression, tension, irritability, loss of confidence and loss of concentration.

Significance of primary dysmenorrhoea

Data from several studies have shown that primary dysmenorrhoea occurs in 40 — 60 per cent of postpubertal females, and 10 — 15 per cent are temporarily incapacitated for 2 — 3 days each month. It accounts for economic losses because of absenteeism from work and reduced working capability. Furthermore, a significant number of schoolgirls miss their classes because they are confined to bed at home. Occasionally, the disorder is detrimental to mental health and it may cause marital disharmony.

Factors associated with the prevalence of dysmenorrhoea

Current evidence indicates that dysmenorrhoea bears some relationship to the factors shown in Table I.

Table 1. Associated factors in dysmenorrhoea

-
- (1) Age
 - (2) Parity
 - (3) Constitutional and emotional make-up
 - (4) Socio-economic status and family history
 - (5) The oral contraceptive pill
 - (6) Cigarette smoking
 - (7) Alcohol intake
 - (8) Exposure to cold
-

Age and parity

The condition is encountered less often in older patients. At age 20, the incidence is about 50 per cent but by age 30, it falls to 10 — 20 per cent. Nulliparous women are more likely to have the disorder; the effect of childbirth is salutary.

Constitutional and emotional factors

Dysmenorrhoea occurs more frequently among teenagers, college girls, nurses and secretaries, probably because they lead a sedentary life-style. It is uncommon among those whose occupations involve strenuous physical work and those who engage in sporting activities. Haman (1944) demonstrated the concept of a diminished pain threshold in dysmenorrheic girls and there is undoubtedly a strong emotional pattern interwoven with the complaint: the introspective, apprehensive girl is more likely to experience painful menstruation than her extroverted sister. Since feminine role conflicts have been found to be more prevalent among women with dysmenorrhoea, psychological factors should be considered in those who have not responded to medical therapy.

Socio-economic status and family history

Girls belonging to a higher socio-economic bracket are more likely to have menstrual pain, particularly if there was a maternal history of this disorder.

The contraceptive pill

Users of the combined pill seldom complain of painful menses.

Smoking and alcohol

There are conflicting reports concerning the influence of these factors on the occurrence of dysmenorrhoea but the evidence in favour of strong positive correlations is convincing.

Exposure to cold

The prevalence of the condition has been recorded to be significantly higher among women whose work environment is very cold, compared to those who work in warm, comfortable surroundings.

Pathophysiology of dysmenorrhoea

The pathophysiology of primary dysmenorrhoea was poorly understood until recently. Many theories have been advanced (Table 2) but with greater understanding of the pharmacology of prostaglandins, the proposal that myometrial hyperactivity is the basis for painful menses, appears to be the most plausible.

Table 2. Aetiology of dysmenorrhoea

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- (1) Endocrine theory
 - (2) Cervical stenosis
 - (3) Membranous dysmenorrhoea
 - (4) Allergic factors
 - (5) Uterine ischaemia
-

Endocrine theory

In general, anovulatory cycles are painless, and the converse is true. In metropathia haemorrhagica, for example, where the endometrium has been stimulated by oestrogens alone, unopposed by progesterone bleeding is painless.

As stated above, dysmenorrhoea is usually absent for a time after the menarche because the cycles are anovulatory. Furthermore, patients taking the combined contraceptive pill, which inhibits ovulation, may have painless menstruation.

An oestrogen-progesterone imbalance therefore has been implicated to explain this phenomenon. Formerly, it was believed that endogenous progesterone secreted from the corpus luteum following ovulation, raises the tone of the muscles of the isthmus and the internal os, so that the cervical “sphincter” is tighter in the luteal phase. However, recent studies have demonstrated that these women have an elevated level of circulating vasopressin which gives rise to exaggerated uterine contractions.

Cervical stenosis

Congenital anatomical defects of the cervix may cause primary dysmenorrhoea. Acquired cervical stenosis may occur as a result of fibrosis from cone-biopsy or laceration of the cervix and is a cause of secondary dysmenorrhoea.

A tight cervical canal retards the outflow of menstrual blood. Consequently, the uterus contracts more forcibly to overcome the obstruction. The observation that passage of large menstrual blood clots is usually followed by cessation of pain supports this theory.

This concept forms the basis for dilatation of the cervix for the relief of painful menses and is supported by the finding that dysmenorrhoea disappears after a vaginal delivery, although the net increase in uterine vascularity which persists after pregnancy, plays a part.

It should be mentioned, however, that the absence of cervical stenosis in most girls with severe dysmenorrhoea has dissuaded most Gynaecologists from accepting this hypothesis. By means of cervicograms, no change in the tightness of the cervical canal has been demonstrated, during the menses.

Membranous theory

The endometrial shedding that normally occurs at menstruation is patchy, fragmentary and irregularly distributed, the process of regeneration proceeding side by side with tissue necrosis and menstrual bleeding.

Occasionally, the endometrium does not disintegrate piecemeal, and instead is shed as a complete endometrial cast.

Allergic theory

It is believed that the smooth muscle of the myometrium may go into spasm in the same way as bronchial muscle, in response to allergens. This theory is closely tied up with the concept of uterine ischaemia.

Uterine ischaemia

It is a well-known fact that prostaglandins can stimulate myometrial hyperactivity, as a result of which, uterine ischaemia occurs. This situation is analogous to intermittent claudication. Hence, the term “uterine angina” is used.

Prostaglandins

The evidence for the role of prostaglandins in the pathophysiology of primary dysmenorrhoea is as follows:

- (1) Prostaglandins $F_2\alpha$ and E_2 are present in menstrual fluid. They originate from the disintegrating endometrium, and they stimulate the myometrium either by local diffusion or by absorption into the circulation.
- (2) Menstrual fluid of dysmenorrheic women has a higher $F_2\alpha/E_2$ ratio than that of non-dysmenorrheic women.
- (3) The maximum amount of prostaglandin $F_2\alpha$ is released during the first 48 hours of menstrual flow, which is consistent with the duration of the clinical picture.
- (4) The use of certain prostaglandin synthetase inhibitors such as indole-acetic acid derivatives and the arylpropionic acid derivatives, has been found to be effective in treating dysmenorrheic patients.

Prostaglandins cause

- (a) elevation of uterine tone
- (b) increase in intra-uterine pressure
- (c) increase in the frequency of uterine contractions
- (d) inco-ordinate uterine activity

These findings implicate prostaglandin $F_2\alpha$ as a major contributor to painful menstruation, and suggest that the pain is due to uterine ischaemia.

The sequence of events is depicted in Figure 1.

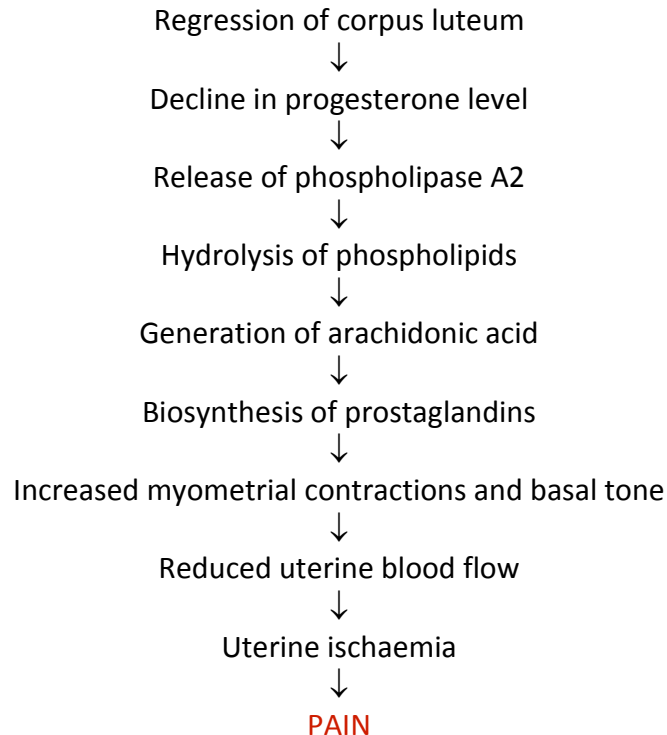


Figure 1. Mechanism of prostaglandin action in primary dysmenorrhoea

Patient evaluation

A sympathetic and reassuring attitude is necessary in assessing an adolescent with dysmenorrhoea. Apart from a routine gynaecological history, it is important to find out

- (a) whether she is missing school or not, and if so, for how long
- (b) if there is associated nausea, vomiting and diarrhoea
- (c) what is the nature of mother-daughter interaction
- (d) what medications have been taken before

Abdominal examination may be reassuring and vaginal examination should be avoided. Medical therapy is often necessary, the most effective being the oral contraceptives and the prostaglandin synthetase inhibitors. If this fails, operative treatment may have to be resorted to.

Medical therapy

Analgesics and antispasmodics

Popular drugs are Codeine, Baralgin and Buscopan compositum. Most patients would have tried one or more of these preparations, 2 tablets four times daily, before consulting the

Gynaecologist. Isoxsuprine (Vasodilan), a smooth muscle relaxant, 10 mg given four times daily, provides good pain relief in some.

Anti-prostaglandins

Salicylates, such as Aspirins, were widely used for treating dysmenorrhoea for a long time but it was not until 1974 that anti-prostaglandins were deliberately employed.

Treatment of primary dysmenorrhoea with prostaglandin synthetase inhibitors is currently in vogue, and is undoubtedly engendered by the new information about the relationship between endometrial prostaglandins and painful conditions of uterine origin, and by the introduction of a wide variety of non-steroidal anti-inflammatory drugs.

Workers have claimed varying degrees of success with these agents. Their effectiveness ranges from 70 to 100 per cent.

Indomethacin may cause gastro-intestinal side-effects.

Ibuprofen and Naproxen sodium have been reported to be highly effective. This latter drug, also called Anaprox, can be administered as a single dose of 1,000 mg on the first day of menstruation; it reduces intra-uterine pressure and decreases the intensity of the pain.

Flurbiprofen (Ansaid) is a new potent analgesic/anti-inflammatory agent. It is administered in a dose of 50 mg, four times daily, at the onset of pain and continued its cessation. It has been ranked as one of the most effective agents for primary dysmenorrhoea.

The anti-prostaglandin selected should be started at the first sign of the onset of the period and continued for the duration of the cramps or the duration of the period.

These compounds have an inhibitory effect on microsomal enzymes involved in the synthesis of prostaglandins from various precursors, thus restoring normal uterine activity and providing pain relief. Contra-indications to the use of anti-prostaglandin agents include the presence of gastro-intestinal ulcers, and hypersensitivity to Aspirin or similar agents. These inhibitors should be taken with milk.

If the patient fails to respond to anti-prostaglandins and analgesics, she should be tried on a course of oral contraceptive pills.

Hormonal treatment

The majority of patients taking the combined oral contraceptive pill have painless periods. The pill should be tried as a temporary measure for 3 or 4 cycles in order to tide the patient over an important occasion. In anovulatory cycles, prostaglandin levels are several times lower than in

ovulatory cycles, and the use of oral contraceptives, unless there is anxiety regarding their side-effects, is a reasonable alternative. In other words, the suppression of ovulation results in the growth of an endometrium with a much lower content of prostaglandin. Uterine tone and activity are decreased and dysmenorrhoea is alleviated.

Calcium antagonist

Nifedipine, in a dose of 30-80 mg daily, decreases uterine contractility by reducing Ca^{2+} transport across the cell membrane. The stimulatory effects of prostaglandins are abolished and pain is relieved.

Surgical therapy

Cervical dilatation is justified in some cases. Under general anaesthesia, the cervix is gradually dilated to No. 12 Hegar, to a point just short of tearing the fibro-muscular tissue of the internal os. Its therapeutic effect is probably due to damage of the cervical sympathetic nerve fibres. Only 40 — 50 per cent of patients are cured and in some, the relief is only temporary, presumably because of regeneration of the postganglionic nerve fibres. The danger of cervical dilatation is deep laceration which may result in cervical stenosis, or cervical incompetence and recurrent miscarriage.

When there has been an unsatisfactory response to prostaglandin inhibitors, oral contraceptives or cervical dilatation, a search for organic pathology is warranted. Laparoscopy is now indicated to diagnose organic lesions, of which endometriosis is the most important in the young age-group, but myomas and occlusive congenital Mullerian abnormalities must be borne in mind.

SECONDARY DYSMENORRHOEA

This type of dysmenorrhoea occurs in older persons after many years of painless menses. The pain begins 4 — 5 days prior to menstruation and is usually relieved on day 4—5 of bleeding. It is due to pelvic congestion as well as to the passage of large clots in the menstrual flow, and is, more often than not, associated with menorrhagia. As the uterus attempts to expel the clots, there is myometrial contractility which results in spasmodic dysmenorrhoea.

The most important causes of this condition are uterine fibroids, chronic pelvic inflammatory disease, pelvic endometriosis, the intra-uterine device (IUCD), fixed retroversion of the uterus, and acquired cervical stenosis.

Table 3. Investigations for secondary dysmenorrhoea

-
1. Pelvic ultrasound — fibroids, IUCD
 2. Laparoscopy — endometriosis, pelvic inflammatory disease
 3. High vaginal swab
 4. Endocervical swab
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Treatment is aimed at the underlying gynaecological disorder. It should be remembered, however, that endometrial prostaglandin levels are higher in the presence of an IUCD, and instead of removing the IUCD, one of the prostaglandin synthetase inhibitors should be tried to relieve the dysmenorrhoea and reduce the accompanying menorrhagia.

Discussion of the treatment of the causes of secondary dysmenorrhoea can be found in the relevant chapters in this text.

KEY POINTS

- 1. Dysmenorrhoea can be considered as either primary or secondary, and congestive or spasmodic.**
- 2. Primary dysmenorrhoea is common among teenagers especially of the upper social class, whereas secondary dysmenorrhoea is typically experienced in the 3 and 4th decades.**
- 3. It is appropriate to recommend a course of the oral contraceptive pill for primary dysmenorrhoea.**
- 4. When a patient presents with secondary dysmenorrhoea, attempts should be made to elucidate the cause.**
- 5. Treatment of secondary dysmenorrhoea is directed to the aetiology.**

CHAPTER SIX

INFERTILITY

Hayden Homer

INTRODUCTION

Fertility-related problems occur frequently affecting about 1 in 7 couples at some stage in their reproductive career, and can be highly distressing for couples who are intent on either starting or extending their family. Fortunately, in many cases, infertility is not insurmountable and many infertile couples can be assisted in achieving a pregnancy with appropriate and timely intervention. The investigation and management of fertility-related problems incorporate a truly multi-dimensional team that routinely extends beyond conventional gynaecology to include other disciplines such as endocrinology, andrology and embryology. The objective of this chapter is to provide a contemporary overview of infertility and its management as well as to give some insight into newer dimensions of fertility-related services.

Reproductive biology

As a foundation for better understanding the rationale underpinning clinical fertility investigation and management, it will be informative to briefly review some key aspects of basic reproductive biology.

Gametogenesis and the effect of female ageing

Gametes are specialised cells dedicated to sexual reproduction. In females, gametes are referred to as eggs whereas in males they are termed spermatozoa. Eggs and spermatozoa share the common feature of containing only half the normal complement of chromosomes (termed haploid). In this way, when an egg fuses with a spermatozoon at fertilisation, the normal (or diploid) chromosome complement is restored.

Both male and female gametes are ultimately derived from repositories of germ-line stem cells (or GSCs) in the gonads. Notably however, unlike GSCs in males which persist indefinitely into post-natal life, the conventional view is that GSCs in females are exhausted by birth. Thus, males have the capacity to sustain spermatogenesis throughout their lifetime, whereas females are born with a finite endowment of immature-stage eggs, termed oocytes. This striking sexual dimorphism results in men retaining fertility potential into old-age whereas for women, their fertility potential expires when their egg supply is exhausted. In females, aging not only results in a relentless attrition in the oocyte reservoir, but also in a decline in oocyte quality.

This so-called female age-effect has far-reaching implications for human reproductive success since the egg contributes virtually all of the cytoplasm to the early embryo and is thus critical for sustaining early embryonic development. Consequently, reduced egg quality translates directly into reduced embryo quality and ultimately, into a higher risk of reproductive failure. For older women, this biological phenomenon makes pregnancy treacherous and is manifested in higher rates of chromosomally abnormal offspring (such as those with Down's syndrome), higher rates of miscarriage, reduced fertility and reduced rates of success with fertility treatments such as *in vitro* fertilisation (or IVF; Figure 1).

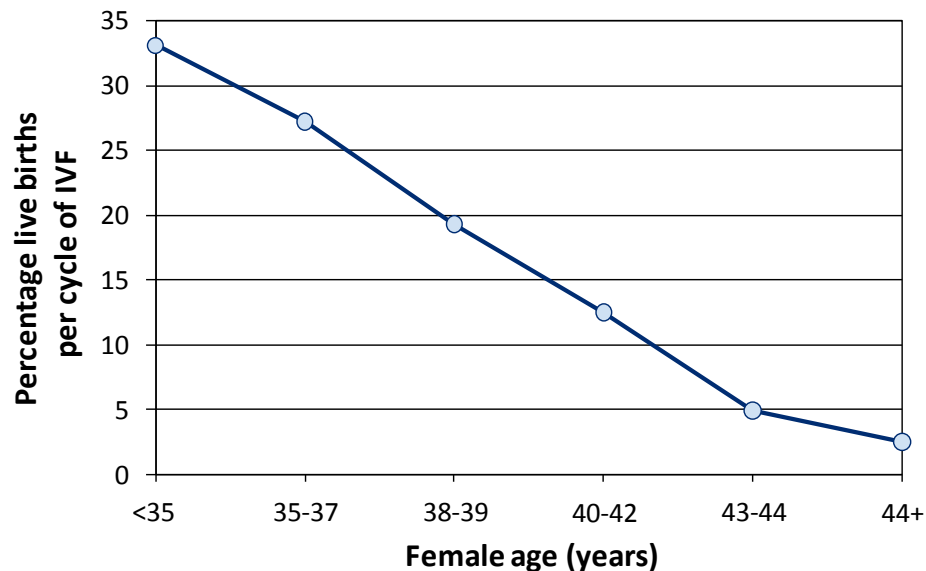


Figure 1. UK IVF success rates by female age in 2008

Physiology of the menstrual cycle: the hypothalamic-pituitary-ovarian axis

The menstrual cycle is the culmination of endocrine signalling pathways involving the hypothalamic-pituitary-ovarian (HPO) axis. The anterior pituitary gland produces follicle-stimulating hormone (FSH) and luteinising hormone (LH) under the influence of gonadotrophin-releasing hormone (GnRH) secreted into the pituitary portal circulation from axonal terminals in the hypothalamic median eminence. FSH and LH in turn act on the ovary to drive the recruitment and development of ovarian follicles which synthesise oestrogen and progesterone, the two pivotal hormones responsible for inducing cyclical changes at the level of the uterine endometrium that ultimately results in episodic vaginal bleeding, the menses.

The first half of the menstrual cycle is FSH- and oestrogen-dominated resulting in proliferative changes within the endometrium. Although about 15-20 follicles are initially recruited to develop, usually only one dominant follicle will emerge and acquire the competence to produce a mature egg at ovulation. High circulating levels of oestradiol at mid-cycle, mostly produced by the dominant follicle, induce a positive feedback effect at the hypothalamic-pituitary level resulting in the so-called LH surge which, through the production of prostaglandins, results in

ovulation about 35-38 h after commencement of the surge. The LH surge also stimulates the completion of first meiotic maturation in the oocyte contained within the dominant follicle along with luteinisation of the follicular granulosa cells leading to formation of a corpus luteum. During the luteal phase, progesterone becomes a prominent feature of the endocrine milieu resulting in secretory endometrial changes in preparation for implantation should fertilisation of the ovulated egg occur. In the absence of fertilisation, the endometrial layer is shed as the menstrual effluent marking the start of a new cycle.

A functional HPO axis will result in regular ovulation and menstrual cycles lasting about 28 days (normal range of 21-42 days). Conversely, as discussed later, any derangement of the axis as occurs for example in cases of polycystic ovarian syndrome (PCOS) will impair ovulation and result in infrequent (oligomenorrhoea) or absent menses (amenorrhoea). Because ovulation is associated with measurable endocrine events such as the LH surge (detectable in serum and urine) and the luteal phase rise in serum progesterone, ovulation can be detected using these parameters.

Definition and interval to fertility assessment

The UK National Institute for Clinical Excellence (NICE) has defined infertility as “failure to conceive after regular unprotected sexual intercourse for 2 years in the absence of known reproductive pathology”. This is based on population data showing that about 84% of couples would conceive within a year of unprotected regular sexual intercourse, rising to 92% by two years of exposure.

Couples should be considered for assessment and clinical investigation if they have been unsuccessful in conceiving after 12 months of regular unprotected sexual intercourse. Cumulative conception rates such as those presented above provide a clear rationale for setting a 12-month threshold for investigation; on the one hand, to avoid unnecessary intervention for those couples for whom spontaneous conception is reasonably high but on the other hand, to identify and instigate timely intervention for those who would benefit from appropriate treatment. That said however, there are sub-groups of patients for whom attention is merited even before the 12-month window of exposure has elapsed. For instance, earlier work-up is indicated in the presence of predisposing factors that suggest an underlying fertility problem. For the female, factors to be considered include irregular or absent menstrual cycles (suggestive of a disorder of ovulation) or a known history of pelvic inflammatory disease (PID) which increases the likelihood of tubal problems whereas for the male, a background of undescended testes or post-pubertal mumps orchitis is significant due to the deleterious effects on spermatogenesis.

As alluded to above, advanced female age is a significant parameter for an increasing proportion of couples seeking fertility assistance. For fertile women aged 35 years, 94% will conceive after 3 years of regular unprotected intercourse whereas by age 38 years, this figure drops to 77%. This inescapable phenomenon is a reflection of the inexorable and irreversible

decline in oocyte quality with advancing female age consequent upon the biology of female gametes outlined previously. Because of this inherent decline in fertility potential with advancing female age, a lower threshold for investigation should therefore also be applied to women older than 35-38 years.

Categories of infertility

Infertility is defined as primary if the female has never been pregnant previously or as secondary if there has been a previous pregnancy. Regardless of whether infertility is primary or secondary, the ability of a couple to achieve a pregnancy ultimately necessitates synergism between key biological events, many of which have been introduced above. Firstly, both parties must produce sufficient numbers of good quality gametes. For women, this requires a stock-pile of eggs that are of good enough quality to both support fertilisation and sustain early embryonic development in conjunction with an intact HPO axis to drive the ovulation of these eggs. For men, spermatogenesis should produce adequate numbers of motile spermatozoa that can propel themselves through the female genital tract and penetrate the oocyte's barriers in order to achieve fertilisation. Even with optimal gametes, fertilisation will only be possible if spermatozoa can gain access to the ovum in the Fallopian tube and pregnancy will only result if the embryo has an uninterrupted transit to the uterine cavity, altogether necessitating healthy and patent Fallopian tubes. Finally, the molecular signature of the endometrium must be one that is receptive to the embryo if implantation and placentation are to be successful.

On the basis of the foregoing, infertility has traditionally been categorised as due to one or a combination of male factor problems, disorders of ovulation or tubal factor problems (Figure 2). Other conditions which can impact on fertility such as fibroids and endometriosis may be grouped together as uterine and peritoneal factors. Endometriosis, which refers to the deposition of endometrial tissue at ectopic sites, is associated with infertility, being found in 30-50% of subfertile women compared with 5-10% of the general female population. In cases of severe endometriosis, it is easy to appreciate how pelvic scarring and ovarian endometriomas (endometriotic cysts in the ovary also known as "chocolate cysts") compromise fertility. Although pelvic distortion is minimal with the mildest forms of the disease, treatment can nevertheless improve fertility.

Fibroids are very common and consequently are often found in infertile women. However, whether fibroids are a *bona fide* cause of infertility remains a highly contentious issue. Recent evidence indicate that the category of fibroids which distort the endometrial cavity (submucous fibroids) is the most important with regard to fertility, perhaps because they are the most likely to impair implantation, and that treatment of such fibroids could improve fertility prospects. It is important to note that there can be considerable overlap between these categories, for instance, tubal distortion is not an infrequent occurrence in severe cases of endometriosis. Moreover, in about 20% of cases, more than one factor co-exists. If none of the above conditions can be identified, infertility is said to be unexplained which is therefore a diagnosis of exclusion.

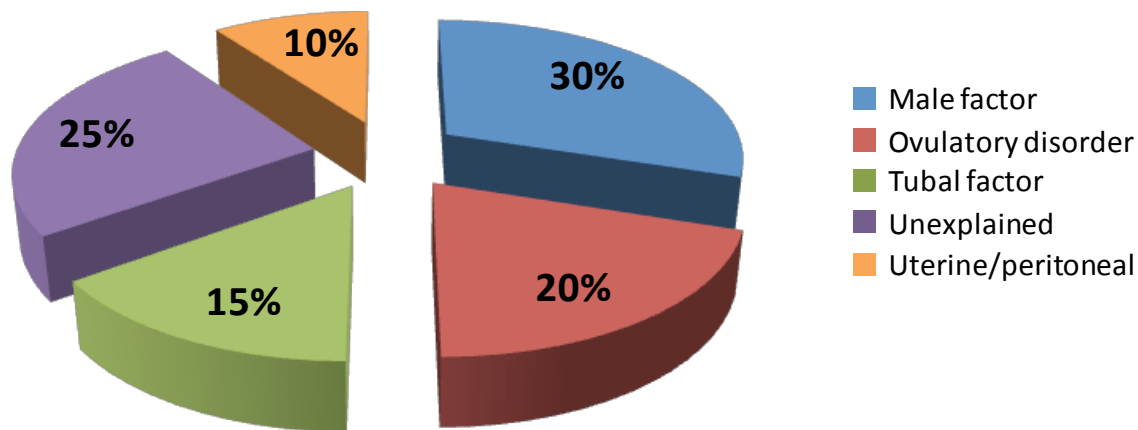


Figure 2. Categories of sub-fertility and their incidence rates

History and examination

Setting

It is clear from the foregoing that male factors make a major contribution to the aetiology of infertility making it imperative that consultations involve both male and female parties. A range of resources are needed for undertaking a comprehensive infertility workup along with any subsequent treatment including biochemical and andrology laboratories, dedicated ultrasound scanning facilities, access to radiological resources for performing tubal patency tests and operating theatres equipped for minimal access surgery. Specially trained clinical nurse specialists are invaluable for managing routine patient care as are fertility counsellors for providing psychological support.

Female history

Important general issues include female age, duration of infertility and previous pregnancy history. Age is of over-arching importance as this constitutes the key determinant of ovarian reserve and oocyte quality. Previous successful pregnancies are a positive prognostic feature whereas a long duration of primary infertility is an unfavourable prognosticator. Another general enquiry pertains to frequency of sexual intercourse which, although not specific for the female history, is included here for convenience. Although often overlooked, inadequate sexual exposure is not infrequently a significant factor especially if one party's occupation leads to extended periods of separation. Given that spermatozoa can survive in the female tract for up to 7 days, intercourse every 2-3 days will increase the likelihood that sperm is present whenever the egg is released thereby obviating the need for timed intercourse guided by ovulation predictor kits, a practice that intensifies the stress surrounding infertility.

Other aspects of the female history should be geared towards identifying specific problems that may be indicative of compromised ovulation and tubal integrity. With regard to ovulation, the most important item in the history is menstrual regularity. Regular menses occurring at monthly intervals are virtually confirmatory of regular ovulation. Other features indicative of ovulation include mid-cycle pain (or Mittelschmerz), mid-cycle spotting as a consequence of a transient fall in oestrogen levels at the time of ovulation and symptoms associated with progesterone production such as mastalgia and premenstrual symptoms.

Conversely, sparse or absent periods allude to ovulatory dysfunction which can be categorised using the World Health Organisation (WHO) classification system (Table 1).

Table 1. WHO classification of disorders of ovulation

WHO Group 1 (Hypogonadotrophic hypogonadism and hypothalamic amenorrhoea) • Low FSH and low oestrogen levels • Examples: Kallmann's syndrome and intracranial tumours (e.g. craniopharyngiomas)
WHO Group II (Hypothalamic-pituitary dysfunction) • Normal FSH and oestrogen • Examples: PCOS and excessive weight gain
WHO Group III (Ovarian failure) • High FSH and low oestrogen • Examples: Chromosomal and genetic abnormalities (e.g. Turner's syndrome and <i>FMR1</i> premutation), immunological causes and iatrogenic (e.g. following cancer treatment).

As the commonest cause for anovulation is PCOS, symptoms of hyperandrogenism often co-exist manifesting as oily skin, acne and excessive facial and body hair. In extreme circumstances, hyperandrogenism could induce male-pattern balding. Weight loss, psychological stress and extreme exercise can also impair ovulation by decreasing hypothalamic GnRH neuronal activity resulting in a fall in the levels of FSH and LH. Menses are unlikely to be regular with a body mass index (BMI) that is less than 19 kg/m², when underlying eating disorders such as anorexia nervosa may be a precipitating factor.

Another cause of anovulation is hyperprolactinaemia, most often originating from a pituitary prolactinoma (40-50% of cases), when galactorrhoea may be an accompanying feature in up to 30% of cases. For larger pituitary adenomas that extend beyond the sella turcica, compression of the optic chiasma could result in visual field defects. Because dopamine-mediated inhibition is the principal means for regulating prolactin levels, concurrent medications that inhibit

dopamine action or production (Table 2) could also culminate in hyperprolactinaemia (1-2% of cases). Due to the stimulatory effect of thyrotrophin-releasing hormone (TRH) on prolactin secretion, high prolactin could also occur in cases of primary hypothyroidism (3-5% of cases).

Table 2. Pharmacological agents associated with hyperprolactinaemia

Phenothiazines
- Perphenazine
- Chlorpromazine
- Thioridazine
- Trifluoperazine
- Prochlorperazine
Cimetidine
Butyrophenones
- Haloperidol
Benzamides
- Metoclopramide
- Clebopride
Methyldopa
Rauwolfia alkaloids
- Reserpine
Pimozide (Orap)

Premature ovarian failure (POF), which is the loss of ovarian function before the age of 40 years, could also present as infertility. Along with the premature cessation of menses many such women experience climacteric symptoms including hot flushes, night sweats and vaginal dryness. The incidence of POF is estimated to be 1-5% of the female population, in at least two-thirds of which the aetiology cannot be identified. In a series of 323 women with POF attending our hospital, 23% of cases were associated with Turner's syndrome, a chromosomal abnormality in which only one X chromosome (45X0) is present instead of the usual two (46XX) leading to gonadal dysgenesis. Although normal numbers of oocytes appear at the genital ridge, accelerated atresia during fetal life results in the typical streak gonads. Other chromosomal configurations affecting X-chromosome dosage and associated with POF include Turner's mosaics (46XX/45X) and trisomy X (47XXX). In some reports of POF, X mosaicisms are the commonest chromosomal abnormality, ranging from 5-40%. Carriers of the *FMR1* (for fragile X mental retardation 1) premutation allele show an increased prevalence of POF, estimated to be 15-24%. POF may also have an immunological basis and occur as a component of polyglandular autoimmune syndromes involving other disorders such as hypothyroidism and type 1 diabetes.

With improving survivorship from childhood cancers, increasing numbers of women are presenting in their reproductive years with POF as a consequence of having been exposed to gonadotoxic chemotherapy and radiotherapy in earlier life. Better integration between cancer services and reproductive medicine units would ideally enable cancer patients to be seen prior to treatment to explore options for fertility preservation.

From the point of view of tubal disease, a history of PID is highly significant. In the UK, infection due to *Chlamydia trachomatis* is the most common curable sexually transmitted infection (STI) and is asymptomatic in about 70% of infected women. In the absence of treatment, 10-40% of women infected with chlamydia develop PID. Although a significant proportion of these may remain asymptomatic, others develop symptoms such as purulent vaginal discharge, chronic pelvic pain, pain with intercourse (dyspareunia) and painful periods (dysmenorrhoea). Other microbiological causes of PID include *Neisseria gonorrhoeae* and *Mycoplasma genitalium*. Pelvic pain symptoms including deep dyspareunia may also be indicative of endometriosis as well as painful defecation (dyschezia). Specific questions should also enquire about previous pelvic surgery for ectopic pregnancy or for conditions that have close proximity to the tubes such as ovarian cysts and ruptured appendices which often result in peri-tubal scarring. Previous Caesarean sections are a relatively common surgical procedure that can also induce pelvic adhesions as can previous terminations of pregnancy if complicated by infection.

It is also important to enquire about fibroid-related symptoms such as heavy periods (menorrhagia), spasmodic dysmenorrhoea and pelvic pressure. In contrast, absent or scanty periods associated with a previous history of uterine instrumentation or endometrial infection, for instance after pregnancy termination, could point to intrauterine synechiae (Asherman's syndrome).

Female examination

BMI and blood pressure should be recorded for all women and an assessment made of secondary sexual characteristics as the latter will provide a guide as to the degree of oestrogenisation. Weight-loss related amenorrhoea is associated with a BMI < 19 kg/m² whereas symptoms of PCOS are often exacerbated by weight gain so that 40-50% of women are overweight (BMI >25 kg/m²). Hirsutism associated with PCOS can be objectively graded using the Ferriman-Gallwey scoring system. Women with Turner's syndrome are typically of short stature and may have other features such as a webbed neck and high-arched palate. They are also susceptible to a number of co-morbidities such as hypertension and a risk of aortic root dilatation.

Examination of the abdomen will reveal surgical scars in cases of previous surgery and a pelvic mass if large uterine fibroids are present. Speculum examination may reveal a yellow discharge in cases of PID or endometriotic deposits at the vaginal vault. Bimanual examination could elicit tenderness in the adnexae as well as reduced mobility of the pelvic organs in cases of PID or endometriosis. In the latter case, a fixed retroverted uterus and or palpable nodules in the recto-vaginal septum are pathognomonic features.

Male history

The male history seeks to elucidate conditions which may compromise sperm quality or the delivery of sperm to the female genital tract. A history of children in previous relationships reduces (but by no means eliminates) the likelihood of major abnormalities of sperm quality. A

childhood history of undescended testes (cryptorchidism) is hugely significant as testes which have not established a low testicular position by 2 years of age will be histologically abnormal and have reduced spermatogenic capacity.

Although vasectomy is the most obvious iatrogenic cause of male infertility, other surgical procedures could also compromise the genital tract including herniorrhaphy, orchidopexy, drainage of a hydrocoele or bladder neck surgery. A history of dysuria, haematuria or urethral discharge is indicative of genital tract infection which could result in fibrosis-mediated obstruction to sperm outflow. Specific enquiry should be made about previous episodes of mumps occurring post-pubertally and whether or not there was accompanying orchitis as 84% of men with a history of bilateral mumps orchitis have abnormal semen and 22% are azoospermic (absence of spermatozoa in the ejaculate). Specific questioning about a family or personal history of cystic fibrosis is also important as congenital bilateral absence of the vas deferens, found in about 2% of men with obstructive azoospermia, is a well-known association of cystic fibrosis gene mutations.

A careful medical history is pertinent as erectile and ejaculatory dysfunction may be caused by psychosexual dysfunction, depression, diabetes mellitus, spinal cord injuries, multiple sclerosis and chronic renal failure. Respiratory symptoms may be significant as conditions such as chronic bronchiectasis can be associated with ciliary dysfunction, for instance in cases of Kartagener's syndrome, and therefore with defects in sperm motility. A drug, alcohol and smoking history is directly relevant as adrenergic blocking anti-hypertensive agents and alcohol abuse affect erectile function whilst alcohol, tobacco and drugs such as Sulphasalazine, Cimetidine and Nitrofurantoin impair spermatogenesis.

As with women, increasing numbers of men attending fertility clinics would have been exposed to gonadotoxic treatment for cancers. Unlike approaches for female fertility preservation which involves procedures that are either invasive or experimental, sperm cryopreservation is a relatively simple, long-established and highly effective means of male fertility preservation. Thus, male cancer survivors have often had sperm frozen prior to cancer treatment making it important to enquire about the existence of such samples so that steps can be taken to acquire them for assisted fertility treatments.

Male examination

An assessment should be made of overall secondary sexual development as well as the presence of surgical scars in the groin area. On urogenital examination, the penis is examined for evidence of phimosis, hypospadias or epispadias. The scrotum should be examined to confirm the presence of testes in the scrotal sacs. As testicular volume is strongly correlated with sperm production, the volume in millilitres of the testes is extremely important and may be estimated with the aid of a Prader orchidometer. Testicular palpation will also determine their consistency. Any tenderness on palpation should be noted as well as the presence of any thickenings or cysts within the epididymis. The vasa deferentia should be palpated to confirm

their presence and patients should be asked to perform Valsalva's manoeuvre so as to make lower grade varicoceles more obvious.

Investigations

The subsequent sections maintain a problem-oriented approach to the infertility work-up and will outline investigations that are aimed at assessing those aspects that are critical to pregnancy success focusing primarily on tests for ovulation, semen quality and tubal patency.

Assessment of ovulation

Luteal phase serum progesterone estimation is currently considered to be the standard test for confirming ovulation with levels >30 nmol/l indicative of ovulation. Results are only interpretable if blood is obtained 7 days before the next anticipated menstrual bleed which, for a typical 28-day cycle, would be day-21. It is important to note however that referring to this test as a day-21 serum progesterone is inaccurate as the appropriate day of sampling will vary depending on cycle length. For instance, sampling should be performed on day-28 of a 35-day cycle and for irregular cycles may need to be repeated at weekly intervals until the next menstrual cycle commences.

A more elaborate means for obtaining direct evidence of ovulation is by tracking the growth and rupture of the dominant follicle followed by corpus luteum formation using ultrasound scanning. Ultrasound follicle tracking may be particularly useful for diagnosing cases of luteinised unruptured follicle (LUF) syndrome in which LH-induced luteinisation of an unruptured dominant follicle occurs. In such cases, serum progesterone estimations are misleading as progesterone levels increase in the absence of ovulation. In practice however, follicle tracking is intrusive and whilst routinely used for monitoring treatment cycles, is rarely used as a first-line diagnostic tool.

Although not confirmatory, other markers provide presumptive evidence of ovulation. Such markers include the consistency and stretchability of cervical mucus (Spinnbarkeit) or a 0.5 - 1.0°C rise in the basal body temperature (BBT), both of which are progesterone-dependent effects, and urinary kits designed to detect the LH surge that triggers ovulation.

Semen analysis

The semen analysis is the cornerstone of male factor assessment and evaluates three principal parameters, sperm concentration, motility and morphology based on standards set by WHO in 1999 (Table 3). Prior to producing the sample, the patient should abstain from intercourse for a minimum of 48 hours but no more than 7 days. Because of the exquisite sensitivity of sperm to temperature changes, samples should ideally be produced in dedicated sperm production rooms within the fertility unit allowing for immediate analysis or else be brought to the laboratory within 30 minutes to 1 hour of production. Because of inter-ejaculate variability,

proper evaluation requires a minimum of 2 separate samples produced not less than 7 days or more than 3 months apart.

Table 3. WHO (1999) reference values for semen parameters

Volume: ≥ 2.0 ml
• Liquefaction time: within 60 minutes • pH: ≥ 7.2 • Sperm concentration: ≥ 20 million spermatozoa per ml • Total sperm number: ≥ 40 million spermatozoa per ejaculate • Motility within 60 minutes of ejaculation: $\geq 50\%$ motile (grades a* and b**) or $\geq 25\%$ or more with progressive motility (grade a) • Vitality: $\geq 75\%$ live • White blood cells: < 1 million per ml • Morphology: 15%***
• *Grade a: rapid progressive motility (sperm moving swiftly, usually in a straight line). • **Grade b: slow or sluggish progressive motility (sperm may be less linear in their progression). • ***Less than 15% normal morphology is associated with reduced fertilisation rates within the context of IVF

Tests for tubal patency

Tests are based on a common principle of instilling a medium through the cervical canal and detecting whether or not that medium fills the tubes and then spills through the fimbrial ends, fill and spill being indicative of patent tubes. Three main tests are regularly used and differ in terms of how fill and spill of medium are detected: X-ray hysterosalpingography (HSG), laparoscopy with dye instillation and hysterosalpingo-contrast sonography (HyCoSy).

Because of the risk of disseminating infection to the upper genital tract with these methods, it is advised that all women be screened for infection prior to having the test performed or, in the absence of screening, to be covered by appropriate broad-spectrum antibiotics active against organisms such as *C. trachomatis*, *N. gonorrhoea* and anaerobes (such as *Gardnerella vaginalis* and *Mycoplasma hominis*). Single-dose regimes are highly effective and offer the highest likelihood of compliance. We advise a combination of Azithromycin 1 g orally and Metronidazole 1 g rectally 2 hours prior to testing.

Hysterosalpingogram (HSG)

In the absence of co-morbidities that increase the likelihood of tubal pathology such as previous PID, ectopic pregnancy or pelvic surgery, HSG is a first-line modality for evaluating tubal patency. The test is performed in X-ray department as an out-patient procedure and should be

timed to occur after the menses but before ovulation within a menstrual cycle in which the couple abstain from intercourse. The cervix is cannulated and either a water- or oil-soluble radio-opaque contrast medium is instilled. Using X-rays for acquiring images at high frequency, the dye can then be traced as it traverses the uterine cavity and Fallopian tubes.

HSG provides an outline of the contour of the uterine cavity, avoids the need for a general anaesthetic and potentially dangerous surgery and if an oil-soluble contrast medium (lipiodol) is used, there is evidence to suggest that HSG could enhance the chance of pregnancy in cases of unexplained infertility. HSG has a high negative predictive value (94%) meaning that the finding of normal tubes on HSG is likely to be correct but a low positive predictive value (38%) meaning that it is not a reliable indicator of tubal occlusion. False positive results could occur due to tubal spasm for which amyl nitrite inhalation might be helpful for inducing relaxation. Because of the low positive predictive value, cases suggestive of tubal blockage should be evaluated further using laparoscopy.

Laparoscopy and dye hydrotubation

Laparoscopic dye instillation is considered the gold-standard for evaluating tubal patency but requires a general anaesthetic and carries the risk of injury to intra-abdominal structures such as bowel, bladder and blood vessels. Laparoscopy is an appropriate choice of test in the presence of known co-morbidities or if findings in the history or examination, such as dyspareunia or pelvic tenderness, point to a higher likelihood of pelvic disease or when HSG results are abnormal.

With the patient asleep, the pelvis is meticulously inspected, usually requiring a minimum of two laparoscopic ports. Using a cervical cannula (e.g. a Spackman cannula), a dye such as methylene blue is instilled into the uterus and each tube is inspected in turn for fill and spill of dye. It is important to remember that the impetus for performing a higher risk intervention for assessing tubal patency is the greater likelihood of there being pelvic pathology. Therefore, both patient and surgeon should be prepared for the possibility of extending the procedure from a diagnostic test into one involving therapeutic laparoscopic surgery. Not infrequently, pathology such as pelvic adhesions (often PID-induced), endometriosis, hydrosalpinges and ovarian cysts are encountered. For adequately treating these conditions, up to 4 laparoscopic ports are required that allow the introduction of scissors, diathermy, atraumatic graspers, suction/irrigation apparatus and newer-generation devices such as the ultrasonic Harmonic® scalpel (Ethicon Endo-Surgery, Inc.), which is especially useful for developing tissue planes during excision of deeply infiltrating endometriotic lesions.

Hystero-contrast-sonography (HyCoSy)

HyCoSy is an outpatient procedure that involves the passage of a small balloon catheter through the cervix. The balloon of the two-channel catheter is inflated by instilling water through one channel in order to form a seal at the level of the internal os and via a separate channel, an ultrasound contrast agent such as Echovist is injected into the uterine cavity.

Ultrasound detection of free contrast flow through the tubes is indicative of tubal patency. One advantage of this approach is that ultrasound can be used to simultaneously assess the contour of the uterus as well as the morphology of the ovaries.

Instead of agents such as Echovist, saline can be instilled, a process referred to as saline sonohysterography. Whilst not amenable to tubal patency testing, saline instillation improves the capability of ultrasound in identifying intrauterine lesions such as endometrial polyps and fibroids that have a significant submucosal component. In our Unit, all patients undergoing IVF routinely have a saline sonohysterography for assessment of the endometrial cavity as part of their pre-treatment workup. Any lesions identified in this way would then be resected hysteroscopically as they would otherwise compromise the success of treatment.

Additional investigations for the female

In anticipation of pregnancy, it is useful to confirm female immunity to rubella and to ascertain blood group and Rhesus status. Both partners should ideally have a virology screen for HIV, Hepatitis B and C, especially if IVF is being contemplated. Specific features in the history and clinical examination might highlight the need for additional investigations as elaborated below. The role of anti-sperm antibodies, natural killer cells (NK cells), the post-coital test and sperm DNA fragmentation tests awaits further clarification and will not be discussed further here.

Biochemistry

In women with oligo/amenorrhoea, LH hypersecretion with normal FSH levels during the early follicular phase is indicative of PCOS; about 60% of women with PCOS have elevated LH levels (>10 IU/l) and up to 95% have an elevated LH/FSH ratio (≥ 2). In contrast, amenorrhoea associated with raised FSH levels is indicative of compromised ovarian function; FSH levels >30 IU/l on two occasions 3-6 months apart are consistent with ovarian failure. In women with amenorrhoea, especially in the presence of galactorrhoea, prolactin levels should be assessed. Persistently elevated prolactin (>1500 mU/l) warrants pituitary imaging for detection of adenomas. Magnetic resonance imaging (MRI) does not incur radiation hazards and is more sensitive than computerised tomography (CT) scanning but takes longer and is more expensive. It should be noted however that in the absence of specific symptoms or indications, tests for thyroid function and measurements of prolactin levels are not routinely indicated. Serum levels of testosterone, sex hormone binding globulin, free androgen index and adrenal androgens including androstenedione, dehydroepiandrosterone, dehydroepiandrosterone sulphate and 17-OH progesterone are indicated in women who exhibit features of hyperandrogenism.

Hyperinsulinaemia due to insulin resistance is an important component of the pathophysiology of PCOS and are exacerbated by obesity. Consequently, glucose intolerance develops in 50% and diabetes in 20% of obese women with PCOS making an assessment of glucose tolerance important in such women.

Pelvic ultrasound scanning

Women with oligo/amenorrhoea should have a pelvic scan to determine whether or not the ovaries are polycystic. Based on the ESHRE/ASRM Rotterdam consensus, ultrasonographic criteria for defining polycystic ovaries (PCO) include the presence of 12 or more follicles measuring 2-9 mm in diameter and/or an ovarian volume that exceeds 10 cm³. A peripheral distribution of follicles (necklace pattern) is no longer a pre-requisite for defining PCO and only one ovary fitting these criteria is sufficient to make the diagnosis. It is important to note that the mere presence of PCO does not necessarily imply the presence of the syndrome (PCOS). Based on the international consensus definition requires the presence of two out of the following three criteria:

1. Oligo-ovulation and/or anovulation (often manifested as oligomenorrhoea or amenorrhoea)
2. Hyperandrogenism (biochemical and/or clinical)
3. Morphologically appearing polycystic ovaries (PCO)

Interestingly, as polycystic ovarian morphology accounts for only one of these three criteria, a diagnosis of PCOS can be ascribed in the absence of polycystic-appearing ovaries.

Women with pelvic pain symptoms also warrant a pelvic scan as this could identify features of endometriosis such as endometriomas which give a typical ground-glass appearance on scan. When there is a past history of PID, there may also be ultrasonographic evidence of hydrosalpinges. Dysmenorrhoea may be associated with fibroids which are readily detected during ultrasound examination. The presence of congenital uterine anomalies such as septate and bicornuate deformities might also become evident. As detailed next, ultrasound is also very useful for gaining insight into ovarian reserve.

Ovarian reserve tests

Ovarian reserve refers to the compliment of oocytes present in the ovary at a given time. It is not feasible to accurately determine absolute oocytes numbers but tests can provide a crude estimate. The longest established measure of ovarian reserve is an early follicular phase serum FSH level usually taken between days 2-5 of the menstrual cycle. Values >10 IU/l indicate reduced ovarian reserve and low fertility. The discriminatory capacity for FSH is limited and because levels vary widely throughout the menstrual cycle, small variations in the timing of sampling could affect interpretation.

More recently, two newer biomarkers have superseded FSH measurements, anti-Mullerian hormone (AMH) and antral follicle counts (AFC). AMH is produced by the granulosa cells of early developing follicles and levels are largely independent of cycle stage. AFC is a biophysical assessment of the total number of small antral follicles (2-5 mm) that are visible on ultrasound during the early follicular phase of the cycle.

Many do not consider ovarian reserve testing to be a routine part of the infertility workup, especially for women less than 35 years of age. For older women however, especially those above 38 years, such information is useful in providing a guide as to the current fertility status, to the urgency with which more sophisticated treatments such as IVF should be pursued and as a prognostic indicator for the likelihood of success following treatment. AMH is used widely in IVF programmes for individually tailoring stimulation regimes and for identifying women at greatest risk of ovarian hyperstimulation syndrome (OHSS).

OHSS is a potentially serious condition brought about by an exaggerated response to ovarian stimulating drugs and characterised by abdominal pain, vomiting, fluid and electrolyte imbalance, ascites and in severe cases, pleural effusions and arteriovenous thrombosis.

Additional investigations for the male

When the semen analysis indicates azoospermia, the question of over-riding importance from the perspective of fertility management pertains to whether or not sperm can be retrieved from the male for assisted fertility treatment. Serum FSH levels are helpful in making this distinction. An obstructive cause for azoospermia, as in the case of congenital bilateral absence of the vas deferens, is consistent with the presence of testes that are of normal volume and consistency and normal FSH levels. In such cases spermatogenesis is ongoing and sperm can often be obtained proximal to the block using surgical sperm retrieval techniques including percutaneous epididymal sperm aspiration (PESA) or testicular sperm aspiration (TESA).

In contrast, the combination of small testes with soft consistency and raised FSH levels points to non-obstructive azoospermia and increases the likelihood that spermatogenesis is severely compromised, reducing the likelihood of being able to retrieve sperm. In such cases, it is often necessary to perform microsurgical testicular biopsies to search for usable sperm with immediate access to cryopreservation facilities for freezing any sperm obtained for use in later treatment.

Male infertility is associated with chromosomal and genetic defects, the frequency of which is correlated with the severity of the sperm defect. Thus, abnormal karyotypes occur in 4% of men who have oligozoospermia (reduced sperm counts) increasing to 15% in azoospermic patients in which the commonest abnormality accounting for about 90% of cases is 47XXY (Klinefelter's syndrome). More recently, a family of genes including *RBM*, *DAZ*, *DFFRY*, *DBY* and *CDY* found in three non-overlapping regions of the Y chromosome, AZF a-b-c (for azoospermia factor a-b-c), have been found to be involved in spermatogenesis. A little over 10% of cases of non-obstructive azoospermia and a smaller proportion with very severe oligozoospermia have deletions affecting these regions, the most frequently reported abnormality being a microdeletion in the AZFc region encompassing the *DAZ* (for deleted in azoospermia) gene.

Men found to have severe sperm defects on semen analysis should therefore have a karyotype performed and increasingly, such men are also being offered genetic analyses for Y-

chromosome microdeletions. Because microassisted fertilisation can assist men to have children even when semen quality is severely reduced, any such chromosomal aberrations and their associated fertility defects will be passed on to male offspring. Thus, whilst our current knowledge indicate that such defects confer no overt health risks, men found to have these abnormalities should be offered genetic counselling.

Treating the infertile couple

The most widely used treatments include ovulation induction, controlled ovarian stimulation, intra-uterine insemination (IUI) and IVF. Treatment is also warranted for co-existing conditions such as endometriosis and uterine pathology including fibroids, endometrial polyps, adhesions and septae.

General measures

Patients should be advised about the importance of regular intercourse (every 2-3 days). Women should be taking 0.4 mg folic acid daily to reduce the risk of neural tube defects unless they are using anti-epileptic medication or have previously had a pregnancy affected by a neural tube defect when a higher dose of 5 mg daily is recommended. Because of the psychological stress associated with infertility, its investigation and treatment, patients should have access to trained fertility counsellors. Women who are obese (BMI ≥ 30) should be advised that weight reduction is advantageous for pregnancy success and also reduces the risk of pregnancy-related complications such as pregnancy-induced hypertension and gestational diabetes. Patients should be advised of the deleterious effects of smoking, excessive alcohol intake and recreational drug usage and should be encouraged to reduce, or ideally to stop, such practices. An important part of the assessment of a couple prior to embarking on treatment must take into account the welfare of any children that may result from treatment and should seek to ascertain whether or not the couple can provide a safe and suitable environment in which to raise a child.

Ovulation induction

Ovulation induction refers to the induction of uni-follicular development and mono-ovulation in women who are anovulatory. The appropriate treatment is dictated by the underlying cause of anovulation. About 85% of anovulation is due to HPO axis dysfunction (WHO Group II; see Table 1), 90% of which are associated with PCO. Weight loss is recommended as a first-line therapy in obese women with PCOS as weight reduction of just 5-10% can restore reproductive function in a significant proportion of patients within 6 months. Conversely, for weight-loss induced amenorrhoea, weight gain that restores the BMI to 19 or above is advocated.

Clomiphene citrate and Tamoxifen

The most widely used ovulation induction agent is Clomiphene citrate, a non-steroidal selective oestrogen receptor modulator (or SERM), which acts primarily as an anti-oestrogen. Clomiphene binds to oestrogen receptors at the hypothalamic-pituitary level thereby blocking

the negative-feedback effect of oestradiol on GnRH secretion resulting in an increase in GnRH pulse amplitude and consequently an increase in gonadotrophins sufficient for stimulating follicle development.

The typical starting dose is 50 mg/day and can be initiated from day 2-5 or from day-5 of the cycle for a total of 5 days. Doses can be increased in 50 mg increments to a total of 150 mg/day or reduced to 25 mg/day in over-responders. At least the first cycle of Clomiphene usage should be monitored using ultrasound scanning in order to objectively evaluate responsiveness thereby minimising the risks of multiple pregnancy due to multi-follicular development and OHSS which can occur in around 10-15% of cases but is rarely severe. Another benefit of ultrasound is to monitor endometrial development which can be compromised in Clomiphene users and reduce the prospects of pregnancy success. Having established an effective ovulatory dosage of Clomiphene, treatment can be continued for 6-12 months as cumulative conception rates continues to increase until it reaches a plateau at treatment cycle 12. There is some concern with prolongation of Clomiphene usage beyond 12 cycles as this may be associated with an increased risk of ovarian tumours. Ovulation rates with Clomiphene are 60-85% with a pregnancy rate of 30-40%.

Tamoxifen, a structurally similar drug to Clomiphene, may be used as a second-line drug for women who experience side-effects with Clomiphene or in whom endometrial development is sub-optimal. The dose of Tamoxifen is 20-80 mg/day from day 2-6 and has comparable efficacy to Clomiphene. Aromatase inhibitors such as letrozole and anastrozole which stimulate endogenous FSH release by inhibiting peripheral oestrogen production may be another option for ovulation induction but experience with its use is limited and further evaluation is needed.

Metformin

Because hyperinsulinaemia is a major aetiological factor in the ovulatory dysfunction associated with PCOS, insulin-sensitising agents have been investigated as ovulation-induction agents. The most commonly used is Metformin, a biguanide oral hypoglycaemic drug shown to reduce serum insulin and androgen levels in PCOS. The most common doses employed are 500 mg three times daily and 850 mg twice daily.

Initial support for Metformin has waned in the wake of recent large randomised controlled trials showing that Clomiphene is superior to Metformin as a first-line agent for ovulation induction. The largest of these, the US Pregnancy in PCOS (PPCOS) trial, reported its findings in the New England Journal of Medicine in 2007 and, using live-birth rate as the end-point, concluded that Clomiphene was superior to Metformin alone and that addition of Metformin to Clomiphene conferred no significant benefit. In line with this, a Scientific Advisory Committee of the UK Royal College of Obstetricians and Gynaecologists (RCOG) very recently concluded that "Metformin is not a first-line treatment of choice in the management of PCOS". Although Metformin is an appealing approach for preventing or ameliorating the adverse metabolic effects associated with PCOS such as glucose intolerance and cardiovascular disease, there is currently insufficient evidence to advocate routine use on this basis.

Gonadotrophins

Gonadotrophins are suitable for Clomiphene-resistant women with PCO or as a first-line for women who are deficient in gonadotrophins, the pathophysiology of which may be related to:

1. Primary deficiency of GnRH (WHO Group I; see Table 1) such as in Kallmann's syndrome, a rare (1:50,000) condition characterised by anosmia and a congenital absence of GnRH neurons.
2. Primary deficiency of gonadotrophins (WHO Group I) for instance when hypopituitarism results from pituitary ischaemia induced by massive postpartum haemorrhage (Sheehan's syndrome).
3. Dysfunction of the HPO axis without an absolute deficiency in either GnRH or gonadotrophins as is the case with weight-loss related amenorrhoea.

There are several commercially available gonadotrophin preparations belonging to one of two groups, either urinary-derived or recombinant, with current evidence indicating similar efficacy for both. The most commonly used preparations are human menopausal gonadotrophins (hMG) such as Menopur[®] (Ferring), a highly purified urinary-derived hMG containing 75 IU each of FSH and LH. The usual starting dose is either 75 IU or 150 IU administered daily as a subcutaneous injection. Due to the increased sensitivity of women with PCO to hMG, a lower starting dose of 37.5 IU may be required. Ultrasound monitoring is imperative due to the heightened risk of multi-follicular development and OHSS and for guiding dose titration. Once a single follicle of ovulatory size (17-20 mm) has emerged, ovulation is induced by simulating an LH surge, generally by administering a bolus of human chorionic gonadotrophin (e.g. 5,000-10,000 IU Pregnyl[®], Organon).

Pulsatile GnRH

An alternative to gonadotrophins in the presence of an intact pituitary is the pulsatile administration of GnRH via a portable mini-pump. The pump administers GnRH, either intravenously or subcutaneously, with a low starting dose of 2.5-5.0 µg/pulse and at a frequency of 60-90 minutes. Pulsatile GnRH is highly effective with reported ovulation and pregnancy rates per treatment cycle of roughly 70-90% and 20-30% respectively. The pump is much less likely than gonadotrophins to induce multi-follicular development and is therefore considerably safer regarding parameters such as multiple pregnancy and OHSS. In spite of its efficacy however, it is quite demanding for patients as the pump must be worn continuously and can be difficult to conceal.

Laparoscopic ovarian drilling (LOD)

LOD refers to puncturing the ovarian capsule using a laparoscopic diathermy needle and is an effective alternative to gonadotrophins for Clomiphene-resistant PCO. Optimal results are

obtained with 4 punctures per ovary using a monopolar setting of 30 W for 5 seconds per puncture. It is particularly suitable for PCOS cases with severely elevated LH levels as it may rectify LH hypersecretion. About one-third of women ovulate following LOD and about 50% will conceive within the first year. LOD carries advantages over gonadotrophins in that there is no increased risk of multi-follicular development and no requirement for intensive monitoring. The disadvantages of LOD pertain to the risks associated with surgery including the possibility of inducing pelvic adhesions. There are also concerns that by destroying healthy follicles, the electrical energy used for LOD could compromise ovarian follicular reserve.

Dopamine agonists

In cases of hypothalamic-pituitary dysfunction secondary to hyperprolactinaemia, first-line treatment usually consists of dopamine-agonists. Bromocriptine is frequently used at a dose of 5-7.5 mg daily in divided doses but is commonly associated with side-effects such as nausea, vomiting and postural hypotension. Carbergoline (from 0.25-1 mg twice weekly up to 1mg daily) is an alternative to Bromocriptine that has less side-effects and is longer acting. Trans-sphenoidal resection of adenomas may be required for those who are intolerant of medication, for non-functioning macroadenomas (>10 mm) or when there is supra-sellar extension unresponsive to medication.

Management of uterine and peritoneal factors: reproductive surgery

The conditions that can potentially impact on fertility and/or pregnancy, which are amenable to surgery include endometriosis, tubal pathology and uterine disorders (fibroids, polyps, intrauterine adhesions and uterine septae).

Endometriosis and PID

Medical management of endometriosis is not conducive to fertility as this generally involves approaches that suppress endogenous ovarian function. Therefore, from a reproductive perspective, surgery is the cornerstone of treatment and wherever possible, laparoscopy is the preferred route. Minimal and mild cases of endometriosis can be treated using either ablation (diathermy or lasers) or excision. For the treatment of endometriomas, cyst excision is generally considered superior to cyst ablation whilst simple fenestration and cyst drainage is to be discouraged due to the high rates of recurrence. In very severe cases, advanced laparoscopic skills are required and should preferably be performed in regional centres with the expertise to manage deeply infiltrating lesions that often involve bowel and encroach on the ureters.

An important sequela of endometriosis, PID and previous surgery is pelvic adhesions, which can envelop the fimbrial ends of the tubes and lead to dilatation of the ampullary regions forming hydrosalpinges. Peri-tubal adhesiolysis along with salpingostomy are effective tube-sparing treatments for some cases of hydrosalpinges. In severe cases however, especially when hydrosalpinges are bilateral, IVF may offer a better chance for pregnancy than reconstructive

tubal surgery. In such circumstances, salpingectomy or laparoscopic tubal occlusion is advisable due to the detrimental impact of hydrosalpinges on IVF implantation rates.

Fibroids, endometrial polyps, intrauterine adhesions and uterine septae

Evidence from IVF treatment shows that treatment of submucous fibroids improves success rates, at least in part by reducing miscarriage. It is therefore recommended that submucous fibroids, especially those with significant intra-cavitary extension (Types 0 and I) be removed prior to fertility treatment.

Transcervical resection of fibroids (TCRF) refers to a minimal access hysteroscopic approach for removing submucous fibroids involving a hysteroscopic resectoscope (Olympus), usually glycine as the uterine distension medium and diathermy as the energy to “carve” out the fibroid from the surrounding myometrium. Hysteroscopy can also be used to treat other uterine defects including polyps, uterine septae and adhesions (Asherman’s syndrome) all of which could impair implantation and predispose to miscarriage. A specially designed narrow-calibre hysteroscope known as the Versascope™ (Gynecare, Ethicon) is particularly well-adapted to treating the latter conditions, and in the case of polyps may even be employed on an outpatient basis without the need for anaesthesia.

Intra-uterine Insemination (IUI) and Superovulation

The aim of IUI is to place a small volume (about 0.2-1.0 ml) of motile sperm high in the uterine cavity beyond the cervical mucus barrier and close to the Fallopian tubes and to coincide this with ovulation. IUI is an out-patient procedure that involves cervical cannulation with a fine catheter through which sperm are injected into the uterine cavity. Sperm preparation procedures, including swim-up and gradient techniques, are employed to isolate motile and morphologically normal spermatozoa as well as to remove prostaglandins, infectious agents and dead spermatozoa thereby improving the overall fertilising capacity of the sample.

IUI is often advocated for cases of mild to moderate male factor infertility, albeit the evidence to support this is limited. Most authors recommend IVF when the total motile sperm count (TMSC) following sperm preparation is less than 1×10^6 per ml. Although this sets a lower limit below which IUI is unsuitable, no clear threshold has been established above which IUI has a proven role. However, on the basis of published results, and given its lower risks and less invasive nature compared with IVF, IUI would seem a reasonable option in cases of male factor infertility with a TMSC of $5-10 \times 10^6$ per ml.

IUI may also be used for treatments involving cryopreserved sperm provided the post-thaw specimen is of adequate quality. This application of IUI is commonly used for men who electively stored sperm prior to having gonadotoxic chemotherapy and radiotherapy for cancer treatment. Another application involves couples who require donated sperm, for instance if the male has non-obstructive azoospermia secondary to failed spermatogenesis. Donor sperm is

also used to help single women and lesbian couples achieve a pregnancy. Prior to using donor gametes, patients must have extensive counselling regarding the implications of such treatment.

Increasingly IUI is being performed in conjunction with superovulation (SO), not only for male factor problems but also as a recognised treatment modality for unexplained infertility and minimal or mild endometriosis. SO refers to ovarian stimulation using either Clomiphene or gonadotrophins with the intention of inducing the ovulation of 2-3 eggs. Notably, unlike ovulation induction which aims for mono-ovulation in anovulatory women, SO is very often employed in ovulatory women and aims to achieve multi-follicular development. Compared with ovulation induction, higher doses of drugs are used for SO with concomitantly higher risks of multiple pregnancy and OHSS making close monitoring with ultrasound mandatory.

In Vitro Fertilization (IVF)

IVF involves the extra-corporeal manipulation and fertilisation of gametes with the intention of replacing a limited number of the resulting embryos into the uterine cavity. Increasingly, as discussed below, IVF-based technology is being applied to women whose fertility potential is threatened. Although the first successful birth from IVF in 1978, Louise Brown, resulted from the retrieval of a single oocyte in a natural cycle, subsequent data have shown significantly higher success rates in stimulated cycles in which multiple eggs and embryos are produced thereby allowing for competitive selection of the best-quality embryos for transfer.

Conceptually, the main components of IVF involve the stimulation of multi-follicular development, egg-retrieval with simultaneous sperm production (or retrieval), laboratory-based fertilisation, embryo culture and selection, embryo transfer and luteal support. As outlined in greater detail below for a typical IVF cycle, a number of refinements are often superimposed on this basic road-map so as to improve overall success rates whilst at the same time minimising the risks:

1. Pituitary down-regulation using GnRH agonists.

GnRH agonist analogues used in IVF carry an amino-acid substitution that reduces their susceptibility to enzymatic cleavage, resulting in a half-life 2.5-6 times longer than native GnRH. After an initial increase in gonadotrophin secretion (flare response), sustained exposure to these stable GnRH agonists induces down-regulation of the GnRH receptors on the gonadotrophs of the anterior pituitary thereby suppressing endogenous gonadotrophin secretion along with premature surges of endogenous LH. GnRH agonists (e.g. Buserelin) are administered by subcutaneous injection or nasal spray and are usually commenced around day-21 of the preceding cycle. Use of GnRH agonists greatly facilitates the timing of egg-collection and confers additional advantages such as increased numbers of retrieved eggs, improved pregnancy rates and reduced cycle cancellation. GnRH antagonists are an alternative to GnRH agonists and have the advantage of inducing immediate pituitary down-regulation without the flare effect thereby facilitating shorter treatment protocols.

However, on the downside, antagonists have been associated with lower clinical pregnancy rates.

2. Controlled ovarian stimulation using gonadotrophins

Ovarian reserve tests involving estimations of FSH, AMH and AFC amongst others are used by many units as a guide for tailoring individual stimulation regimes. A typical regime might start with a daily dose of 225 IU (3 ampoules) of hMG, which can then be increased (usually to a maximum of 6 ampoules) or decreased depending on the response.

3. Close monitoring of follicular development

The cornerstone of monitoring is transvaginal ultrasonography for tracking follicle growth (by measuring the diameters of individual follicles in two dimensions) and follicle numbers. Ultrasonography is often used in conjunction with serum oestradiol measurements. Large numbers of follicles (≥ 20) in combination with high serum oestradiol are harbingers of OHSS and require very close supervision for the onset of symptoms.

4. Induction of oocyte maturation using hCG as a surrogate LH surge.

Follicles usually grow at a rate of about 2-3 mm/day. When the majority of follicles are in the region of 18-20 mm, signifying that the contained oocytes are competent to resume and complete meiotic maturation into fertilisable eggs, a bolus of hCG is administered. A pre-requisite for progressing to the stage of egg-collection is usually a minimum of 3-4 ovulatory sized follicles.

5. Ultrasound-guided transvaginal egg collection (TVEC)

TVEC is timed to occur about 36 h post-hCG as this gives sufficient time for oocyte maturation but pre-empts ovulation into the peritoneal cavity. TVEC does not necessitate a general anesthetic and is commonly performed using conscious sedation. Using a transvaginal ultrasound probe mounted with a needle guide, a 16-gauge aspiration needle is advanced under ultrasonographic vision directly into each follicle in turn and the egg aspirated by applying suction, usually from a foot-operated pump. TVEC is a relatively safe procedure with the main risks being that of bleeding (>100 mls in 0.8%) and infection (0.6%).

6. Sperm production.

On the day of TVEC, a fresh sperm sample is obtained, usually by ejaculation, and analysed. In cases of obstructive azoospermia, such as after previous vasectomy, sperm can be surgically retrieved using techniques such as PESA (see above). In cases in which the likelihood of retrieving sperm is reduced, as in non-obstructive azoospermia, sperm retrieval techniques are not usually left for the day of TVEC in the event that so sperm are obtained. Often, because of the real possibility that sperm cannot be obtained using surgical retrieval, preparations would have been made for a back-up of donor sperm to be available, provided the couple are amenable and have been extensively counselled beforehand regarding the implications of donor treatment.

7. Insemination.

Provided the sperm sample is adequate, conventional insemination procedures involving the addition of sperm (5000 sperm per oocyte) to oocytes maintained within droplets of specialised culture medium will usually suffice. For sub-optimal semen parameters, fertilisation might require the use of ICSI (intracytoplasmic sperm injection) which involves the introduction of a single immobilised spermatozoon using a micro-needle that traverses the zona pellucida and the oolemma under direct vision provided by high-magnification optics. A relatively recent adjunct to ICSI that shows promise for enhancing success rates has been the use of very high magnification microscopy (over x6000) to pre-select only morphologically normal sperm for micro-injection (intracytoplasmic morphologically selected sperm injection or IMSI).

8. Assessment of fertilisation and embryonic development.

About 18-20 hours following insemination, the presence of two pronuclei provides evidence of fertilisation. Over the subsequent days, further embryonic development is closely monitored and embryo quality is graded. Embryos that have the capacity to develop to the blastocyst stage within the laboratory (usually by days 5-6) have higher implantation rates than earlier cleavage-stage embryos.

9. Embryo transfer.

The best quality embryos are selected (usually a maximum of two) for transfer. With the patient awake, the cervix is traversed using a catheter (preferably a soft catheter) that has been pre-loaded with embryos suspended in culture medium. With the catheter tip about 10 mm from the uterine fundus, embryos are very gently released into the mid-cavity. Accurate placement of the catheter can be aided by ultrasound.

10. Luteal support.

Following embryo transfer, luteal support in the form of progesterone administered either vaginally (e.g. Cyclogest 400-800 mg/day) or intramuscularly (e.g. Gestone 50-100 mg/day) is continued for 14-16 days, when a pregnancy test is performed.

In the UK, the most recent data available from the national regulatory body known as the HFEA (Human Fertilisation and Embryo Authority) shows an average live birth rate of about 24% per cycle of IVF started. IVF is known to be associated with a number of risks such as that of ectopic pregnancy (2-5% versus 1% following natural conception) and heterotopic pregnancy (0.1-0.3% versus 1:30,000). Another risk is that of OHSS, which occurs in its mild form in up to one-third of cycles and in moderate to severe forms in 3-8% of cycles. Multiple births and the attendant risk of prematurity is a profound risk for children born following IVF treatment. Recognising this, the HFEA set out a policy in January 2009 aimed at reducing the incidence of IVF-related multiple births to 10% over a period of years. Integral to this policy is a move towards transferring only one embryo at a time, elective single embryo transfer (eSET), whenever feasible. As a consequence of this campaign, eSET rates increased from 4.8% in 2008 to 14.7% in 2010 with a concomitant fall in multiple pregnancies from 26.7% to 22%.

In some cases, IVF may involve the use of donated eggs or donated sperm. Conceptually, the principles of treatment remain the same with the biggest challenge being in cycles involving donor eggs when the production of eggs from the donor has to be carefully synchronised with the endometrial preparation of the recipient.

New dimensions in fertility services

The reach of fertility services and the profile of patients who seek to benefit from fertility-related technology are expanding. Socioeconomic changes in the past four decades have seen increasing numbers of women entering the work-place and deferring attempts to conceive until their mid-thirties or later. This has resulted in the emergence of a significant cohort of patients in whom infertility is the consequence of the poor oocyte quality that is associated with advanced female age and which cannot be redressed by current forms of ART. In order to circumvent this biological phenomenon, some younger women are turning to pre-emptive measures by electively cryopreserving oocytes as a form of "fertility insurance policy". Whereas this approach was of limited use when cryopreservation involved slow-freezing techniques, advances in technology such as vitrification now make this a clinically viable option. Although at present, the only IVF-related option for women whose remaining eggs can no longer support normal pregnancy development is to use donor eggs, in the future, it may be possible to generate oocytes *de novo* in the lab using stem cell technology, a feat which has already been accomplished in the mouse model.

The opportunity for manipulating embryos offered by IVF has led to the development of preimplantation genetic diagnosis (PGD) which can prevent the transmission of genetically inherited disorders including single-gene defects (e.g. sickle cell disease), sex-linked disorders (e.g. haemophilia) and chromosomal disorders (e.g. translocations). During PGD, IVF is used to create embryos from which 1-2 cells are removed at the 8-10-cell stage and subjected to chromosomal/genetic analysis thereby enabling unaffected embryos to be identified and transferred. Unlike PGD which is used for defined genetic or chromosomal disorders, a related technique that uses much the same technology called preimplantation genetic screening (PGS) has been pursued with couples lacking a defined disorder but who possess risk factors that predispose to chromosomally imbalanced embryos, most notably advanced maternal age. In principle, by pre-selecting only chromosomally normal embryos for transfer and excluding aneuploid ones, PGS might be expected to improve pregnancy outcome. In practice however, this has not been borne out by randomised controlled data which do not therefore support the routine use of PGS as an adjunct to IVF.

Although fertility treatment has traditionally centred on approaches for by-passing or rectifying established fertility problems, a rapidly evolving facet of fertility management pertains to the preservation of reproductive potential in fully fertile individuals whose fertility is threatened, for instance by impending treatment for malignancy. With improving cancer survival rates, fertility preservation for young adults is becoming increasingly important. With regard to female fertility preservation, the current options are embryo

cryopreservation for those in stable relationships and more recently, oocyte vitrification for single women. Other techniques such as cryopreservation of ovarian biopsies and whole ovaries combined with either subsequent transplantation or follicle isolation for *in vitro* growth and maturation of oocytes are experimental.

KEY POINTS

- Infertility affects 1 in 7 couples.
- Investigation is warranted if conception fails to occur after 12 months of regular unprotected sexual intercourse.
- Earlier investigation is justified in the presence of features which suggest an underlying fertility problem such as irregular or absent menstruation.
- Categories of infertility include ovulatory disorders, male factor, tubal disease or uterine/peritoneal factors. Infertility is considered unexplained if none of these conditions is present.
- The commonest cause of ovulatory dysfunction is PCOS.
- PID, endometriosis and previous surgery are the main causes of tubal disease.
- Although many male factor problems are idiopathic, there is an increasing association with chromosomal and genetic defects.
- Investigations primarily involve tests for ovulation, tubal patency and semen quality.
- Treatments are tailored to the underlying condition and involve ovulation induction for ovulatory disorders; either tubal surgery or IVF for tubal disease; either IUI or IVF (with ICSI) for male factor problems, and; either SO + IUI or IVF for unexplained infertility.
- Endometriosis in its minimal and mild forms should be treated surgically with or without subsequent SO + IUI. More advanced disease can be treated surgically but may also

require IVF due to irreversible pelvic distortion.

- The role of uterine fibroids in infertility is controversial but those of the submucous variety should be removed prior to embarking on fertility treatments such as IVF.
- IVF and related technology, such as PGD and embryo/oocyte cryopreservation, have extended the reach of fertility-related services.

CHAPTER SEVEN

MISCARRIAGE

N Patravali, G Mires

Introduction

Miscarriage defined as the loss of a pregnancy before 24 weeks (28 weeks in the Caribbean) of gestation, is a common complication of early pregnancy with rates reported at up to 20% of pregnancies. The incidence may however actually be higher due to under reporting. Most miscarriages occur before 12 weeks of gestation and the risk reduces beyond this stage of pregnancy. This chapter considers the types and terminology of miscarriage, approaches to confirming a diagnosis via history, examination and investigation and current approaches to management.

Definition

A miscarriage can be defined as a spontaneous loss of intrauterine pregnancy that occurs prior to 24(or 28) weeks of gestation.

Types of miscarriage

Types of miscarriage can be defined according to:

1. Gestation at which it occurs,
2. Stage of miscarriage at the time of detection,
3. Frequency of miscarriage.

Gestation of miscarriage:

Most miscarriages occur before 12 completed weeks of gestation and are defined as *early* miscarriage. *Late* miscarriages are those that occur between 12 and 24 completed weeks of gestation. Pregnancy loss after 24 weeks is defined as pre term delivery with 24 weeks being the gestation at which a fetus becomes viable i.e. has a chance of survival.

Stage of miscarriage at the time of detection:

Miscarriage can also be defined according to the 'stage' at which it has occurred or has been detected. The following are types of miscarriage defined in this way:

Missed / silent miscarriage: Miscarriage detected incidentally with no prior symptoms. There is no fetal cardiac activity and often an empty gestation sac is seen on ultrasound scan.

Threatened miscarriage: Warning symptoms of bleeding and / or discomfort are present but the fetus is still alive on ultrasound. This type of miscarriage may or may not progress to

inevitable, incomplete or complete miscarriage. The pain and bleeding however may also settle and the pregnancy continues.

Inevitable miscarriage: Confirmed miscarriage accompanied by heavier bleeding and dilated cervix. There is a high possibility the pregnancy will be expelled spontaneously but even if this does not happen in this type of miscarriage there is no hope that the pregnancy will continue.

Incomplete miscarriage: In this type of miscarriage, there is partial expulsion of products of conception from the uterus. Again in this type of miscarriage the pregnancy cannot continue.

Complete miscarriage: In this type of miscarriage there is complete expulsion of products of conception from the uterus. The cervical internal os is closed and the uterus may be normal in size and the pregnancy test may revert back to negative.

Septic miscarriage: In this type of miscarriage usually following an incomplete miscarriage secondary infection of retained products can result in septicaemia and systemic symptoms e.g. fever, rigours.

Frequency of Miscarriage

As indicated above miscarriage is common and investigation of cause not always undertaken. However, if spontaneous miscarriage occurs on 3 consecutive pregnancies it is defined as recurrent miscarriage with a greater chance of there being an underlying recurring cause. This will be dealt with in a separate section of this chapter.

Physiology of Early Pregnancy

In order to understand the pathophysiology, investigation and management planning of miscarriage it is important to understand the basic physiology in the early weeks of pregnancy.

The fertilized ovum releases the hormone β - Human chorionic gonadotrophin (β -hCG), which prevents the corpus luteum from involuting and initiates the anatomical, endocrinological and physiological changes that occur in pregnancy. In the post conception period, at day nine post fertilisation (day 23 of a 28 day cycle), the β -hCG reaches a level of approximately 25 IU/l. Urine pregnancy kits can detect β -hCG at this level. First void morning samples of urine are preferred for testing as they potentially have higher concentrations of β -HCG. Serum levels can also be measured and for all practical purposes a doubling of β -hCG over a 48 hour period, is considered an indirect marker of viable intrauterine pregnancy.

An ultrasound scan can assess site, cardiac activity and gestational age in early pregnancy. The preferred method is transvaginal ultrasound to achieve easy access to the pelvic organs. An ultrasound scan at 4 weeks of gestation would be expected to demonstrate a gestational sac. By 5-6 weeks a yolk sac would be identifiable. Fetal cardiac activity can be detected by 6 weeks.

By 7-8 weeks a viable intrauterine pregnancy should demonstrate a gestation sac, yolk sac and fetal cardiac activity.

Serum progesterone levels can be useful adjunct in predicting the nature and outcome of the pregnancy. If the serum level is more than 25 nmol/L, it is likely that the pregnancy will be viable and intrauterine. Levels of progesterone lower than 5 nmol/L may suggest poorer prognosis or a failing pregnancy.

Table 1 Correlation of ultrasound and β -hCG in a normal pregnancy

Pregnancy in Weeks	Trans-abdominal ultrasound	Trans-vaginal Ultrasound	β-hCG (IU/L)
4- 5 weeks	---	GS	>1500
5-6 weeks	GS	GS+YS	3000- 5000
6-7 weeks	GS+YS+/-Fetal heart	GS+ YS+ fetal heart	> 20000

GS= Gestational sac, YS= Yolk sac. All the values and ultrasound findings are approximate and change according to individual cases, laboratory references and type of ultrasound machine used.

History

History taking is an important step in management of patients with miscarriage as a potential diagnosis.

Background before commencing:

There is a significant psychological impact for patients undergoing miscarriage. If a patient is suspected of undergoing a miscarriage, prior to the consultation session begins it is important to ensure that sufficient time is given to support the patient. Preferably such patients should be seen in secluded environments with a relative, friend and /or a nurse to support the patient. They should be allowed to stay with the patient during the history and examination if the patient wishes. It is also important to organise the necessary equipment e.g. light source, speculums etc. at the outset.

Content of history taking:

Age:

It is important to know the age of patient. The relevance of this is that the rate of miscarriage is higher in the very young and the older age group. Older women have an increased risk of chromosomal abnormalities such as Down's syndrome which are associated with early pregnancy loss.

Address:

This would help to organise care and understand the implications of day care management as opposed to inpatient management.

Gestational Age:

This is calculated from the first day of the last menstrual period. A gestational age will indicate whether the pregnancy is early or late and also allow assessment of findings e.g. HCG levels and ultrasound scans to be interpreted i.e. are levels lower than expected for the gestation. As the management for both is slightly different it is very important to calculate the age accurately.

Presenting complaint:

‘What symptom/s the patient came in with?’

In silent miscarriage there may not be any symptoms, and the diagnosis may be made as an incidental finding e.g. on routine ultrasound for booking purposes. Commonly patients may present with bleeding or pain in the abdomen. It is important to explore the nature of the bleeding and pain in terms of duration, quantity and nature of bleeding and site, severity, aggravating and relieving factors for pain in addition to any associated symptoms e.g. urinary, bowel, nausea and vomiting. If the bleeding is very heavy the patient may need inpatient admission and urgent management. Likewise ongoing abdominal pain may suggest an incomplete miscarriage or an ectopic pregnancy. In complete miscarriage the pain and bleeding may have subsided at the time of presentation and the patient may indicate that she has passed some ‘tissue’ (products of conception).

Obstetric History:

A detailed past obstetric history should be obtained. This should include details of all previous pregnancies including dates and outcomes such as gestation at delivery/pregnancy loss, antenatal, intrapartum, postpartum complications and details of babies i.e. birthweight and sex. The number of previous pregnancy losses may help to rule out recurrent miscarriages. A history of ectopic pregnancy may increase the chance of having a repeat ectopic pregnancy in future pregnancy by 10%. Ectopic pregnancy may present with similar symptoms to that of a miscarriage and will be included in the differential diagnosis.

Gynaecological History:

Certain conditions e.g. *polycystic ovarian syndrome* may cause erratic cycles and also have increased propensity of miscarriage. A previous history of STI’s may increase the risk of ectopic pregnancy. Fibroids which may present with menstrual symptoms can result in an increased risk of miscarriage.

Past Medical, Surgical History: Chronic conditions e.g. diabetes, lupus and antiphospholipid syndrome are associated with increased rates of miscarriage. Pre-conceptual management e.g. tight control of diabetes can reduce the risk.

Personal and Social History:

For effective management of the patients, it is important to understand their background. This is mainly to understand whether they are well supported at home and if any further management can be implemented effectively. Rehabilitation may be an issue depending on their occupation and social support.

Patients Ideas, Concerns, and Expectations:

In the consultation process it is important that patient's views and understanding is explored at the outset. This facilitates understanding background fears or actual expectations of the patients for the session. It is important to address these issues during the consultation to avoid dissatisfaction. Some patients may only want to know what is happening, but may not want any further treatment.

Examination:

General examination:

It is important to consider the general health of the patient at presentation. Assessment of pain is important and measurement of pulse, blood pressure and respiratory rate will provide an indication of the presence of shock. An elevated temperature may be an indicator of a septic miscarriage. Conjunctiva should be examined for signs of anaemia and checking capillary refill time and pallor act as guide checking for anaemia due to acute blood loss. Patients with very high or very low body mass index (BMI) are more prone for miscarriages⁵.

Abdominal examination:

In a patient presenting in early pregnancy with pain and/or bleeding it is important to exclude other diagnoses than miscarriage e.g. ectopic pregnancy or coincidental pathology e.g. appendicitis or torsion/haemorrhage into an ovarian cyst. Abdominal palpation may also help elicit the size of the uterus. It will also identify other masses in the abdomen and also tenderness and its localisation. This helps to differentiate different causes of pain which may not be related to gynaecology.

Vaginal and speculum examination:

Prior to embarking on a speculum and vaginal examination it is important to obtain consent from the patient. A chaperone should be present. Equipment should be made available prior to commencing the examination. This avoids delay and added embarrassment to the patient. Essentially this should include a speculum, lubricating gel, a charcoal swab (to rule out infection if necessary) and a light source.

It is important to inspect the vagina, cervix and the vulval area at the examination.

A speculum examination will allow visualisation of the cervix. This will confirm the presence of vaginal bleeding and exclude any local causes e.g. cervical erosion. It will also allow an assessment of whether the cervical os is open and whether products of conception are protruding through it.

A bimanual vaginal examination will allow an assessment of uterine size. This should be compared to the gestation. This examination is useful if ultrasound facilities are not available. In a missed miscarriage for example the uterus may be smaller than expected for gestation and in molar pregnancy larger. As assessment of tenderness can also be made particularly in the adnexae and on moving the cervix (cervical excitation) present in ectopic pregnancy. Bimanual examination will also help rule out any masses in the pelvis e.g. ovarian cyst.

Examination findings will depend on the stage and type of miscarriage. In early threatened miscarriage, the cervical os will be closed and uterus an appropriate size for gestation. In missed miscarriage the uterus may be smaller than expected with no active bleeding. In incomplete miscarriage the cervical os will be dilated and there may be products of conception protruding through it. These may be removed to prevent further bleeding. In septic miscarriage the vaginal loss will be offensive smelling and may be purulent. In molar pregnancies there may be 'grape-like' products of conception seen in the vagina.

In late miscarriages the cervical os may be dilated with bulging membranes. There may not be any associated symptoms in some cases of cervical incompetence.

It is important to send any specimen collected for a histopathological diagnosis to confirm pregnancy and to rule out any molar changes and in cases of suspected infection send an endocervical swab for culture and antibiotic sensitivity. Some of the typical findings are illustrated in the Table 2.

Table 2: Clinical findings in various types of miscarriage

<i>Type of Miscarriage</i>	<i>Findings on Per- Speculum Examination</i>
Silent	Cervical os closed , No bleeding
Threatened	Cervical os closed, some bleeding, No POCs
Inevitable /Incomplete	Cervical os open, Bleeding + POCs
Complete	Cervical os closed /slightly open,minimal bleeding +/- POCs in the vagina
Septic	Any of the above + offensive vaginal discharge
Molar Pregnancy	Any of the above +/- 'grape' like POCs in the vagina

Investigations

Initial investigations will include:

Full Blood count (FBC): It is important to determine the haemoglobin status. If there has been bleeding, this may help manage patients with iron therapy or using blood transfusion.

Coagulation studies: In cases of significant blood loss e.g. in ruptured ectopic with associated hypovolaemic shock coagulation studies will rule out developing disseminated intravascular coagulation.

Blood grouping: This will be of value to indicate not only the need for Anti –D injection, but also for cross matching blood if necessary. Such sample should be preserved in the laboratory, until the need for acute management of the patient is complete. It is a good practice to inquire from individual laboratories to determine their policy for grouping and saving blood samples.

Ultrasound scan: This is an invaluable investigation to assess the status of pregnancy. Preferably a trans-vaginal ultrasound scan should be performed in early pregnancy, as the uterus is still a pelvic organ at this stage.

An ultrasound scan is used to determine not only the site of pregnancy, but also to assess the cardiac activity (Figure 1). It can also determine the type of miscarriage. Retained products of conception can indicate incomplete miscarriage. If a clear midline is seen with clinical evidence of tissues being passed, then a diagnosis of complete miscarriage can be made. Ultrasound is also a useful tool to differentiate miscarriage from ectopic pregnancies as the symptoms of both conditions can overlap.

Table 3. Ultrasound features in various types of miscarriage

<i>Type of miscarriage</i>	<i>Ultrasound findings</i>
<i>Silent miscarriage</i>	<i>GS +/- YS + Fetal pole with no heart beat</i>
<i>Anembryonic pregnancy</i>	<i>GS > 20 mm with no fetal pole</i>
<i>Incomplete /threatened miscarriage</i>	<i>POCs in uterus (+ Clinical evidence of tissues or bleeding)</i>
<i>Complete miscarriage</i>	<i>Clear midline (+ Clinical evidence of tissues passed or previous ultrasound finding of Intrauterine pregnancy)</i>

GS= Gestational sac, YS= Yolk sac, POCs= products of conception

Screening for infection: It is important to screen for infections like Chlamydia if the patient were to require a surgical treatment option. Early treatment will reduce the chance of pelvic inflammatory disease by reducing the risk of ascending infection ⁶. Swabs for bacterial vaginosis can be taken if clinically indicated.

Additional investigations: These can be performed to find the cause of recurrent miscarriage, and is dealt in the relevant section.

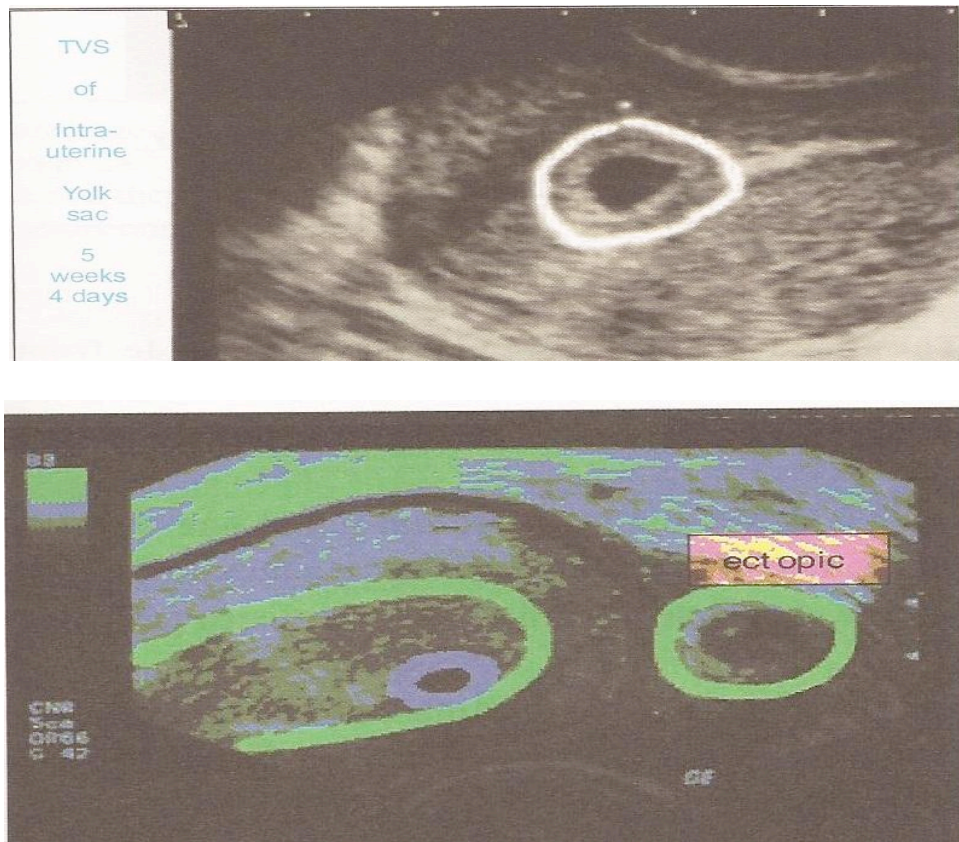


Figure 1. Pelvic ultrasound in early pregnancy

Treatment:

The treatment of a patient undergoing a miscarriage depends on several factors. It depends on their general condition at the time of presentation, the type of miscarriage, patient choice and the amenities available with the treating hospital. A comprehensive plan can only be made after considering these points along with any other influencing factors.

Treatment options can be classified into three categories:

1. *Expectant management*
2. *Medical management.*
3. *Surgical management*

1. Expectant management:

This approach can be adopted if the patient does not wish medical or surgical intervention for the management of her miscarriage. It is not appropriate for all patients and the clinical situation must be taken into account i.e. if she is bleeding heavily with shock this would not be an appropriate approach. Serum progesterone levels can be used to predict the pregnancies that are more likely to resolve spontaneously with low levels indicative of greatest success. In selected cases expectant management is an effective treatment option e.g. for an incomplete miscarriage, although patients with an intact gestation sac, should be warned that complete resolution may take several weeks. The efficacy is rated anywhere from 25% to 100% in the literature. Patients however should be warned about the unpredictability of the treatment and that they may still require surgical or medical intervention. Expectant management in addition to avoiding the need for intervention has been shown to reduce the chance of pelvic inflammatory disease and does not affect future fertility. This treatment should however be used only where there is an easy access to a 24 hour service and an emergency treatment can be offered. Adequate counselling and confirming patient understanding are essential prior to offering expectant management.

Medical management:

Medical management is a useful option in intact sacs or incomplete miscarriages.

It is facilitated with the use of two medications:

1. *Antiprogesterone- Mifepristone*
2. *Prostaglandin analogues- Misoprostol/ Gemeprost.*

Antiprogesterone(Mifepristone)

Mifepristone is used as a cervical priming agent prior to using prostaglandins in selected cases. It is usually used in intact gestational sacs in cases like silent / missed miscarriage, anembryonic pregnancy or threatened miscarriage. Most women complete the miscarriage without the use of further prostaglandin. Mifepristone is given as a single dose oral agent. The patient is allowed home for 48 hours. The patient is seen after, and prostaglandins are given orally or vaginally if required.

Prostaglandins (Misoprostol / Gemeprost):

Prostaglandins and their analogues can be used solely or in conjunction with Anti-progesterone in the medical management of miscarriage. Prostaglandins alone can be used in cases of incomplete or inevitable miscarriage. These are used sequentially at three to four hourly intervals according to the local policy, until the miscarriage is complete. A maximum of five doses are given. Most pregnancies miscarry at these doses.

There is a high acceptability factor for medical management of miscarriage ¹⁵. However increased pain and bleeding may be influencing factors for its acceptability. These may last

from two to three weeks following the treatment and patients should be warned about this during counselling for the treatment.

Surgical management

Surgical intervention is the first line treatment where the patient's condition is unstable due to haemorrhage. Vacuum aspiration is the method of choice where the sac is intact. Prostaglandins can be used prior to the procedure to help soften the cervix. This helps to prevent undue force being used to dilate the cervix and hence preventing serious injuries to the cervix and perforation of the uterus. Such procedures can be carried out under local or general anaesthetic or intravenous sedation.

The procedure involves dilatation of the cervix using graduated Hegar's dilators. This is followed by vacuum aspiration of the products of conception using suction cannula. Routine use of metal curette is not required. Where infection is suspected such procedures should be delayed for 12 hours to allow the administration of intravenous antibiotic cover.

Anti-D immunoglobulin injection:

It is important to offer patients with Rhesus negative status an Anti-D injection to prevent Rhesus isoimmunisation in future pregnancies. The recommended dose is 250 µgm for gestations below 20 weeks and 500µgm for gestations above 20 weeks gestation. It is recommended that all pregnancies undergoing surgical interventions and pregnancies over 12 weeks should get Anti-D. Complete miscarriages below 12 weeks do not require Anti-D. It should be only give to pregnancies below 12 weeks if associated with heavy bleeding or pain.

Recurrent miscarriage

Definition: The consecutive loss of three pregnancies is defined as recurrent miscarriage. The theoretical risk of 3 consecutive losses is approximately 0.34 %.

Investigations and treatment

1. Genetic tests

Products of conception should be sent for cytogenetic studies to identify a chromosomal abnormality as a potential cause of the recurrent miscarriages. In 3- 5 % of couples presenting with recurrent miscarriage at least one of the partners carries a balanced chromosomal translocation anomaly. One of the common anomalies is balanced reciprocal or Robertsonian translocations.

Treatment: Genetic counseling should be offered with a clinical geneticist. The chance of having a fetus with unbalanced translocation is around 5- 10 % in such cases and the chance of having a healthy normal pregnancy with natural conception in future is around 40- 50 %.

2. Assessment of anatomical factors

The reported prevalence of uterine anomalies (e.g. uterine septum) in recurrent miscarriage population ranges between 1.8% and 37.6%. The prevalence of such anomalies is higher in women with late miscarriages as compared to early miscarriages. Imaging including ultrasound and surgical techniques including hysteroscopy and laparoscopy can be used to diagnose uterine anomalies.

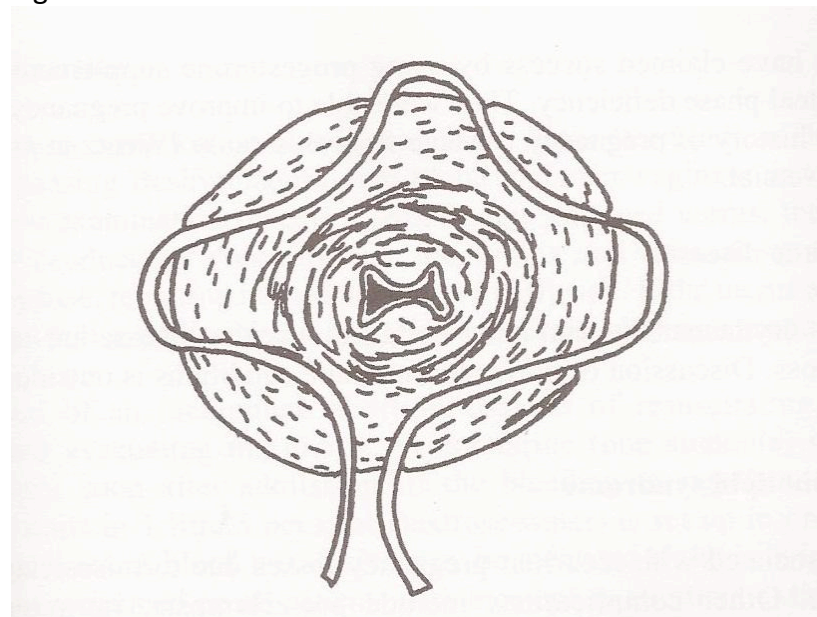
Treatment: Treatment using hysteroscope (telescopic surgery through the cervix) for such anatomical anomalies has a higher success than open techniques and should be offered at the first instance.

3. Cervical weakness

Some women with recurrent miscarriage have an element of cervical weakness. There are no current criteria to diagnose weakness in a non-pregnant state. It is diagnosed on the basis of history of late miscarriage preceded by painless dilatation of cervix or spontaneous rupture of membranes. The clinical diagnosis of cervical incompetence can be made from a history of three or more spontaneous miscarriages which typically occur in the second trimester. These miscarriages are usually painless, complete, rapid and are associated with minimal bleeding. There may also be progressive painless cervical dilatation and effacement in early pregnancy. This may be confirmed with a transvaginal ultrasound. Outside of pregnancy, the diagnosis may be confirmed with the Hegar dilator test or cervicography.

Treatment: Transvaginal ultrasound may be useful in diagnosing such cases in subsequent pregnancy. Selected cases may benefit having a cervical or a trans-abdominal cerclage. During this procedure mersilene tape is inserted around the cervix in an attempt to prevent its dilatation (Figure 2). The common vaginal method is the Macdonald stitch less commonly one may use the Shirodkar approach especially with a very short cervix.

Figure 2. Macdonald stitch



4. Antiphospholipid syndrome

Primary antiphospholipid syndrome (APS) refers to the association between antiphospholipid antibodies (aPL) and adverse pregnancy outcome or vascular thrombosis²⁴.

Adverse pregnancy outcomes include:

- a. Three or more consecutive miscarriages before ten weeks of gestation,
- b. One or more morphologically normal fetal deaths after the tenth week of gestation and
- c. One or more preterm births before the 34th week of gestation due to severe pre-eclampsia, eclampsia or placental insufficiency.

APS causes inhibition of trophoblastic function and differentiation. Later in pregnancy it causes thrombosis of utero-placental vasculature.

Investigation: APS is diagnosed if the women are positive on two occasions, six weeks apart with either medium to high titres of Anti-cardiolipin antibodies. (aCL) of IgG or IgM class or the presence of lupus anticoagulant.

Treatment: Thromboprophylaxis in the form of low dose aspirin and low dose heparin in the pregnancy significantly reduce the chance of miscarriage and improve live birth rates in women with recurrent miscarriage due to APS.

5. Inherited thrombophilic defects

Inherited thrombophilic defects, include activated protein C resistance (most commonly due to factor V Leiden gene mutation), deficiencies of protein C, S and antithrombin III, hyperhomocysteinaemia and prothrombin gene mutation. Thrombosis of the utero-pacental vasculature is considered the probable cause of recurrent miscarriage.

Treatment: Thromboprophylaxis may be considered in such high risk women as a form of treatment option.

There are certain conditions that are discredited as a cause of recurrent miscarriage.

Investigations for such conditions should not be performed as a routine work up for recurrent miscarriage.

Conditions not associated with recurrent miscarriage

- Occult diabetes
- Well controlled diabetes
- Occult hypothyroidism
- Polycystic ovarian syndrome

- Hyperprolactinaemia
- TORCH infection (toxoplasmosis, other [congenital syphilis and viruses], rubella, cytomegalovirus and herpes simplex virus)

Molar Pregnancy (Hydatidiform Mole)

Molar pregnancy belongs to a group of conditions known as gestational trophoblastic disease (GTD). The other conditions that are included in GTD are invasive mole, choriocarcinoma and the very rare placental site trophoblastic tumour (PSTT).

Molar pregnancy occurs in approximately one in 700 pregnancies. Higher incidences are seen in Asian ethnic populations.

Types and causes of Molar Pregnancy:

1. Complete Mole: This is caused by duplication of a sperm after fertilisation of an 'empty' ovum (80%). All the chromosomes are androgenic in origin. Rarely a dispermic fertilisation of an empty ovum may cause complete mole (20%).
2. Partial Mole: these are a result of dispermic fertilisation of an ovum (Figure 3). Most of partial moles are triploid (90%). The remainder may be tetraploid or mosaic.

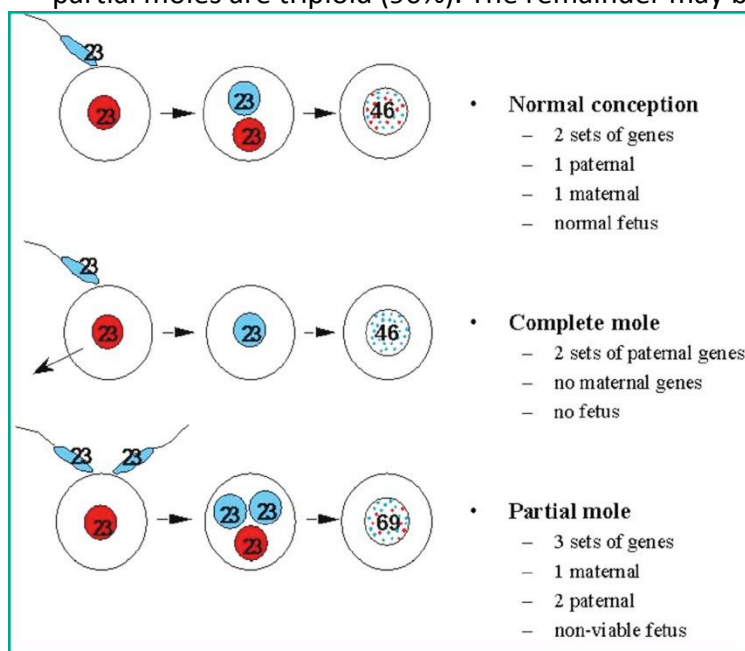


Figure 3: Genetic basis of molar pregnancy

Clinical presentation

Patient may present with clinical features of miscarriage e.g. irregular vaginal bleeding, hyperemesis, excessive uterine enlargement and failed pregnancy.

Investigations

Diagnostic investigations

Investigations that may be useful in diagnosis of molar pregnancy are:

Serum Beta-HCG: The serum level is raised, inconsistent with the gestational age. Levels usually greater than two multiples of median may be indicative of the condition.

Ultrasound scan: It is a very useful test in diagnosing molar pregnancy (Figure 4). Complete moles are easier to diagnose as compared to partial moles. Complete moles have a 'snowstorm' appearance of the tissues on ultrasound scan.

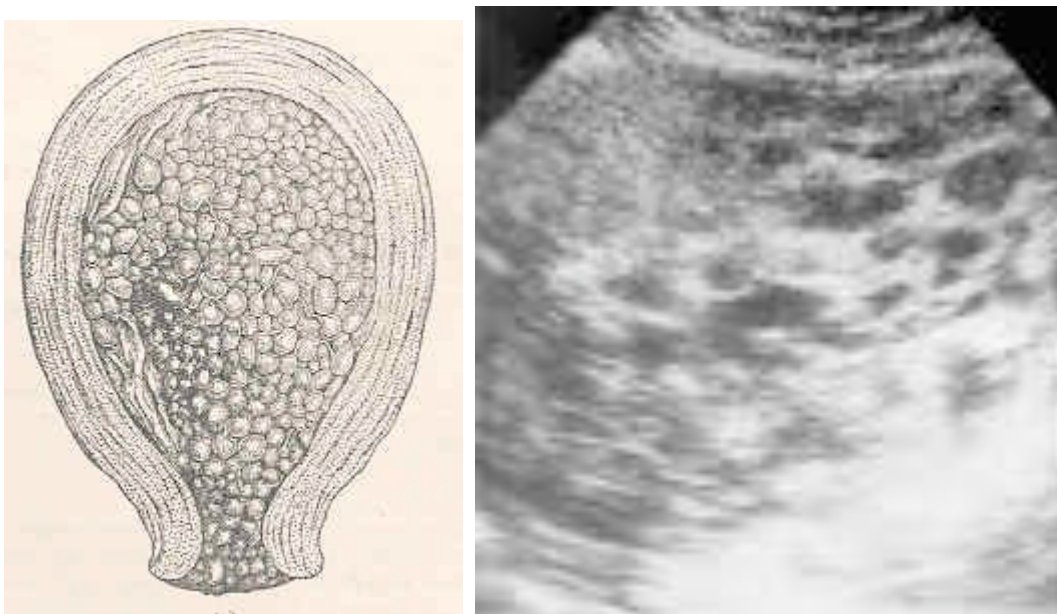


Figure 4. Complete Mole in the Uterus 'Snow storm appearance' on Ultrasound

Other investigations

Haemoglobin: Will be useful for assessing anaemia.

Thyroid status: Hyperthyroidism rarely may present with similar symptoms of hyperemesis.

Blood Group: These patients are more prone for bleeding during definitive management including evacuation of uterus. It is useful to know the blood group for organising blood prior to

evacuation. Also patient with rhesus negative blood type will benefit with an Anti-D injection to prevent isoimmunisation in future pregnancies.

Treatment

Surgical evacuation is the treatment of choice for complete and partial molar pregnancies. It should be performed by a senior professional as the risk of perforation and complications like haemorrhage is high. Exceptionally, partial moles with a large fetal size may benefit from medical management.

Ripening of cervix prior to evacuation is safe. Oxytocin infusion should not be used prior to evacuation due to the risk of embolisation of trophoblastic tissue. Ultrasound guidance during evacuation can be used as an adjunct to avoid complications and assess completeness of evacuation.

Follow up

Patient should be followed up closely following evacuation. In United Kingdom there are specialised centres that follow up women with gestational trophoblastic disease. These women are followed up using assessment of urine or blood HCG levels. If these levels return to normal within 56 days, then these women are followed up for up to six months.

If the HCG levels do not return to normal within 56 days then, they are followed up for up to six months from the date these level return to normal.

It is important to follow these women up to assess if the condition resolves completely or develops into gestational trophoblastic neoplasia (GTN). GTN are chemo-sensitive tumours and resolve completely with a good prognosis using chemotherapy. Either a single agent using Methotrexate or a Multiagent chemotherapy can be used with agents such as Methotrexate, Dactinomycin, Etoposide, Cyclophosphamide and Vincristine (EMACO).

KEY POINTS

- Miscarriage is defined as spontaneous loss of pregnancy prior to 24 weeks (28 weeks in the Caribbean) gestation.
- Types of miscarriage depend on gestation, stage and frequency of occurrence.
- Doubling of β -hCG over 48 hours may suggest viable intrauterine pregnancy.
- A viable intrauterine pregnancy should demonstrate a gestational sac, yolk sac and fetal cardiac activity by 7-8 weeks of gestation on ultrasound scan.
- Serum progesterone level may help predict outcomes of early pregnancy.
- There is a significant psychological impact in patients undergoing miscarriage.
- Good history taking will help in the management.
- It is important to understand patient's ideas, concerns and expectations in a consultation.
- A chaperone should be present during a pelvic examination.
- Specimens collected should be sent for histopathological diagnosis.
- A trans-vaginal ultrasound should be performed to assess the type of miscarriage.
- Chlamydia screening should be carried out to prevent PID in surgically treated patients.
- Patients can be managed by expectant, medical or surgical management protocols.
- Medical management uses prostaglandin analogues with or without anti-progesterone.
- Vacuum aspiration is the preferred method of choice for surgical evacuation.
- Anti -D should be offered to patients with Rhesus negative blood group.
- Specific investigations need to be carried out to assess the cause of recurrent miscarriage.
- Certain conditions are discredited as a cause of recurrent miscarriage and should not

be routinely investigated.

- **Molar pregnancies can present as a miscarriage.**
- **Molar pregnancies are either complete or partial.**
- **Complete moles are androgenic in origin.**
- **Ultrasound scans are useful in diagnosis of molar pregnancies.**
- **Complete moles have a 'snow storm' appearance on ultrasound scan.**
- **Surgical evacuation is the treatment of choice for molar pregnancy.**
- **Patients should be followed up for six months following the return of HCG level to normal.**
- **Gestational trophoblastic neoplasia can cause persistent high level of HCG.**
- **GTN has a good prognosis, either using single agent or multiagent chemotherapy.**

CHAPTER EIGHT

ECTOPIC PREGNANCY

Neelam Potdar, Justin C Konje

DEFINITION

Ectopic pregnancy is defined as the implantation of the fertilised ovum in a site other than the uterine endometrium. It is derived from the Greek word *ektopos*, meaning out of place.

HISTORY

Ectopic pregnancy was first described in the 11th century by the famous Arabic physician Abulcasis (936-1013) and until the 19th century it was known as a universally fatal event. In 1884, Robert Lawson Tait (1845-1899) reported successful treatment of tubal pregnancy with laparotomy and salpingectomy.

INCIDENCE

The incidence has been reported to vary between 0.25% and 1.5% of all pregnancies (including live births, medical termination of pregnancy and ectopic gestations). More recently, the incidence and prevalence of ectopic pregnancies are thought to have increased because of risk factors including advanced maternal age, tubal surgery, pelvic inflammatory disease (PID) and assisted reproduction techniques.

In the United Kingdom, there were 11.3 ectopic gestations per 1,000 pregnancies reported in the 8th report of the Confidential Enquiry into Maternal Deaths in the United Kingdom, (CMACE, 2011). In the USA, the incidence of ectopic pregnancy is about 2% of all pregnancies, having increased 5-fold since the 1970's.

MORBIDITY AND MORTALITY

Ectopic pregnancy is a cause of significant morbidity and mortality worldwide. In the United Kingdom, although the number of deaths from ectopic pregnancies has declined, most occur due to failure to make a diagnosis and the death rate per 100,000 estimated ectopic pregnancies was 16.9 for the 2006-2008 triennium (CEMACE, 2011). In the USA, ectopic pregnancy accounts for 10 % of all maternal deaths and is the leading cause of maternal mortality in the first trimester.

AETIOLOGY

Physiologically, the myoelectrical activity of the Fallopian tube allows the process of fertilisation and migration of the zygote into the uterine cavity for implantation. Oestrogens increase smooth muscle activity and progesterone decrease muscular tone. It is believed that delay in the transport of the fertilised ovum is involved in the pathogenesis of most ectopic pregnancies. As a result, the fertilised ovum becomes too large to pass through certain areas of the Fallopian tube and implantation occurs in the tube.

The cilia of the tubal epithelium are involved in transportation of oocytes towards the uterine cavity. Salpingitis results in loss of ciliated epithelium and subsequently delayed propulsion of the zygote/embryo towards the uterine cavity. Additionally, oestrogens stimulate epithelial cell hyperplasia and ciliogenesis, whereas high progesterone levels are associated with de-ciliation and epithelial atrophy. An imbalance in favour of high progesterone levels may explain the increased incidence of tubal pregnancy as occurs in women who take the progesterone-only pill or have a progesterone-containing intrauterine contraceptive device (IUCD) in-situ.

RISK FACTORS

All sexually active women are at risk of an ectopic pregnancy. Associated factors may be present only in 25%-50% of cases (Table 1).

Table 1. Risk factors for ectopic pregnancy

Previous tubal pregnancy
Previous tubal surgery
Pelvic inflammatory disease
Medical termination of pregnancy
Current IUCD users
Assisted conception
Salpingitis isthmica nodosa (SIN)
Smoking
Diethylstilbestrol (DES)

Previous tubal pregnancy

A woman who has experienced one ectopic pregnancy has a 10-20% chance of recurrence in subsequent pregnancy.

Previous tubal surgery

Sterilization

The risk of an ectopic is 10 times higher when a pregnancy occurs as a result of failure of tubal sterilization. The risk varies with the sterilization technique and the woman's age. Probable reasons are fistula formation and re-canalization of the proximal and distal stumps of the sterilised Fallopian tube. The cumulative probability of ectopic pregnancy for all methods of tubal sterilisation is 7.3 per 1000 procedures.

Reversal of sterilization

Following reconstruction of a cauterized tube approximately 15% of women who conceive have an ectopic pregnancy. The risk is reduced to 5% when the reversal is performed following Pomeroy's method or clip sterilization as these are less damaging to the tube.

Tubal reconstruction and repair

It remains unclear whether the increased incidence of ectopic pregnancy following reconstructive tubal surgery is due to the surgical procedure itself or to the pre-existing pathology of the ciliated tubal epithelium. Following microsurgical operations, 5% of patients presented with an ectopic pregnancy whereas 35% achieved an intrauterine pregnancy.

Pelvic inflammatory disease (PID)

The relationship between PID, tubal damage and ectopic pregnancy is well documented. Infections of the tube result in damage of the ciliated epithelium and formation of intraluminal adhesions and pockets. Consequently, there is entrapment of the zygote and ectopic implantation of the embryo. The incidence of tubal damage increases with successive episodes of PID. About 50% of Fallopian tubes removed for an ectopic pregnancy have histological evidence of salpingitis.

Chlamydia trachomatis infection can cause silent tubal damage and is a known cause of ectopic pregnancy. Patients with an anti-*Chlamydia trachomatis* titre of $\geq 1:64$ are three times more likely to have an ectopic pregnancy compared with negative controls.

Use of intrauterine contraceptive device (IUCD)

IUCDs prevent pregnancies; however, a woman who conceives with an IUCD in-situ is seven times more likely to have a tubal pregnancy compared to conception without contraception. With copper IUCDs, 4% of all accidental pregnancies are tubal whereas with progesterone-coated IUCDs 17% of failures are ectopic.

Medical termination of pregnancy

There is an association between the number of previous induced abortions and ectopic pregnancy (odds ratio 1.4 for one previous induced abortion and 1.9 for two or more). It is uncertain whether this is related to the spread of asymptomatic *Chlamydia trachomatis* infection or to the procedure itself.

Assisted conception

Ovulation induction with either Clomiphene citrate or human menopausal gonadotrophin is a predisposing factor for tubal implantation. The incidence of ectopic pregnancy is about 1-4% with ovulation induction and 4-5% following in-vitro fertilisation. These women may already have risk factors for ectopic pregnancy such as tubal pathology, tubal surgery or previous ectopic pregnancy.

Salpingitis isthmica nodosa (SIN)

SIN is diagnosed from histological evidence of tubal isthmic diverticula and may be suspected on hysterosalpingogram. It is seen significantly more in women with ectopic pregnancy (49%). The exact reason for its association with ectopic pregnancy is not known, although defective myoelectrical activity over the diverticula and entrapment of the embryo is a possible mechanical explanation.

Smoking

The risk of ectopic pregnancy is significantly higher in women who smoke. It is two times higher for women who smoke 10-12 cigarettes per day compared to non smokers. Nicotine induced defective ovum pick-up by the fimbrial end, and a reduction of ciliary beat frequency are possible explanatory mechanisms.

Diethylstilboestrol (DES)

Exposure to DES in-utero significantly increases the risk of ectopic pregnancy; 13% compared to 4% for women who were not exposed in-utero.

PATHOLOGY

Sites of ectopic pregnancy

Fallopian tubes are the commonest site for ectopic pregnancy. Figure 1 shows the various sites and their frequencies (Table 2). In recent years, ectopic Caesarean scar pregnancies are on the increase in keeping with the increasing rates of Caesarean sections.

Table 2. Sites of ectopic gestation

Fallopian tube	
Ampullary segment	80%
Isthmic segment	12%
Fimbrial end	5%
Interstitial and cornual	2%
Abdominal	1.4%
Ovarian	0.2%
Cervical	0.2%
Caesarean scar pregnancy	

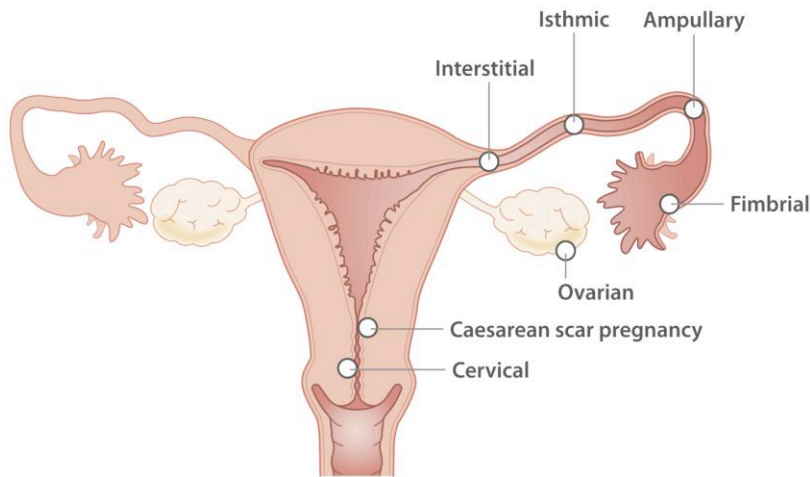


Figure 1. Sites of Ectopic Pregnancy

Natural progression of a tubal pregnancy

Spontaneous resolution

In a significant number of cases, the embryo dies at an early stage without any notable symptoms or signs of an ectopic pregnancy.

Tubal miscarriage

Here the products of conception/embryo are extruded from the fimbrial end into the peritoneal cavity and subsequently absorbed.

Unruptured tubal pregnancy

Initially the tube has normal macroscopic appearances but as the pregnancy progresses enlargement of the tube occurs at the site of implantation and it gets distended with a discoloured, dark red or purple appearance (Figure 2).

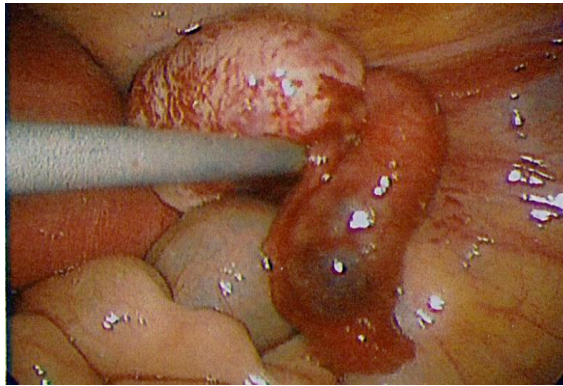


Figure 2. Unruptured tubal ectopic as seen at Laparoscopy

Tubal rupture

Invasion of the trophoblast within the Fallopian tube, results in destruction of blood vessels, local haemorrhage and thinning of the tubal wall. Rupture of the thinned tubal wall results in intraperitoneal bleeding and this is more frequently seen at the anti-mesenteric border. If the rupture involves the inferior and mesenteric part of the tube, blood can track between the two layers of the broad ligament.

Timing and site of tubal rupture

Rupture of tubal ectopic pregnancy depends to some extent on its implantation site. The ampullary segment is the widest segment of the tube and is the site for 80% of all tubal ectopic pregnancies. Rupture of an ampullary pregnancy occurs at 8-12 weeks of gestation. The isthmic segment is narrow and less distensible; hence, ectopic pregnancies at this site rupture early, at 6-8 weeks of gestation. The interstitial segment of the Fallopian tube is surrounded by the uterine myometrium, which can undergo hypertrophy to accommodate the enlarging conceptus (Figure 3). Ectopic pregnancy at this site can survive longer and present at a relatively late gestation of 12-14 weeks. Interstitial rupture can cause damage of the highly vascularized cornual end of the uterus, resulting in severe intra-abdominal haemorrhage.

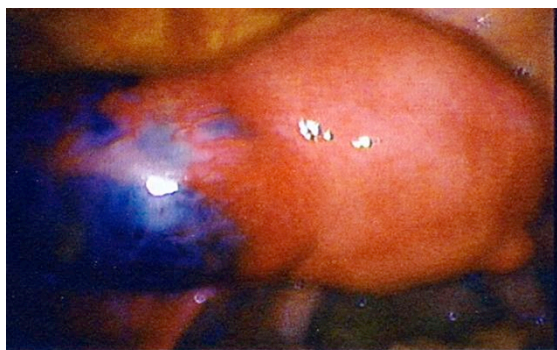


Figure 3. Unruptured interstitial pregnancy as seen at Laparoscopy

Histology

In the presence of ectopic pregnancy, the uterine endometrial glands demonstrate a histological pattern known as the Arias-Stella phenomenon. This is characterised by hyperplasia of glandular cells with large irregular hyperchromatic nuclei, cytoplasmic vacuolation and loss of cellular polarity. This reaction is non-specific and can be seen with an intrauterine pregnancy.

Simultaneous with the glandular changes, the endometrial stroma is converted into decidual tissue containing large polyhedral cells with hyperchromatic nuclei. As the pregnancy fails to continue the decidua undergoes gradual disintegration, giving rise to the intermittent, occasionally heavy, vaginal bleeding that occurs in ectopic pregnancy. In some cases the decidua may be detached abruptly and passed as a flat, triangular, reddish-brown piece of tissue called a 'decidual cast'.

DIAGNOSIS

Symptoms and signs

Ectopic pregnancy remains a diagnostic challenge. It should be considered as an important differential in any woman of reproductive age group who presents with the triad of amenorrhoea, abdominal pain and irregular vaginal bleeding. The index of suspicion should be high if the patient has any of the risk factor(s) identified in Table 1. Symptoms and signs associated with an ectopic pregnancy are shown in Table 3.

Table 3. Symptoms and signs of ectopic pregnancy

Symptoms and signs	Cases (%)
Abdominal pain	99
Generalised	44
Unilateral	33
Radiating to the shoulder	22
Abnormal uterine bleeding	74

Amenorrhoea ≤ 12 weeks	68
Syncopal symptoms	37
Adnexal tenderness	96
Normal size uterus	71

Abdominal pain

This is the most common symptom, can be generalized or localized pain in the pelvis (unilateral or bilateral) and/or radiate to the shoulder. The generalized abdominal pain is usually due to rupture of an ectopic pregnancy and intraperitoneal haemorrhage. Shoulder tip pain (Kehr's sign) is also an indirect indication of intraperitoneal haemorrhage. Accumulation of blood in the sub-diaphragmatic region stimulates the phrenic nerve and creates shoulder tip pain. Localized pain may be due to distension of the Fallopian tube. The pain may be sudden or progressive and continuous or intermittent.

Amenorrhoea and abnormal uterine bleeding

Most patients present with amenorrhoea of at least 6 weeks duration. Abnormal uterine bleeding occurs in 75% of women with an ectopic pregnancy. The bleeding is often light, recurrent and results from detachment of the uterine decidua. Even with a history of a recent scanty or light period, a pregnancy test should be performed.

Other symptoms

Patients may present with syncopal attacks, which could be due to the severe pain, haemorrhage, hypovolemia or anaemia. Other atypical symptoms include diarrhoea or vomiting, which should not be ignored and a pregnancy test should always be performed in a woman of reproductive age group presenting with these symptoms.

Physical examination

Examination should include early assessment of vital signs and examination of the abdomen and pelvis. The *general* condition of the patient can vary from slight pallor to a state of haemodynamic shock. *Abdominal* examination may reveal generalized or localized tenderness with or without guarding and rebound tenderness. A typical finding with a ruptured ectopic pregnancy is tachycardia without pyrexia.

An unusual feature is the Cullen's sign. This is bluish discoloration of the skin around the umbilicus caused by a considerable quantity of free blood in the peritoneal cavity. However, this sign is rare and its absence does not exclude massive intraperitoneal haemorrhage.

Pelvic examination findings vary from a completely normal pelvis to the presence of a large, fixed, soft and tender mass. An adnexal mass may be palpable in up to 55% of cases. The uterus

may be slightly enlarged but its size does not normally correspond to the gestational age. Cervical motion tenderness may or may not be present. A tender boggy mass in the pouch of Douglas, when present, usually represents collection of intraperitoneal blood.

Clinical presentation

The presentation of an ectopic pregnancy can be acute or sub-acute.

Acute presentation

This is usually a consequence of rupture and the ensuing intraperitoneal haemorrhage with or without haemodynamic shock. The patient is often pale, hypotensive and tachycardic. She may complain of shoulder tip pain or urge to defecate. Abdominal examination commonly reveals generalized and rebound tenderness. Vaginal examination may reveal tenderness in both adnexae and associated cervical motion tenderness. However, vaginal examination in patients who present with an acute abdomen due to a ruptured ectopic pregnancy is generally considered unnecessary and potentially dangerous for the following reasons: (a) generalized haemoperitoneum and pain often mean that specific information cannot be elicited, (b) the patient is often very uncomfortable and the assessment is thus difficult and inadequate and (c) it could result in total rupture of the ectopic pregnancy and delay in management of the patient. In recent years, acute presentation is becoming less common due to increased patient awareness and early referral to the hospital for evaluation of 'suspected ectopic' pregnancies.

Sub-acute presentation

When the process of tubal miscarriage or rupture is gradual the presentation of ectopic pregnancy is sub-acute. According to Stabile (1996a, b), this is the group of women who are symptomatic but clinically stable. A history of a missed period and recurrent episodes of light vaginal bleeding may exist. The circulatory system adjusts and the patient is haemodynamically stable. On bimanual examination there may be localized tenderness in one of the fornices and cervical motion tenderness is often present. Sub-acute presentation occurs in 80-90% of ectopic pregnancies. In cases with such a presentation the establishment of an accurate diagnosis becomes more difficult hence the need for further investigations.

INVESTIGATIONS

Biochemical tests

Serum β -hCG

Human chorionic gonadotrophin (hCG) is a glycoprotein with a molecular weight of 36,700 dalton and it is secreted by the syncytiotrophoblasts. It is heterodimeric, composed of two non-covalently linked α and β subunits. The α -subunit is identical to that of follicle stimulating

hormone and luteinizing hormone, whereas the β -subunit is specific to hCG. In response to β -hCG, the corpus luteum produces increasing concentrations of progesterone and other steroid hormones required to maintain early pregnancy, until the placenta takes over by the 8th to 12th week of gestation. Serum concentrations of β -hCG are approximately 1000 IU/L (third international standard) at 4 weeks of pregnancy, and it increases exponentially during the first six weeks of gestation, reaching 20,000 to 250,000 IU/L at 10 weeks of gestation.

Most commercially available urine pregnancy tests can detect β -hCG concentrations above 25 IU/L, which corresponds to day's 24-25 of a regular 28-day cycle. Using highly sensitive assays (detection limit 0.1-0.3 IU/L) serum β -hCG can be detected in the maternal circulation around the time of implantation. The non-pregnant level of serum β -hCG are <5 IU/L. A single measurement of β -hCG may be useful to exclude the diagnosis of pregnancy.

Serial β -hCG measurements

Since the normal doubling time of β -hCG is 2.2 days and its half-life is 32 to 37 hours, serial measurements of β -hCG help to distinguish normal from abnormal pregnancies. With a normal intrauterine viable pregnancy, serum β -hCG doubles in 48 hours. Kadar *et al* (1981) first reported a method of screening for ectopic pregnancy based on β -hCG doubling time. An increase in serum β -hCG of <66% over 48 hours is suggestive of an ectopic pregnancy or a failing intrauterine pregnancy (using an 85% confidence interval for β -hCG levels). Serum β -hCG levels can plateau or fall sub-optimally (<15%) in 48 hours. This should again raise the suspicion of an ectopic pregnancy or a failing pregnancy. Fifteen percent of normal pregnancies can have abnormal doubling time and 13% of ectopic pregnancies can have a normal doubling time. Hence, serum β -hCG levels should not be used on their own for diagnosis of ectopic pregnancy, but should be interpreted in conjunction with clinical presentation and ultrasound scan findings.

Serum Progesterone

Progesterone is produced entirely by the corpus luteum until 6 weeks of gestation and thereafter, the contribution from the trophoblast begins. In failing pregnancies, whether an ectopic or miscarriage, progesterone concentrations are expected to be low compared to values in healthy ongoing pregnancies. Progesterone levels >25 nmol/L are 'likely to indicate' and >60 nmol/L are 'strongly associated with' pregnancies subsequently shown to be normal. A single serum progesterone measurement is good at predicting a viable intrauterine or failed pregnancy, but it is not useful for locating the site of pregnancy.

Other protein and steroid markers

Various steroid and protein markers have been studied but none of these has been found to make a significant impact on clinical practice.

Ultrasound Scan

Ultrasound has become an essential tool in the assessment and diagnosis of suspected ectopic pregnancies. The transvaginal approach is superior to the abdominal approach as the proximity of the vaginal probe to the pelvic structures and use of high-frequency transducers (5-7 MHz) significantly improve resolution. The operator should systematically check the uterus, both adnexae including ovaries and the pouch of Douglas for evidence of free fluid (indicative of intraperitoneal bleeding). The ultrasound features of an early pregnancy including gestational sac, yolk sac, fetal pole and fetal heart activity should be checked and documented.

In 20% of patients with an ectopic pregnancy, collection of fluid within the uterine cavity results in the formation of a pseudo-gestational sac which mimics a true intrauterine sac. A true gestational sac is eccentrically placed within the uterine cavity with the *double ring* sign and an intact midline endometrial echo. Whereas, a pseudosac is surrounded by a single layer of tissue, tends to follow the contour of the cavity and in a longitudinal section of the uterus the midline endometrial echo cannot be seen because of the fluid. Ultrasound scan features of an ectopic pregnancy are as shown in Table 4.

Table 4. Ultrasound scan features of ectopic gestation

An empty uterus (28%) (Figure 4)

Characteristic adnexal mass (35%): seen as hyperechogenic tubal ring (doughnut or bagel sign) with empty uterus (Figure 5)

An intrauterine sac or pseudo-gestational sac (25%)

Ectopic gestation sac (12%) with or without a yolk sac and fetal cardiac activity and an empty uterus

Varying amount of fluid in the pouch of Douglas (25%)

Corpus luteum may be present (85%)

In summary, ultrasound scan in early pregnancy may show (a) an early intrauterine pregnancy (cardiac activity, miscarriage or uncertain viability), (b) ectopic pregnancy (viable or non-viable), and (c) neither an intra nor extrauterine pregnancy otherwise referred to as pregnancy of unknown location (PUL), which is discussed later (Figure 4).

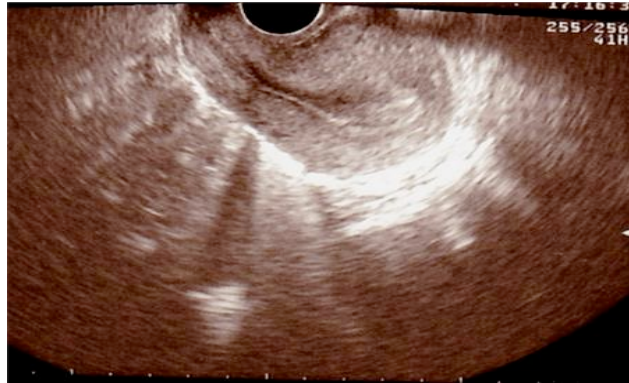


Figure 4. Transvaginal ultrasound image of an empty uterus

Serum β -hCG and Ultrasound scan interpretation

The absence of an intrauterine gestational sac on transabdominal scan (TA) with serum β -hCG levels of ≥ 6500 IU/L has an 86% positive predictive value for the presence of an ectopic pregnancy. Failure to visualize an intrauterine gestational sac by transvaginal (TV) ultrasound if β -hCG concentrations are ≥ 1000 -2000 IU/L indicates an abnormal intrauterine pregnancy, a recent miscarriage or an ectopic pregnancy (Figure 5).

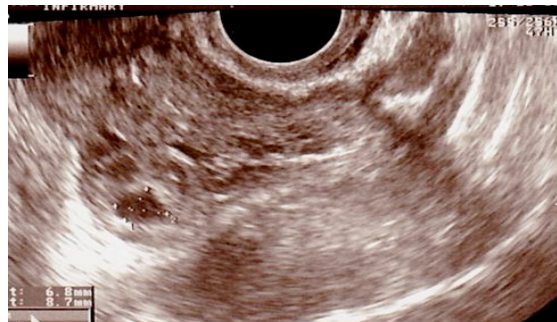


Figure 5. Transvaginal ultrasound image of adnexal mass (doughnut sign)

Doppler ultrasound

This is a non-invasive biophysical method of investigating patterns of blood flow. In ectopic pregnancies, Doppler flow imaging shows the presence of high vascularity at the periphery of an adnexal mass.

Diagnostic laparoscopy

With the availability of sensitive quantitative serum β -hCG assays and high resolution transvaginal ultrasound, diagnostic laparoscopy is seldom used as a primary diagnostic tool.

Culdocentesis

Culdocentesis is rarely used for the diagnosis of ectopic pregnancy. However, in parts of the world where facilities are lacking, aspirating blood from the pouch of Douglas which does not clot on standing is most likely suggestive of an ectopic pregnancy.

DIFFERENTIAL DIAGNOSIS

The differential diagnosis of an ectopic pregnancy includes ruptured corpus luteum, threatened or incomplete miscarriage, pelvic inflammatory disease and degenerating fibroids.

PREGNANCY OF UNKNOWN LOCATION (PUL)

In recent years, PUL has been used as a diagnosis in patients with a positive pregnancy test with no evidence of intra or extra-uterine pregnancy or retained products of conception on an initial TV scan. PUL is identified in 8-31% of women referred for ultrasound assessment in early pregnancy.

Subsequent follow-up in woman with PUL results in the diagnosis of either (a) an intrauterine pregnancy that was too early to be visualised on the initial ultrasound scan, (b) an ectopic pregnancy that was too early to be visualised or was missed on the initial ultrasound scan (7-20%), (c) a spontaneously resolving pregnancy known as a failing PUL (50%-70%), which would include both intrauterine and extrauterine pregnancies, or (d) persistent PUL, where biochemical findings mimic an ectopic pregnancy with suboptimal serum β -hCG levels that fail to decline, and there is still failure to localize pregnancy on further ultrasound scan or laparoscopy (Figure 6).

For clinically stable women expectant management has been demonstrated to be safe. PULs are followed up with serial hormone measurements, repeat TV scans and possible laparoscopy.

Management of Ectopic Pregnancy

Treatment of ectopic pregnancy can be conservative with preservation of the affected tube or radical with the removal of the affected tube. Depending upon the clinical presentation and findings three treatment options are available. These include expectant, medical or surgical management.

Expectant Management

The essential prerequisite for successful expectant management is appropriate patient selection as shown in Table 5. Women managed expectantly should be followed up twice weekly with serial β -hCG measurements and weekly TV scans to assess reduction in the size of an adnexal mass. Thereafter, weekly β -hCG concentrations and TV scans should be undertaken until β -hCG concentrations are less than 10 IU/L (Table 5). Women should be counselled about the importance of compliance and should be within easy access of medical help.

Table 5. Criteria for Expectant and Medical Management

Expectant management	Medical management
Clinically stable asymptomatic patient	Clinically stable asymptomatic patient
Baseline serum β -hCG concentrations <1000-2000 IU/L	Pre-treatment serum β -hCG concentration <3000 IU/L
Tubal mass <2 cm	Tubal mass <3 cm
Absence of recognizable fetal parts on ultrasound scan	Absence of fetal heart on ultrasound scan
Absence of haemoperitoneum	Minimal haemoperitoneum <50-100 mL

Medical Treatment

The desire to retain fertility, minimizing surgical morbidity and cost are the main reasons for medical management of ectopic pregnancies. The most frequently used drug for medical management is Methotrexate (MTX). Hreschchyshyn *et al* (1965) were the first to report on the use of MTX in the management of ectopic pregnancy.

Mode of action and pharmacokinetics

MTX, formerly known as Amethopterin, is a potent folic acid antagonist. It inhibits the action of the enzyme dihydrofolate reductase and 5-methyl tetrahydro-folate coenzyme. These enzymes are essential for the conversion of dihydrofolic acid into tetrahydrofolic acid and the conversion of deoxyuridylic acid into thymidylic acid, which is critical for the synthesis of DNA during the S phase of the cell cycle. MTX is ideal for inhibition of rapidly growing cells such as trophoblasts. The peak serum concentrations are reached within 30-60 minutes. Up to 90% of the drug is secreted unchanged in the urine within 48 hours.

Side-effects of Methotrexate

The main side-effects are from suppression of bone marrow and epithelial cells of the gastrointestinal tract. The possible side-effects include: abdominal pain, conjunctivitis, stomatitis, gastritis, diarrhoea, alopecia (reversible), myelosuppression, liver toxicity, photosensitivity (skin reaction), pneumonitis, renal toxicity and anaphylaxis. Single-dose regimens, currently used for the non-surgical treatment of ectopic pregnancy are associated with minimal side-effects.

Guidelines for the prescription of Methotrexate

Before prescribing MTX, baseline full blood count, serum liver function tests and serum renal function tests should be done to prevent and achieve early identification of serious side-effects.

Caution should be exercised with concurrent use of non-steroidal anti-inflammatory drugs as these may adversely affect renal function and cause a further increase in the serum MTX concentrations. MTX morbidity can be reduced by the use of the 'antidote' folinic acid which is then converted to 5-methyltetrahydrofolate.

Criteria for use of Medical Treatment (MTX)

Table 4 shows the inclusion criteria for appropriate patient selection. Medical treatment is contraindicated in symptomatic haemodynamically unstable patients, presence of haemoperitoneum, initial pre-treatment β -hCG concentration >3,000 IU/L, tubal diameter >3 cm, presence of fetal heartbeat on ultrasound scan and evidence of liver disease, renal insufficiency or myelosuppression. Prior to commencing medical treatment, women should be given clear information about the possible need for further treatment, adverse effects and they should be able to return easily for assessment at any time during follow-up. Patients should be advised regarding adequate compliance, the need to avoid alcohol, aspirin, non-steroidal inflammatory drugs or folic acid supplements, avoidance of sexual intercourse and sunlight exposure and delay of pregnancy for at least three to six months after treatment because of the teratogenic effect of MTX.

Routes of administration and dose

MTX can be administered systemically (intravenously or intramuscularly) or locally using a laparoscopic or TV ultrasound-guided approach. Protocols for systemic administration of MTX include:

- variable single dose regimen (1 mg/kg or 50 mg/m² of body surface area)
- standard single dose (60-200 mg IM)
- fixed multiple dose regimen of MTX with folinic acid

The success rates of all three regimens are comparable. MTX in a single IM dose of 50 mg/m² body surface area is associated with a complete reabsorption in 92%, a subsequent intrauterine pregnancy rate of 58% and a recurrent ectopic pregnancy rate of 9%.

Follow up with single dose regimen

Serum β -hCG levels should be checked on days 4 and 7 after the first dose of MTX and a further dose given if β -hCG levels have failed to fall by more than 15% between days four and seven. The level of serum β -hCG is likely to increase between days 1 and 4 after treatment as the destruction of the trophoblast cells releases more β -hCG into the systemic circulation. About 15% of women require more than a single dose of methotrexate and 7% experience tubal rupture during follow-up.

Abdominal pain occurs in 30-50% of patients 6-7 days after the administration of MTX, which could be due to imminent tubal rupture, pre-existing tubal rupture or post-MTX pain. Post-MTX

pain is likely due to the destruction of the trophoblasts and a resultant bleeding leading to peritoneal irritation. Close monitoring of patients is essential to avoid unnecessary surgical intervention. The pain usually subsides within 24-48 hours. Routine ultrasonography is not necessary after MTX treatment. There is no correlation between resolution of β -hCG levels and sonographic findings.

SURGICAL TREATMENT

Factors which determine either a laparoscopic or laparotomy approach to surgical treatment include:

- the haemodynamic condition of the patient
- the size and the location of the ectopic pregnancy
- the laparoscopic experience of the surgeon, and
- the availability of appropriate equipment for laparoscopic surgery

Laparotomy should be considered where the patient is haemodynamically unstable, there is a suspicion of extensive abdominal and pelvic adhesions making laparoscopy difficult or certain cases of cornual pregnancy. A ruptured ectopic pregnancy does not necessarily require laparotomy; the main aim of treatment should be patient safety and the use of the most expedient method. Table 6 shows the advantages of laparoscopy.

Table 6. Advantages of laparoscopic approach

Less intraoperative blood loss
Lower analgesic requirements
Shorter duration of hospitalization and convalescence
Fewer adhesions at the surgical site compared with laparotomy
Cost effective as early recovery, less time off work

Procedures for surgical treatment

These include fimbrial evacuation, salpingotomy, salpingostomy or salpingectomy.

Fimbrial evacuation

This procedure referred to as 'milking' is easily performed at laparotomy. When the pregnancy lies in the fimbrial segment of the Fallopian tube, it can be gently squeezed out of the tube. It is associated with a two-fold increase in the recurrence of ectopic pregnancy.

Linear salpingostomy/salpingotomy

Here an incision is made on the anti-mesenteric border of the Fallopian tube with a needle-point monopolar cutting diathermy, scissors or laser and the tubal lumen is exposed. The products of conception are removed with grasping forceps. Sometimes the laparoscopic suction/irrigation probe may be used for aqua dissection and removal of the products. Irrigation of the tubal lumen after removal of products of conception is essential. The incision can either be closed with a fine non-absorbable suture or left open (salpingotomy) for healing by secondary intention. Either approach results in satisfactory healing of the tubal wall.

Linear salpingotomy should be the procedure of choice in the presence of contralateral tubal disease and where the patient wishes to retain her potential for future fertility. Conservative tubal surgery is associated with a small risk of persistent trophoblastic disease, therefore serial β -hCG should be measured starting 3 days post surgery. If the levels fail to drop by >35% of pre-surgery levels, MTX can be used for further treatment. The possibility of repeat ectopic pregnancy in the conserved tube should also be discussed if salpingotomy is being considered.

Salpingectomy

Removal of the Fallopian tube is the most common method for the treatment of ectopic pregnancy. It can be performed via laparotomy or laparoscopy. Laparoscopic salpingectomy can be performed with a pretied loop ligature, bipolar diathermy and scissors or disposable stapling devices.

Salpingectomy is indicated in the presence of a ruptured ectopic pregnancy, a recurrent ectopic pregnancy in a tube previously treated conservatively, a grossly diseased or distorted tube (irrespective of whether it has been previously operated upon or not). In the presence of a healthy contralateral tube there is no clear evidence that salpingotomy should be used in preference to salpingectomy.

Reproductive outcome

Table 7 shows the comparative reproductive outcome following treatment for ectopic pregnancy.

Table 7. Reproductive outcomes with different treatments

	Expectant	Salpingotomy	Salpingectomy
Intrauterine pregnancy rate	83.7%	55%	49.3%

Recurrent pregnancy	ectopic	8%	14%	10%
Persistent trophoblast disease		-	4.8-11%	none

NON TUBAL ECTOPIC PREGNANCY

Interstitial pregnancy

Although it is a variety of tubal pregnancy, it is discussed in this section. It occurs in 1:2500-5000 live births and comprises 2% of all ectopic pregnancies. The signs and symptoms are similar to those of tubal ectopic pregnancy. Rupture of the uterine wall is the most frequent outcome and haemorrhage is usually severe. Ultrasound diagnosis is made by visualisation of the interstitial line adjoining the gestational sac and the lateral aspect of the uterine cavity, and continuation of the myometrial mantle around the ectopic sac. Differential diagnosis includes cornual myoma and pregnancy in one horn of a bicornuate uterus.

Cervical pregnancy

A cervical pregnancy implants entirely within the cervical canal. The incidence ranges from 1:2500 to 1:18000 deliveries. Predisposing factors include previous instrumentation of the endocervical canal, previous Caesarean delivery, Asherman's syndrome and exposure to DES.

Cervical pregnancy causes profuse vaginal bleeding without associated cramping pain. On bimanual examination, the external os is usually dilated and the cervix feels distended or globular with a small uterus. Ultrasound scan shows no evidence of an intrauterine gestational sac, a ballooned cervical canal/barrel shaped cervix, a gestational sac in the endocervix below the uterine arteries and a closed internal os. A Doppler ultrasound will show blood flow around the sac.

The differential diagnoses include an intrauterine pregnancy with a low implantation site (isthmico-cervical pregnancy), spontaneous miscarriage, cervical carcinoma, cervical submucosal myoma and trophoblastic tumour.

Caesarean scar ectopic pregnancy

The incidence is not well known as there are only a few case reports of first trimester caesarean scar ectopic pregnancies described in the literature. Risk factors include previous Caesarean section, myomectomy, adenomyosis, previous dilatation and curettage and manual removal of the placenta. Clinical presentation can be painless vaginal bleeding or hypovolemic shock and

uterine scar rupture. Ultrasound diagnostic criteria include an empty uterine cavity, a gestational sac located anteriorly at the level of the internal cervical os covering the visible or presumed site of previous lower segment Caesarean scar, evidence of functional trophoblastic circulation on Doppler examination and a negative 'sliding organs sign'. In some cases hysteroscopy and MRI have also been used to aid in the diagnosis.

Ovarian pregnancy

Ovarian pregnancy occurs in 0.2-1 % of all ectopic pregnancies. The incidence ranges from 1:4000 to 1:7000 deliveries. Risk factors include current use of an IUCD and assisted reproduction techniques.

The symptoms and signs of an ovarian pregnancy are similar to those of a tubal pregnancy. The diagnosis is usually made at laparoscopy or laparotomy. The classic diagnostic criteria of Spiegelberg (1878) are:

- 1) Gestational sac must occupy a portion of the ovary.
- 2) Gestational sac must be connected to the uterus by the ovarian ligament.
- 3) Ovarian tissue must be identified in the wall of the sac.
- 4) Fallopian tube on the affected side of the pelvis must be intact.

TV scan appearances are of a cyst on or within the ovary with a wide outer echogenic ring. They are usually mistaken for a corpus luteum. Three-dimensional ultrasound imaging can be useful in differentiating a corpus luteum from an ovarian pregnancy. Differential diagnoses include tubal pregnancy and an ovarian cyst.

Abdominal pregnancy

There are two types of abdominal pregnancy:

- 1) primary peritoneal implantation of the embryo and
- 2) secondary peritoneal implantation of the conceptus following a tubal miscarriage or rupture, or less often, after uterine rupture with subsequent implantation within the abdomen.

Abdominal pregnancy represents approximately 1.4% of all ectopic pregnancies and its incidence varies from 1:3000 to 1:10000 deliveries. These are associated with high maternal and perinatal mortality. The incidence of congenital fetal abnormalities ranges from 35% to 75%.

The classic criteria for the diagnosis of primary abdominal pregnancy are:

- 1) No evidence of uteroplacental fistula

- 2) Presence of normal tubes and ovaries with no evidence of a recent or past pregnancy and
- 3) Pregnancy attached only to the peritoneal surface.

In the first and early second trimesters, the symptoms are those of an ectopic pregnancy. As the pregnancy advances, unexplained abdominal pain, occasional vomiting, diarrhoea or constipation may occur. Fetal movements are very marked or painful and felt high in the abdomen. Braxton-Hicks contractions are absent. Abdominal palpation may reveal a persistently abnormal fetal lie, abdominal tenderness and easily palpable fetal parts. The fetus gives the impression that it lies just under the skin of the abdomen. On bimanual examination the uterus is usually normal in size and the cervix is long, firm and displaced. Although an abdominal ultrasound often provides an unequivocal diagnosis, in some cases, this may not be possible, although computerized tomography and magnetic resonance imaging have also been used to confirm the diagnosis. A transvaginal ultrasound scan showing a normal/bulky uterus and a normal endometrial echo adjacent to a gestational sac will clinch the diagnosis (especially in obese women where a scan may not be very reliable)

Cornual pregnancy

The term cornual pregnancy indicates pregnancy in one horn of a bicornuate uterus or in the rudimentary horn of a unicornuate uterus. It is very rare (1:100 000 pregnancies) but carries a 5% maternal mortality. Asymmetrical enlargement of the early pregnant uterus should suggest uterine anomaly. Ultrasound features include a single interstitial part of the Fallopian tube in the main uterine body, a gestational sac which is mobile and separate from the uterus, surrounded by myometrium and a vascular pedicle adjoining the gestational sac to the unicornuate uterus.

Heterotopic pregnancy

Heterotopic pregnancy is defined as the simultaneous occurrence of an intrauterine and an extrauterine pregnancy. In spontaneous conception, the incidence has traditionally been quoted as to 1:30,000 pregnancies. However, the incidence has risen to 1 in 3,889 pregnancies due to the increased incidence of genital infections and the use of ovulation induction agents. With assisted conception techniques, particularly in-vitro fertilization and embryo transfer, 1-3% of all clinical pregnancies are heterotopic.

An unusual abdominal pain in the presence of a spontaneous miscarriage or profuse uterine bleeding with signs of peritoneal irritation should suggest a heterotopic pregnancy. The diagnosis depends on maintaining a high index of suspicion, particularly for the high-risk group and early TV scan. If the diagnosis is early, the prognosis of the intrauterine pregnancy is more favourable. After a surgical procedure, the intrauterine pregnancy continues in approximately 75% of patients.

Management of Non Tubal Ectopic Pregnancy

Management should be tailored depending upon the clinical presentation and investigations (Table 8).

Table 8. Management options for non tubal ectopic pregnancy

Ectopic pregnancy	Management Options
Interstitial	Medical with TV scan guided MTX local injection or Surgical with cornual resection and salpingotomy. In the past wedge resection and hysterectomy were performed.
Cervical	Medical with MTX: systemic or ultrasound guided local injection (MTX or potassium chloride) in the pregnancy sac. Surgical with suction curettage and tamponade using 30 mL foley catheter or cervical suture. Occasionally need uterine artery or internal iliac artery ligation, rarely hysterectomy
Caesarean Scar	TV scan guided local injection of MTX (25 mg) in the ectopic gestation (70%-80% success rate). Surgical as cervical pregnancy management, uterine artery embolisation can be done to achieve tamponade
Ovarian	Surgical with removal of ectopic and conservation of ovary, rarely salpingo-oophorectomy. Medical management with systemic MTX can be considered.
Abdominal	Early diagnosis: MTX or laparoscopic removal. Late diagnosis: laparotomy followed by removal of fetus +/- placenta left to resorb or removed depending upon its attachment. Risk of haemorrhage and infection very high.
Cornual	Medical or Surgical with excision of the rudimentary horn and the tube on affected side.
Heterotopic	Manage ectopic pregnancy as above: either surgical or expectant management if very small ectopic.

Anti-D IMMUNOGLOBULIN

All non-sensitised, rhesus negative women with suspected or confirmed ectopic pregnancy should receive 250 IU (50 µg) of anti-D immunoglobulin.

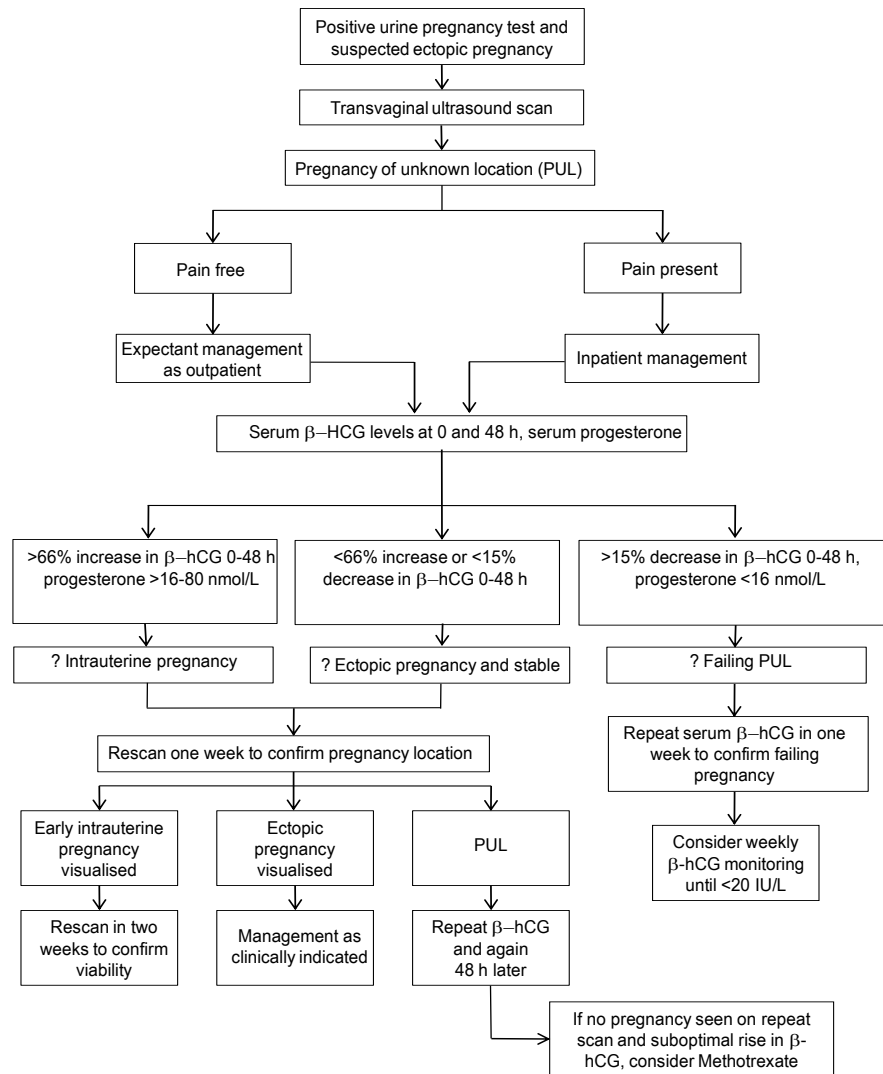


Figure 6. Algorithm for management of PUL and Ectopic Pregnancy

KEY POINTS

- The incidence of ectopic pregnancy is increasing mainly due to the increased incidence of PID, assisted conception and increased maternal age.
- Ectopic pregnancy can be present at different sites in the pelvis, but tubal pregnancies are the commonest.
- Diagnosis of ectopic pregnancy is challenging and there should be a high index of suspicion in reproductive age group in women presenting with a positive pregnancy test, abdominal pain and vaginal bleeding.
- Serum hCG and ultrasound scan features should be used in conjunction with clinical findings to confirm diagnosis and plan management.
- Treatment can be expectant, medical or surgical and patients should be selected appropriately for successful outcome.
- All rhesus negative women should receive Anti-D immunoglobulin.
- Amongst non-tubal ectopic pregnancies, Caesarean scar pregnancies are on the rise

CHAPTER NINE

UTERINE FIBROIDS

Horace Fletcher

Uterine fibroids (leiomyomata uteri) are common tumours arising from the uterine muscle. They have been found in one study in 80% of uteri at postmortem in Caucasian women but are more common in women of African origin.

Only 50% of women with fibroids will have symptoms as they can vary in size and number and position in the uterus.

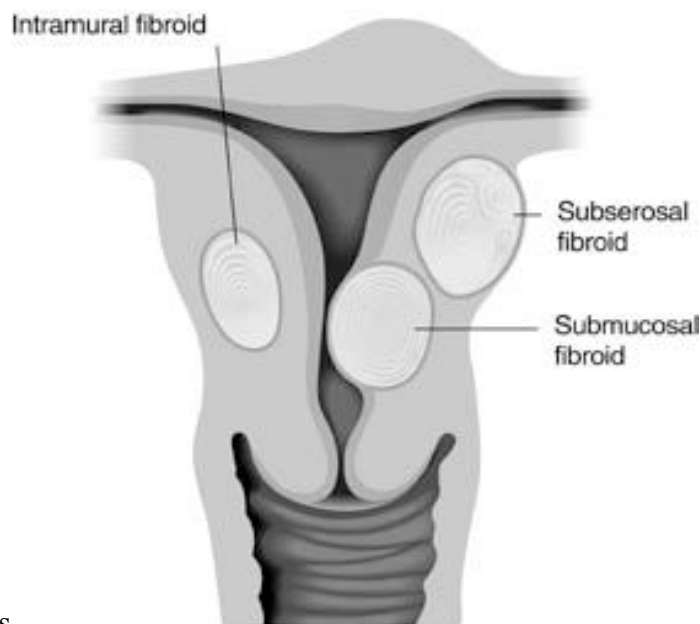


Figure 1. Fibroids

Clinical findings

Women with fibroids are usually in the reproductive age group and are usually nulliparous or had their last pregnancy more than 5 years prior to presentation. The usual complaints are heavy, prolonged, painful menstrual periods – with subsequent symptoms of anaemia such as dyspnoea, easy fatiguability, palpitations, headache, orthopnoea, and fainting spells. When the anaemia is severe they can present with congestive cardiac failure. Other symptoms include a distended abdomen, pressure on the bladder with frequency of micturition. Constipation is a rare complaint. However, complications can arise which may lead to other symptoms.

Large low posterior fibroids can cause urinary retention if they are wedged in the pelvis and kink the urethra. Large fibroids can also obstruct the ureters leading to hydronephrosis and hydroureter, pelvic/ureteral obstruction with renal failure, if both kidneys are affected. Deep vein thrombosis is also a problem that can occur from obstruction of the pelvic veins.

Fibroids can also have other complications such as degeneration – the most common being red degeneration, which occurs commonly in pregnancy as the fibroids outgrow their blood supply and the center becomes ischemic and necrotic. They may also become calcified and this calcification has been known to cause intestinal obstruction.

Fibroids which are pedunculated can undergo torsion leading to acute abdominal pain. Malignant transformation with leiomyosarcoma is very rare occurring in less than 1% of fibroids. In these patients the diagnosis is suspected if there is sudden rapid growth of fibroids in perimenopausal women.

Subserous fibroids are believed to be associated with infertility and recurrent spontaneous abortion.

Examination findings

Fibroids are usually found as a mass arising from the pelvis. The common differential diagnosis includes a full bladder, pregnancy, an ovarian or tubal mass. Gastro-intestinal tumours and retroperitoneal masses arising from blood vessels, nerves, cartilage, bone, connective tissue and large lymph nodes can also cause some confusion. In 15% of cases of suspected uterine fibroids the patient has an ovarian tumour. Full examination of patient is therefore needed to rule out severe anaemia and congestive cardiac failure. One should also check the lungs for crackles, breasts for tumour (spread to ovaries), abdominal hepatosplenomegaly and ascites. Assessment of the characteristics of the mass such as size, character, mobility must be performed, and most importantly ability to get below it and also if it moves with the cervix on bimanual abdominal vaginal examination. (abdominal-rectal bimanual examination in virgo intacta patient).

Investigations: complete blood count (anaemia, thrombocytosis and polycythemia are all possible), urea, creatinine and electrolytes (for renal impairment) and in patients where the diagnosis is in doubt a CA 125 may be useful. Ultrasonography may also be useful to determine ureteric obstruction and rule out an ovarian mass.

Definitive Treatment

No treatment is needed in most patients who have no symptoms. In fact, treatment in some patients may result in more harm than good. Patients who have significant symptoms such as heavy menstrual bleeding, pain or compression mass effect, need treatment.

Conservative Treatment

This includes analgesics for painful fibroids. Iron replacement in anaemic patients (make sure that the anaemia is due to iron deficiency before starting). Some patients with severe anaemia and cardiac decompensation will need blood transfusion but the cause of the anaemia must be established before starting iron or blood transfusion, as fibroids are very common and anaemia can be due to other reasons in patients with asymptomatic fibroids.

Conservative drug treatment also includes Depo Provera and the more expensive gonadotrophin releasing hormone antagonist (GnRH). Both will decrease bleeding and allow the patient to improve the haemoglobin (Hb) prior to surgery. GnRH also shrinks fibroids making them smaller and decreasing the need for non cosmetic sub-umbilical incision. However, GnRH cannot be used as definitive treatment because it can only be given 3-6 months as it causes osteoporosis. One year after stopping GnRH fibroids usually return to original size.

Surgical Treatment

1. Conservative surgery

Myomectomy involves the removal of the fibroids and uterus is left intact. This can be done by hysteroscopy for submucous (fibroids), laparoscopy for subserous fibroids and laparotomy for all types of fibroids. The main complications of myomectomy are haemorrhage primary (during surgery) or secondary (after surgery) and postoperative adhesion formation leading to infertility in 50% cases or intestinal obstruction in about 1% myomectomies. Haemorrhage is prevented by use of haemostatic agents such as peri-vascular vasopressin and also by careful surgical technique ensuring that all dead space is closed at myomectomy. Prevention of adhesions is also dependent on attention to detail at surgery, avoiding too many incisions especially posteriorly on the uterus and those near to the Fallopian tubes and ovaries. Adhesion barriers such as Interceed® are also of some value. Conversion to hysterectomy is needed occasionally due to bleeding. So when doing myomectomy patients should always be warned of the possibility of hysterectomy especially if they are nulliparous.

Hysterectomy for fibroids

Patients who have completed their family are usually better off having a hysterectomy for fibroids. This can be total or subtotal abdominal, laparotomy or laparoscopic. Vaginal hysterectomy is becoming established for large fibroids but this should only be attempted by very experienced vaginal surgeons.

Hysterectomy results in less bleeding and results in permanent cure of fibroids. However, with hysterectomy there is a risk of damage to surrounding structures, such as the bladder, ureters and bowel also there is a risk of postoperative vault prolapse in total abdominal hysterectomy.

KEY POINTS

- 1. Fibroids are very common – 50% of patients are asymptomatic.**
- 2. Patients who have symptoms usually complain of painful heavy periods or a mass effect, large abdomen, urinary frequency or retention.**
- 3. Complications include: anaemia, venous thrombo-embolism (DVT), renal failure, spontaneous abortion, ureteric obstruction, and red degeneration.**
- 4. Treatment can be conservative, medical or surgical.**
- 5. Myomectomy is useful and women who wish to have children.**
- 6. Hysterectomy is definitive treatment in women who have completed their family.**
- 7. Complications of the treatment include anaesthetic, intra-operative bleeding or damage to surrounding structures with infertility and intestinal obstruction and post operative wound dehiscence, DVT, vault prolapse, bleeding, and fistula.**
- 8. Newer treatments include uterine artery embolization, radiofrequency and MRI focussed high energy ultrasound waves.**

CHAPTER TEN

ENDOMETRIOSIS

Natasha Mohammed

Introduction

Endometriosis is a common cause of morbidity in women. It has been associated with pain and infertility and can impact on a woman's physical, mental and social well being. It mainly affects women between the ages of 25 – 40 years old. Endometriosis affects two million women in the UK and during 2008-2009 it was responsible for 16,643 episodes of admitted patient care, accounting for 22,024 occupied bed days (1). In the USA it is the third leading cause of gynaecological hospitalization. The precise incidence and prevalence remains elusive and the condition still arouses much controversy with regard to its treatment and management. It is considered a disease of enigmas.

Endometriosis is defined as the presence of endometrial like tissue outside the uterus.

- The endometrial tissue is morphologically and biologically similar to normal endometrium.
- The ectopic endometrial glands and stroma are functional and active and respond to hormones and drugs in a similar manner to the eutopic endometrium.
- The ectopic endometrium undergoes cyclical changes and bleeding contributing to a local inflammatory reaction and fibrous adhesion formation.
- The content of the ectopic endometrial glands is dark, altered, tarry menstrual blood.

Prevalence/Incidence

This varies with studies done in different countries and ethnic groups. Japanese women have been reported to have twice the incidence of Caucasian women. It is also more common in the USA and UK than in the Caribbean. Although endometriosis is seen primarily among women of reproductive age (10%), this disease also can affect postmenopausal women and adolescents, especially those with uterine abnormalities.

Aetiology

The exact aetiology of endometriosis is unknown but several hypotheses about its origin exist.

1. Sampson's theory of retrograde menstruation:

In 1927 Sampson proposed that endometrial tissue during menstruation passes through the Fallopian tube and then attaches and proliferates at ectopic sites in the peritoneal cavity. However, studies using laparoscopy have demonstrated that retrograde menstruation is a nearly universal phenomenon in women with patent Fallopian tubes. Halme et al (3) found bloody peritoneal fluid in 90% of women having laparoscopy near the time of menses. Liu and Hitchcock (4) also reported retrograde spill of menstrual blood in 76% of women undergoing laparoscopy during menstruation. Subsequent animal studies have confirmed that the endometrial tissue could implant and grow within the abdominal cavity (5). Findings of endometriotic foci in a laparotomy scar, Caesarean section scar or within an episiotomy scar following delivery further supports this metastatic theory. It is not surprising that patients with Mullerian anomalies and obstructed menstrual flow through the vagina have an increased risk of endometriosis. Sampson's theory is the most widely recognised theory on the genesis of endometriosis but it is likely that it oversimplifies the disease process.

2. Meyer's theory (Metaplastic transformation):

Meyer in 1919 proposed that metaplasia of the coelomic epithelium is the origin of endometriosis. This is based on embryologic studies demonstrating that Mullerian ducts, germinal epithelium of the ovary, and pelvic peritoneum are all derived from the same embryologic precursor. This theory has been difficult to prove experimentally and attempts to transform peritoneal cells into endometrial like tissues have been unsuccessful. However, recently it has been suggested that stem cells may be a source of endometriosis and recent studies have suggested that circulating cells from bone marrow may differentiate into endometriosis at peritoneum and other sites.

3. Lymphatic or haematogenous spread distant to pelvis (Halban's theory):

An extensive communication of lymphatics has been demonstrated between the uterus, ovaries, tubes, pelvic and vaginal lymph nodes, kidney, and umbilicus. Metastasis of endometrial cells through the lymphatic system to these areas is therefore anatomically possible. Vascular or lymphatic metastasis may explain rare cases of endometriosis occurring in locations remote from the peritoneal cavity such as the lungs, bone, biceps muscle, peripheral nerves, and the brain. However, this theory does not explain why ectopic tissue is able to become established and prosper in these locations.

Immunology of Endometriosis

Since retrograde menstruation is a nearly universal phenomenon in women with patent Fallopian tubes, other factors are likely to be involved in encouraging or allowing endometrial cells to escape from the uterus and to attach and proliferate outside of the uterine cavity. Susceptibility to endometriosis is thought to depend on the complex interaction of genetic, immunologic, hormonal and environmental factors.

Increasing evidence points to the role of immunologic factors and angiogenesis in the disease pathogenesis. Women with endometriosis appear to have altered function of peritoneal macrophages, natural killer cells and lymphocytes, as well as changes in growth factors and inflammatory mediators in the peritoneal fluid. This failure of the ability of the immune system at the uterine, systemic or peritoneal level to clear retrograde viable menstrual debris and to protect the peritoneal cavity from the invasion and adhesion of endometrial tissue may be involved in the onset of endometriosis.

Increased knowledge about immune aspects of the pathogenesis of endometriosis may allow for the development of novel medical therapies for this debilitating condition.

Genetic and risk factors

Endometriosis appears to be a multifactorial genetic disorder, in which allelic variants of many genes (including cancer susceptibility genes and genes coding for cytochrome P450 enzymes, nuclear receptors and immunologic mediators) can pre-dispose women to develop endometriosis, depending on environmental conditions.

First degree relatives of women with endometriosis have been reported to have endometriosis 4 to 8 times that of the general population.

Earlier age of menarche, shorter cycle lengths, heavy prolonged menstrual flow have been associated with increased risk of developing endometriosis.

Decreased risk has been associated with smoking, greater parity (>4), lactation and exercise.

Pathology and sites

The disease may occur anywhere in the body including the kidney, ureter and pleura. Commonly affected are the organ and tissues of the abdomen and pelvis including the omentum.

Visceral lesions are often multiple. The ovary is the most common site (30-40%) of cases especially the posterior surfaces and is almost always bilateral.

The uterus, pelvic peritoneum, uterosacral ligaments and recto-vaginal septum are other common sites. The pelvic colon, rectum, appendix, bladder, round ligament, umbilicus, vagina and perineum are other sites.

The extent of the disease varies from a few, small lesions on otherwise normal pelvic organs to large, ovarian endometriotic cysts (endometriomas) and/or extensive fibrosis and adhesion formation causing marked distortion of pelvic anatomy.

History

Patients with endometriosis may present with many different symptoms. Endometriosis is the underlying cause in 15% of cases of pelvic pain. The variety of symptoms and their different presentations lead to difficulties in diagnosing the condition. Some patients may have no symptoms and the disease may be coincidentally seen at laparoscopy or laparotomy. Likewise patients with minimal disease may experience severe debilitating symptoms. Hence, the symptomatology is not always reflected by the severity of the disease.

Pain

- Chronic pelvic pain is one of the common symptoms. It is estimated that up to 70-90% of women with chronic pelvic pain have endometriosis. Pelvic pain may be expressed as lower abdominal pain or back pain, pain of defecation (dyschezia), loin pain, pain on exercise, during micturition or at the time of ovulation. Pain can be severe enough to cause significant incapacitation.
- Dysmenorrhoea: this often occurs with pelvic pain. Women commonly describe a background of constant dragging, aching pain which may be exacerbated by the menses. The pain can occur before, during or even after menstruation. Although the dysmenorrhoea is mostly secondary, primary dysmenorrhoea in an adolescent can be due to endometriosis. It is typically congestive in nature.

Pelvic pain or dysmenorrhoea is thought to be related to both the direct and indirect effects of endometriotic lesions. Direct painful effects are due to growth into surrounding tissues with cyclic bleeding of an endometriotic lesion and the formation of accompanying fibrosis especially in the areas of the body with rich nerve supply. It may also be due to stretching of tissues by the menstrual process. Indirect effects include the production of cytokines and other substances (e.g. nerve growth factors) by the endometriotic cells and activated macrophages which may be increased by bleeding from the lesion. The pain also relates to tissue damage and fixation of organs from scar and adhesion formation.

Dyspareunia: Deep dyspareunia results from stretching at intercourse of the involved pelvic tissues such as the fixed retroverted uterus, the uterosacral ligaments, rectovaginal septum or pressure on an enlarged ovary or endometrioma.

Menorrhagia: Menorrhagia may occur with adenomyosis. When irregular bleeding from the uterine endometrium occurs, it is possible that ovarian endometriosis has so disrupted the

usual ovarian hormone production, that the endometrium is reflecting this abnormality by irregular bleeding in the form of pre- or post-menstrual spotting or intermenstrual bleeding. Postcoital bleeding can result from endometriosis in the vagina or the cervix.

Bowel symptoms: Increasing constipation or discomfort during defecation (dyschezia) may occur when endometriosis involves the rectosigmoid area. Rarely extensive involvement of the gut can lead to subacute intestinal obstruction. Endometriosis of the bowel is essentially a serosal involvement but ulceration into the lumen can occur and cause cyclical bleeding from the bowel (Figure 1).



Figure 1. Endometriosis

Bladder symptoms: Rare involvement of the bladder may cause cyclical haematuria.

Table 1. Summary of symptoms

• Dysmenorrhoea
Infertility
• Pelvic pain
• Lower abdominal pain or back pain

Endometriosis is present in 20-40% of women with infertility and 30-50% of women with endometriosis are infertile (Table 1). In normal couples the probability of achieving a pregnancy in any single month is 15-20% (monthly cycle fecundity rate (MFR) of 0.15-0.2), whereas untreated women with endometriosis conceive less frequently with MFR less than 0.05. In more severe cases of endometriosis it is obvious that the disease is a cause of the patient's infertility. In such cases fertility is compromised as a consequence of the anatomical distortion of the pelvis caused by adhesions or endometriotic ovarian cysts, or both. However, in minimal or mild cases of endometriosis, the exact nature of the relation between the disease and

infertility is unclear. For example, it is unclear how superficial peritoneal lesions would give rise to infertility, especially if they were distant from the frimbria of the tubes or ovaries.

Endometriosis can also occur alongside pelvic inflammatory disease to cause infertility. However, unlike an infection, endometriosis does not damage the luminal epithelium of the Fallopian tube and thus surgery is more likely to be successful. It is now universally accepted, however, that there is no place for medical treatment with drugs that are also potent contraceptives in the treatment of patients who are infertile.

Examination findings

Abdominal examination: It is possible to demonstrate local tender lesion in a Caesarean section or laparotomy scar. Lesions may also be palpable at the umbilicus or other laparoscopy port sites.

Speculum examination: One may visualize lesions on the vagina. These will be seen as clear red or as bluish cysts or nodules and are most commonly found in the posterior fornix or on the cervix.

Bimanual examination: The finding of pelvic tenderness, a fixed, retroverted uterus, tender uterosacral ligaments or enlarged ovaries is suggestive of endometriosis. The diagnosis is more certain if deeply infiltrating nodules are palpated on the uterosacral ligaments or in the pouch of Douglas.

The detection of nodules is improved by performing the clinical examination during menstruation, although patient acceptance may be an issue.

It is worth remembering that the clinical examination may be normal. Thus if the symptoms are convincing then despite normal clinical findings it will be wise to arrange further investigations to exclude or confirm the presence of endometriosis. Table 2 lists some possible differential diagnosis of endometriosis.

Table 2. Differential diagnosis of endometriosis

-
- Chronic pelvic inflammatory disease
 - Pelvic adhesions
 - Uterine fibroids
 - Torsion or haemorrhage into an ovarian cyst
 - Bowel: irritable bowel syndrome, diverticulitis, inflammatory bowel disease (Crohn's disease or ulcerative colitis), bowel obstruction
 - Musculoskeletal
-

Investigations

1. **Serum markers e.g. CA 125:** Elevated serum CA 125 levels are usually associated with ovarian malignancy but can be reported in association with endometriosis. Many studies have investigated serum CA 125 levels in women with surgically confirmed endometriosis and the conclusion is that the test's performance in diagnosing all disease stages was limited. Compared with laparoscopy, measuring serum CA 125 levels has no value as a diagnostic tool. However, in individuals in whom the disease has been confirmed and treated, an increase in serum levels above 35U/l may well be a useful marker of recurrence of the disease.
2. **Imaging:** has limited utility in the diagnosis of endometriosis as it lacks adequate resolution to identify adhesions or superficial peritoneal implants.
 - *Ultrasound* is inexpensive and easy to perform but user dependent. With an experienced sonographer, transvaginal ultrasound of the pelvis may be useful in delineating the presence of ovarian endometriomas. Walls of an endometrioma are irregular compared to the smooth wall of the simple ovarian cyst.
 - *MRI* is expensive but may help guide surgical approaches for women with suspected endometriosis especially deep infiltrating endometriosis. MRI is superior to ultrasound in diagnosing rectosigmoid lesions and endometriosis of the bladder.
3. **Laparoscopy:** For a definitive diagnosis of endometriosis, visual inspection of the pelvis at laparoscopy is the gold standard investigation. The advantages of laparoscopy are:
 - It provides direct visualisation of the endometriotic lesions and the presence of adhesions.
 - Areas can be biopsied.
 - The disease can be staged. Classification systems for endometriosis are subjective and correlate poorly with pain symptoms but may be of value in infertility prognosis and management.

Figure 1 shows the American Society for Reproductive Medicine revised classification of endometriosis which is the most common classification system currently used.
 - It provides an opportunity for concomitant surgical treatment.

Morphology of endometriotic lesions seen at time of laparoscopy

- Endometriosis typically appears as superficial 'powderburn' or 'gunshot' lesions on the ovaries, serosal surfaces and peritoneum – black, dark-brown, or bluish puckered lesions, nodules or small cysts containing old haemorrhage surrounded

by a variable extent of fibrosis.

- Atypical or 'subtle' lesions are also common, including red implants (petechial, vesicular, polypoid, haemorrhagic, red flame-like) and serous or clear vesicles.
- Other appearances include white plaques or scarring and yellow-brown peritoneal discoloration of the peritoneum. Circular defects (peritoneal pouches) of the pelvic peritoneum, uterosacrals or broad ligament can also be seen.
- Endometriomas usually contain thick fluid like tar or chocolate coloured fluid; such cysts are often densely adherent to the peritoneum of the ovarian fossa and the surrounding fibrosis may involve the tubes and bowel.
- Deep infiltrating endometriotic nodules can extend >5 mm beneath the peritoneum and may involve the uterosacral ligaments, vagina, bowel, bladder or ureters. The depth of infiltration is related to the type and severity of symptoms.

Treatment

Treatment should be individualised taking into account the woman's age, desire for fertility, severity of symptoms and extent of disease. It is also important to consider the degree of functional impairment which may be reflected by time off from work, psychosexual issues as a result of dyspareunia and emotional distress from long standing pain and or infertility.

A. Medical Treatment

The aim of medical therapy is to relieve pain and bleeding. It is rarely offered to women wishing to conceive because treatment may further compromise fertility in women.

All medical treatments seem to be equally effective in managing endometriosis. About 80-85% of patients have improvement in their symptoms. The difference between various medical treatments is in their side effects, with some treatments being more acceptable than others.

1. Pain relief:

Non-steroidal anti-inflammatory drugs (NSAID) may be effective in reducing endometriosis-associated pain. Transcutaneous electrical nerve stimulation (TENS) may also be used to relieve pain.

2. Hormonal therapy:

- *Combined oral contraceptive pill (COCP)*: The symptoms of endometriosis are known to ameliorate during pregnancy and after the menopause suggesting a hormonal dependence. Symptomatic improvement with the use of the COCP is presumed to be due to is pseudopregnancy effects. Current low-dose formulations appear to be effective as higher dose pills but with minimal side effects. The use of the COCP may be considered in women with mild symptoms who are not attempting to conceive. Although data are lacking, clinical experience of bi-, tri-cycling and continuous use

regimens (reduction or avoidance of the pill-free interval) suggests that they are the best for cycle-related symptoms.

- *Continuous progestogen treatment:* Pseudopregnant state can be achieved by continuous administration of progestogenic preparations. The progestones used are derivatives of progesterone (dydrogesterone or medroxyprogesterone acetate) or testosterone derivatives (norethisterone acetate and norgestrel). Apart from oral preparations, an intramuscular depot formulation is available and can be administered for 6-9 months. The results are comparable to that of the COCP but use can be limited due to the undesirable side effects such as weight gain, breakthrough bleeding, abdominal bloating, acne and mood changes.
- *Danazol:* is an isoxazole derivative of 17-ethinyl testosterone. It inhibits multiple steroidogenic enzymes in the ovary and adrenal gland. It also binds to progesterone, androgen and glucocorticoid receptors. Danazol has a mixed agonist/antagonist effect at the progesterone receptor and an agonist effect at androgen and glucocorticoid receptors. It decreases sex-hormone binding globulin (SHBG) production, displacing testosterone from SHBG, thus increasing free testosterone concentrations. The net effect is the creation of a high-androgen low-oestrogen environment which induces endometrial atrophy. Side effects (mainly due to its androgenic properties) are weight gain, oily skin, acne, fluid retention, muscle cramps, hot flushes, depression, headaches, mood changes. It is important to forewarn women to discontinue treatment if they develop rare side effects such as hirsutism, deepening of the voice or a skin rash. Danazol is administered in a dosage of 400mg per day (200mg twice daily). In the case of mild-moderate endometriosis there is a highly effective symptomatic improvement in over 85% of cases. Recurrence rates of up to 40% have been reported in the 36 months after completion of a course of danazol. Women should be advised to use additional barrier methods of contraception as it causes androgenization of a developing female fetus if used in the first trimester of pregnancy.
- *Gonadotrophin-releasing hormone (GnRH) agonists* eg goserelin, triptorelin or leuprorelin: Continuous administration of GnRH results in down-regulation of follicle stimulating hormone (FSH) and luteinising hormone (LH) with suppression of oestradiol to postmenopausal levels. This results in induction of a medical menopause which causes marked regression but not disappearance of the endometrial implants and improvement in symptoms.

Side effects resemble those of a pseudomenopause. These include hot flushes, headaches, atrophic vaginitis, vaginal dryness, reduced libido and osteoporosis. Dual photon absorptiometry has shown a statistically significant but reversible decrease in spinal bone mineral density after six months of GnRH therapy. These are often given with add-back hormone replacement therapy (eg Livial or tibolone) to minimise their

hypoestrogenic adverse effects without apparent reduction in efficacy. This strategy allows extension of the treatment period of use beyond the 6-month product license by providing protection against osteoporosis, although bone density should be monitored with long-term usage.

- *Levonorgestrel-releasing intrauterine system (MIRENA)*: The steady slow release of levonorgestrel appears to have a direct effect on endometriotic deposits through peritoneal fluid, possibly by a haematogenous spread. Pilot studies have shown a great improvement in pain control in women with endometriosis confirmed by prior laparoscopy. The studies have shown a reduction in ultrasonographic size of rectovaginal nodules and an improvement in the American Fertility Society (AFS) staging of the disease at second look laparoscopy.
- Aromatase inhibitors: aromatase p-450/prostaglandin E2 (PGE2) pathway appears to be involved in the genesis of endometriotic deposits. Aromatase is the enzyme responsible for the transformation of androgens, androstenedione and testosterone into oestrogens, oestrone and oestradiol respectively. Aromatase activity is not detectable in normal endometrium but is expressed inappropriately in endometriosis and in the eutopic endometrium from women with endometriosis. Aromatase inhibitors (AIs) are now being used as adjunctive therapy to GnRH, the pill and progestogens in women with resistant endometriosis-related symptoms. Recent data include reports of letrozole used with norethisterone acetate and anastrozole with a 20 micrograms ethinylestradiol/levonorgestrel contraceptive pill. A reported reduction in laparoscopic AFS scores and 90–93% pelvic pain relief has reported in recent studies. Another study randomised 80 women with resistant severe endometriosis to receive goserelin (gonadotrophin-releasing hormone, GnRH) + anastrozole or GnRH alone for 6 months. A significantly greater improvement in pain scores was achieved in the combined arm ($P < 0.0001$) with minimal adverse effects and no significant bone loss. More RCT data are required to confirm the findings from these studies and to look at the efficacy and safety of other combinations of oestrogen/ovulation suppressing regimens with AIs.
- Presently there is insufficient evidence to recommend the usage of progesterone antagonist (mifepristone), selective oestrogen receptor modulators or statins in the treatment of endometriosis.
- Targeted therapies in research and development: selective progesterone receptor modulators, oestrogen receptor ligands (or selective oestrogen receptor beta agonists), angiogenesis inhibitors, immunomodulators/inflammatory modulators.

B. Surgical treatment

Surgical treatment can be conservative or radical depending on the woman's choice or desire to retain fertility. The indications for surgical treatment are:

- Women with severe endometriosis who do not respond to medical treatment
- For the treatment of ovarian endometrioma
- To improve fertility

Surgical options include:

- Laparoscopy: Minimal access surgery has become the more popular treatment option as laparoscopic instruments are widely available and there is an increase in the experience among gynaecologists in laparoscopic techniques. Laparoscopic excision of endometrial deposits, division of adhesions and treatment of endometrioma are now undertaken.

Recurrence of disease

Recurrence after surgery remains a formidable challenge.

About 40-45% of women have relapse of the disease 5 years after the primary surgery and would engender further surgery. The likelihood of hospital re-admission for endometriosis within 4 years after surgery for additional surgical treatment is 27% and re-operation occurs in over half of women with endometriosis. About one quarter of women will require three or more operations.

KEY POINTS

- 1. Endometriosis is a leading cause of disability in women of reproductive age. It is debilitating and costly.**
- 2. Examination findings may be normal and there should be a low threshold to further investigate if symptoms are present.**
- 3. It is present in up to 50% of women with infertility.**
- 4. Medical and surgical treatment should be individualised according to the severity of the disease and the wishes of each women.**

CHAPTER ELEVEN

INFECTIONS in GYNAECOLOGY

Horace Fletcher

The anatomical relations of the female genital organs make them particularly prone to many different types of infections. In most women the genital tract directly communicates the peritoneal cavity with the external environment. As a result infectious agents from outside can gain entry to the different organs of the genital tract and to the peritoneal cavity itself causing an infection. The internal organs are also in direct contact with other organs which are potential sources of infection such as the appendix and the colon. Infections in these organs can spread easily to the genital tract.

The external genital tract is directly adjacent to the rectum and anus, as a result coliform organisms are a frequent cause of genital infections.

Normal activities such as coitus and pregnancy also predispose women to infections of the genital organs.

Iatrogenic infections are also common with the use of intra-uterine devices or other surgical procedures and also postoperatively when the peritoneal cavity is opened by gynaecologists or surgeons.

The normal protective mechanism of the genital tract includes the vaginal acidity maintained by the presence of lactobacillus acidophilus commensal bacteria. The external genitalia, vulva, vagina and ectocervix in adult women are lined by cornified stratified squamous epithelium which is resistant to the entry of most infections.

In neonates, the vaginal epithelium is resistant similar to what is seen in adults, as the baby is under the influence of maternal oestrogen. Neonates are however susceptible to candidiasis as their vagina is not yet colonized by acidophilus. In children from age one month to puberty the vaginal lining is very thin and infections can gain entry through vaginal epithelium. These girls are susceptible to not only gonococcus and Chlamydia, but also to Shigella species, Group B Streptococcus and pin worms. All of these can cause vaginal infections but it is also important to rule out a foreign body (retrieved with a cotton tipped swab). Also important are the cervical mucus and endometrial glandular secretions which prevent infections from non virulent organisms which commonly enter the uterus during coitus.

The presence of immunoglobulins and complement as well as other innate immunity are also protective. The presence of a gonococcal vaginitis (GC) in a child is quite often not due to sexual abuse. This may become evident from history and if on examination there is no evidence of introital penetration. Very often the infection comes from an adult who is looking after the child's domestic needs. The infection is passed by insanitary practices by the adult harboring the infection not washing their hands before handling the child. If a child is confirmed to have GC or Chlamydia, the parents and the nanny should also be tested for the infection.

All of these protective mechanisms can be overcome by a large enough bolus of non virulent organisms such as streptococcus or coliforms usually iatrogenically via the cervix or peritoneal cavity, or a small dose of virulent organism such as gonococcus or Chlamydia or other STD during coitus. Women are also vulnerable to infections after abortion or delivery when the endometrial lining is absent because of the recent pregnancy.

Very rarely infections in the genital tract arise from haematogenous spread from a distant source such as the lungs (e.g. tuberculous infection).

Sexually transmitted infections

The genital tract is very prone to sexually transmitted infections because of intimate contact between two humans, when organisms in one person can easily pass to the other. Infections also spread easily because during coitus there are usually micro-abrasions through which organisms can gain entry. The female genital tract is particularly susceptible to infections because during coitus the male ejaculate semen containing motile spermatozoa along with any attached bacteria, viruses or protozoans, gain entry into the female peritoneal cavity. When non virulent bacteria and viruses gain entry, these are easily overcome by the woman's host defenses. However virulent bacteria are more difficult to eradicate and can cause an infection in the woman.

If the woman is immunocompromised or postmenopausal, where cornification of the epithelium is less, infections occur much more easily. Infections also can enter through breaks in the skin. Any ulceration of the genital tract can lead to acquisition of an infection. The presence of ulcerated STDs makes it much easier to become infected with HIV and HPV.

Mucopurulent Discharge

Vaginal discharge is one of the most common reason women will visit their doctor. These infections can be diagnosed using clinical evaluation (syndromic diagnosis). Treatment is therefore not delayed as the patient is given empirical treatment at the first visit after history examination and samples sent to the laboratory.

A malodorous fishy discharge is strongly associated with bacterial vaginosis. A profuse green frothy discharge is associated with Trichomoniasis (the bubbles caused by the flagellates swimming around). A yellow purulent discharge is associated with Chlamydia or gonococcus.

In all of these cases a history of unprotected intercourse is usual and in many cases if probed, the patient will admit to a history of a new partner. However, it is important to remember that even women in stable monogamous relationships can present with STD as their partner may have had casual sexual contact outside the relationship. This is a very sensitive issue and must be handled with great tact. Patients should never be told that their partner is being unfaithful based on syndromic evaluation. However it is also important for them to know that this is a possibility, especially with confirmation of the diagnosis by laboratory tests. Treatment of the partner is also mandatory to avoid reinfection.

Gonococcus

This is caused by an obligate intracellular gram negative diplococcal organism. The organism has a predilection for glandular cells and causes a purulent discharge a few days after unprotected sexual contact. The organism causes infection in the cervix, Bartholin's and Skene's glands and sometimes the anorectal canal. It may also cause a urethritis with intense burning on micturition. Examination of the patient should reveal the pus like discharge. Swabs should be taken from the endocervix, and transported to the laboratory in Stuart's medium and plated on Thayer Martin medium. Treatment is possible with most common antibiotics. Culture and sensitivity of the bacteria to the available antibiotics is important as resistant strains are common. The patient must be advised to take a full course of the prescribed antibiotics to avoid development of resistance. Shorter course of more powerful antibiotics are best used to ensure compliance but cost is also important.

Chlamydia

This organism is not a bacterium however it causes a very similar pattern of disease as gonococcus. The symptoms are also very similar. The main difference is that it is more difficult to diagnose using simple culture. Modern testing requires use of PCR based testing. The newest PCR test kits test for GC at the same time since in many cases both coexist. Treatment is also different and is with Doxycycline or Tetracycline. Many patients who harbor Chlamydia are asymptomatic after a few weeks of infection. Erythromycin replaces Tetracycline in pregnancy.

Trichomoniasis

The causative organism in this case is a protozoan, *Trichomonad vaginalis*. These are multicellular organisms which are oval and flagellated. They are easily visible using light microscopy in vaginal secretions swimming around. Treatment is with Metronidazole or Tinidazole. Treatment of the asymptomatic male partner is important to avoid "ping pong" infections where the organisms are passed back and forth from the untreated to the treated partner leading to a frustrating cycle what appears to be persisting infection with many visits to many doctors in a vain attempt to get a cure. Both partners are best treated at the same time with a short course of drugs.

Bacterial vaginosis

This is vaginal condition that can produce a malodorous vaginal discharge and results from an overgrowth of certain normal bacteria in the vagina. In the past, the condition was called Gardnerella because it was thought that the causative agent was solely *Gardnerella vaginalis*. However, the newer name, bacterial vaginosis, reflects the fact that there are a number of species of bacteria that naturally live in the vaginal area and may grow to excess under certain conditions

The cause of the condition is now believed to be due to a reduction in the number of the normal hydrogen peroxide-producing lactobacilli in the vagina. This results in an environment that favours an increase in concentration of other types of bacteria, especially anaerobic bacteria.

The condition appears to be associated with certain conditions. These include multiple or new sexual partners, intrauterine devices for contraception, recent antibiotic use, douching, and cigarette smoking. However, the role of sexual activity in the development of the condition is not fully understood, and bacterial vaginosis can still develop in women who have not had sexual intercourse.

Bacterial vaginosis can be confirmed on microscopy by the presence of Clue cells which are vaginal cells covered with bacteria. Microscopy should also reveal fewer than normal lactobacilli. A vaginal pH greater than 4.5 is also suggestive of bacterial vaginosis. Cultures of bacteria are generally not useful in establishing the diagnosis of bacterial vaginosis.

When a drop of potassium hydroxide is placed on the discharge from a woman with bacterial vaginosis, a fishy odor can result also called the “whiff test”.

Treatment is with Metronidazole, which can cause nausea (especially with alcohol the antabuse effect) and a metallic after taste. Tinidazole causes less side effects than Metronidazole and is easier to take. Clindamycin is also useful as treatment and can be prescribed as oral medication or as a vaginal gel.

Ulcerative lesions

Syphilis (*the great imitator*)

All patients with genital ulceration or an unusual skin rash should have a VDRL (venereal disease research laboratory) test. This is because secondary syphilis can mimic many conditions. All pregnant women should have VDRL test as well because early treatment will prevent

congenital syphilis which can be a lethal condition or at least cause significant morbidity in the child.

Primary syphilis is usually evident about 9-90 days after sexual exposure. The ulcer and nodes palpated are painless (Figure 1).



Figure 1. Syphilitic ulcer (chancre)

Nodes are unilateral, regional, firm, mobile and non-suppurative. The primary chancre (ulcer) is small and painless and may not be noticed and thus the patient may pass through this stage without knowing.

Secondary syphilis occurs 6 weeks to 6 months after sexual contact and can mimic many skin condition lesions. These lesions are also very infectious harder to treat than primary disease.

Tertiary syphilis occurs 2 years to 30 years after sexual contact. About 20% patients have neurosyphilis and 30% patients will have cardiovascular syphilis. Tertiary syphilis is hardest to treat of all the types.

Diagnosis is made on a smear of the lesion using dark field microscopy the spirochaetes *Treponema pallidum* may be visible.

Confirmation is done using the VDRL test. However, this is non specific and can be negative in primary syphilis.

False positive VDRL tests are common in certain conditions (collagen vascular disease, yaws, leprosy, malaria, drug addiction, old age, even pregnancy). Fluorescent Treponema Antigen (FTA) and Treponema Pallidum Haemagglutination Antibody Test (TPHA) are more specific tests.

Treatment of Syphilis

- Primary 2.4 Mega Units (MU) Benzathine Penicillin (Penadur) IM stat
- Secondary 2.4 MU Penadur weekly x 3
- Neuro syphilis 2-4 million U Crystalline Penicillin IV Q4H for 10-14 days
- Penicillin allergy Tetracycline 500 qid 14 days

Lympho granuloma venereum

This is caused by *Chlamydia trachomatis*. It causes a painful shallow transient ulcer which results in no scarring. Typically it results in enlarged groin lymph nodes with groove sign (occurring on either side of Cooper's ligament), and in two-thirds of cases nodes are bilateral. Treatment is with Tetracycline or Doxycycline.

Granuloma inguinale

This results from infection with *Calymmatobacterium granulomatis* which is a gram negative bacterium (GNB). The pathognomonic feature of significance is the painless beefy red ulcer. Enlarged nodes are uncommon but subcutaneous granulomata may occur which also ulcerate to give the typical beefy look. Autoinoculation is common with this condition but systemic infection is rare. This is more common in women who have cervical infection and can lead to bone and joint infection.

The infection is very difficult to diagnose and it is often mistaken for a cancer. It is hard to culture this organism and usually the diagnosis is confirmed by observation of Donovan bodies in histological samples. This however is not easy to prepare as paraffin prepared samples may not show these bodies and normal haematoxylin and eosin stains may not show them either. The Donovan bodies are best seen if the tissue is fixed with glutaraldehyde and embedded in plastic and if the slides are stained with Warthin Starry or Giemsa's stain. Alternatively tissue samples can be crushed and air dried and stained with Giemsa's stain to show the Donovan bodies. Treatment is with Tetracycline, Gentamicin, Ampicillin or Chloramphenicol however protracted treatment is needed as recurrence is common.

Chancroid

This results from infection with *Haemophilus ducreyi* gram negative bacillus GNB. This causes a very specific painful necrotic irregular ulcer with tender lymphadenopathy. The tender nodes are unilateral and may rupture as a bubo. Treatment is with Erythromycin, and Sulphonamides.

Herpes simplex

This is caused by a DNA virus related to the chicken pox virus *herpes zoster*. There are two types, HSV 1 and HSV 2 and are the most common cause of genital ulcers.

Most cases are sexually acquired and most are asymptomatic. Transmission can occur even with no lesion present.

Vesicles occur at first, these rupture leaving multiple shallow painful ulcers which are erythematous around their edge. The infection may be primary, latent or recurrent. Primary infections occur 3-7 days after contact, leaving burning and paraesthesia. The vesicles and ulcers are small <0.5cm diameter and usually occur in the same area with recurrence.

The painful ulcers leave no scarring and also occur with tender lymph nodes. Recurrences occur during stress, menses or other illness with or without fever.

Diagnosis is confirmed with a swab transported in Stuart's medium. The diagnosis can be confirmed with direct examination using Giemsa Stain. Tissue culture and antibody testing are also alternatives although the IgG test is often positive from exposure at an indeterminate time in the past.

Treatment is with Acyclovir, Famcyclovir Valcyclovir which are all used to suppress the virus as the condition is incurable.

Human Papilloma Virus (HPV)

This is caused by DNA viruses of which there are now over 300 serotypes identified. These viruses cause STD warts and malignancies. The warts are soft fine fleshy growths (condylomata acuminata) which may be flat on the cervix. These viruses usually get passed from person to person during sexual contact. This can occur without penetration such as during fondling or oral contact. The virus enters the skin through micro-abrasions and invades the epithelial cells. The virus is incorporated into the cell and the infection causes mainly a local cellular immune response. Serum antibodies to HPV from normal infection are therefore usually at a very low level leading to repeat infections. Most women who get an infection with any type of HPV do not get persistence of the virus as the immune system will eliminate it. The presence of the antibodies in genital fluids is however very common with reports of up to 80% sexually active women being infected in studies done in Jamaica and the United States. In about 1% of women a persistent infection occurs, leading to a clinical infection (wart or intra-epithelial neoplasia or cancer).

These viruses are implicated in squamous cell cancer of the vulva, vagina, cervix, anus and head and neck. Certain types such as HPV 6 & 11 cause warts and HPV 16 & 18 cause cancer. When found, treat regardless of the serotype. Treatment of warts or intra-epithelial neoplasia requires local destruction with podophyllin, interferon alpha, cautery, cryotherapy or laser. Invasive cancer is treated with radical surgery or radiotherapy.

Vaccines are now available to prevent HPV infection. These vaccines are made from the inactive shell of the virus types 16, 18, which cause 70% of cervical cancers and 6 and 11 which cause

90% genital warts. The vaccines are best given to adolescent girls before sexual debut and are very effective. They cause high levels of serum antibodies to the four types of HPV and also give some protection to other related HPV types (Figure 2).

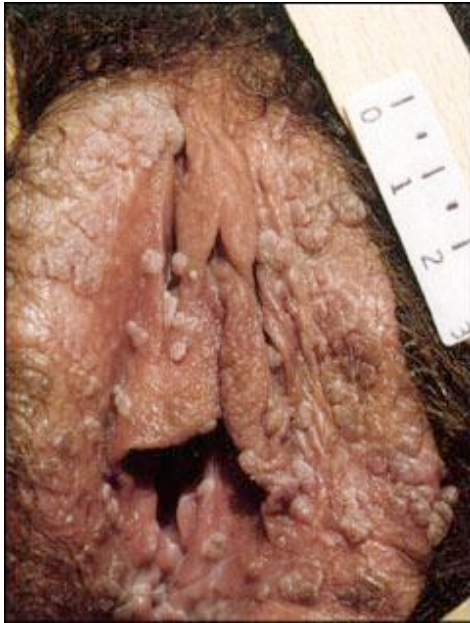


Figure 2. HPV warts (condyloma acuminata)

Bartholinitis

The Bartholin's glands are a pair of organs on either side of the introitus, at the lower third of the labia majora (posteriorly). These glands provide some of the normal vaginal secretions. They are prone to being infected during coitus leading to Bartholin's abscess. The infecting agent is usually Chlamydia or gonococcus but most microbiological studies show mixed infection.

The infection causes blockage of the gland opening leading to a painful swelling at the introitus. Chronic infection of the gland leads to a Bartholin's cyst. Acute infections are treated with antibiotics to cover gonococcus, Chlamydia and anaerobic bacteria. Drainage of the abscess gives immediate relief of the pain but recurrences are common unless a formal procedure is done. Treatment is best done by marsupialisation, where a large opening (1.5 cm) is made and the cyst wall is sutured to the vaginal epithelium using either a continuous purse string suture or interrupted sutures. More recently surgeons have used a Word or Foley catheter which is inserted through a small aperture and inflated with about 3 cc saline. The catheter is allowed to drain the fluid for 4 weeks and then removed leaving an epithelialised fistula between the gland and the introitus. This is a simple office procedure and shows promise in replacing formal marsupialisation.

Pelvic Inflammatory Disease

Pelvic inflammatory infection arises when bacteria infect the Fallopian tubes and ovaries. This is often as a result of sexually transmitted infection with a virulent organism such as chlamydia or gonococcus. This infecting agent enters the uterus and peritoneal cavity with spermatozoa as they swim to find the ovum. These infectious agents themselves are not motile and cannot travel all the way into the peritoneal cavity on their own. In the peritoneal cavity they can traverse all the way to the liver causing a perihepatitis. The chronic state of this condition results in violin string adhesions between the diaphragm and the liver (Fitz Hugh Curtis syndrome). Systemic infection is also possible and infections in remote areas such as a joint may lead to a migrating poly-arthritis. Young female patient who presents with arthritis not responding to non steroidal anti inflammatory drugs, should have an aspirate of the joint tested for Gonococcus. A single hot swollen joint is usually present in patients with non gonococcal bacterial infection.

PID can however be caused by other means. The notable examples are due to infections post partum or post-abortion, iatrogenically after dilatation and curettage or after hysterosalpingogram or hysteroscopy, or post-insertion of an intrauterine contraceptive device. In some cases it arises after a laparotomy.

In most cases iatrogenic infection arises from procedures being done under less than sterile conditions. However women may also become infected if they are susceptible to infection (diabetic or HIV). Most procedures do not require antibiotic coverage, however antibiotic prophylaxis should be considered in immunocompromised women. In those where there is evidence of contamination antibiotic treatment should be given.

Modern intrauterine contraceptive devices are rarely associated with PID. Older IUCDS such as the Dalkon shield were notorious as a cause of PID and were eventually withdrawn from the market as a result. It was eventually realized that the multifilament IUCD strings acted like a lamp wick and pulled bacteria up into the uterus. Rare cases of fungal PID (actinomycoses) are usually associated with IUCD in Immunocompromised women.

Infections from other sites can spread to the pelvis and Fallopian tubes. This is most common from a ruptured appendix. The same thing can also theoretically occur from a ruptured diverticulum (colonic diverticulosis or small bowel Meckel's diverticulum) or rarely from another ruptured organ such as the gall bladder or perforated duodenum or after colonic anastomotic surgery rupture. Very rarely infection may spread to the Fallopian tube haematogenously. This is most often tuberculous spread from the lungs.

Diagnosis of PID is done from history (sexual history, pregnancy, recent surgical procedure, other intraperitoneal event). On examination the patient is febrile with lower abdominal guarding and rebound tenderness most marked bilaterally in both iliac fossae. Vaginal examination may reveal a malodorous purulent discharge, with marked cervical motion tenderness. In chronic cases the pyosalpinges or hydrosalpinges may be readily palpable.

Investigations include swabs are done for chlamydia and gonococcus. Imaging may be helpful if masses are felt to rule out ovarian disease.

Treatment requires hospitalization with severe disease, analgesics and antibiotics. There is little evidence that short wave diathermy is of any value.

The antibiotics should include coverage for chlamydia, gonococcus and anaerobes.

In some cases surgical intervention is needed. This is especially relevant for tubo-ovarian abscess that are not responding to intravenous antibiotics. Surgical treatment may require salpingo-oophorectomy or in some cases total abdominal hysterectomy and bilateral salpingo-oophorectomy. For more details on pelvic inflammatory disease refer to Chapter Twelve.

HIV

This is a RNA virus which is passed by contact with body fluids such as blood and semen. Most people contract HIV during unprotected sexual contact. The epidemic is also as a result of use of illicit drugs especially from exchange of hypodermic needles and from prostitution. HIV may also arise from blood transfusions, as most Caribbean countries still test donated blood for HIV antibodies and not HIV antigens, as is done in some developed countries. It takes about three months for an infected person to seroconvert and become antibody positive after exposure. This so called window period of being seronegative but having the infection, can result in a blood donor passing the infection to another person. The disease was at first more common in homosexual men however heterosexual contact is now the leading mode of transmission accounting for 65% of cases in the Caribbean.

The HIV ELISA test usually becomes positive at about 4 months. The Western blot confirmatory test is always done if ELISA is positive, to ensure true seropositivity. A rapid test has been introduced which gives results in a few hours and is of great value for surgical patients who present in emergencies.

Doctors are advised to however take universal precaution of protecting themselves from contact with body fluids when taking care of patients even with a negative antibody test.

Patients who have HIV later develop acquired immunodeficiency syndrome, where they are prone to opportunistic infections. The gynaecological manifestation of this is the presence of a greater chance of being infected with all other STDs. Cervical intra-epithelial lesions is also more common and more severe and recurrence after treatment often occurs. Vaginal candidiasis is also more common and more intractable.

The presence of recurrent, exotic, intractable infections in the genital tract should lead gynaecologists to look for HIV.

Monitoring of infected women is done using their CD4 count and viral load and this determines the type and dose of antiretroviral therapy given.

Treatment is with the antiretroviral agents of which there are four groups.

1. Nonnucleoside Reverse Transcriptase Inhibitors (NNRTIs), such as nevirapine and efavirenz.
2. Nucleoside Reverse Transcriptase Inhibitors (NRTIs), such as zidovudine, competitively blocks protein HIV needs to make more copies of itself.
3. Protease Inhibitors (PIs), such as lopinavir, ritonavir, disable protease, needed for HIV reproduction.
4. Fusion Inhibitors, such as enfuvirtide, block HIV entry into cells.

Resistance to drugs is very common especially if patients do not take their drugs properly.

It is very important to remember that any ulcerated STD can make it much easier for patients to acquire HIV.

Non STD infections

Many cases of vaginitis are not due to sexually transmitted disease. Some may be due to the use of tampons (especially if forgotten), spermicides, cervical caps and douching. Douching removes the normal protective acidophilus and leaves the vaginal environment prone to infectious agents.

Candidiasis

Candidiasis is caused by several spp of candida fungi. The most common is *Candida albicans*. These fungi are commensals which are usually found in humans on skin or in the vagina. They do not usually lead to any problem as the vaginal acidity prevents over growth. Vaginal acidity is maintained by lactobacillus acidophilus another commensal organism which inhabits the female vagina from childhood. Any change in the vaginal environmental milieu to make the vagina more alkaline will eventually lead to over growth of fungi. Some of the factors implicated include excess moisture from sweating (swimming, saunas, exercise, tight clothing, panty shields or very hot days). Fungal infections also arise if the acidophilus are killed (antibiotics, excess douching), or if the woman is Immunocompromised.

Diagnosis is based on history (syndromic) where the patient should be asked about the risk factors above and also the main complaints such as a thick curdy green or off white discharge, with no odour and intense itching.

On examination there is notable hyperaemia, scratch marks and the typical discharge. Swabs can be taken for culture and sensitivity and empirical treatment started. The partner does not have to be treated as this is not regarded as a sexually transmitted infection. The most important thing to prevent recurrence is to ensure that the woman removes the risk factor and also get rid of the spore forms of the fungus by placing infectious clothing (underwear, gym clothes, socks and shoes) in the sun after washing or by soaking the clothes in boiling water. Washing alone will not kill the spores and this is a common cause of what appears to be a resistant persistent infection.

Treatment is usually successful with topical Clotrimazole or oral Fluconazole. Quick relief of the symptoms is obtained with topical Clotrimazole with Hydrocortisone.

Surgical Infections

Infections after gynaecological procedures are very common. These occur as a result of contamination through the skin incision, the vagina at hysterectomy or from inadvertent perforation of large bowel which is adjacent to the internal gynaecological organs.

Preoperative preparation of patients requires that they be fit for surgery to avoid infections patients at risk such as diabetics should have good glucose control. Where contamination is likely such as if bowel may be damaged, preoperative bowel preparation may be helpful. This is done routinely for ovarian cancer surgery and in other patients who may be immunocompromised or who are having emergency surgery for infectious conditions such as a tubo-ovarian abscess may need to get full antibiotic coverage during and after the procedure.

At the time of surgery cleaning of the skin where the incision is to be made is important. The standard cleaning solutions include alcohol, iodine, or chlorhexidine. Iodine if used should be allowed to dry and fix as in this state it is bactericidal. However, iodine and alcohol should not be used on the delicate tissue of the vulvo-vagina. Chlorhexidine is safer to use there. Surgeons should also remember to scrub properly, wear sterile gloves and gowns and drape the patient in a sterile fashion.

Prophylactic antibiotics are useful especially at vaginal surgery and are best given 20-30 minutes prior to the skin incision. True prophylactic antibiotics consist of a single dose of drug given in this fashion.

At surgery the main principles are gentle tissue handling, low blood loss, avoid perforation of bowel and minimal tissue damage. Collections of blood in the pouch of Douglas and excess dead and devitalized tissue can lead to a pelvic abscess. Closure of the layers of the abdominal wall to eliminate dead space also will avoid sub fascial collections which can lead to wound abscesses.

The use of drains to the pelvis or subfascial area in the case of possible collection of blood or other fluids is of great value.

Patients with post operative pyrexia may have an infection in the pelvis or in the wound. The history and examination may be all that is needed to make the diagnosis. An ultrasonogram may help to determine where the problem is. The main differential is pelvic thrombophlebitis, thromboembolism or a chest infection.

Management of wound infections usually requires opening the wound and allowing it to drain and then heal by secondary intention. Occasionally a clean granulated wound can be closed secondarily. Treatment of a pelvic collection can be done with antibiotics intravenously or if the collection is large aspiration through the vagina using ultrasound guidance. Patients who are not settling require re-laparotomy and formal drainage.

In patients who have large bowel injury where the bowel is unprepared, small perforations may be closed and the patient covered with antibiotics. Major perforation with excess soiling may require colostomy. Patients with severe infections at surgery should not have the incision closed until clean granulation is apparent.

KEY POINTS

- 1. Pruritus vulvae and vulval ulcers may be due to an infective cause or cancer.**
- 2. Sexually transmitted diseases are on the rise and it is not surprising that a single woman may harbor numerous infections.**
- 3. Always consider a VDRL and HIV screening when one presents with a vulval infection.**
- 4. In many cases of these infections the partner(s) must also be investigated and treated.**
- 5. In some countries these patients are referred to special clinics such as genitor-urinary medicine (GUM) clinics which allow for contact tracing.**

CHAPTER TWELVE

PELVIC INFLAMMATORY DISEASE

Bharat Bassaw

Pelvic inflammatory disease (PID) is defined as an infection involving the Fallopian tubes, one or both ovaries, the uterus and the pelvic peritoneum. Acute salpingitis has become very common during the last two or three decades, affecting young women particularly in the age group 15-20 years. Nearly 300,000 women are hospitalized annually in the United States for PID and in Jamaica, a significant number of gynaecological patients attending outpatients clinics have PID. Studies conducted to evaluate the economic costs of PID and PID-associated ectopic pregnancy and infertility in the USA, revealed that the total costs in 1984 exceeded 2.6 billion dollars.

Some of the long-term complications of acute PID are recurrent PID, chronic pelvic pain, ectopic pregnancy and infertility. Dyspareunia, pelvic adhesions, pyosalpinx and tubo-ovarian abscess may also follow. Overall, such complications occur in about 15-20% of cases and they often require surgical intervention.

Aetiology

PID affects sexually-active women during their childbearing years. Most cases of PID are thought to develop from sexually-transmitted diseases. It is usually the result of an ascending infection which leads to cervicitis, endometritis, salpingitis, parametritis, oophoritis and peritonitis.

Chlamydia trachomatis is now the commonest cause of PID; it is implicated in almost two-thirds of cases. Gonorrhoea is still an important cause. In many cases many micro-organisms are involved. *Mycoplasma genitalium*, enterobacteria and anaerobes such as *bacteroides fragilis* are also associated with PID. There seems to be a geographic variation since chlamydia is more common in the Scandinavian countries whereas gonorrhoea is more common in the USA and in developing countries such as the Caribbean (Table 1). This may be as a result of failure to recognize chlamydial infection due to the absence of routine screening using sensitive tests such as nucleic acid amplification tests e.g. PCR.

Tuberculosis of the genital tract is now making a resurgence with the prevalence of HIV infection in women of the reproductive age. Pelvic actinomyces, was considered rare until recently. This organism is a normal commensal of the mouth and gut. It may be associated with

the presence of an intra-uterine device. Cervical colonization is most likely when a non-copper containing device is used, especially if it is in-situ for more than two years.

Table 1. Aetiology of pelvic inflammatory disease

-
1. Chlamydia trachomatis
 2. Neisseria gonorrhoea
 3. Streptococcus (aerobic and anaerobic)
 4. Staphylococcus (aerobic and anaerobic)
 5. Bacteroides fragilis
 6. Actinomyces Israeli
 7. Mycoplasma hominis
 8. Clostridia
 9. Klebsiella
 10. Proteus mirabilis
 11. Escherichia coli
 12. Mycobacterium
 13. Trichomonas vaginalis
-

Pathogenesis

Pelvic inflammatory disease develops in a two-stage process.

Firstly, the organisms infect the cervix. In about 1 out of 10 women with a cervical infection, the organisms ascend to the uterus, tubes or ovaries.

Organisms can also reach the upper genital tract after termination of pregnancy (post-abortal sepsis) or by insertion of an intra-uterine device.

Vectors

(a) Trichomonads

These motile organisms can ascend from the vagina to the level of the Fallopian tubes and thus, they may carry infective organisms. Trichomonads have been cultured from the Fallopian tubes and pouch of Douglas in patients examined by laparoscopy to confirm the clinical diagnosis of acute salpingitis. Trussel provided evidence to show the association of trichomonads with gram-negative and anaerobic bacteria.

(b) Spermatozoa

Gonococci, mycoplasma, toxoplasma and cytomegalovirus have been shown to be associated with sperms. A casual role for spermatozoa in the carriage of bacteria and subsequent development of salpingitis, has been put forward. Patients who use barrier methods of contraception have a lower incidence of PID. The levonorgestrel intra-uterine releasing device (Mirena) is said to protect against PID since it causes the cervical mucus to become thickened which inhibits the ascent of bacteria.

Passive transport

Evidence in support of this mechanism is based on the observation that particulate matter, carbon or carmine particles in suspension, have been shown to ascend passively through the uterus and into the Fallopian tubes.

Table 2. Risk factors for PID

- 1. Age**
- 2. Sexual activity and number of sexual partners**
- 3. Frequency of intercourse**
- 4. Sexual life-style and sex trade (promiscuity)**
- 5. Copper-containing IUCD**

Risk factors

1. Age
PID is a disease confined to the reproductive age group and the highest incidence occurs between 15 and 20 years (Table 2).
2. Sexual activity
Sexual activity is a pre-requisite in most cases of PID. This condition is rare in celibate nuns and may be consequence of tuberculosis or extension of appendicitis. The presence of several multiple partners appears to be an important risk factor. Divorced and separated women have a higher incidence than married women.
3. Frequency of sexual intercourse
Increased frequency of sexual activity bears a direct relationship to the occurrence of PID.
4. Sexual life-style
The age of first intercourse is also important. Patients with PID are younger at their first coital experience but the factor of multiple sexual partners is also a contributory factor.

5. Intra-uterine contraceptive device (IUCD)
PID occurs 3-9 times more often in copper-containing IUCD-users than in non-users. The oral contraceptive pill may be protective probably because the cervical mucus is denser, thus inhibiting sperm and bacterial migration into the uterus.

Clinical features

The most common features of acute pelvic inflammatory disease are

- (a) lower abdominal pain
- (b) pyrexia (low grade)
- (c) purulent and offensive vaginal discharge
- (d) abnormal uterine bleeding
- (e) nausea and vomiting
- (f) dyspareunia and dysuria
- (g) menorrhagia
- (h) dysmenorrhea

Some of the clinical signs are fever, tachycardia, lower abdominal tenderness, guarding and rigidity, and marked cervical excitation pain on digital pelvic examination.

A low threshold for the diagnosis of PID is suggested.

When the diagnosis is in doubt, or when the condition fails to improve with antibiotics, laparoscopy is indicated. This may show hyperemia and edema of the Fallopian tubes, a purulent exudate and possible evidence of previous tubal disease such as peritubal filmsy adhesions or a hydrosalpinx. In about 15-20%, there may be no laparoscopic evidence of acute infection, despite organisms being identified from the Fallopian tubes.

Other conditions which cause pelvic pain should be borne in mind (Table 3).

Table 3. Differential diagnosis

-
1. Acute appendicitis
 2. Ruptured ectopic pregnancy
 3. Torsion of an ovarian cyst
 4. Degeneration of a fibroid
 5. Endometriosis
-

Table 4. Investigations of pelvic inflammatory disease

-
- 1. Haematological**
 - 2. Culdocentesis**
 - 3. Microscopy of vaginal discharge**
 - 4. High vaginal swab**
 - 5. Ultrasound**
 - 6. Serology**
 - 7. Laparoscopy**
-

Investigations (Table 4)

1. Haematological

An elevated C-reactive protein, erythrocyte sedimentation rate (ESR) and leukocytosis in excess of 10,000/ml, are highly suggestive of acute infection but these are non-specific..

2. Culdocentesis

Fluid aspirated from the pouch of Douglas is gram-stained. A positive needle culdocentesis for bacteria is usually specific for PID. The absence of peritoneal fluid does not rule out the possibility of salpingitis since adhesions or a misplaced needle can give to a 'dry tap'. Culdocentesis is particularly useful in ruling out a ruptured ectopic pregnancy.

3. Microscopy

This reveals inflammatory cells. The absence of pus cells on a wet-mount smear has a good negative predictive value but not a good positive predictive value.

4. Swab of discharge

High vaginal and endocervical swabs must be taken. In addition, if a sexually-transmitted disease is suspected, swabs should also be taken from the rectum, urethra and the throat. Other samples are self-taken vulvovaginal swab, and first urine. All swabs must be placed in a special transport medium containing antibiotics for identification of chlamydia and gonorrhea. Culture or nucleic acid amplification test (PCR) may be used to identify chlamydia and gonococcus.

5. Laparoscopy

Routine laparoscopy in patients suspected of having salpingitis is not practiced. This is due to the fact that the majority of women with mild disease are not admitted to hospital. However, as stated above, when the diagnosis is in doubt, there is a place for laparoscopic scrutiny of the pelvic organs. A useful classification of the disease is shown in Table 5. Perihepatic adhesions (Fitz-Hugh-Curtis syndrome) may be seen by turning the laparoscope around. This is due to chlamydial infection (Figure 1).



Figure 1. Pelvic inflammatory disease

6. Transvaginal ultrasound

In acute PID, the Fallopian tubes can be identified when they are distended by inflammatory exudate especially if power Doppler is also used. When a large, dilated, convoluted tube is present, and the ovary is not identifiable as a separate mass, then the diagnosis of tubo-ovarian abscess is likely.

7. Estimation of anti-chlamydia IgG antibody may confirm the presence of chlamydial disease when the results of culture have been negative.

Severity of PID is shown in Table 5.

Table 5. Severity of PID by laparoscopy

Mild	Erythema and edema
	No purulent exudates
	Tubes freely mobile
Moderate	Erythema and edema
	Gross purulent exudates

	Tubes not freely mobile
Severe	Inflammatory exudates
	Abscess
	Pyosalpinx

Management

The aims of management of patients with PID are

- (a) Early accurate diagnosis
- (b) Institution of appropriate antibiotic therapy
- (c) Frequent follow-up evaluation
- (d) Treatment of sexual partners
- (e) Consider the possibility of STI's

The criteria for admission of patients to hospital with PID to hospital are shown in Table 6.

Table 6. Criteria for hospital admission

-
1. Suspected pelvic abscess
 2. Temperature $>38^{\circ}\text{C}$
 3. White cell count $> 15,000/\text{ml}$
 4. Pregnancy
 5. Uncertain diagnosis
 6. Upper abdominal pain
 7. Failure to respond to oral antibiotics in 48 hours
-

Medical Management

Treatment should be started immediately and not delayed until full culture reports are available. Intra-uterine device may need to be removed especially if the response to treatment is slow. Pregnancy should be excluded. A curettage for septic incomplete miscarriage is indicated but this procedure must be done under antibiotic cover by a senior officer.

There are many antibiotic regimes for the treatment of PID. Parenteral broad-spectrum antibiotics should be used for at least 48-72 hours with severe disease after which oral medication is prescribed for 14 days.

The Centre for Disease Control (CDC) recommendations are as follows:

- (a) 1. Cefoxitin 2.0 g IV qidx 2 days
2. Doxycycline 0.1 g IV bd x 2 days
and 100 mg po bd x 10 days
- (b) 1. Clindamycin 0.6 g IV qid x 2 days
2. Gentamycin 2 mg/kg IV stat
and 1.5 mg/kg IV tid x 2 days
3. Oral Clindamycin 0,45 mg/kg qid x 10 days

RCOG Green Top Guidelines state:

- 1. Ceftriaxone 2g IV daily plus intravenous Doxycycline 100mg twice daily, followed by oral Doxycycline 100mg twice daily and
- 2. Oral Metronidazole 400mg twice daily for 14 days.

Cephalosporins have a lesser role as they are not effective against chlamydia. Penicillinase-producing *Neisseria* may be managed with Cefoxitin in an intramuscular dose of 1-2 g, six hourly. If the patient is pregnant, Erythromycin should be substituted for Doxycycline.

Mild acute salpingitis may be treated as an outpatient with Doxycycline 100 mg twice daily for 14 days and Metronidazole 400 mg two or three times daily for 7-14 days. The patient should be reviewed in 48 hours to assess the effectiveness of the treatment.

Surgical management

1. Ruptured tubo-ovarian abscess

The patient with a ruptured tubo-ovarian abscess is seriously ill and requires urgent operative intervention. Older studies stressed the need for total abdominal hysterectomy and bilateral salpingo-oophorectomy. However, recent reports indicated that a sub-total hysterectomy or no hysterectomy at all is associated with comparable results. At surgery, all intra-peritoneal purulent material should be removed and a pelvic drain should be left in-situ. Broad-spectrum antibiotics should be administered. A senior surgeon should be in attendance as the adjacent viscera are at high risk for injury. This patient may need to be nursed either in a High Dependency Unit (HDU) or an Intensive Care Unit (ICU) after the surgery.

2. Unruptured tubo-ovarian abscess

Treatment may be via immediate laparotomy or by a conservative approach with a combination of broad-spectrum antibiotics. If the patient's condition does not improve within 48 hours, then surgery is indicated.

3. Pelvic abscess

This may be drained either through a colpotomy incision or at laparotomy or transperitoneally with ultrasound guidance. Laparoscopy may help early resolution of the disease by division of adhesions and drainage of pelvic abscess.

4. Treatment in pregnancy

A pregnancy test should be done in all cases of women to exclude the possibility of a pregnancy especially an ectopic. PID is rare in intrauterine pregnancy except in the case of septic abortion. Tetracycline should be substituted with Erythromycin. A combination of Cefotaxime, Azithromycin and Metronidazole for 14 days may be used.

5. PID with an IUCD in situ

If the patient's condition does not improve with antibiotics, then consideration should be given to remove the IUCD. Some other form of reliable contraception should be recommended.

6. Women who also have HIV infection should receive the same antibiotics as those without HIV.

7. If a woman requests the IUCD and she is at high for PID, then the Mirena is a good choice.

KEY POINTS

- 1. Pelvic inflammatory disease is common during the reproductive period and is associated with sexual activity.**
- 2. Chlamydia trachomatis is now the most common aetiological agent but other rarer causes may be seen in some high groups.**
- 3. The clinical features include abdominal pain, dysmenorrhoea, menstrual abnormalities, dyspareunia (deep), a purulent discharge, dysuria and post-coital bleeding.**
- 4. Laparoscopy may be required to distinguish between pelvic sepsis and endometriosis since the clinical features are similar and also these two conditions may co-exist.**

- 5. A high index of suspicion must be maintained for PID.**
- 6. Any opportunity to screen for Chlamydia must be utilised since early recognition and appropriate treatment will reduce morbidity and lead to major economic benefit.**
- 7. Mild cases of PID can be treated with oral antibiotics but severe cases require in-hospitalization and the use of broad-spectrum antibiotics.**
- 8. The male partner(s) must be also treated simultaneously.**
- 9. Acute PID may lead to chronic PID, chronic pelvic pain and infertility.**
- 10. Surgical intervention is required for the patient with chronic PID especially with a ruptured tubo-ovarian abscess.**

CHAPTER THIRTEEN

CONTRACEPTION AND STERILIZATION

Adesh Sirjusingh

Within recent times family planning has been recognised worldwide as being a human right, basic to human dignity. In 1994, at the International Conference on Population and Development held in Cairo, Egypt the concept of reproductive health and reproductive rights were clearly defined and the linkages between population, sustained economic growth and sustainable development were articulated.

In countries with limited resources, unrestricted population expansion leads to further strain on national income, developmental policies, health care, education, housing, and crime levels. Adolescent pregnancy, high parity and elderly women are more prone to complications. Increased illegal abortions with resulting morbidity and mortality are also related to contraceptive practice and availability. In the Caribbean it is essential for Governments to understand that family planning policies should be elevated on their lists of priorities.

There is a wide array of contraceptive choices available (Table 1). The question of proper patient selection is important as compliance is poor and discontinuation rates are high when the woman is persuaded to use a method against her own inclination. Some are less safe than others and it is important for the doctor to obtain a full medical history to identify any contra-indications.

PEARL Index

This is a measure of the efficacy of the contraceptive method and is the risk of pregnancy per 100 woman years. If the PEARL Index (PI) is 3, then of 100 women using that contraceptive for a year, three will become pregnant. Compliance affects the PI.

Table 1. Contraceptive Methods

Natural	Withdrawal Rhythm/Periodic Abstinence Lactational Amenorrhoea Method (LAM)
Barrier	Male Condom Female Condom Diaphragm Cervical Cap Spermicides (used with barriers)
Intrauterine Contraceptive Device (IUCD)	Inert (Lippes loop) no longer manufactured Copper-containing (Cu-T, Multiload, Nova T) Hormone-containing (Mirena®)
Oral contraceptive pills	Progestogen-only pill (POP/Mini-pill) Combined oral contraceptives (COC) Post-coital pill
Injectable hormones	Oestrogen/progestogen (Mesigyna ®) Depot progestogen (Depo-Provera®, Noristerat®) Implants (Norplant®/Jadelle®) Implanon, Nexplanon
Others	Vaginal hormone ring (Nuvelle®) Transdermal hormone patches (Ortho-Evra®)

Natural methods

This is of course less reliable than other methods and does not offer any protection against sexually transmitted diseases (STDs) unless the couple is exclusively monogamous. The Catholic Church accepts these methods.

Periodic abstinence (rhythm) method

As it implies, this is based on avoiding sexual intercourse during the fertile days of a woman's menstrual cycle. The scientific basis of this method is the short time span in which the ovum can be fertilised (less than 24 hours) and the fertilising capacity of the sperm in the female reproductive tract (2-5 days). The problem, however, lies with the variable menstrual cycle length and the need to adjust sexual behaviour. Even with regular cycles, as many as 8 to 16 days may be considered as fertile days.



Figure 1. Measurement of LH

Several indices can aid in determining the fertile period and can be used in conjunction with this method, measuring LH (Figure 1).

Basal body temperature (BBT) rise provides indirect evidence that ovulation occurred as a result of the effects of increased progesterone. By charting basal body temperatures for a few cycles, a more reliable idea of the fertile period can be obtained.

Cervical mucus changes (Billing's method) can be monitored. As ovulation approaches, the mucus becomes thin and stretchy as a result of the effects of oestrogen. After ovulation, the mucus rapidly thickens and fries due to the effects of progesterone.

Motivated women can learn to examine for changes in the cervix. After menstruation, the cervix is lower in the vagina, the os is closed and firmer in texture. As oestrogen secretion increases the cervix moves upwards, becomes softer and the os begins to open, until at ovulation it will admit a fingertip. After this there is an abrupt reversal due to the high levels of progesterone.

Newer computerised techniques are now available to determine **urinary LH changes** and therefore the most fertile days. This will increase the number of days that are safe for sexual intercourse.

Failure rates vary considerably with these methods (2 – 20 pregnancies per 100 women years after the first year of use).

Withdrawal (coitus interruptus)

This method has been attempted by many a couple with varying degrees of success.

Spermatozoa may still be present in the pre-ejaculatory fluid. There is a high failure rate (PI 15-30) as the method is highly user-dependent.

Lactational amenorrhoea method (LAM)

This can be an efficient interim contraceptive method in the first six months postpartum. The baby should get little or no food or drink except breast milk and should breastfeed often, day and night. If menstruation occurs in the time period, alternative contraception should be advised. The PI is 0.5 – 2 pregnancies per 100 women in the first six months after childbirth.

a. Spermicidal agents

These are used in conjunction with some of the barrier methods and come in foaming tablets, jelly and cream preparations. They should be placed at least 10 minutes before ejaculation but not more than an hour. They can be messy, and may cause irritation but are inexpensive. They do not protect against STDs. There is a high failure rate if used alone but they may be useful in the perimenopausal woman (Figure 2).



Figure 2. Spermicidal agents

b. Condoms

Both male and female condoms are now available. They protect against STDs and are associated with a reduction in the incidence of cervical cancer.

Most male condoms are lubricated and contain a spermicidal agent. They are available in different shapes, sizes, colours and textures. They are inexpensive (can be obtained freely in many centres) and with the increased incidence of HIV, their use should be increasing. Improper use accounts for a higher than expected failure rate and care must be taken while withdrawing the sheathed penis from the vagina. PI is 3-14.

Female condoms have recently been introduced and are made of polyurethane. They are less likely to be allergenic than the latex male condom. They are more impervious to virus particles. Problems with the female condom include noisy crackling sounds during intercourse, a period of time is needed to master its insertion and removal, and they are more expensive and less widely available than the male condoms. PI is similar to male condom (Figure 3).



Figure 3. Condoms

c. Diaphragm and cervical caps

Diaphragms are fitted before intercourse with spermicidal lubrication and must remain inside at least six hours after. They are cumbersome to use and there is a lack of spontaneity. They are reusable. Urinary tract infections are more common with their use. Diaphragms come in various sizes and need to be fitted by trained personnel to provide a snug fit but not be too large so as to cause awareness by the wearer or dislodging during intercourse. The diaphragm has a spring around a latex dome and comes in sizes 55-100 mm. PI is 6-20.

Cervical and vault caps are not widely used and are placed over the cervix. Removal can be sometimes difficult. PI is 9-35 (higher in parous women) (Figure 4).



Figure 4. Cervical caps and diaphragm

Intrauterine contraceptive device (IUCD, IUD)

Modern IUCDs offer long-term reversible contraception with some approved for at least 10 years of use. The device is usually a small, flexible plastic frame which has strings attached. They are radio-opaque (Figure 5).

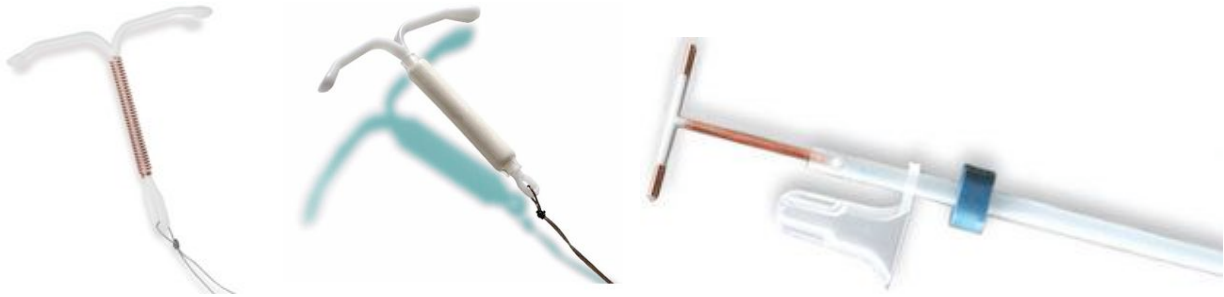


Figure 5. IUCDs

Insertion

Trained personnel insert the IUCD through the cervical os using a ‘no-touch’, aseptic technique in the outpatient setting. Anaesthesia is not usually required. It can be inserted at any time if the woman is not pregnant but insertion can be easier during or just after the menses. It can also be inserted post-miscarriage or post-delivery but the expulsion rate is higher.

A pelvic examination is done to assess the uterine size and position. A bivalve speculum is inserted after antiseptic is used to clean the vagina and cervix at least twice. The anterior lip of the cervix is steadied with a tenaculum. A uterine sound or a graduated tube included usually with the device is used to measure the distance from the fundus to the external os. A sliding outer collar on the tube is pushed flush with the external os. Depending on the device different techniques are used (push, pull, withdrawal). Follow the manufacturer’s instructions carefully. The threads are trimmed so that about 2-3 cm are visible past the external os. The strings can be felt by the woman after each menstruation to ensure the IUCS remains *in situ* and to allow removal.

Steps that can reduce the risks of ascending infection include

- a. Clean the cervix thoroughly
- b. Cut the string very short
- c. Recommend an antibiotic (e.g. Augmentin) for 3 – 5 days

Mode of action

The main mode of action appears to be through reducing the ability of the sperm to move through the reproductive tract. IUCDs exhibit a marked inflammatory response in the uterus causing an increased concentration of macrophages and leucocytes, prostaglandins and various enzymes in both the uterine and tubal fluid. These effects are thought to be toxic to the sperm and oocyte. Should a fertilised egg be produced however, the endometrium will be unreceptive for implantation. In order to improve efficacy, copper, silver and hormones have been added to the modern day devices.

Types of IUCDs

a. Lippes Loop

This was previously commonly used but is now no longer manufactured. They are of various sizes and are inert. The PI is 3 per 100 women in the first year of use.

b. Copper -containing IUCDs

The addition of copper to the device was found to improve the contraception efficacy and enabled smaller devices to be developed over the years. A smaller device is associated with a reduced risk of menorrhagia and dysmenorrhoea. Copper is spermicidal and bactericidal.

The first generation devices were the Copper 7 and Copper T 200 (2-3 year use). The second-generation devices were the Nova T and the Multiload Cu 250 (approved for 5-year use).

Modern day devices are the Multiload Cu 375 (5 years) and the Copper T 380 A series (approved for 10 year use). The Cu T 380 series has two copper collars crimped on the horizontal stem near the ends and copper wound around the vertical stem to give a total surface area of 380 mm². These smaller devices are suitable for use in nulliparous women. The PI is 0.6 to 0.8 per 100 women in the first year of use, which makes it a very effective contraceptive.

In the 1970's the progesterone containing *Porgestasert* was introduced but in the 1990's the levonorgestrel-intrauterine device (Mirena®) was a significant improvement and introduced a variety of additional non-contraceptive benefits.

Mirena consists of a T-shaped frame carrying a core in the main stem of 52 mg levonorgestrel. Approximately 20 µg are released locally per day with little systemic absorption and side-effects. Thinning of the endometrium occurs as well as anovulation in some women. After one year of use the median menstrual loss falls to 10 ml.

It is one of the most effective of the reversible methods of contraception with a pregnancy rate of 0.1 per 100 woman years. Additionally health benefits with this device include a significant reduction in menstrual blood flow, and it has been used to reduce the rates of hysterectomy for menorrhagia.

Complications

1. **Menstrual disturbances, pain and dysmenorrhoea**

The effect of local prostaglandins in particular is thought to cause both an increased menstrual flow and secondary dysmenorrhoea with the non-hormonal IUCDs. Prostaglandin synthetase inhibitors (e.g. Mefenamic acid) are effective in treating these problems and can reduce discontinuation rates. Intermenstrual bleeding also can occur. Pain or even cervical shock (vagal response) can complicate insertion.

2. **Perforation, absent threads**

This is rare and is not device-specific. Rates vary from 0.6 to 1.3 per 1000 insertions. Operator skill is the most important factor. A routine visit 6 weeks after insertion should detect most perforations, as these tend to occur at insertion. Absent threads may occur by migration into the uterine cavity or from perforation. An IUCD thread locator or an IUCD hook may be able to retrieve the device if it is still in utero.

An ultrasound is helpful in localization or an X-ray can be done with dye or a sound in the uterine cavity.

Removal of a recently perforated IUCD may be possible with the laparoscope but significant adhesions can occur with time (especially with the copper devices) and a laparotomy may become necessary. Hysteroscopy and curettage may be necessary to remove a lost IUCD.

3. **Expulsion**

This is highest in the first three months after insertion and in nulliparous women and those less than 30 years of age. The larger devices are more liable to be expelled.

4. **Pregnancy and ectopic pregnancy**

Although pregnancy rates are low, an intrauterine pregnancy can occur rarely. If the strings are visible before 13 weeks of pregnancy, it is best to remove the device. There will be a risk of miscarriage. If the strings are not visible there is a risk of more serious infection, miscarriage, rupture of membranes and preterm labour.

Patients with IUCDs are protected from both intra- and extra-uterine pregnancies. **The risk of ectopic pregnancy is 80-90% less than in women who do not use contraception. If a pregnancy occurs, however, it is more likely to be an ectopic than a normal pregnancy.** All IUCD users should be appropriately counselled on the warning symptoms and signs of ectopic pregnancy.

5. **Pelvic infection**

Infection is more likely to occur in the first 20 days after insertion. Screening for PID at insertion is performed in some centers and appropriate antibiotic treatment is offered at that time especially in populations with high prevalence of STDs. There appears to be a reduced risk of PID with the hormonal IUCD due to prevention of ascending infection by thickening of the cervical mucus.

Pelvic actinomycosis can rarely occur in IUCD-users. The device should be removed if there is PID that does not respond to treatment with antibiotics.

Contra-indications

- a. Distorted uterine cavity e.g. large submucosal fibroids, uterine abnormality
- b. Active PID
- c. Past ectopic pregnancy (relative)
- d. Pregnancy (can be inserted for post-coital contraception up to five days after unprotected sexual intercourse)
- e. Undiagnosed abnormal bleeding, suspected malignancy
- f. After hydatidiform mole
- g. Valvular heart disease (relative, antibiotic prophylaxis can be given pre-insertion)
- h. Patient at high risk of STDs, HIV
- i. Rarely, patients with copper allergy, Wilson's disease

Advantages

1. Effective, long-term contraceptive
2. No need to take tablets daily
3. No interference with sex
4. Increased enjoyment as no need to worry about pregnancy, no preparation needed
5. Easily reversible
6. Does not affect milk production
7. Can be used until the menopause
8. No drug interaction, side-effects
9. Prevents ectopic pregnancy
10. Less bleeding with hormonal IUCD

11. Can be used by most women, few contra-indications

Disadvantages

1. Initial irregular bleeding
2. Longer and heavier periods
3. More period cramps
4. Perforation risk
5. Pelvic infection (especially in the first 20 days)
6. Expulsion risk
7. Need to check for strings regularly
8. Medical procedure needed for insertion
9. Less protection against ectopic than for intrauterine pregnancy

Oral contraceptives

a. Progesterone only pill (mini-pill)

This is often considered as a 'poor relative' of the combined oral contraceptive pill (OCP) but still represents about 25% of women who use oral pills. There are several preparations available but 350µg norethisterone and 75µg norgestrel are commonly employed (this is even lower than the dose in the OCP).

These pills are taken at the same time every day without any break. A pill is considered missed if more than three hours delay occurs from the time that the pill is normally taken. These pills are frequently prescribed to women who have contra-indications to the use of oestrogen-containing OCP. Some of the contra-indications are migraines, increased risk of thromboembolism, cardiovascular risks and nursing mothers.

Mechanism of action

Sperm permeability is affected by the thickening of the cervical mucus; in 40% of women ovulation does not occur; the endometrium becomes unreceptive for implantation. There is a variable effect on the epithelium of the Fallopian tubes.

Contra-indications

These are few but care is advised in women with previous ectopic pregnancy, with functional ovarian cysts, severe liver disease, severe arterial disease, pregnancy, undiagnosed vaginal bleeding, recent molar pregnancy, and breast cancer.

Side effects

These include irregular bleeding, persistent ovarian follicles (follicular cysts), bloating, weight gain, headache, nausea and skin changes (with the more androgenic preparations).

Efficacy

When used correctly the failure rate is 0.5 per 100 women years. The usual failure rate in practice is 3.

b. The combined oral contraceptive pill (OCP)

In 1961 Pincus, Chang and Rock pioneered the original OCP. Over the years there have been several improvements but the basic principles of contraception have remained the same. There has been a significant reduction in the hormonal dosage and a parallel reduction in side effects. It is one of the most popular of the contraceptive choices in the developed world.

The OCP contains a synthetic oestrogen (usually Ethinyl oestradiol) and a progestogen. The progestogens fall into three groups: First – (norethindrone), second – (levonorgestrel) and thirds – general (gestodene, norgestimate and desogestrel). Third generation progestogens produce a more favourable lipid profile but are associated with a slight increase in venous thromboembolism over the early generations. However, the risk is still less than half the rate of thromboembolism in pregnancy. Two additional compounds are also used as they have additional properties beside the progestational effect. Cyproterone acetate is an anti-androgen and is useful in the treatment of women who also have acne. Drospirenone is an analogue of Spironolactone which helps to reduce excess fluid and hence bloating and premenstrual tension.

There are numerous preparations and there is little sense in trying to learn all, as there are frequent name changes and new developments.

The initial preparations contained doses of 50 µg Ethinyloestradiol. Nowadays, there are low dose of OCPs with 20 µg (Alesse®, Femaine®) and 30-35 µg (Gynera, Cilest, and ultra-low dose preparations with 15 µg (Mirelle®). Contraceptive efficacy is not compromised and they are potentially safer, however, breakthrough bleeding and cycle control problems are common in the first few months of use.

New developments on the horizon are a once-a-week pill based on a steroidal compound, Centchroman, which is free from the side-effect of oestrogen and progesterone. It is already in wide use in India.

The pill should be stopped one month before any major surgery (alternative contraception is advised) but can be safely taken up to the time on minor surgery.

They are available in monophasic (all pills same), biphasic and triphasic preparations. In the latter two, the dose of steroid changes once or twice in each cycle. These changes were made to mimic the natural cycle and, theoretically, provide better cycle control but this had not been seen in practice. It can also be confusing to women to cope with missed pills.

Biphasic pills are no longer available.

The pill is usually taken for 21 consecutive days followed by seven days off or seven inert tablets. Withdrawal bleeding usually starts at this time. The pill should be taken at the same time each day. Vomiting, diarrhoea, antibiotics, and drugs that are 'liver-enzyme inducing' such as Rifampicin, Griseofulvin, Phenytoin and Carbamazepine can interfere with the bioavailability (Figure 6).



Figure 6. Combined oral contraceptive pill

Missed pills

If less than 12 hours have passed then the pill can still be taken with no loss of efficacy, If more than 12 hours late, then she should take the most recently delayed pill and use extra precautions for the next 7 days. If less than 7 pills are left in the packet, then the next packet of pills should be started without a pill-free break between them.

Post-pill amenorrhoea

Most women who cease using the pill regain normal fertility three months later. The syndrome of post-pill amenorrhoea is most likely the return of menstrual abnormalities that were present prior to starting the pill (e.g. polycystic ovaries).

Mode of action

The principle role would be interference with the normal development of follicles by inhibiting the pituitary follicle stimulating hormone and the luteinising hormone surge which lead to anovulation.

The endometrium becomes unreceptive for implantation.

Alteration in cervical mucus affecting sperm permeability and also altered tubal mobility may also occur.

Health benefits of the OCP

1. Very reliable contraceptive
2. Menstrual period are shorter and lighter
3. Periods are less painful
4. Periods are regular and predictable
5. Less risk of iron-deficiency anaemia
6. Less premenstrual syndrome
7. Reduction in ovarian cancer
8. Reduction in endometrial cancer
9. Decreased incidence of functional ovarian cysts
10. Decreases fibrocystic breast disease
11. Decreases the incidence of PID
12. Reduces endometriosis and fibroids
13. Treats hirsutism and acne
14. Protects against ectopic pregnancy

Contra-indications

1. Arterial thrombosis
2. Venous thrombosis and known predisposition to this condition
3. Valvular heart disease
4. Ischaemic heart disease
5. Previous cerebrovascular accident and brain vascular malformation
6. Uncontrolled hypertension
7. Focal and crescendo migraine
8. Liver diseases, tumours, gallstones
9. Pregnancy
10. Porphyria
11. Oestrogen-dependent neoplasia
12. Undiagnosed genital tract bleeding
13. Conditions affected by steroid use e.g. Hydatidiform mole (unless the serum HCG is normal)

Relative contra-indications include

1. Obesity
2. Increasing age
3. Diabetes mellitus (low dose pills can be used)
4. Hyperprolactinaemia
5. Smoking (they should be advised to stop)

Side effects

Oral contraceptive steroids are metabolised by the liver and affect the metabolism of carbohydrates, lipids, plasma proteins, amino acids, vitamins and the clotting factors. Table 2 lists the common 'minor' side effects of the two major components of the OCP. Most side effects improve with continued use but, occasionally, it is worthwhile trying alternative dose oestrogen and type of progestogen products before deciding on discontinuation. The newest pills are not associated with weight gain (Table 2).

Table 2. Side effects (there is overlap because of their common steroid origin)

Oestrogenic	Progestogenic
Nausea	Depression
Headaches	‘Premenstrual Tension’ – like symptoms
Increased mucus	Irregular bleeding
Fluid retention and weight gain	Acne
Increase blood pressure	Breast discomfort
Bleeding	Weight gain
Cervical ectropion	Reduced libido
Chloasma	
Suppression of lactation	

Serious side-effects are related to the vascular system and include venous and arterial thromboembolism. This appears to be related to the dose of oestrogen, which alters the clotting factors to create a hypercoagulable state. Obesity, hyperlipidemia, hypertension, elderliness and cigarette smoking increase the risk to cardiovascular disease.

Table 3. Risk of non-fatal thromboembolism for users of OCP

User Category	Incidence per 100,000 women per year
All women not using the pill	5-11
Pregnant women	60
Women on low dose of OCP (1 st and 2 nd generation progestogen)	15
Women on low dose pill with gestodene and desogestrel	30
Pill users who smoke	60

The risk of thromboembolism in pregnancy is 60 per 100,000 women per year and this must be considered when interpreting the risk of thromboembolism from the OCP (Table 3).

Cancer and the pill

Published data are difficult to interpret and change over time. In 1992 the World Health Organization concluded that there was no overall increased risk, but there was a weak association with OCP use up to the age of 45 years. In 1996 The Collaborative Group on Hormonal Factors in Breast Cancer published a meta-analysis, which included 90% of published studies, and this concluded that there was a small increase in breast cancer that persists for 10 years after stopping the OCP.

Similarly, data on cervical cancer is also difficult to interpret, as pill-users are more likely to undergo screening for cancer. From the most recent publication on a 25-year follow-up on a cohort of 46,000 women, there appears to be a small increase in squamous cell carcinoma and an increased risk of the adenocarcinoma.

Benign hepatic adenoma is a rare complication of OCP use.

Practical prescribing

As always, a full history and examination are especially important when prescribing and following-up patients who are on the pill with particular attention to the blood pressure, breast abnormalities, Pap smear-screening, migraines and liver abnormalities.

The lower dose preparations can be safely used from the young to the older non-smoking, well woman. They can be used to treat conditions such as functional cysts, abnormal cycles, menorrhagia, and premenstrual symptoms. Continuous OCP can also be used in women for medical treatment of endometriosis (no pill free breaks).

After pregnancy, it should be started six months postpartum or after she stops breast feeding, whichever comes first.

Efficacy

When correctly used the PI is 0.1. In practice, the failure rate is 6 to 8 pregnancies per 100 women years.

Injectable Contraceptives

This group is very effective and safe and are designed for long-term use but are frequently associated with menstrual abnormalities and weight changes plus a delay in return to fertility. The progestogens can be used in a variety of women with health problems in whom other contraceptive use may be hazardous.

a. Monthly Depot oestrogen/progestogen

One such preparation is Mesigyna®, which contains 50 mg Norethisterone enanthate and 5 mg Oestradiol valerate.

b. Depot-progestogen

Depo-Provera® contains 150 mg Medroxyprogesterone acetate and is given every three months. Noristerat® contains 200 mg Norethisterone enanthate and is recommended every two months. Depot-progestogens work by interfering with ovulation, and thickening of cervical mucus. With regular use, at the end of one year 80% of women will have infrequent scanty menses or amenorrhoea. These preparations are very effective (PI 0.3 per 100 women at the end of one year of use) and also help prevent iron-deficiency anaemia, ectopic pregnancies, fibroids and endometrial cancer. They may also reduce ovarian cancer. Additionally benefits include a reduction of seizures in epileptic women, and less frequent and less painful sickle cell crises in sicklers.

c. Implants

The newer version of Norplant® (from 1992) consists of six small soft rods containing 36mg of Levonorgestrel that are implanted under local anaesthesia subdermally in the skin of the upper arm. They last for at least 5 years and are very effective (1.6 pregnancies per 100 women over 5 years of use). A newer similar implant is Jadelle®, which consists of two rods. Implanon® is a single implant containing 68 mg Etonorgestrel designed for three years of use. It was recently replaced by Nexplanon.

Others

NuvaRing® is a vaginal ring, which is left in place for one month. It contains Etonorgestrel and Ethinyl oestradiol. Ortho-Evra® is a weekly contraceptive patch, which contains Norlegestromin and Ethinyl oestradiol. These are suitable for use in women who can use the OCP and have the advantage of there being no need for the woman to remember to use daily pills.

Emergency Contraception

These methods are designed to impair implantation or to alter ovulation in some way.

The copper IUCD can be inserted up to 5 days after the expected date of ovulation, which could be more than 5 days after unprotected sexual intercourse. The failure rate is 1%.

PC4® is a 'morning after' pill with 100µg Ethinyl oestradiol with 0.5 mg Levonorgestrel; given 12 hours apart for two doses. They must be started within 72 hours of unprotected intercourse. Nausea and vomiting along with menstrual cycle abnormalities are frequent. This is also called the *Yuzpe* method. Failure rate is up to 7%. Postinor, which contains 0.75 mg of Levonorgestrel, can be used twice daily, 12 hours apart after unprotected coitus. It is also available as a single dose treatment. Mifepristone (the abortion pill) is an anti-progesterone, which can be used for post-coital contraception.

Female Sterilization

Table 4. Female sterilization

Ligation	Fallopian tubes are divided
Salpingectomy	Tube are removed
Electrocautery	One or more areas are cauterised
Falope Rings	Silicone rubber loop
Clips	Filshie clip (titanium and silicone) Hulka-Clemens (stainless steel, polycarbonate)
Laser	Laser is used to divide the tubes
Sclerosing agents	Essure® (a tiny spring-like coil) Quinacrine transcervical pellets
Hysterectomy	With coexistent disease such as symptomatic fibroids, this may be offered (uncommon)

There are many methods available for female sterilization (Table 4).

Twenty-five percent of couples worldwide depend on male or female sterilization for contraception. The minimum that needs to be done is to interrupt the tubes so that sperm and eggs cannot meet.

Sterilization is associated with moral, cultural and emotional concerns. It is important that the following be discussed with both partners before performing the procedure:

1. That the procedure is permanent (reversal is possible but this is a complicated process with variable success rates)
2. The method to be used (see below)
3. The risk of failure
4. The risk of an ectopic pregnancy if failure occurs
5. That alternative long-term contraceptives are available
6. Verbal and written information should be provided.

In addition, the incidence of regret is higher in women before the age of 30 years, and for sterilization after a recent pregnancy.

Sterilization can be performed post-abortion, postpartum, at Caesarean section, and as an interval procedure (after the puerperium).

New information has suggested that there is a reduction in the incidence of breast cancer and ovarian cancer following sterilization. There is a slight increase in hysterectomy rates but there is no effect on the menses if the procedure is performed after 30 years of age. Menstrual problems do occur after sterilization but may usually be related to cessation of the OCP or physiological hormonal dysfunction in later life.

Efficacy

The failure rate for all methods of tubal sterilization is 2 to 3 per 1000. The rates are higher for immediate post-pregnancy procedures and for certain methods, especially electrocautery.

In the developed world this is the approach commonly employed. The Filshie clip is also one of the commonest methods used. Falope rings and Hulka-Clemens clips were commonly used in the past. The procedure can be done as a day-case under local or general anaesthesia. The contra-indications and complications of laparoscopy apply to this procedure. Bipolar electrocautery to occlude the tube should not be done routinely as there is a higher recanalization rate and there are risks of thermal damage to adjacent structures. Care must be taken to apply the device across the entire lumen of the tube and to not mistake the round ligament for the tube.

This is still frequently performed in our setting due to economic factors and operator experience with the procedure as compared with laparoscopy. A small supra-pubic transverse incision is made and the tubes are identified by the fimbriae. A variety of methods are used but the Pomeroy technique is commonest. A small loop of each tube is grasped with a Babcock's forceps. An avascular portion is then ligated with an absorbable suture (catgut) and the loop divided to reveal two cut ends. Over time the ends become separated and fibrosis occurs. For immediate postpartum women the incision will be at the umbilicus due to the uterine size. Other methods to tubal ligation include

- a. Irving
- b. Uchida
- c. Fimbriectomy
- d. Madlener

Colpotomy (incision through the pouch of Douglas to access the tubes) and culdoscopy are not frequently used and are not recommended as a first-line approach for tubal sterilization except in obese women.

Methods to occlude the tubal ostia have been under investigation for some time now. Recently a small metallic implant (Essure®) was approved for placement into the Fallopian tubes of women who wish to be permanently sterilised. Unlike other currently available surgical sterilization procedures for women, placement of the device does not require an incision or general anaesthesia. During the implantation procedure, the physician inserts one of the

devices into each of the two Fallopian tubes. This is done with a special catheter that is inserted through the vagina into the uterus, and then into the Fallopian tube. The device works by inducing scar tissue to form over the implant, blocking the Fallopain tube and preventing fertilisation.

During the first three months, women cannot rely on the Essure® implants and must use alternate contraception. At the end of three months, a radio-opaque dye is placed in the uterus and an X-ray is taken to confirm proper device placement. Once placement is confirmed, the alternate contraception can be discontinued.

Intratubal plugging devices (ITD). These devices are placed into the uterotubal junction and led in place by lateral “wings” at the distal tip and a loop-size extension of the proximal part.

A special type of these devices is the P-block developed by Brundin. The P-block consists of a nylon skeleton with a 4-mm long hydrogel body and two nylon, anchoring wings. After insertion, the hydrogel expands and occludes the intramural part of the tube.

Two monthly Quinacrine transcervical pellets have also been shown in a number of third world centres to be very effective and avoid the surgical approach. The ar inserted using the same instruments and technique as for an IUCD and cause permanent sclerosing.

Male sterilization

The conventional method involves vasectomy. A new “no scalpel” approach is recommended under local anaesthesia. The failure rate is 1 in 2000. Confirmation of sterility is recommended by the absence of spermatozoa (asoospermia) and this takes several months to occur. Complications include postoperative haematoma and infection but these are reduced with the new technique.

KEY POINTS

- 1. When discussing contraceptive choices with a couple, one must consider several factors, for example, age, parity, further reproductive ambition, past medical history and previous contraceptive use.**
- 2. The combined oral contraceptive pill has a low Pearl Index and also confers many other non-contraceptive benefits.**
- 3. The adverse effects of the combined pill include thromboembolic phenomena, a slight increase in the incidence of cervical and breast carcinoma, and cardiovascular disease especially in elderly obese women who smoke.**
- 4. The intra-uterine device also carries a high efficacy rate but certain steps must be instituted to reduce the problem of pelvic inflammatory disease.**
- 5. It is important to be very clear on the mechanisms of action of the OCP and IUCD.**
- 6. Postpartum and post-abortal female sterilization have a higher failure rate than the interval procedure.**
- 7. Always counsel the couple on failure rate, possibility of an ectopic pregnancy, irreversibility when dealing with sterilization.**
- 8. There are newer methods of female sterilization which do not require either a laparoscopy or a laparotomy.**

CHAPTER FOURTEEN

THE MENOPAUSE

Bharat Bassaw, Garth McIntyre

The menopause is the cessation of menses and is defined as the date of the last menstrual period. The mean age of the menopause has been constant for at least the last century. The average age in Trinidad is about 51—52 years.

No correlation has been established between the age of the menopause and the menarche. The postmenopause commences six months after the last menses. The climacteric, on the other hand, is a variable interval beginning with physiological failure of ovarian function prior to the cessation of menstruation. It is a gradual process occurring generally after the age of forty and is characterised by anovulatory cycles. Its onset varies from woman to woman and may be influenced by heredity, environment and socio-economic status.

Vasomotor symptoms as well as those of an emotional, sexual and musculo-skeletal nature are common at this time.

Menopause is the last menstrual period.

Aetiology

It will be recalled that the process of oogenesis is complete at the time of, or very soon after birth, and the stock of primordial follicles in the ovary is at its maximum during fetal life after which it gradually diminishes. During reproductive life, some follicles develop, and only a few will actually proceed to ovulation; the majority will disappear by a process known as atresia. This results in a decline of follicles. The remaining ones become less and less responsive to stimulation by the pituitary gonadotrophins. As a result of the follicular exhaustion, oestradiol secretion falls, and menstruation subsequently ceases. It is this decrease in ovarian hormone secretion that is responsible for the profound somatic changes of the climacteric.

Hormonal changes

The most obvious hormonal changes of impending ovarian failure are:

- a. A marked reduction in oestrogen levels; oestradiol drops below 20 pg/ml and oestrone to below 40 pg/ml.
- b. A marked fall in progesterone levels because ovulation and hence corpus luteum formation does not take place.
- c. A progressive increase in LH and FSH levels, as the negative feedback effects of oestradiol are lost. This is thought to represent a compensatory effort by the anterior pituitary to sustain declining ovarian function.

The atrophic ovaries become greyish-yellow, the tunica thickens and the surface becomes smooth.

There is evidence that the stroma of the postmenopausal ovary retains the ability to carry out some degree of hormonal function. Appreciable amounts of testosterone are secreted and become the major steroids.

The adenohipophysis:

Most postmenopausal women exhibit evidence of increased pituitary secretory activity. However, the number of gonadotrophic basophils does not increase following a natural menopause. In contrast, after bilateral oophorectomy in younger patients, the gonadotrophic basophils undergo hypertrophy, a change which is suppressed by replacement therapy with oestrogen.

The urogenital tract:

The urogenital tract shows regressive changes during the climacteric. There is a decrease in the size and weight of the pelvic organs as well as atrophic changes in their epithelial lining.

The Fallopian tubes:

They gradually shrink in size and there is diminished motility of the musculature. Ciliary and secretory functions are lost as the lining epithelium becomes gradually flattened.

The uterus:

The uterine muscle is gradually replaced by fibrous tissue and the myometrium shrinks. The endometrium may show:

- (a) Swiss-cheese pattern
- (b) An atrophic senile type
- (c) Active hyperplasia

The active hyperplasia may be due to extra-ovarian sources of oestrogen. The ligamentous uterine supports atrophy and uterine descent is common. Fibroids usually shrink and become calcified.

The cervix shrinks until it becomes almost flush with the vaginal vault. The cervical mucus in the premenopausal phase shows good arborisation indicating an oestrogenic effect unopposed by progesterone. After the menopause, the fern test becomes negative. The squamo-columnar junction recedes upwards into the endocervical canal.

The vagina gradually becomes pale because of decreased vascularity, and less distensible owing to an increase in submucosal connective tissue. The rugae become less prominent, and the epithelium appears smooth.

The supporting structures atrophy, predisposing to cystocele and rectocele. The cytological changes of the vagina smear reflect the decline in ovarian function. Initially, in the premenopausal phase, there is a good oestrogenic effect, i.e. predominantly superficial cells. As ovarian function wanes, there is a shift to the left in the maturation function. As a result of the progressive oestrogen deficiency, the glycogen content and vaginal acidity gradually decrease. The vaginal pH rises from 4.5 to 7.0 and Doderlein's bacilli are replaced by commensal organisms, thus increasing the susceptibility to infection. Dyspareunia and vaginitis are commonly noted.

The labia majora and minora lose some subcutaneous fat and become shrivelled. The atrophic changes extend to the perineal body, so that the whole introitus gives the impression of having receded.

The urethral epithelium becomes thinned and the bladder sphincter loses tone, giving rise to urgency, frequency and urge incontinence. The mucosa of the urethra may prolapse. The breasts at first become tender and turgid probably because of the unopposed oestrogenic effect. Later, they appear small and flaccid, and the nipples regress.

Menstrual changes:

The menopause is normally preceded by irregular cycle length. The cycle is anovulatory: later it becomes longer because a longer stimulus is now required for endometrial regeneration. During this transitional phase therefore, the periods become scantier or irregular and then menstruation ceases entirely. Occasionally, menstruation may cease abruptly without any alteration in flow or regularity. If menopause occurs before the age of 40 years, it is classified as premature.

A woman is regarded as being postmenopausal if menstruation has ceased for six months.

The most characteristic symptoms of the menopause are the vasomotor disturbances, commonly referred to as hot flashes, often accompanied by sweats, chills, irritability, depression and headache. A “hot flash” is a sensation of heat, usually confined to the chest, neck and face, and lasts for a few seconds to a few minutes. This is followed by a “flush” in which the area appears hyperaemic and then there is profuse sweating. These disturbances are generally precipitated by some emotional experience but may occur during sleep. Occasionally, there are palpitations or episodes of paroxysmal tachycardia (Table 1).

Table 1. Climacteric symptoms

-
1. Hot flashes
 2. Flushes
 3. Night sweats
 4. Mood swings
 5. Insomnia
 6. Vaginal dryness
 7. Loss of libido
 8. Dyspareunia
 9. Tiredness
 10. Impaired concentration, attention and memory
 11. Headache
-

The aetiology of the vasomotor symptoms is not known with certainty, but oestrogen i lack and/or gonodotrophin excess may be responsible.

There is a distinct possibility, however, that these disturbances are associated with instability of the vasomotor centre in the hypothalamus, probably related to changes in levels of brain catecholamines.

Other disturbances include emotional instability, insomnia, irritability and depression. Psychological disorders ensue. The woman fears approaching old age she realises that her sex life will change and that degenerative changes are inevitable. Her children have grown up and are more or less independent and her husband’s attention may be elsewhere. The woman may

withdraw from society. She may complain of headaches due to nervous tension. She may develop dyspareunia and loss of libido, due to a combination of diminished hormonal secretion and the psychological influence.

Complications of the postmenopause are shown in Table 2.

Table 2. Complications of the postmenopause

1.Climacteric symptoms
2.Osteoporosis
3. Coronary artery disease
4: Dogenita1 atrophy
5. Utero-vaginal prolapse
6. Postmenopausal bleeding
7. Endometrial hyperplasia and adenocarcinoma
8. Dementia

Treatment

The menopausal syndrome responds specifically to hormone replacement therapy. There is no doubt that oestrogens alleviate the vasomotor symptoms such as hot flushes and they correct the deficiency state.

The use of oral oestrogens is debatable. They may induce breast lumps and irregular uterine bleeding. Proponents of oestrogen therapy suggest conjugated oestrogens such as Premarin 0.3 to 0.625 mg in a cyclical fashion, adding an oral progestogen such as Provera during the last 12 to 14 days of treatment, to prevent hyperplasia and to ensure regular and normal withdrawal bleeding. Menopausal patients prefer to “see their periods” than to suffer the symptoms.

The progestogen was formerly given for only 5 to 7 days. However, morphological and biochemical studies of the endometrium suggest that 12 to 14 days of progestogen are needed during oestrogen cycling to provide maximal maturation of the endometrium and eliminate any hyperplastic changes.

Alternatively, combined oestrogen and progestogen preparations such as Angeliq can be prescribed which will diminish the possibility of any vaginal bleeding. Another non-bleeding form of hormone therapy is Tibolone which appears to be at least as effective for climacteric symptoms as other forms of therapy. Other preparations are gaining popularity in recent times. These include a nasal aerosol spray and oestradiol gels.

Table 3. Routes of administration of hormone therapy

1.Orally e.g. Oestradiol
2. Transdermally e.g. Estraderm, Evorel
3.. Vaginal cream e.g. Premarin
4. Nasal spray e.g. Aerodiol
5. Vaginal ring e.g. Estring
6. Oestradiol gel
7. Implant

Therefore, the addition of a progestogen can be said to be protective to the endometrium. The incidence of hyperplasia is significantly reduced and oestrogen- related hyperplasia can be reversed to normal endometrium. If a patient develops break- through bleeding when on hormone replacement therapy, this situation demands immediate investigation.

A new innovation in administering oestrogen is the Estraderm, oestradiol transdermal system. Estraderm comes in the form of a patch which, when applied to the skin, allows continuous diffusion of the hormone at a relatively controlled rate. It reduces the frequency of hot flushes and significantly increases the percentage of vaginal superficial cells.

Some of the options for the treatment of climacteric symptoms are mentioned in Table 4.

Table 4. Treatment of climacteric symptoms

1. Oestrogen
2. Clonidine
3. Tibolone
4. Selective serotonin re-uptake inhibitors
5. Megestrol acetate
6. ? Soy products

Atrophic vaginitis may be treated with Premarin vaginal cream. Small amounts of the oestrogen are readily absorbed through the vaginal epithelium and may stimulate the endometrium to produce hyperplasia, if given for prolonged periods. Therefore, uterine bleeding may occur during topical therapy as it does during therapy with oral oestrogens.

Avoid prolonged unopposed oestrogen with an intact uterus.

In most patients with the menopausal syndrome, reassurance and superficial psychotherapy are an essential part of management. For example, she is no longer inhibited by the fear of an unwanted pregnancy and she is free from the discomfort and the nuisance of a period. She should be advised to reactivate old interests and to accept new challenges. Mild tranquillisers may be used to relieve anxiety and depression and supportive discussion of other problems are indicated.

Several studies have established an accelerated rate of bone-loss after the menopause. The skeleton becomes fragile and about 25 per cent of postmenopausal women can be expected to develop symptoms from the complications of osteoporosis. It is due to oestrogen deficiency, but the mechanism of action of oestrogen remains in doubt. One in three women will sustain an osteoporotic fracture in their lifetime, either in the proximal radius, the spine or the neck of the femur.

The criteria for diagnosis of osteoporosis are shown in Table 5.

Table 5. Diagnosis of osteoporosis

1.	Loss of height
2.	Low backache (below T7)
3.	Normal serum calcium, phosphorus and alkaline phosphatase
4.	X-ray of the spine shows: decreased trabeculation, increased cortico-trabecular contrast, vertebral deformity — biconcavity, compression, absence of cortical erosions .

Treatment with oestrogen reduces the severity of bone-loss in osteoporosis. The spine and pelvis are most commonly involved, the long bones less frequently and the skull almost never, in contrast to hyperparathyroidism. The woman complains of backache, loss of height and kyphosis. The differential diagnosis is metastatic carcinoma, myeloma, osteomalacia, hyperparathyroidism and prolonged corticosteroid therapy.

Osteoporosis is characterised by increased translucency of the skeleton in the presence of normal serum calcium levels and alkaline phosphatase.

Oestrogens can be administered with the usual safeguards as they have the same side-effects as when used for vasomotor symptoms. Other non-oestrogenic treatments for osteoporosis include weight-bearing exercises, calcium, vitamin D, bisphosphonates (e.g. Alendronate), calcitonin and selective oestrogen receptor modulators such as Raloxifene.

Cardiovascular disease:

It is well known that cardiovascular disease before the age of 40 is less common in women than in men. In postmenopausal women, the incidence increases, and an oestrogen lack has been postulated.

It has been found that in women subjected to bilateral oophorectomy, there is an increase in the occurrence of atherosclerosis. Furthermore, the plasma cholesterol level and the cholesterol/phospholipid ratio have been found to be lower in young women than in young men. After the menopause, cholesterol and triglyceride levels increase.

The role of oestrogen to reduce cardiovascular disease is however controversial and the present consensus is that long term oestrogen therapy is of doubtful value in preventing or treating atherosclerosis in postmenopausal women.

Oral oestrogen has beneficial effects on serum lipids, a neutral or slightly beneficial action on blood pressure and a modest but adverse effect on clotting factors. While epidemiologic data have suggested a beneficial effect of oestrogen in the primary prevention of coronary artery disease, several randomised trials, such as the Women's Health Initiative (WHI) and the Heart and Estrogen/Progestin Replacement Study (HERS) have demonstrated that oestrogen progestin therapy for primary and secondary prevention is not cardioprotective and may increase the risk slightly. WHI also found a small increased risk of breast cancer, stroke and thrombo-embolism. On the other hand, osteoporosis and colonic cancer are less common with hormone therapy versus placebo.

Postmenopausal bleeding

Postmenopausal bleeding is bleeding from the genital tract occurring after an interval of more than six months following the menopause. The bleeding may be in the form of slight spotting or may be more substantial, recurrent and bright-red in colour. All women with postmenopausal bleeding should be thoroughly evaluated in order to arrive at an accurate diagnosis and to rule out a malignancy of the genital tract.

Aetiology

Postmenopausal bleeding may occur from a lesion situated on the epithelial surface anywhere in the genital tract (Table 6).

Table 6. Causes of postmenopausal bleeding

Urethral orifice: caruncle, prolapsed urethral mucosa
Vulva: carcinoma, granuloma inguinale, ulcer
Vagina: atrophic vaginitis, decubitus ulcer, retained pessary, trauma from coitus, carcinoma (rare)
Cervix: carcinoma, polyp, erosion, cervicitis
Uterus: endometrial carcinoma, submucous polyps, adenomatous hyperplasia, atypical endometrial hyperplasia, atrophic endometritis
Fallopian tube: carcinoma (very rare)
Ovaries: granulosa-cell tumours: they produce oestrogens which cause endometrial hyperplasia
Hormonal replacement therapy
Bleeding diathesis
Anticoagulant therapy

Diagnosis

After a thorough examination, the basic method of investigation includes a Papanicolaou smear, a diagnostic curettage and biopsy of any cervical lesion.

A thorough evaluation is required because of the increased incidence of malignancy that occurs after the menopause. Since endometrial lesions may be missed by a single-stage endometrial biopsy done on an outpatient basis, we prefer to subject the woman, despite anaesthetic risks, to a thorough formal curettage. A transvaginal ultrasound scan for the measurement of endometrial thickness is a useful screening procedure. If the endometrium is greater than 4mm, curettage or hysteroscopic-directed biopsy is warranted to exclude hyperplasia or carcinoma.

Other diagnostic methods that have been employed to recover carcinomatous cells include the Gravlee jet washer and the Vabra vacuum aspirator. Hysteroscopy is a valuable procedure in delineating the extent and location of tumours, and through it, a biopsy can be taken.

Treatment

If exogenous oestrogen therapy is not the cause of postmenopausal bleeding, treatment consists essentially of correcting the underlying cause.

Briefly, a urethral caruncle and prolapsed urethral mucosa are excised; granuloma inguinale is treated with antibiotics; vulval carcinoma requires a vulvectomy; atrophic vaginitis and

ulceration of the vagina are treated with an oestrogenic cream; cervical carcinoma requires radical hysterectomy with pelvic lymphadenectomy or radiotherapy depending on the stage; carcinoma of the endometrium is generally treated by a total hysterectomy and bilateral salpingo-oophorectomy followed by postoperative radiation, and if there is metastatic disease, high dose progestogens alleviate the symptoms and produce objective remission; and ovarian tumours require debulking surgery and chemotherapy.

It should be stressed that postmenopausal bleeding is a most important symptom and may be due to an underlying carcinoma of the genital tract. Prompt investigation is required even in the presence of an obvious benign lesion, such as atrophic vaginitis. This includes a Pap smear and a cervical biopsy to rule out cervical carcinoma and a hysteroscopy or curettage to rule out an endometrial carcinoma. Since postmenopausal atypical endometrial hyperplasia may progress to malignancy, such patients should be treated as for an endometrial carcinoma.

Some possible complications of oestrogen therapy are

- 1) Gall-bladder disease
- 2) Breast tenderness and breast neoplasia
- 3) Endometrial hyperplasia and malignancy
- 4) Thrombophlebitis
- 5) Thrombo-embolism

KEY POINTS

- 1. Humans are unique among the animal kingdom in that the exhaustion of oocytes occurs relatively early in life.**
- 2. The climacteric is a transitional phase which bridges the gap between normal reproductive function and ovarian senescence.**
- 3. Many of the clinical features of the climacteric are believed to result from a decline and eventual failure of ovarian function.**
- 4. The problems frequently seen are hot flushes, irritability, atrophic vulvitis and vaginitis, trigonitis and urethritis, insomnia, breast regression, low backache and osteoporosis.**
- 5. Basic concepts of management include psychotherapy designed to educate and to reassure the women and hormonal therapy designed to correct oestrogen insufficiency.**
- 6. The undoubted benefits of unopposed exogenous oestrogens, given cyclically, can be achieved only at a risk of excessive stimulation of the endometrium in an unacceptably high number of patients. The addition of a progestogen for 12 to 14 days each month or the use of a combined oestrogen/progestogen preparation is protective to a certain extent, to the endometrium.**
- 7. Hormone therapy is an effective therapy for climacteric symptoms as well as for the prevention of osteoporosis.**
- 8. Hormone therapy should be prescribed on an individual case-by-case basis. Individualising therapy entails a careful assessment of each patient's unique benefit and risk profile.**
- 9. Treatment should be re-evaluated over time and generally therapy should be limited to the shortest duration possible.**
- 10. Hormone therapy is not cardioprotective.**
- 11. For oestrogen-alone users, there remains a risk for stroke and thromboembolism; however, breast cancer is lower and there is a protective effect on hip fracture.**
- 12. After discontinuing oestrogen, there remains a lower risk of breast cancer and the risk of stroke and blood clots disappears.**

CHAPTER SIXTEEN

PELVIC ORGAN PROLAPSE

K. Muthukumarappan

Introduction

Prolapse is a relatively common problem encountered in the gynaecological clinic and was previously termed uterovaginal prolapse. This term which appears to emphasise the prolapse of the uterus has been largely superseded in order to reflect the more varied manifestations of prolapse. The majority of patients have a prolapse of the bladder (cystocoele) with or without accompanying uterine prolapse. This may be accompanied by a urethrocoele which is a prolapse of the urethra in the distal anterior compartment. Prolapse through the posterior compartment of the vagina will involve the rectum (rectocoele) or small bowel (enterocoele) or a combination of the two (enterorectocoele). The term 'pelvic organ prolapse' is more accurate and is currently accepted by the International Continence Society as well as the International Urogynaecology Association and is the term used in the literature.

The extent of the problem is not known as it is very much dependent on the patient seeking medical help. There are few studies that have evaluated the incidence and prevalence of prolapse and similarly, few that have looked prospectively at the natural history of prolapse. It has been estimated that 40-60% of women may have some form of pelvic organ prolapse and about 11% of 20 year olds will need surgery for prolapse by the time they are 60 years old. Most prevalence studies are hampered by selection bias in the form of hospital based studies in women presenting with other gynaecological symptoms or in those having surgery for pelvic organ prolapse.

Some evidence is available from the Womens Health Initiative Study. Annual pelvic examinations were performed on 412 menopausal women as part of the oestrogen- progestin trial and the mean followup was 5.7 years. Using a non-validated classification system, 31.8% of women had pelvic organ prolapse at baseline. The annual incidence of new onset pelvic organ prolapse was 9, 5 and 7 per 100 women for cystocoele, rectocoele and uterine prolapse, respectively. Rates of progression and regression were also looked at. The rate of progression was 9.5, 13.5 and 1.9 per 100 women years, respectively. Of greater interest were the figures for spontaneous regression which were 23.5, 22 and 48 respectively. These data suggest that pelvic organ prolapse is common and spontaneous regression occurs in many cases.

A further inference that can be drawn from this is that the majority of patients with pelvic organ prolapse are asymptomatic and only look to medical assistance if the condition impacts on quality of life issues or there are associated symptoms such as urinary incontinence.

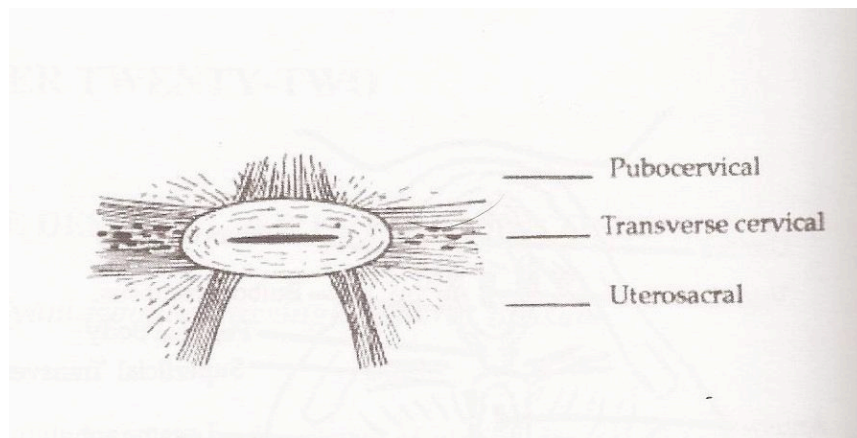
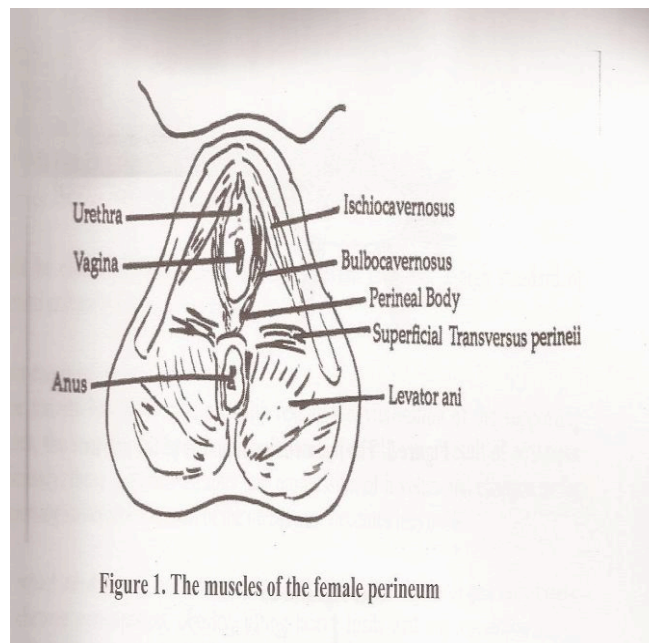


Figure 2. Ligamentous support

Predisposing factors

Age

This is the only universally accepted risk factor for prolapse and is borne out in studies regardless of the definition adopted for pelvic organ prolapse. Most estimates show approximate doubling of the risk of prolapse with every completed decade of life. In the postmenopause, both the muscular and ligamentous support of the uterus becomes deficient (Figures 1 and 2).

Menopausal status and hormone replacement

The association between menopausal status and the use of hormone replacement with prolapse is complicated by the previous risk factor i.e. age. Menopausal women are the older age group and when age is taken into account, the menopausal state becomes non-significant. Assessment of the effect of hormone replacement is complicated by the various dosing schedules that are available and no clear association is discernible between ever and never using hormone replacement. Despite this, hormone replacement has a role in management of these patients as some response is seen and this may reduce the need for surgery. It also improves the skin texture in those going to have surgery.

Pregnancy and parity

This relationship has been extensively studied. The pathophysiology is assumed to be direct damage to the pelvic nerves, fascia and muscles as the fetus passes through the birth canal. This results in eventual relaxation of the pelvic floor musculature and causes pelvic organ prolapse. Increasing parity appears to increase the risk of prolapse. There is anything between a 4-11-fold increase with increasing parity. The mode of delivery has not been shown to have a consistent effect. When women delivered by caesarean only were compared to those having had a vaginal delivery, the data was not suggestive of a protective effect. There is inadequate data on instrumental vaginal delivery.

Data relating to birth weight shows greater consistency, with most studies showing an increasing risk of prolapse with increasing birth weight. Macrosomic babies appear to confer the greatest risk.

Pregnancy and macrosomia are therefore identified as risk factors but there is debate as to whether caesarean deliveries or the avoidance of assisted deliveries has a protective effect on the future incidence of pelvic organ prolapse.

Medical illnesses

This centres mainly on conditions that increase intra-abdominal pressure, the collagen vascular diseases as well as those conditions that have vasculopathic and neuropathic consequences. There does not appear to be a strong relationship between pulmonary disease and prolapse if smoking is taken out of the equation. There is inconsistent data with collagen vascular diseases such as Marfan's and Ehlers-Danlos syndrome as well as prolonged use of steroids. Similarly, no association is seen with hypertension and diabetes mellitus.

Constipation has a more complex relationship with pelvic organ prolapse being both a cause and effect of the condition. Patients who are chronically constipated are more likely to develop prolapse as a result of straining and this may be seen in those with habitual constipation, hypothyroidism and those on long term opioid treatment.

Race

There is observational data that Hispanic women are at increased risk of pelvic organ prolapse when compared to Caucasian women. African-American women seem to have the lowest risk of these three groups whilst the data on Asian women is inadequate.

Lifestyle factors

Occupations that require a lot of lifting appear to have a greater risk and this has been documented with nursing assistants as well as with housewives and manual workers.

Obesity is also commonly quoted as an independent risk factor as is smoking but the evidence is not conclusive.

Pelvic surgery

It is well known that up to 30% of women who have had surgery for pelvic organ prolapse will require a second operation for recurrent prolapse. This group however may already have other predisposing factors for prolapse and it would be difficult to quantify risk. Patients who have had a hysterectomy for other indications are more prone to pelvic organ prolapse and the pathophysiology is the disruption of the supporting ligaments and structures to the vaginal apex. Most studies that have evaluated hysterectomy suggest that there is an increased risk of pelvic organ prolapse.

Patients who have had a hysterectomy or other surgery for prolapse are at up to five-fold increased risk of recurrent prolapse.

Relevant anatomy and pathophysiology

Pelvic organ prolapse occurs when there is a weakness or disruption of the pelvic floor musculature. The main structure of the pelvic floor is the levator ani muscle which is made up of two parts, the iliococcygeal muscle and the pubovisceral muscles.

The iliococcygeus arises from a fibrous band on the pelvic wall (the arcus tendineus fasciae pelvis) and inserts into the iliococcygeal raphe that is connected to the inner surface of the coccyx and sacrum. The pubovisceral muscle is a thick U-shaped muscle that consists of several parts named after the insertion points. These are the puborectalis and the pubococcygeus which form the main bulk. There is also the puboperineal muscle that inserts into the perineal body, pubovaginalis and the puboanalis that inserts in the space between the internal and external anal sphincters. The urethra, vagina and rectum pass through the urogenital hiatus of the levator ani.

The secondary supports are the endopelvic fascia and the ligaments. The main support is the transverse cervical (cardinal) ligaments with further suspensory support from the broad ligaments and the uterosacrals.

The levator ani is in a state of constant contraction and this activity keeps the urogenital hiatus closed thereby squeezing the structures against the pubic bone. The contracted muscles form a horizontal shelf on which the pelvic organs are supported. If it does not function properly, the secondary supports are called upon and are put under tension. These ligaments are soon overloaded and eventually fail to hold the vagina in place. This results in prolapse.

Presenting symptoms

Of those patients who seek help, the majority of patients will present with symptoms attributable to the protrusion. They will describe a sensation of fullness within the vagina or a 'dragging feeling' down below. Other descriptions would be of 'something coming down' or the feeling of sitting on a ball. Low backache may be a feature of stretching of the uterosacral and cardinal ligaments. This will usually be minimal soon after waking but progressively get worse throughout the day reflecting the effect of gravity in the upright position. Increased vaginal discharge may be a feature.

A patient may present because of the effect of these symptoms on their occupation, housework, personal relationships and psychological wellbeing. The associated features may be of urinary incontinence, sexual dysfunction, constipation and faecal incontinence.

In some cases, there may be trophic ulceration of the exposed prolapse and in extreme situations, urinary retention due to bladder outlet obstruction leading to retrograde flow to the ureters and kidneys.

On the whole, in pelvic organ prolapse, mortality is nil and morbidity is low. This is reflected in the history, examination, management and counselling.

History

A detailed and comprehensive history is important. The presenting history should always be taken in the patient's own words. The majority of these patients would have had the condition for a length of time and one should try to define the trigger that has led to the patient seeking assistance. This is critical as the treatment can then be catered to the patient's needs. The history should also shed light on predisposing factors, if any.

Direct questioning will then elucidate other areas that may be related. As the primary problem is a weakness of the pelvic floor, questions should be directed at the four main domains affected by pelvic floor dysfunction namely:

1. Symptoms of prolapse
2. Symptoms of urinary incontinence and bladder dysfunction
3. Symptoms of bowel dysfunction
4. Sexual dysfunction

Unless one looks for these areas, patients tend to shy away from talking about bowel and sexual dysfunction as they find it awkward and embarrassing. Indeed, many medical professionals find it difficult to discuss these issues.

A list of common symptoms is shown in Table 1.

Table 1. Common symptoms

Prolapse	Symptoms
Uterine prolapsed	Pressure symptoms Backache Impairment with coitus
Cystocele	Fullness, pressure symptoms Hesitancy, incomplete voiding Overflow and coital incontinence Urinary infections
Rectocele	Fullness and pressure symptoms Constipation May have alternating diarrhoea Incontinence of flatus and faeces
Enterocoele	Fullness

A detailed obstetric and medical history is important as discussed previously- this should include history of prolonged labours, difficult deliveries and birth weights. The family history will shed light on inherited conditions and familial tendency to pelvic organ prolapse. Social history is needed not just for occupational hazards but to assess the type of treatment which will be required as well as the timing and recovery requirements. Obviously, a full drug, smoking and allergy history is vital.

Clinical examination and investigations

A full general examination should be performed. The respiratory system should be assessed for conditions that may cause chronic coughing. One should look for abdominal and pelvic masses especially uterine fibroids and ovarian cysts.

The pelvic examination should be performed in the left lateral position with a Sims speculum. This will allow a full examination of the cervix followed by sequential assessment of the anterior and posterior wall of the vagina to look for the various types of pelvic organ prolapse as the patient strains. There is no hard and fast rule on the actual classification of the severity or grade of prolapse. In most instances, it is a clinical assessment to explain the symptoms the patient has presented with i.e. is there any prolapse present. One should also try to define the maximal protrusion through the vaginal wall and this may necessitate examining the patient whilst she is standing.

A simple clinical assessment of the severity of uterine prolapse is

Grade 0: no descent of the cervix

Grade 1: cervical descent within vagina

Grade 2: cervical descent to the introitus

Grade 3: cervical descent outside the vagina

This is combined with a subjective description of small/moderate/ large cystocele and/or rectocele. Complete extrusion of the uterus is termed procidentia (Figure 2).

From the point of research and clinical reporting for direct comparison, older literature will use the Baden-Walker classification which is the vaginal profile and the various prolapses are classified as

Grade 0: no descent

Grade 1: descent halfway to the hymen

Grade 2: descent to the hymen

Grade 3: descent halfway past the hymen

Grade 4: maximal descent

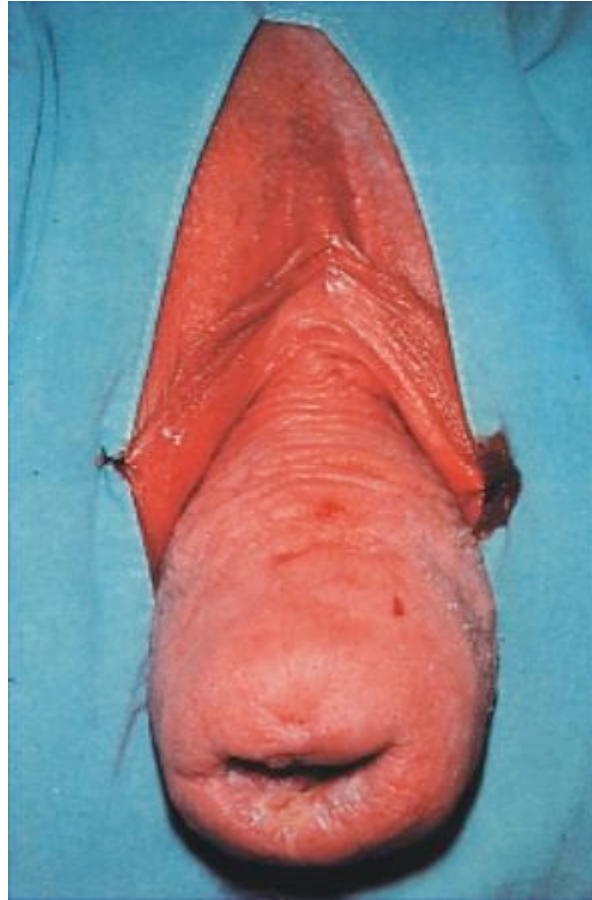


Figure 2: Showing uterine prolapse.

Since 1996, the International Continence Society has adopted the POP-Q which is a physical recording of defects relative to the hymenal ring in centimetre gradients. The strength of this system is that it uses precise anatomic measurements that are useful for research but it can be cumbersome for general clinical use.

There are very few investigations that are needed in the management of pelvic organ prolapse. A mid stream urine specimen is indicated for urinary symptoms and an ultrasound to rule out abdominal and pelvic masses. Other investigations will be guided by the presence of medical co-morbidity and the need for pre operative assessment if surgery is planned.



Figure 3. Cystocoele

Figure 3 shows a large cystocoele.

Treatment

The treatment modalities are conservative and definitive. Conservative treatments are defined as lifestyle interventions, physical therapies, rings and pessaries. Definitive treatment is surgical reduction of prolapse.

There are several important considerations. As mentioned previously, pelvic organ prolapse has minimal morbidity and one is often treating a subjective problem. It is vital to have a clear and frank discussion with the patient and decide on the goals that are to be achieved. The goals will have to be realistic and achievable.

The primary problem is pelvic floor dysfunction and the only measure that directly addresses this is pelvic floor muscle toning. All other treatments are aimed at subjective cure and each patient will have their idea with regards to what they would term 'cure'. It is important to make it clear that no one single measure is guaranteed to provide long term success. The best outcomes are seen in patients who take on board all aspects of management i.e. a combined regimen.

Lifestyle interventions

This will include general measures such as weight loss, the cessation of smoking and reducing exacerbating activities such as lifting, coughing and constipation. The use of local oestrogens will reduce vaginal atrophy.

Physical therapies

The role of pelvic floor muscle toning is well established in the management of urinary incontinence but its role is less understood in pelvic organ prolapse. This is probably a result of the varying nature of subjective awareness of the symptoms as well the higher incidence of spontaneous regression of clinical symptoms.

Success of pelvic floor training programmes is very dependent on patient motivation and the degree of response. The better the initial improvement, the more likely the patient is to continue with the exercises. The response itself is dependent on the ability to perform a correct pelvic floor muscle contraction and up to a quarter of patients may actually be performing a valsalva manoeuvre. It is therefore, important that the patient is examined and taught how to perform the necessary contractions by a person with specialist training in this area i.e. a womens health physiotherapist. The patient should be examined whilst performing the contraction to ensure that the correct muscle groups are being targeted and the regimen is catered to individual needs.

All patients should be encouraged to perform pelvic floor muscle toning in addition to all other planned interventions. If surgery is the definitive treatment, then patients need to learn pelvic floor toning prior to embarking down this line.

Rings and other pessaries

Pessaries have been reportedly in use for centuries and they fell out of favour as modern surgical techniques became safer. However, with current understanding of surgical risks and failure rates, pessaries are increasingly popular again. They can be used in two ways, either as primary therapy or as an interim measure till surgery can be performed. Primary treatment is for the patients who are medically unfit or too frail for surgery as well as those who do not wish to have an operation.

There are many types of pessaries but they can be grouped into those that are supportive, in which the levator ani muscle tone is needed to keep the pessary in place or space occupying pessaries which keep the prolapse reduced by filling the vagina. An example of the first is the ring pessary and the second is the shelf pessary. Modern pessaries are made of medical grade silicone which is non allergenic and non toxic compared to the older rubber pessaries.

Pessaries will need to be replaced regularly and this allows for regular pelvic examinations to ensure there are no problems with the devices. Commonly encountered problems include

trophic ulcers, vaginal discharge, urinary retention and occasionally, unmasking of urinary incontinence as the protective kinking of the bladder neck is reduced.

The patients will need to use regular intra vaginal oestrogen creams to reduce atrophy and this is usually twice weekly. They are seen every three to four months for the change of pessary. Wu et al reported up to 74% of patients can have a pessary fitted successfully and in those who persist, 66% still use it after one year. Of those with symptomatic pelvic organ prolapse, only 3% of women reported still feeling a bulge after two months and other symptoms that improved included pressure effects, vaginal discharge and voiding difficulty¹⁵.

Surgery

Repair of the anterior compartment

Anterior colporrhaphy is aimed at plicating the vaginal muscularis and the adventitia overlying the bladder to reduce the protrusion of the bladder and vagina. One needs to identify the defect prior to deciding on how lateral to take the dissection. A small midline defect can be reduced by a midline plication whereas a large of a lateral defect may need mobilisation of the fascia laterally or a vaginal paravaginal repair. In the latter, the prevesical fascia and vaginal muscularis are approximated to the arcus tendineus fasciae pelvis (the white line as it is called). The repair is combined with a plication of the urethrovesical junction to minimise the risk of stress urinary incontinence.

In the more complex cases, an abdominal or retropubic approach is sometimes required.

Complications include bleeding, bladder injury, urinary sepsis, voiding difficulty and new onset stress incontinence. The effect on sexual function is variable. Ureteric injuries are rare.

Repair of the posterior compartment

Rectocoeles can be repaired either vaginally or transrectally depending on whether the surgeon is gynaecologist or a colorectal surgeon. The initial step involves deciding on how much approximation of the introitus is needed followed by fascial placcation after dissecting the vaginal mucosa. The final step is the reconstruction of the perineal body. The operation is therefore termed a posterior colpoperineorrhaphy.

Complications include bleeding, dyspareunia, rectal injury and rectal urgency.

An enterocoele arises when there is a large Pouch of Douglas which is open allowing bowel to settle into the space. Surgical treatment involves obliterating the rectovaginal pouch usually in addition to major pelvic floor surgery. Several procedures are described (such as the Moschowitz and Halban's operations) and the complications include ureteral kinking and small bowel obstruction.

Uterine prolapse

If treatment is required, a vaginal or abdominal hysterectomy accompanies correction of compartmental defects. The uterus is usually an innocent bystander and its removal leads to an increased incidence of recurrent prolapse.

Recurrent prolapse

The so called definitive management of pelvic organ prolapse has one major drawback- it is not a long term cure. Surgical repair corrects the effects of the pelvic floor weakness but does not treat the primary problem. It is well known that 30% of patients undergoing surgery for prolapse will need a second procedure for recurrence. Again, these figures reflect the symptomatic patients who seek help.

The risk factors for recurrent prolapse include

- Previous hysterectomy
- Obesity
- Smoking
- Failure to persevere with pelvic floor toning
- Constipation
- Respiratory illness leading to chronic cough

Recurrence can happen within months of the primary surgery to several years later. The assessment and treatment plans are not different to that at initial presentation.

Further surgery can be done with a vaginal approach (sacrospinous vault suspension, iliococcygeal fixation, posterior intravaginal sling, mesh repairs) or with an abdominal approach (sacrocolpopexy using mesh).

The role of mesh in pelvic reconstructive surgery

Synthetic mesh has been developed due to a lack of satisfaction and safety issues with biologic materials. Various materials have been tried and tested but the ideal mesh is yet to be developed. The best mesh currently available is the monofilament macroporous non-absorbable polypropylene mesh which is non allergenic, non toxic and light weight.

The majority of practising surgeons still reserve the use of mesh for recurrent prolapse although its use in primary repair is increasing. Mesh can be used as an inlay where the mesh is cut into

the required shape and size to fit the defect or as the new mesh kits which come with introducers to enable fixation to defined anatomical landmarks.

Infection is a worrying problem with any foreign body but this is minimised with the larger pore size which enables leucocytes to traverse the mesh. There is a 10-11% risk of mesh erosions or extrusions which is usually through the vaginal mucosa and presents with vaginal bleeding and discharge. The mesh will progressively retract and this will lead to deformity of the vaginal shape and capacity. Combined with altered sensation due to the mesh being sited beneath the vaginal mucosa, this will lead to impairment of sexual function and de novo dyspareunia.

Prevention

Several of the predisposing factors are not changeable i.e. race, ethnicity, parity, genetic defects. Certain factors can be optimised such as medical illnesses.

Obesity and smoking are lifestyle changes that can be advocated. The use of topical hormone replacement is somewhat more controversial and needs a risk-benefit analysis that has to be individualised to the patient.

Increasingly, in the United Kingdom, pregnant women are being counselled and shown pelvic floor muscle toning in the antenatal period when the women are more receptive to advice.

KEY POINTS

- 1. Pelvic organ prolapse is a common problem easily affecting 30% of the female population but the actual prevalence is not known**
- 2. Only the symptomatic patients seek treatment. It does spontaneously regress in some patients.**
- 3. Age and parity appear to be the most consistently linked predisposing factors.**
- 4. Assessment of the degree of prolapse is mainly a clinical impression. POP-Q is a validated tool to precisely measure the extent of prolapse with reference to the hymenal ring.**
- 5. Initial management involves addressing obesity, smoking, chronic cough and constipation.**
- 6. Pelvic floor muscle toning plays an important role in minimising progression.**
- 7. Prolapse can be reduced with vaginal pessaries or with surgery catered to the**

affected vaginal compartment.

- 8. 30% of patients having surgery will need further treatment in the future.**

CHAPTER SEVENTEEN

URINARY INCONTINENCE

K. Muthukumarappan

Introduction

Urinary incontinence is the involuntary loss of urine. Continence itself has been defined as the ability to hold the urine until one is in an appropriate and socially acceptable time and place to void. Incontinence can therefore be defined as a lack of continence. This is a problem that significantly disrupts the lives of about 5% of the adult population. It can happen at any age but mainly affects the elderly. The Continence Foundation estimates that one in four women and one in nine men will suffer from incontinence that is significant enough to affect their quality of life, at some stage in their lives.

A Gallup survey in 1994 showed that 70% of patients put up with the problem attributing it to 'getting old'. They did not seek medical help as they were reluctant to talk about the problem. On average, patients put up with the leakage for about 4 years before they seek medical help.

The predominant effect is on the quality of life and the patients are embarrassed that they leak. They worry about needing to change clothes and the possibility of being smelly. They avoid going out for fear of leaking or not getting to toilet on time. This leads to social withdrawal and affects personal relationships and may cause problems with their ability to work.

The bladder

The urinary bladder is a hollow muscular organ situated entirely in the pelvis and comprises of a potential space, the bladder cavity surrounded by the bladder wall. The wall itself has several layers. The detrusor muscle is a complex network of muscle fibres and connective tissue. The trigone is a small triangular muscle on the posterior wall which prevents urinary reflux into the upper renal tract. The ureters open into the superior angles of the trigone and the lowermost angle is the bladder neck and urethra. The bladder is lined by transitional epithelium known as the urothelium.

It is innervated by the sympathetic nerves originating in the thoracolumbar region and running in the hypogastric nerves. The parasympathetic supply is from S2-4. The pontine micturition centre (PMC) in the brainstem switches the bladder from capacitance to micturition mode. Cortical control is inhibitory over the PMC and is critical to delay voiding until it is socially convenient.

In babies and infants, the bladder is an automatic organ that fills to a certain extent and empties- this goes on till the toddler is potty trained i.e. has achieved cortical control.

The mechanism of continence in women

Continence is determined by four main factors which are

- The ability to store urine
- An efficient urethral sphincter
- Efficient urethral support
- Urethral mucosal seal

The ability to store urine without any rise in the intravesical pressure is known as compliance. The bladder is a capacitance organ that is able to store urine without an increase in detrusor pressure during the filling phase until full capacity is achieved. This depends on the inhibitory neural pathways and the arrangement and stretching potential of the detrusor muscle.

The bladder has a sphincter that has a smooth muscle and striated muscle component. The striated muscle has a concentric arrangement that covers the proximal two thirds of the urethra and its contraction closes the urethra. The smooth muscle lies deep to this and is arranged in an inner longitudinal and outer (much thinner) concentric fashion and is thought to help shorten the urethra during micturition. The sphincter is deficient posteriorly where it sits on the pelvic floor.

The urethra is supported by the pelvic floor and passes through the urogenital hiatus with the vagina (see chapter on pelvic organ prolapse). If the supporting layer is firm, the abdominal pressure keeps the urethra closed. Should this supporting layer become lax, the occlusive action is diminished.

Within the urethra lies a surprisingly well developed venous plexus for a small organ and this is thought to provide a watertight mucosal seal due to the swelling of the vessels. These vessels can be damaged by surgery, radiotherapy and a lack of oestrogen.

Types of urinary incontinence

- Stress incontinence- leakage of urine with an increase in intra-abdominal pressure.
- Urge incontinence- leakage of urine preceded by urgency.
- Overflow incontinence- leakage of urine from a maximally filled bladder usually a result of damage to the pelvic innervations. This is akin to trying to fill a full jug as the excess will not be able to be stored.
- Automatic bladder- a bladder that fills to capacity and empties i.e. the infantile bladder. This is seen in cerebrovascular accidents, mental retardation, dementia, psychiatric disorders.

- Neurogenic bladder-this could mimic any of the last three and is seen in neurological disorders such as multiple sclerosis and in medical conditions that cause a neuropathy.
- Fistulae: continuous persistent leakage with no predisposing symptoms. This can be developmental and present to the paediatric urologist. It can also occur as a result of obstetric injury, complication of bladder and prolapse surgery or long term complication of radiotherapy.

In urogynaecologic practice, the first two conditions account for almost 95% of the patients seeking medical help.

Terms that are often used are

Urgency: this is a sudden and compelling desire to void that is difficult to defer.

Urge incontinence: the leakage of urine that is preceded by urgency.

Frequency: usually accompanied by urgency with or without incontinence and is a subjective complaint by a patient that she voids too often during the day.

Nocturia: again accompanies by urgency with or without incontinence when a patient has to get up at night on one or more occasions to void.

Hesitancy: a difficulty in either initiating or completing a void or both.

These terms are used mainly in the context of the overactive bladder which will be discussed later in the chapter.

Predisposing factors

Race

Several American studies have suggested that Caucasian women are more susceptible to stress incontinence than African-American or Asian women. The study designs are often wanting being based on hospital populations i.e. there is selection bias.

Family history

There appears to be greater likelihood of developing stress incontinence if first degree relatives are also affected.

Pregnancy, childbirth and parity

Urinary incontinence is common during pregnancy and is usually self limiting. Up to 28% of women will develop incontinence and the majority will resolve during the puerperium. Pregnancy related incontinence is thought to be a predictor of future urinary incontinence.

Instrumental delivery carries the highest risk of future urinary incontinence and caesareans appear to be protective. This led to the recent controversy in the United Kingdom when a survey of female obstetricians showed that they would prefer an elective Caesarean delivery. The relative risk of urinary incontinence of assisted delivery compared to normal vaginal and caesarean delivery is 1.5 and 3.1. Macrosomia appears to predispose to incontinence.

It appears to be more common in women with four or more children but the association is reduced with increasing age.

Age

Most studies indicate that the risk of urinary incontinence increases with age and this may be related to several factors. The older women are more likely to be menopausal and more likely to have co-morbidities. This leads to confounding factors that make it difficult to attribute risk to increased age alone.

Obesity

This is an established risk factor and the increased weight on the pelvic floor as a result of increased intra-abdominal and intra-vesical pressure causes weakening of the muscles and supporting structures. Obese women are 4.2 times more likely to develop stress urinary incontinence and 2.2 times more likely to develop urge incontinence. Weight reduction in morbidly obese women has been shown to reduce stress incontinence from 62% to 12%.

Increased intra-abdominal pressure

Patients with chronic constipation and chronic cough are associated with stress incontinence as a result of pudendal nerve neuropathy from straining and persistently increased intra-abdominal pressure.

Pelvic organ prolapse and anterior repair

Pelvic floor disorders can cause four major effects on the woman and these are prolapse, incontinence, bowel dysfunction and sexual dysfunction. Patients with anterior compartment prolapse can have bladder neck hypermobility with stress leakage. As prolapse and incontinence are common, it would not be surprising to find co-existing morbidity in some patients. Cystocele is known to demonstrate detrusor overactivity on urodynamic testing.

The presence of prolapse can often occlude the bladder neck and mask stress incontinence and the urinary leakage starts after reduction of the prolapse with either pessaries or surgery.

History and examination

A detailed history regarding the leakage can often give a true representation of the final diagnosis. In an audit carried out in Chesterfield Royal Hospital, all patients who give a history of pure stress incontinence symptoms had urodynamically proven stress leakage whilst those with pure overactive bladder symptoms were shown to have overactive bladder. Sixty per cent of patients with mixed urinary incontinence were found to have stress incontinence only with the others showing a true mixed picture.

The history should be able to define predisposing factors, if any and the social and occupational history is important.

A very relevant part of the history, which is often neglected, is the impact of the problem on the patient's quality of life. There are validated questionnaires that the patient can fill whilst waiting to be seen which can be very informative. The Kings Fund Quality of Life Questionnaire is one such example.

The physical examination should consist of a thorough general examination and a calculation of the patient's body mass index. An abdominal examination to exclude masses and previous surgery followed by a pelvic assessment for the presence or absence of pelvic organ prolapse should be undertaken. Vaginal examination will look for bladder neck hypermobility and stress leakage. The pelvic floor muscle tone is assessed including the patient's ability to perform a pelvic floor contraction. A rectal examination is an important part of the full assessment.

Investigations

All patients should have a urine dipstick performed and a mid stream urine culture if indicated. Chronic or recurrent urinary infections would mimic overactive bladder. Or greater worry is the presence of microscopic haematuria which warrants further investigations for more sinister causes.

Biochemical tests to assess renal impairment should be performed in those at risk of renal disease.

Post-void residual urine gives information about incomplete emptying of the bladder. A single measurement is inadequate and this should be done on two or more occasions. It is easily performed with an ultrasound assessment but if this is not available, bladder catheterisation is needed. It is considered significant if the urine remaining exceeds 30% of the total bladder capacity.

The pad test was used to quantify urine loss and the assessment could either be over one hour (short test) or a 24-hour test. A pad is weighed and the patient then wears it. A series of provocation manoeuvres are then performed (climbing stairs, walking, running, coughing or

whatever it is that makes the patient leak) over the period of an hour and then the pad is removed and reweighed. A gain of one gram is a positive test. Similarly, the 24-hour test is positive if the gain is 4g or more. This test is more representative and reliable as the patient just carries on over the 24 hour period as she would on any normal day. This test is not usually performed any more as other more informative tests are available.

The cotton swab or the Q-tip test was used to assess urethral hypermobility but is not recommended anymore.

Urodynamic testing

In its simplest form, this test is used to differentiate between the two commonest causes of urinary leakage in women i.e. are we dealing with stress incontinence or the overactive bladder. Urodynamic investigations allow us to precisely define bladder and urethral function, to guide potential treatment strategies and to identify and predict possible complications.

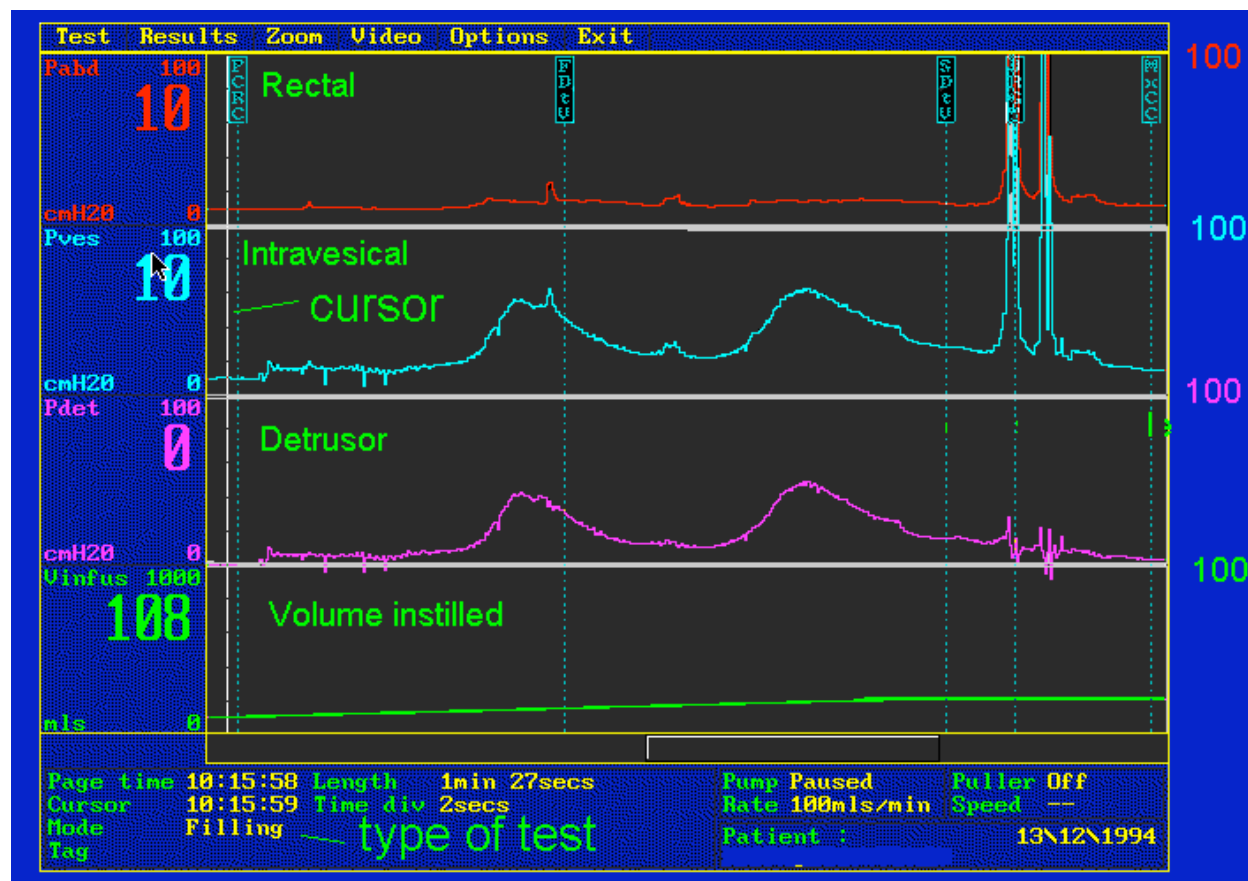


Figure 1. Cystometry

All patients are encouraged to complete a diary of fluid intake and output for at least three days prior to the test. Those on anticholinergic medication need to be advised to stop treatment for four weeks before the test.

Uroflowmetry

This is a simple but useful test that measures the rate and pattern of urinary flow during micturition.

The flow rate depends on detrusor contractility, abdominal straining, sphincter resistance, initial bladder volume and posture. The patient is asked to empty the bladder as normal in a situation that mimics normality i.e. relative privacy, sitting comfortably on a commode with a relatively full bladder. The rate of flow is assessed by either weighing the urine (scale under the commode) or by the rotating disc method, which looks at the urine flow maintaining the inertia of the rotation.

Several parameters are measured

- Flow rate (Q) is the volume of urine expelled via the urethra per unit time, expressed in ml/sec.
- Voided volume is the total volume that has been expelled.
- Maximum flow rate (Q_{max}) is the maximum measured value of the flow rate after correcting artefacts.
- Flow time(TQ) is the total time taken to void.
- Time to maximum flow (TQ_{max}) is the time from initiating a void to achieving the Q_{max} expressed in seconds.

Care should be taken with the analysis if the total voided volume is less than 150mls. A normal flow rate is a smooth curve and the TQ_{max} should be the initial third. The Q_{max} should be >15ml/sec.

An abnormally high flow rate indicates detrusor overactivity whilst an abnormally long pattern suggests outlet obstruction or reduced detrusor contractility. Intermittent flow usually reflects straining or changing posture that may be due to cystocoele or detrusor-sphincter dys-synergia (loss of coordination).

Cystometry

Cystometry is the measurement of pressure changes within the bladder. There are two aspects to this.

- Filling cystometry studies the relationship of detrusor pressure and volume during the bladder filling (Figure 1).

- Voiding cystometry looks at the relationship between detrusor pressure and urine flow rate during bladder emptying.

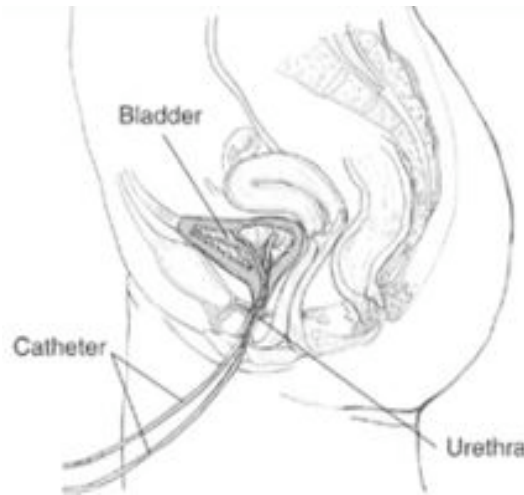


Figure 2. Assessment of detrusor pressure

There is continuous measurement of intra-abdominal and intra-vesical pressure during the process (Figure 2). The pressure within the bladder is the sum of intra-abdominal pressure and that exerted by the detrusor muscle (should be zero in a normal bladder). Subtracting the intra-abdominal pressure from the intra-vesical pressure gives the detrusor pressure (also called the subtracted detrusor pressure).

Two catheters are inserted into the bladder- one for filling and the other carrying a pressure transducer. Intra-abdominal pressure is measured with a pressure transducer in the rectum or in some centres, in the upper vagina.

The bladder is filled at a rate of 50ml/min which is the current standardised recommendation and pressure changes are recorded continuously in the form of a graph. Water or normal saline is used as the medium. The test should be interactive with the patient kept informed as the test progresses. The patient is asked to inform the team when she has first sensation of filling, normal sensation when she would void if she were at home and strong sensation or urgency/pain.

When the bladder is filled to a comfortable level, provocation manoeuvres are carried out to look for leakage. These are

- Coughing
- Straining/Valsalva
- Walking on the spot
- Jumping on the spot
- Any activity the patient says makes her leak

- Response to running water

The test is then completed with voiding while the measurements are recorded.

Normal cystometry is

- Residual urine of less than 50 mls.
- First desire between 150 and 200 ml.
- Bladder capacity of 300 to 600 mls.
- Little or no detrusor pressure rise on filling.
- No detrusor contractions during filling.
- No leakage on provocation.

Pitfalls of cystometry

- It is non physiologic as the bladder is filled rapidly.
- The patient finds it daunting and embarrassing.
- Artefacts may affect readings eg air in the catheters or failure to zero initial readings.

Indications for urodynamic studies include

- Patients being considered for bladder neck surgery
- Previous unsuccessful incontinence surgery
- Mixed urinary incontinence to identify predominant issues
- Suspected voiding disorders
- Neurogenic bladder disorders
- When conservative and pharmacologic measures have failed

More complex investigations

Ambulatory urodynamics over a 24 hour period may be more physiological.

Urethral pressure profiles to assess the pressures at different levels of the urethra- this helps to study urethral closure pressures and pressure at which leakage occurs.

Electrophysiologic testing of the sphincter muscle and perineal muscles (EMG).

Imaging techniques to look at how bladder shape changes with micturition and to look at proximal urethral changes.

Stress urinary incontinence

This has been described in the writings of physicians since Egyptian times and many corrective procedures have been described with varying success. The theories behind the pathophysiology of the condition have led to the most successful treatments.

Pathophysiology

Enhornings theory

Enhorning suggested that the bladder and the functional urethra were totally intra-abdominal organs. Any increase in intra-abdominal pressure would be equally transmitted to both the bladder and the proximal urethra and this maintained continence. Stress incontinence results from descent of the bladder neck so that the proximal urethra is no longer within the intra-abdominal cavity i.e. the increased pressure transmitted to the bladder was not opposed by a corresponding increase in pressure on the urethra. Leakage results and this led to surgical procedures designed to elevate the bladder neck into the intra-abdominal cavity- the colpourethroxies. An example of this procedure is the Burch colposuspension which has been the gold standard surgical option till the 1990s.

Delanceys hammock theory

DeLancey proposed that continence was due to the urethra being compressed against the subcervical hammock created by the levator ani muscle. When there is a weakness in the support, the compression is inadequate to prevent leakage of urine. This led to placement of slings that offered resistance against which the urethra could be compressed.

The integral theory

The mid urethra has been found to be the site of maximal intra-urethral pressure and Petros and Ulmsten proposed that this midurethra is the key to continence. This has led to the development of the suburethral tension-free tapes that are now replacing the Burch colposuspension as the procedure of choice.

History and examination

Leakage is seen with any increase in intra-abdominal pressure and the typical complaints would be incontinence with coughing, sneezing, lifting and straining. Leakage would also be seen with exertion such as walking, running, climbing stairs, bending and changing posture. There will be no urgency or frequency.

It is important to ask for hesitancy and poor or incomplete voiding as this carries implications for treatment.

On examination, one needs to look for concomitant prolapse as well as rule out abdominal masses.

Investigations

The urine dipstick should be normal in pure stress incontinence and there will be no biochemical changes.

Urodynamics should essentially be normal with leakage being seen on provocation without any detrusor pressure changes. If there is difficulty in initiating, maintaining or completing a void on uroflowmetry, one should be very cautious in the event of surgical management as these problems will be aggravated.

Management of stress urinary incontinence

In terms of nomenclature, the term stress urinary incontinence is used to denote a confirmed diagnosis of bladder neck weakness whilst stress incontinence is but a symptom that needs further elucidation.

Once the diagnosis is made, lifestyle issues need to be addressed first. Weight reduction is essential in those who are obese. Patients do struggle to lose weight and they need to be properly motivated with referral to a dietician if necessary. Constipation and chronic cough should be treated.

All patients should be referred to a specially trained physiotherapist in order to have supervised pelvic floor muscle toning. If pelvic floor muscle toning is performed correctly and faithfully, over 40% improvement in symptoms is noted. For a large number of patients, this in itself provides adequate symptomatic relief and further intervention is not required.

Pharmacologic treatment

Duloxetine is a combined norepinephrine and serotonin reuptake inhibitor and has been shown to increase sphincteric muscle activity during the filling phase. It has not gained wide acceptance as it is poorly tolerated. The commonest side effect is nausea (23%) and dry mouth, fatigue, constipation, somnolence and headache.

Oestrogen therapy appears to have little effect.

Surgery

Colpourethropexies were the most performed operations until the new suburethral tapes were launched. These were the Burch colposuspension, needle procedures and laparoscopic procedures. The Burch colposuspension is the most widely used and has the best long term results- up to 85% satisfaction rates after 5 years. The vagina and bladder are elevated by two or more sutures through the lateral vagina secured to the iliopectineal ligament. It carries the risks of

- Low pelvic pain and dyspareunia
- Post operative detrusor overactivity (up to 30%)
- Pelvic organ prolapse with a high incidence of enterocoele (up to 20%)

The tension free vaginal tapes (TVT) have revolutionised the treatment and are now the procedure of choice. They are placed with a vaginal approach and elevate the mid urethra as per the integral theory. They are easier procedures with less intra-operative and post-operative morbidity. The two main approaches are the retropubic TVT and the transobturator tapes (TOT, TVT-O).

Figure 3. TVT-O



Slings are rarely performed these days as the morbidity is higher and the operations more complex. Slings can be fashioned out of autologous material (eg fascia lata) or other biologic and synthetic materials.

Various types of urethral bulking agents have been injected into the urethra to improve the urethral mucosal coaptation (seal effect).

Other options include artificial urinary sphincters.

Overactive bladder

Several terms have been used in the past including detrusor overactivity, detrusor instability and irritable bladder. It is now accepted that detrusor overactivity is a feature of this condition and not the diagnosis. The term overactive bladder (OAB) is now the diagnosis. If this is associated with urinary incontinence, it is called OAB-wet and the patients who do not have urinary leakage are said to have OAB-dry.

Pathophysiology

The pathophysiology behind the condition is poorly understood and is believed to be either neurological or muscular in origin.

Clinical features

The predominant symptoms are daytime frequency, nocturia and urgency with or without urge incontinence. The prevalence is not known as those who present are the group with OAB-wet but it has been estimated that up to 12% of the population may have the symptoms.

Investigations

Urinary infection and other bladder pathology must be ruled out with urine dipstick, mid stream urine culture, urine cytology if indicated and possibly cystoscopy.

Urodynamics will usually show a contracted non-compliant bladder with unprovoked detrusor contractions. Stress leakage with the provocation tests is not a feature.

Management

Lifestyle factors need to be addressed. Obesity needs to be treated and one has to look at the patients daily intake-output charts, paying particular attention to the amount of caffeine and/or

alcohol. The patients tend to drink less and must be encouraged to drink normal amounts and avoid dehydration.

Pelvic floor muscle toning is combined with bladder drill. This is targeted behaviour designed to increase the time between successive voids and needs to be supervised with relevant positive feedback. The process involves setting increasing targets of time prior to bladder emptying eg if the patient voids every half an hour, then they should be encouraged to go 45 mins between voids. When this is achieved, the time is increased to 60mins and so on.

Most patients will need pharmacologic treatment. One of the most effective anticholinergics is oxybutynin but this is a nonselective muscarinic antagonist and has side effects. Most patients will complain of a dry mouth, blurring of vision, constipation and cognitive effects.

More bladder selective agents have been developed and these target the M3 receptors. Drugs commonly used are solifenacin and tolterodine. Trospium is used in the older population as it does not have cognitive effects.

All these agents have been shown to improve cystometric capacity and reduce urgency, frequency as well as nocturia.

The patients with refractory OAB can be offered intra-vesical injections of Botox into the detrusor muscle. A downside is that a large proportion of them will need to learn to self catheterise intermittently. Neuromodulation has been shown to be useful eg peroneal or tibial nerve stimulation.

Surgery

This involves augmenting the bladder with bowel (loop of ileum) as in clam ileocystoplasty or by-passing the bladder altogether eg ileal conduits (draining ureters into a urostomy bag).

Mixed urinary incontinence

The aim is to treat the predominant problem as the relief of one problem may give the patient adequate relief. This is made easier by the fact that the general and lifestyle measures show considerable overlap.

In the patients who suffer with both stress incontinence and OAB, the general treatment strategy is to control the OAB symptoms first and consider bladder neck surgery at a later date.

KEY NOTES

1. **Urinary incontinence is a common problem but patients do not seek treatment either due to embarrassment or ignorance.**
2. **The two main causes in women are stress urinary incontinence and overactive bladder.**
3. **The mechanism of continence and pathophysiology of stress incontinence are based on hypotheses that guide treatment.**
4. **Pathophysiology of OAB is poorly understood.**
5. **Urodynamic testing is the definitive diagnostic method but may not always be necessary.**
6. **A large proportion of symptoms can be improved with pelvic floor muscle toning, bladder drill, weight reduction and avoidance of caffeine. Patients must be motivated.**
7. **Definitive treatment for stress incontinence is surgical and the current procedure of choice is the tension free suburethral tape.**
8. **Anticholinergic medication for OAB is effective but has side effects. This can be minimised by using M3 selective blockers.**

CHAPTER EIGHTEEN

DISEASES OF THE VULVA

Horace Fletcher

The vulva (or pudendum) is the female external genitalia. It consists of the labia majora, the labia minora, the mons pubis, the clitoris, the fourchette and the vestibule.

The vestibule has two main orifices, the vagina and the urethra. It is also adjacent to the anus. Also opening into the vestibule are the Bartholin's glands which are covered by erectile tissue, the bulbs of the vestibule.

The vulva receives blood supply and is drained mainly by the internal iliac vessels (via the vaginal, uterine, pudendal and middle rectal branches), and the nerve supply is from the ilioinguinal nerve anteriorly and the pudendal nerves (S 2,3,4) postero-laterally. The lymphatics drain primarily to the inguinal nodes in the groin then via Cloquet's node in the femoral canal to the iliac nodes in the pelvis and the para-aortic nodes in the abdomen.

The vulva is covered by skin (stratified squamous epithelium). The labia majora and mons pubis are also covered by hair, similar to axillary hair responding to testosterone and appearing around the time of the menarche.

The diseases of the vulva are related to its function, structure and relationship with other nearby structures such as the urethra and the anus. The main function of the vulva is coital and some diseases are related to this. However, it is useful to remember that since the vulva is covered by skin, any dermatological condition can arise in this area in a similar fashion to the rest of the body. These problems are best described in a textbook on dermatology and will not be discussed here.

The common gynaecological problems seen in the vulva are:

1. Sexually transmitted disease (STI)
2. Malignancy
3. Dystrophies

Management of vulval lesions

It requires a history enquiring about the symptoms, onset and duration of the condition and past medical history. Examination and investigation findings should be systemic as well as specific. Definitive treatment is aimed at symptom relief as well as cure.

In the history, it is important to find out if the lesion is painful or painless or if there is pruritus vulvae. One should also ask about sexual contact, whether there was barrier contraceptive use or not and whether the patient is promiscuous or not. This is to determine if there is a possibility of a STI. Always ask about extra-genital symptoms as well.

On examination the main lesions of the vulva appear as rash, an ulcer or a mass. As in the case of the history, one starts with extra-genital signs and then looks for specific signs on the vulva. If there are masses or ulcers, look to see if they are multiple or single, tender or non tender and if there is associated lymphadenopathy in the groin.

Investigations should be directed to general conditions firstly and then come in on the Vulval lesion. Some of the common tests include VDRL serology, swabs of lesions, biopsy of obvious lesions and blood glucose and blood counts. Since all ulcerated lesions aid in the transmission of HIV then this test should also be done if a STI is suspected.

Sexually Transmitted Infections

The common lesions of the vulva are caused by the following:

Herpes simplex

Syphilis

Lymphogranuloma venereum

Granuloma inguinale

Chancroid

Human papilloma virus

Herpes simplex

This is caused by a DNA virus of which there are two types HSV 1 and HSV 2.

This is the most common cause of genital ulceration. Most are due to sexual contact either coitus or by oral sex. In most cases the symptoms are mild or absent and transmission can occur

even in the absence of symptoms. The lesions arise as vesicles at first then become multiple shallow painful ulcers which are erythematous around the paler centre. Lesions can arise of the cervix and will not be visible except to the Gynaecologist, where vesicles and or erosions are seen.

The Infection may be primary, latent or recurrent. Primary infections occur 3-7 days after sexual contact with burning paresthesia and vesicles. They then become ulcers which are small <0.5cm in diameter which heal without scarring. The groin lymph nodes are tender and do not ulcerate. The virus remains latent between attacks and survives in the dorsal nerve ganglia causing recurrences in the same dermatome when ever the body's immune system is stressed or during the menses or with systemic illness.

The diagnosis is made clinically but swabs of the ulcers can be collected in Stuart's medium and examined directly with Giemsa stain. Samples can also be subjected to tissue culture. Antibody testing is non specific and is only useful for primary infections with acute and convalescent samples.

The aims of treatment are for symptom relief and reduction in the frequency of recurrences. The mainstay of treatment is the antiviral drugs acyclovir, famcyclovir and valcyclovir.

Syphilis

This is sometimes called the "the great imitator" as the lesions may resemble many other skin conditions. In fact it is now accepted that all patients with genital ulceration or unusual dermatological conditions on the vulva as well as other parts of the skin should have a serological test (VDRL) to rule out syphilis. This test should also be done in all pregnant women to detect and treat maternal disease to prevent congenital syphilis in the neonate.

Syphilis is usually described in three stages, primary, secondary and tertiary. These stages are associated with the length of time the individual has had the infection and the longer the disease is present, the more destructive it is and the more difficult to treat.

Primary syphilis is usually evident 9-90 days after exposure. The ulcer (also called a primary chancre) and nodes which occur are painless and may go un-noticed. The lymph nodes are unilateral, regional firm mobile and non suppurative.

Secondary syphilis occurs 6 weeks to 6 months after sexual contact and can mimic almost any skin condition. These lesions may occur as papules on the hands or feet or as an annular form usually noticeable in people with dark skin. The typical lesion on the vulva at this is called a condyloma lata. This consists of multiple flat macules.

These lesions are very infectious and are more difficult to treat than primary syphilis.

Tertiary syphilis occurs 2-20 years after initial contact. This affects the central nervous system (neuro-syphilis in 20% cases) or the cardiovascular system in 30% cases.

In the brain the condition can lead to meningitis, cranial nerve palsies, dementia and generalized paralysis (tabes dorsalis) with an organic brain syndrome.

In the cardiovascular system it may cause an aortic aneurysm which usually erodes through the sternum and leads to exsanguination of the patient. The usual vulval lesion at this stage may be a gumma which may start as a nodule and later becomes an indolent painless ulcer. These ulcers can occur on other parts of the body as punched out ulcer 0.5-1 cm in diameter which heal leaving tissue paper scars. They may also affect bone and other internal organs.

Congenital syphilis affects neonates of untreated women and results in linear fissured scars around the mouth (Rhagedes), notching of the teeth (Hutchinson's teeth), scarring of the cornea, interstitial keratitis with a ground glass appearance of the cornea and a perforated palate due to gummatous destruction of the tissue.

The diagnosis is made from a smear of the lesion with dark field microscopy to look for the typical cork-screw like spirochaetes, *treponema pallidum*. The VDRL (venereal disease research laboratory) test should be done in all suspected cases. This may however be negative in the early stages of primary syphilis and may also be false positive (weakly reactive) in many conditions such as collagen vascular disease, yaws, leprosy, malaria, old age and even pregnancy. Confirmatory tests include the fluorescent treponema antigen (FTA) and the *treponema pallidum* haemagglutination test (TPHA).

The treatment of primary syphilis is Procaine Penicillin 2.4mega units intramuscularly units as a single dose. Secondary syphilis requires procaine penicillin 2.4 mega units intramuscularly weekly for 3 weeks. Tertiary syphilis is most difficult to treat requiring intravenous Crystalline Penicillin 2-4 million units four hourly for 10-14 days. Those patients who are allergic to Penicillin can be given Tetracycline 500mg four times daily for 14 days.

Lymphogranuloma Venerum

The causative agent is *Chlamydia trachomatis* and this causes painful shallow transient ulcerations on the vulva. The ulcers appear about 5-21 days after sexual exposure and heal without scarring. The pathognomonic sign is the appearance of enlarged lymph nodes in the groin on either side of Cooper's ligament. This results in a groove between the two groups of nodes and is aptly described as the "groove sign". These enlarged nodes are found to be bilateral in 2/3 of the cases. The enlarged nodes occur about 10-30 days after the primary lesion. These glands may enlarge to form a bubo which if it is not aspirated may suppurate and drain leaving a large ulcer in the groin. Lymphatic blockage may result in elephantiasis of the vulva. Sometimes in chronic cases, vaginal or rectal strictures can occur and rarely recto-vaginal fistulae. The treatment is Tetracycline 500mg every 6 hours for 10 days. This works only for the acute lesions, while the chronic conditions may need surgical intervention.

Granuloma Inguinale (Donovanosis)

This is a sexually transmitted infection caused by a gram negative bacterium *calymmatobacterium granulomatis*. The condition is a chronic progressively destructive genital infection. The organism invades the mononuclear endothelial cells and when mature they make metachromatic bars the so called "Donovan bodies". The primary lesion begins as an indurated nodule 8-80 days after sexual contact. This nodule soon erodes to form an exuberant beefy red heaped ulcer. The lesions may spread by autoinoculation and these multiple ulcers may coalesce to form a large lesion. Cervical involvement may go unrecognized and lead to long standing untreated infection often mistaken for cervical cancer. These long standing infections can lead to haematogenous spread affecting bones, joints and liver. Lymphatic involvement is rare but pseudo buboes may occur in the groin which is really a subcutaneous granuloma. These can ulcerate leaving similar beefy red ulcers in the groin. The condition poses several challenges.

It is difficult to diagnose in the laboratory as the organism is not readily seen on slides fixed with formalin and wax embedded or with normal haematoxylin eosin stain. To see the characteristic "Donovan bodies" one must fix the specimen with glutaraldehyde and embed in plastic. They are also easily seen in fresh specimens which are crushed and air dried and stained with Giemsa or Wright's stains. One of the most accurate modern ways to make the diagnosis is with the polymerase chain reaction. The condition is also difficult to diagnose clinically when it is systemic with the radiographic features in bones mimicking many other conditions. Treatment is also difficult because although the organism is sensitive to many antibiotics (except Penicillin), recurrence is common unless treatment is done for long enough (3 weeks).

Chancroid

This is a sexually transmitted infection caused by a gram negative bacillus called *Haemophilus ducreyi*. The incubation period is 3-5 days and the initial lesion is a vesico/pustule which breaks down to form a necrotic saucer shaped ulcer. These ulcers can be multiple from autoinoculation and are painful but soft. The vulval lesion may be extensive due to coalescence of the ulcers and the lymphadenitis is tender and present in over 50% of cases. The lymph node can form a bubo which may subsequently rupture.

The diagnosis is confirmed by culture of *H. Ducreyi* from the ulcer or lymph node. Vancomycin-containing selective media are needed for culture. The treatment of this condition is Erythromycin 500mg q6h for 7 days, Ceftriaxone 250mg IM stat or Sulphonamides.

Human Papilloma Virus

This is a DNA virus with over 300 serotypes. It is usually sexually transmitted and is responsible for STD warts on the vulva as well as malignancies of vulva, vagina and cervix. The warts appear as soft fleshy growths, condylomata acuminata which must not be confused with condylomata lata (secondary syphilis) which are more flat. HPV serotypes 6 and 11 have been implicated in

condylomata while types 16, 18, 36 (South America) and 45 (Africa and Jamaica) have been implicated in genital tract cancers. Treatment of the condylomata is done with locally destructive agents such as podophyllotoxin, cryotherapy, electrocautery, Laser or with interferon α . However, destruction of visible lesions results in a high recurrence rate because the viral infection may be present in adjacent normal skin. The failure rate of treatment is also not reduced by treating the patient's partner.

Vulval Malignancy

Malignancies of the vulva are usually found in postmenopausal women of a low socioeconomic class. The lesion is usually squamous cell carcinoma in origin and the main aetiological agent is Human Papilloma virus. Very rarely other malignancies may occur such as Paget's disease of the vulva or malignant melanoma. Some malignant change may occur after longstanding vulval dystrophies (Figure 1).

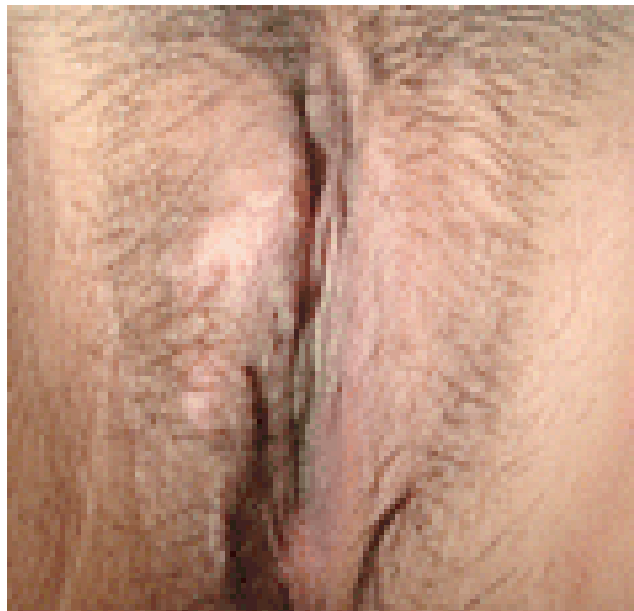


Figure 1. Vulval dystrophy

The main complaint in these patients is usually pruritus vulvae. This when found in postmenopausal women needs investigation. Other symptoms may be a mass, an ulcer with a bloody malodorous discharge (Figure 2). The patient may also notice swollen groin nodes. Other signs may be the presence of white lesions or plaques on the vulva. Diagnosis is usually confirmed by histology of biopsy specimen of any suspicious area on the vulva.



Figure 2. Vulval cancer

Staging of Vulval Carcinoma

Stage 1 confined to the vulva 2cm or less in diameter

Stage 2 confined to the vulva > 2cm diameter

Stage 3a any size lesion with spread to lower vagina, lower urethra, anus or perineum

Stage 3b any size lesions with mobile palpable nodes

Stage 4 Tumour any size with spread to upper vagina, upper urethra, rectum or distant metastases.

Staging is clinical and correlation between the presence of palpable nodes and those positive for malignancy is not very accurate.

The treatment is surgical with modern surgeons attempting to be less radical than those in the past.

Vulval intra-epithelial neoplasia (VIN, precursor to invasive cancer) is treated with simple vulvectomy. Earlier stages such as VIN1 may be with local destruction with 5- Fluorouracil or Laser therapy.

Invasive vulval carcinoma is treated with radical vulvectomy with groin node dissection. This operation used to be done as an en-bloc dissection removing the vulva and the bilateral nodes as one specimen (the Butterfly incision). However, this extensive operation results in difficult healing, wound breakdown and a long hospitalisation with the risk of deep vein thrombosis. Modifications of this to improve outcome include the three incision technique, with the vulvectomy and the node dissection done through separate incisions. Further modifications are being investigated with hemi-vulvectomy for unilateral small tumours and clitoris sparing procedures for posterior tumours.

The five year survival for squamous cell carcinoma is 70-80% with no node involvement and 45-50% if nodes are involved.

Radiotherapy and chemotherapy are adjunctive treatment options to surgery if there is node involvement or if the tumour is extensive.

Vulval dystrophies

This includes a number of conditions localized to the vulva. The lesions may vary in clinical presentation; however the hallmark complaint is pruritus. The colour of the lesions varies but is typically pale or white or may appear as plaques. The disease states are defined by histology and the diagnosis is made by biopsy.

Hyperplastic dystrophy without atypia

This appears as a dusky red to white lesion and may have associated ulceration from constant scratching. Histologically, there is hyperkeratosis (irregular thickening), acanthosis (elongation of the epithelial folds), and parakeratosis (retention of the nuclear material in the keratin layers). There is no evidence that this lesion has any malignant predisposition. Treatment is with local corticosteroid creams.

Hyperplastic dystrophy with atypia

This comprises 10% of the hyperplastic lesions and has an increased risk of malignancy. Lesions if macroscopically are seen as a well delineated slightly elevated rough white lesion. Histologically, there is variable nuclear size and shape with scattered mitotic figures. The more severe conditions have marked pleomorphism and loss of polarity ultimately appearing as carcinoma in situ. The treatment of this condition may be as in the case of squamous cell Ca-in-situ with 5- FU or Laser destruction or as a last resort a simple vulvectomy.

Lichen sclerosis

This is more commonly found in postmenopausal women but may occur in young patients as well. There appears to be some association with autoimmune disease and it sometimes is familial. In pre-menarcheal girls it may go away with the menarche. However in most cases it is incurable and tends to flare up during times of stress.

The lesion starts off as a flat white or translucent macule, while adjacent lesions may be speckled, purple or occur as blisters. Over time the lesion develops a smooth waxy appearance as lichenification occurs. This may result in a constriction of the introitus.

Histologically there is hyperkeratosis but with a thin epidermis and flat rete pegs. Deeper tissues have an infiltrate of chronic inflammatory cells. Treatment is with topical steroids such as 0.1% Triamcinolone twice daily for 1-2 months then daily two months then three times weekly for another two months, then once weekly until clear. Then switch to a milder steroid such as weekly hydrocortisone for maintenance and when symptoms flare up switch back to triamcinolone. Testosterone cream is no longer used for this condition. There is a 1-5% chance of malignant change in these lesions however the tumours are usually low grade with early stages not needing groin node dissection.

Lichen Planus

This is an inflammatory condition of mucosal and keratinized skin. Usually found in younger patients age between 30-60 years. Usually results in difficulty with sexual intercourse because of vaginal stenosis with dyspareunia. When there is vaginal involvement there may be a thick yellow green discharge. Other symptoms include itching, burning, raw sensation and vaginal bleeding. The vagina may become obliterated and unless this is very longstanding the adhesions can be broken down easily. Gray white lacy stranding of the skin erosion is described as Wickhams Striae and this may be evident on the perineum. A biopsy is usually needed with one sample for immunohistochemistry and one for histological evaluation. The differential diagnoses include a plasma cell lesion (plasma cell variant), Pemphigoid, Behcet's disease or a contact dermatitis.

Treatment is with steroids oral prednisone 40-70 mg daily for 2-6 weeks, tapering down as symptoms improve. Patients may also use 25 mg vaginal hydrocortisone pessaries half inserted daily.

Some patients may need vaginal dilators and also vaginal lubricants during sexual intercourse.

Miscellaneous lesions of the vulva

Acute vulvitis may arise as a response to irritation or from allergy. The most common cause is due to fungal infections such as monilia. This may occur as a result of immunocompromise as in diabetics but may also be found in relation to vaginal candidiasis in very active women (exercise

induced) or in women who wear panty shields constantly. The fungus thrives in a moist environment and re-infection is common because the spores are hardy and not killed by washing alone. All exercise clothes, underwear, towels, shoes and bag must be placed in the sun to dry after washing. Treatment is with ketoconazole or clotrimazole cream or oral fluconazole.

Bartholin's swelling

This is common in women in the reproductive age group who are sexually active. The acute condition occurs as an abscess which is as a result of irritation and blockage of the Bartholin's duct. This abscess is found as a painful swelling at the introitus involving the posterior third of the labia minora. Long standing infection leads to a Bartholin's cyst which is usually non tender. Treatment of both conditions includes marsupialisation of the gland. This involves, making an opening about the size of a cent coin and suturing the cyst wall to the vaginal skin. Aspiration or incision and drainage alone lead to recurrence. Swabs of the abscess usually grows mixed organisms

Rarely, when one attempts to marsupialise a Bartholin's swelling one find a firm tumour. Invariably this is a leiomyoma of the vulva which requires excision. These tumors however tend to recur and may need repeated excision.

Labial fusion

This occurs commonly in infant girls and appears as an absent vaginal opening. This usually responds to oestrogen cream.

Haematocolpos

This is due to an absence of the vaginal opening. Usually regarded as being due to an imperforate hymen, it is now regarded as being due to a defect in vaginal development and the transverse membrane occluding the introitus is not the true hymen. This results in primary amenorrhea but is more correctly cryptomenorrhoea as the patient has menstrual periods but the menstrual effluent is retained in the vagina, uterus and Fallopian tubes. Longstanding unrecognized haematocolpos and haematometra results in an abdominal mass and can lead to permanent damage to the reproductive organs. Treatment is by a cruciate incision into the membrane, and this results in a gush of altered blood as the vagina and uterus drain. The vagina is then sutured open in a similar fashion to the Bartholin's marsupialisation.

KEY POINTS

1. **The vulva is a specialized organ and as such it is exposed to special disease processes.**
2. **It is however covered by skin and is subject to the same conditions as the rest of the skin on the body.**
3. **Any unusual lesions which persist must be probed further by history, examination and investigation to ensure that we are not missing systemic disease or a premalignant or an early malignancy which is curable.**

CHAPTER NINETEEN

CERVICAL CANCER

Vanessa N Harry

Introduction

Despite being a potentially preventable disease, cervical cancer remains the second most common malignancy among women worldwide. Almost 80% of cases occur in less developed countries where a lack of effective screening accounts for the disproportionate number of cervical cancer deaths. The highest incidence rates are observed in Africa, Latin America and the Caribbean. Until recently, the greatest strides in reducing cervical cancer mortality have occurred with the development and implementation of screening programs. However, there have now been important advances in the prevention as well as treatment of cervical cancer.

Epidemiology

Cervical cancer is most commonly diagnosed in the fifth decade of life, with the median age being 45 to 59 years. This is several years earlier than the median age at diagnosis of breast, lung and ovarian cancers. Older patients, the economically disadvantaged, and those who do not participate in screening programmes represent the majority of women who develop and ultimately die from this disease.

Aetiology

Approximately 80% of cervical cancers are squamous cell carcinomas, while the remainder are adenocarcinomas (15%) and other uncommon subtypes (5%) such as clear cell, adenosquamous and small cell (neuroendocrine) carcinomas. Although there are concerns that patients with adenocarcinomas have a worse prognosis, there are no current data showing that they should be managed differently.

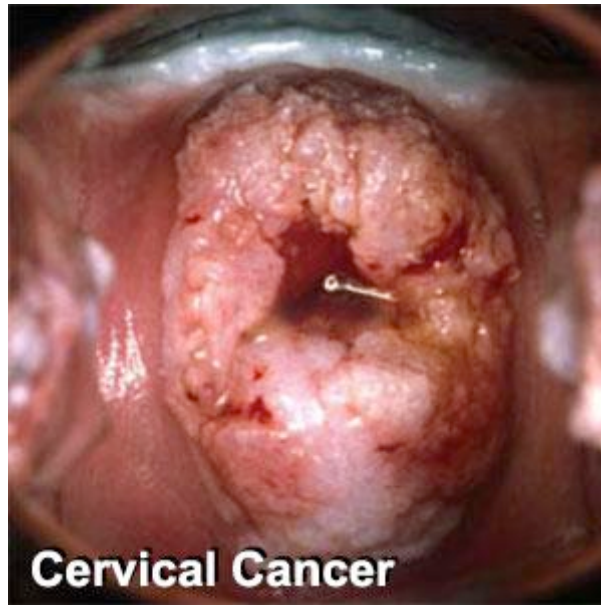


Figure 1. Cervical cancer

Squamous cell cancers arise from the squamo-columnar junction (SCJ) which under normal circumstances exhibit metaplastic changes (Figure 10). However, when the process of metaplasia is influenced by oncogenic substances this results in cellular dysplasia.

Human papillomavirus (HPV) is now accepted to be the most important cause of cervical cancer and more than 90% of squamous cell cervical cancers contain HPV DNA, with HPV types 16 and 18 having the strongest causal relationship. Other HPV types associated with anogenital neoplasia include types 31, 35, 39, 45, 51, 52, 56 and 58. HPV 6 and 11 are associated with genital warts.

The estimated risk of acquiring an HPV infection, which is mainly through sexual contact, is thought to be in the range of 80-85%, but it is the persistence of the high risk HPV types and failure of immunosurveillance that is eventually responsible for malignant change.

Most women infected with HPV will clear the virus before any increase in cellular proliferation and the development of atypia occur.

Predisposing factors

These include:

Young age at first coitus

Multiple sexual partners

High parity

A history of other sexually transmitted diseases

A history of genital warts

HIV positive patients

Patients receiving immunosuppressive agents

Cigarette smoking

Among women with a single lifetime partner, high risk sexual behaviour by the male partner contributes to the development of cervical cancer. Conversely, circumcision has been found to be associated with a reduced prevalence of penile HPV infection and a reduced risk of cervical cancer among current partners.

Education is paramount as cervical cancer is often subjected to stigma because of supposed unacceptable sexual behaviour, with fear having a negative impact on women participating in screening. With increasing knowledge of the natural history of HPV infection, cervical cancer should be considered as a complication of oncogenic HPV infection rather than a sexually transmitted disease, and any negative connotations disregarded.

Cigarette smoking is an independent risk factor for significant cervical dysplasia and invasive cancer. Carcinogens and polycyclic aromatic hydrocarbons found in cigarette smoke are concentrated in cervical mucus and may interfere with local immunity.

Management

A practical approach to any cancer management involves 4 steps:

1. Establish the diagnosis
2. Define the extent of disease
3. Determine and implement treatment
4. Follow-up of patient for evidence of recurrence and /or treatment-related complications

Clinical Features

History

Asymptomatic in microinvasive disease

Heavy/prolonged/irregular vaginal bleeding

Post-coital bleeding

Intermenstrual bleeding

Vaginal discharge which may be bloody and malodorous

Pressure symptoms – in the case of advanced disease – urinary or bowel symptoms, nerve symptoms such as sciatica, and lower limb pain and swelling

Physical examination

Cervical abnormality on speculum/bimanual examination – this may appear as an exophytic fungating tumour, or an ulcer replacing the cervix or an infiltrative tumour which expands the cervix.

Diagnosis

In the event of microinvasive or early stage disease, (Stage 1a) women may be asymptomatic. Cervical cancer may then be detected by an abnormal Papanicolaou (Pap) smear or when performing colposcopy to evaluate an abnormal Pap smear. In these cases, as well as in the case of a clinically suspicious cervix, a biopsy should be taken from any suspicious lesion.

Indications for a diagnostic cone biopsy

Pap smear shows high grade abnormalities but equivocal colposcopy
Biopsy shows microinvasive disease, but this is not grossly apparent on colposcopy
Cervical lesion cannot be fully visualised

When confronted with a clinically evident tumour, (by definition – consistent with at least Stage 1b disease) cervical conisation is inappropriate as it may lead to significant bleeding and a punch biopsy is sufficient.

Staging

The current International Federation of Gynecology and Obstetrics (FIGO) staging system (Table 1) is based on clinical evaluation – mainly on the size of the tumour and its extension into the pelvis. Table 2 lists the staging procedures for cervical cancer. In developing countries, most cervical tumours are diagnosed in the third and fourth stage.

For smaller lesions (stage 1a and stage 1b1), the stage is determined after measurement of the depth of tumour invasion (on cone biopsy) and pelvic examination to assess the tumour size or both. For more advanced tumours, pelvic examination under anaesthesia is usually necessary to allow thorough assessment of the parametrial tissues adjacent to the cervix and uterus.

The additional tests performed for clinical staging are restricted to modalities used in most countries. Although the results of MRI, CT and PET cannot strictly be used for FIGO staging, the information gathered from such studies have been used to more accurately assess the extent of pelvic disease and lymph node metastasis which may affect recommendations for treatment.

Table 1. FIGO staging of Cervical Cancer

Stage I	Tumour confined to the cervix Microscopic lesions
---------	--

la1	Stromal invasion $\leq$ 3mm, extension $\leq$ 7mm
la2	Stromal invasion $>$ 3mm but $\leq$ 5mm, extension $\leq$ 7mm
	Clinically evident lesions
lb1	Tumour $<$ 4cm diameter
lb2	Tumour $>$ 4 cm diameter
Stage II	Invasion beyond uterus, but not to pelvic side wall
Ila	No parametrial involvement, spread to the upper vagina
Ilb	Obvious parametrial involvement
Stage III	Extension to pelvic side wall. No cancer free space between tumour and pelvic wall on rectal examination
IIla	
IIlb	Involves lower 1/3 of vagina. No extension to pelvic sidewall
	Extension to pelvic sidewall and/or hydronephrosis or non-functioning kidney
Stage IV	Extension beyond pelvis
IVa	Spread to adjacent organs (bladder and rectal mucosa)
IVb	Spread to distant organs

Table 2. Staging procedures for cervical cancer

Physical examination

Palpation of lymph nodes (supraclavicular, inguinal)
Vaginal examination
Rectovaginal examination (usually done under anaesthesia)

Radiological investigations

Chest X-Ray
Intravenous urogram (IVU)
Barium enema

Procedures

Cervical biopsy
Cervical conisation
Colposcopy
Hysteroscopy
Cystoscopy
Proctoscopy/sigmoidoscopy
Endocervical curettage

Other studies (not considered for assignment of clinical staging)

MRI
CT
Ultrasound of pelvis/kidneys
PET
Bone scan
Lymphangiography
Laparoscopy

Treatment

This is dependent on the stage of disease:

Stage Ia1

Where the focus of microinvasion extends no deeper than 3mm beyond the basement membrane, the risk of nodal involvement is less than 1%. In these cases, cervical conisation is a reasonable treatment option for women who want to preserve their childbearing potential, while simple hysterectomy can be done for those who have completed childbearing.

Stage Ia2

The risk of lymph nodes metastasis increases to 2 to 8% and most gynaecologists would advise surgery, although the extent of surgery is open to debate. In view of a low risk of parametrial spread, some advocate a modified radical hysterectomy in which less parametrium and vagina is removed, minimising bladder-related dysfunction. Pelvic lymphadenectomy includes removal of common iliac, external iliac, internal iliac and obturator nodes. A plausible management scheme can include pelvic lymph node assessment which if negative, a non-radical hysterectomy is performed.

Stage Ib1

Generally, for Stage Ib1 cervical tumours, either radical hysterectomy (Wertheim's hysterectomy) or pelvic radiotherapy is equally effective. Advantages of surgery include preservation of ovarian function in younger women and avoidance of radiotherapy effects such as vaginal stenosis. Surgical morbidities include bladder dysfunction, nerve injury, pulmonary embolism, lymphocele and fistula formation.

Currently of immense interest is the use of increasingly conservative surgery in women who desire preservation of their fertility, particularly in Stage Ib1 tumours. Radical vaginal or abdominal trachelectomy and pelvic lymph node dissection has been proposed for a select, well-counselled group of women, and there have been successful pregnancies following this procedure, although with a late miscarriage rate of 25%.

Stage Ib2

These tumours pose a particular challenge due to the increased size and the higher likelihood of pelvic and para-aortic lymph node metastases. Several regimes have been proposed such as radical hysterectomy followed by adjuvant radiotherapy, radiotherapy then hysterectomy, and primary radiotherapy with concomitant chemotherapy. A contentious point has been the increased morbidity associated with the combination of surgery and pelvic radiation with no effect on overall survival. Therefore, the superiority of combined radiotherapy with radio-sensitising chemotherapy is usually preferred for this stage of disease (chemoradiotherapy). This includes external beam radiotherapy as well as intracavitary therapy (brachytherapy) given concurrently with cisplatin chemotherapy. Cisplatin is the agent of choice, acting as a radio-sensitiser, and is as effective as and less toxic than combination regimes with fluorouracil and hydroxyurea. This has been supported by a systematic review and meta-analysis, over the use of neoadjuvant chemotherapy followed by radiotherapy as well as radiotherapy alone.

Stage IIa

Most women should receive combined chemoradiotherapy, although treatment should be on an individual basis. Rarely, radical hysterectomy, lymphadenectomy and upper vaginectomy can be used in the event of a very small amount of tumour extension in to the vaginal fornix.

Stages IIb-IV

Generally, chemoradiotherapy in the treatment of choice, although in the event of metastatic disease as in stage IVb, palliative radiotherapy or palliative chemotherapy may be used.

Prognosis

Prognostic factors include:

Tumour stage

Tumour grade

Lymphovascular space invasion

Deep cervical stromal invasion

Positive vaginal or parametrial margins

Nodal metastases

Tumour volume

Patient age/comorbidities

Table 3. Five-year survival rates

Stage Ia	100%
Stage Ib1	70-85%
Stage Ib2	50-85%
Stage II	50-70%
Stage III	30-50%
Stage IV	5-15%

Recurrent disease

Recurrent disease is almost always incurable, with the majority being identified within 3 years of diagnosis. If radiotherapy was the primary treatment, patients can be offered chemotherapy, with cisplatin the most active single agent in recurrent disease. Various trials have evaluated combination of cisplatin with paclitaxel as well as topotecan, but with no improvement in overall survival.

Pelvic recurrence following radical hysterectomy may be treated with radiotherapy, once this has not been previously used. Some women with a central recurrence after radiotherapy may be suitable for pelvic exenteration. Improvements in pelvic reconstruction, rectal anastomosis and urinary diversion have contributed to a reduction in the immediate complications of this procedure.

Cervical Cancer Prevention

Unfortunately, the vast majority of women with cervical cancer in the developing world have not benefited from the significant advances made in the management of this disease. One of the most important aspects of reducing this discrepancy is primary prevention. In order to combat this major problem of cervical cancer there needs to be action on several fronts such as increased spending by major institutions and a greater will on the part of decision makers, research and audit, education and greater participation on the part of women to access screening, changes in lifestyle behaviour especially in relation to age of coitarche and the number of sexual partners, use of condoms, quality control in screening programmes with Pap tests and availability of the HPV vaccine. Colposcopy services must be enhanced.

HPV Vaccine

Introduction of the first-ever anticancer vaccine is a significant milestone for public health. The HPV virus is a DNA virus which consists of an outer capsid coat made of 2 structural proteins L1 and L2. The viral genome has 8000 base pairs as well as early (E) and late (L) regions or genes. Two of these transcriptional units, E6 and E7 encode for proteins necessary for viral replication.

- E6 oncoprotein exerts its effect by binding to and inactivating tumour-suppressor gene TP53
- E7 oncoprotein similarly inactivates products of the retinoblastoma gene, pRb.

These actions ultimately lead to normal cell-cycle disruption and progression of cells infected with HPV 16 or 18. This clear understanding of the virology and the oncological behaviour of HPV was the stimulus that led to the development of the HPV vaccine.

There are currently 2 available HPV vaccines:

Gardasil (Merck & Co., Inc) - a quadrivalent vaccine which protects against HPV types 6, 11, 16 and 18 (Figure 2).



Figure 2. Gardasil

Cervarix (GlaxoSmithKline) – a bivalent vaccine which protects against types 16 and 18 (Figure 3).



Figure 3. Cervarix

Both vaccines are based on the recombinant expression and self-assembly of the major capsid protein, into virus-like particles (VLPs) that resemble the outer capsid of the whole virus, L1. These HPV VLPs contain no DNA and are not live/attenuated viruses, but stimulate an antibody response.

This intramuscular vaccine is administered as a 3-dose regimen with the first injection followed by subsequent doses at months 2 and 6. It is recommended that young girls at 11 to 12 years have routine vaccination which has been found to be well tolerated with sustained immunogenicity. Minor side effects of local pain, tenderness and fever have been reported.

With the difficulties faced in instituting effective screening in low income areas, the HPV vaccine appears to hold the most promise for the future in these settings. Undoubtedly, the cost of the vaccine, estimated at 300 to 500USD, would limit its impact in the areas of the world where it is most needed unless substantial subsidies are offered. It is hoped that with the combined and co-operative effort of health agencies such as WHO and UNICEF, as well as pharmaceutical companies and philanthropic efforts, that this vaccine will be made available to those in developing countries.

KEY POINTS

1. **Cervical cancer remains a global cause of cancer morbidity and mortality particularly in less developed countries due to lack to screening.**
2. **HPV has been found to be the causative agent in the majority of cervical cancers.**
3. **Many women with microinvasive disease are asymptomatic.**
4. **Cervical cancer is clinically staged.**
5. **Treatment of cervical cancer depends on the stage of disease, with surgery more applicable for early stage disease and radiotherapy for locally advanced stage disease.**
6. **The HPV vaccine has significant potential for reducing the incidence of cervical cancer if made available to areas where it is most needed.**

CHAPTER TWENTY

ENDOMETRIAL CARCINOMA

Bharat Bassaw

Introduction

In many developed countries, endometrial carcinoma is the most frequent female gynaecological malignancy. For example, in the United States of America, it is more common than cancer of the cervix and ovary combined. As many as 1200 deaths result from endometrial carcinoma each year in the United Kingdom. It is the fourth most common cancer in the UK with 7536 cases diagnosed in 2007, accounting for about 5% of all female cancers. While the incidence of uterine cancer remained stable between 1975 and 1993, it has increased by more than 40% between 1993 and 2007. The largest increase has been observed in the age group 60 and 79 years. In Trinidad and Tobago, approximately 75 new cases are diagnosed annually.

Epidemiology

The incidence of endometrial carcinoma peaks in the sixth decade, with 75-80% of women being postmenopausal and only 3-5% are less than 40 years old. The median age is 61 years. The majority of young women who develop endometrial carcinoma are obese and they have polycystic ovarian syndrome with anovulation. This results in a relative hyperoestrogenic state.

In terms of ethnicity, white women are most prone.

Predisposing factors and aetiology

Although the aetiology of endometrial carcinoma is unknown, it is accepted that this malignancy is oestrogen-dependent. The typical patient's profile with this carcinoma is a Caucasian woman who is obese, nulliparous or of low parity (possible history of anovulatory infertility), and had a late menopause i.e. after 55 years. However, it is also recognized an atypical pathogenic type is seen in the black female who is thin and multiparous. The latter carries a poor prognosis.

The association between obesity and endometrial carcinoma may be explained by:

- (i) Adipose tissue stores oestrogen.
- (ii) Aromatisation of weak androgens such as androstenedione to oestrone in peripheral tissues especially in the fat.

- (iii) Continuous exposure of the endometrium to oestrone which is unopposed by progesterone in the postmenopause.
- (iv) Increased nuclear uptake and protein synthesis and cell growth.
- (v) A reduced level of sex hormone-binding globulin and therefore, more free oestrogen to bind to their receptors in the endometrium.

Unopposed oestrogen in a woman with an intact uterus is associated with a significantly increased risk for endometrial cancer. The type of oestrogen, the dose and the duration are related to the development of this disease. When oestrogen is combined with a progestogen, the incidence of endometrial cancer is reduced. Furthermore, the modern low-dose oral contraceptive pill also protects against endometrial carcinoma. The same is observed with ovarian cancer. It is estimated that the woman who uses the combined pill for one year receives protection against endometrial carcinoma for at least ten years.

There may be a link between tamoxifen which is used as part of the armamentarium against breast cancer, and endometrial carcinoma. One must be cognizant of the fact that a woman with breast cancer is at a higher risk for endometrial cancer as well.

Although convincing evidence which shows that tamoxifen causes endometrial cancer is lacking, it is prudent that any woman who is on tamoxifen who has abnormal uterine bleeding should be investigated in order to exclude endometrial carcinoma.

Oestrogen-producing tumours such as granulosa-theca cell malignancies are associated with endometrial hyperplasia (50%) and carcinoma (10%). Women with a history of BRCA 1 and 2 gene mutation hereditary non-polyposis colorectal cancer are at risk for malignancies involving the colon, breast, endometrium, ovary and stomach.

Diabetes mellitus and hypertension may also be related to endometrial cancer although this association may be casual not causal since many of these women with endometrial cancer are obese and postmenopausal, and thus prone to these medical disorders. The corpus cancer syndrome comprises endometrial carcinoma, obesity, diabetes mellitus and hypertension.

Fibroids co-exist with endometrial cancer in approximately 30% of cases, but this frequency is no more than would be expected in patients of low parity. Fibroids like endometrial cancer are oestrogen-dependent.

Table 1. Risk factors for endometrial carcinoma

-
1. Obesity
 2. Nulliparity
 3. Unopposed oestrogen
 4. Late menopause
 5. Oestrogen-producing ovarian tumours
 6. Family history of breast, ovarian and colonic cancer
 7. Ethnicity
 8. Polycystic ovarian syndrome
-

Molecular genetics

It is believed that many human cancers develop from damage to DNA in cellular genes. This insult may lead to inhibition of tumour suppression genes or activation of oncogenic genes. Some workers have demonstrated that endometrial carcinoma is associated with mutations of the p 53 tumour suppressor gene.

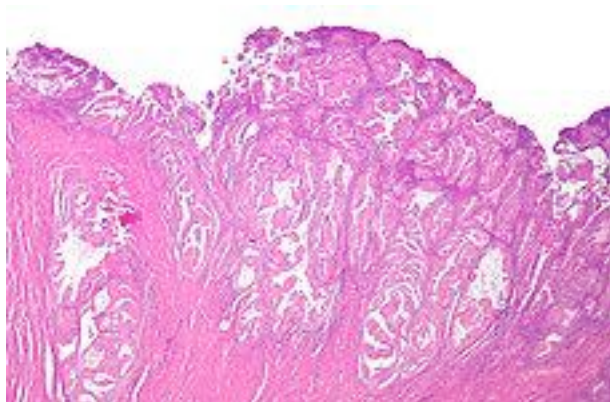


Figure 1. Endometrial hyperplasia

Endometrial hyperplasia

The endometrium is less amenable for screening for pre-malignant pathology as compared to the cervix, vagina and vulva. The effect of prolonged, excessive unopposed stimulation of the endometrium by oestrogen leads to the development of endometrial hyperplasia (Figure 1). Progesterone protects the endometrium from this effect by oestrogen.

Oestradiol, though present at half of the concentration in the plasma when compared to oestrone, is the most important of the oestrogens metabolically. In the postmenopause,

oestradiol is predominantly derived from the adrenal glands and there is a small contribution from peripheral and ovarian conversion of oestrone and testosterone.

Oestrone is the most abundant oestrogen in the postmenopausal woman. Most of the oestrone in the circulation is derived from peripheral conversion of androstenedione in adipose tissue as well as in the muscle, liver, skin, kidney, brain and adrenals. This rate of peripheral conversion is proportional to the body mass index.

There are three types of endometrial hyperplasia viz:

- (i) Cystic glandular
- (ii) Adenomatous
- (iii) Atypical

Cystic glandular hyperplasia

This is the commonest type of endometrial hyperplasia. Formerly referred to as metropathia haemorrhagica, it has a characteristic 'swiss cheese' pattern. This is a relatively benign condition with a low premalignant potential (5%). The endometrium is proliferative and some glands are dilated and focally crowded.

Adenomatous hyperplasia

This type of hyperplasia is composed of highly complex, crowded glands with intervening stroma. Epithelial stratification and mitotic activity generally parallel the architectural complexity. Adenomatous hyperplasia has a premalignant potential of approximately 10%.

Atypical hyperplasia

There is an increase in the number of endometrial glands lined by cells which display cytological atypia with a complex glandular pattern. This type of hyperplasia has the highest premalignant risk (25%).

Any form of hyperplasia must be considered suspicious and further investigation and treatment are mandatory. Some cases of adenocarcinoma of the endometrium may co-exist with hyperplasia. The progression to endometrial cancer from cystic hyperplasia may take 10 years whereas when atypia exists carcinoma may develop in only 2-4 years.

Treatment

Medical

More than 80% of patients with hyperplasia without atypia will revert to normal with medroxyprogesterone acetate therapy compared with only 50% of those with cytological atypia. In addition with atypical hyperplasia, 25% will develop cancer.

Medical treatment for atypical hyperplasia includes:

- a. Medroxyprogesterone acetate 20-40 mg orally daily for 6 months
- b. Depo-provera 150 mg intramuscularly monthly
- c. Levonorgestrel intrauterine releasing system (Mirena)

Following medical treatment, endometrial curettage should be performed. The patient should be informed to return to the clinic if abnormal uterine bleeding recurs.

Surgical

Surgical treatment entails a total abdominal hysterectomy and bilateral salpingo-oophorectomy or if the necessary expertise and facilities are available, a laparoscopic-assisted vaginal hysterectomy with removal of both ovaries and Fallopian tubes. Surgical treatment is reserved for women who have completed their family or for recurrent or resistant cases of atypical hyperplasia.

Pathology of endometrial cancers

Malignancies of the endometrium may arise from the glands (adenocarcinoma) or from the stroma (sarcoma). By far the most common malignancy of the endometrium is adenocarcinoma (Table 2). This usually appears as a raised, rough papillary projection most often from the fundus (Figure 2). This tumour is soft and maybe bulky involving at least half of the endometrial cavity. There may be invasion into the myometrium which maybe obvious on gross inspection of the uterine specimen.

Table 2. Subtypes of endometrial carcinoma

-
1. Adenocarcinoma
 2. Adenosquamous carcinoma
 3. Adenoacanthoma
 4. Clear cell carcinoma
 5. Papillary serous carcinoma
 6. Squamous cell carcinoma
-

Adenosquamous carcinoma is a combination of malignant changes in the glands as well as in the stroma. When adenocarcinoma co-exists with squamous metaplasia, this type of malignancy is referred to as adenoacanthoma. Uterine papillary serous carcinoma is a highly aggressive carcinoma of the uterus which resembles that of the ovaries. Extra-uterine spread and nodal disease are common with papillary serous carcinoma even when the disease appears to be early. It is thus not surprising that this tumour as well as clear cell carcinoma have poor survival rates.

The level of differentiation or grading of endometrial cancer is based on architectural pattern as well as nuclear changes. Well-differentiated (Grade 1) adenocarcinoma carries a very good prognosis. On the other hand, poorly differentiated tumours which are associated with myometrial invasion and pelvic and para-aortic nodal disease have poor prognosis. It was reported in the FIGO annual report (1998) that pelvic node metastasis in Grade 1 lesions is seen in less than 1% when the endometrium alone is involved but this figure is over 15% if deep myometrial spread is present.

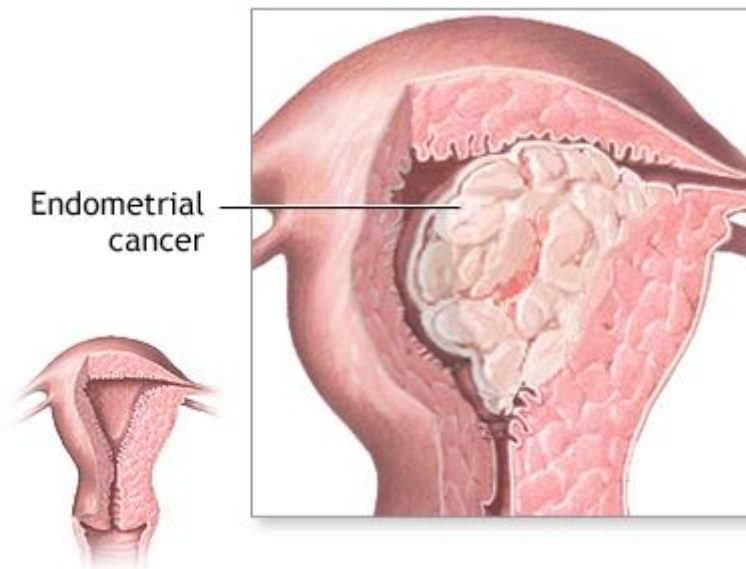


Figure 2. Endometrial cancer

Grading

Grade 1: Well-differentiated, glands are conspicuous with moderate amount of mitotic figures

Grade 11: Moderately-differentiated, glands are less conspicuous and there are deposits of solid masses or infiltrating single cell column

Grade 111: Poorly-differentiated, consists of varying sizes and shapes of solid masses of malignant cells, mitoses are numerous, little or no stroma exists

Staging

In 1988, FIGO changed the staging of endometrial cancer from clinical to surgical (Table 3). For adequate staging, a vertical midline incision is made and then peritoneal cytology is obtained. Careful inspection is carried out in the pelvis and abdomen and the intra-abdominal organs. Particular attention is paid to the omentum, liver, cul-de-sac, adnexal organs, as well as the pelvic and para-aortic lymph nodes.

Table 3. FIGO staging for endometrial cancer (1989)

Stage 1A	Tumour limited to the endometrium
1B	Spread to less than half of the myometrium
1C	Spread to more than half of the myometrium
Stage 2A	Endocervical glands involvement
2B	Cervical stroma involvement
Stage 3A	Spread to the serosa and/or adnexae and/or positive peritoneal cytology
3B	Vaginal metastases
3C	Involvement of pelvic and/or para-aortic nodes
Stage 4A	Tumour involves the mucosa of the bladder and/or the rectum
4B	Distant spread including intra-abdominal disease and/or metastases to the inguinal node

Clinical Features

History

Postmenopausal bleeding which is observed in 70-80% of cases of endometrial carcinoma is by far the leading symptom. Other clinical features include abnormal vaginal discharge which maybe bloody. The postmenopausal bleeding is usually bright red and odourless. More often it is slight but on occasions the bleeding may be heavy with clots. Abdominal pain is uncommon and is usually a feature of late or metastatic disease. The passage of pus from a pyometra is seen with squamous cancer of the endometrium. A history of constitutional symptoms such as loss of appetite, symptomatic anaemia or weight loss must be elicited in the history. It is also prudent to enquire about risk factors for endometrial cancer such as unopposed oestrogen use or family history of cancer of the breast, ovary, colon and endometrium. A thorough gynaecological history must be obtained. Results of any recent Pap smears must be reviewed.

In the premenopausal woman, the typical history includes intermenstrual or irregular uterine bleeding, abnormal menstrual bleeding or a history of menorrhagia which is observed in about one third of case. Nulliparity, infertility from anovulation and a history of polycystic ovarian syndrome are common among women under 40 with endometrial cancer.

Physical examination

In early disease, there may not be any specific signs. The patient may be obese, hypertensive and diabetic. There may be anaemia, and with advanced disease, the left supraclavicular node may be enlarged. Palpation of the abdominal masses arising from the pelvis should be carried out.

Speculum examination may reveal blood emanating through the cervical os in an otherwise normal cervix, vagina and vulva. The uterus may be symmetrically enlarged (bulky). There may be accompanying fibroids, pyometra or ovarian tumours. A pelvi-rectal examination is imperative to exclude extra-uterine disease posteriorly.

Screening and diagnosis

Whereas cervical smear or the Pap test has proven itself as a highly effective screening test for cervical cancer with a high sensitivity and specificity, no single effective test for screening for endometrial carcinoma is available. This is primarily due to the inaccessibility of the endometrium for ease of screening.

There are several methods for endometrial sampling such as the Pipelle de Cournier, Vabra aspirator and Endo-sampler (Figure 3). These carry a sensitivity of about 70-80%. The main advantages are these are outpatient procedures which are relatively inexpensive, easy to perform but in about 30-40% no endometrial tissue is obtained for histological analysis. Another issue is the screening test cannot exclude cancer and hence, if the result reveals no malignancy then other diagnostic investigations are still necessary.

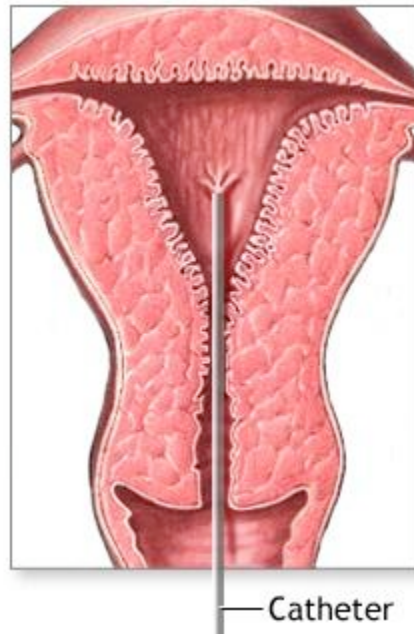


Figure 3. Endometrial sampling

Sometimes a Pap smear may identify malignant cells originating from the endometrium. This occurs in about 50% of patients with symptoms.

Hysteroscopic-directed biopsies may be considered the 'gold standard' or diagnostic tool of choice for endometrial cancer since biopsies are taken under direct visualization from the most abnormal areas. However, some concern remains concerning the possibility for dissemination of abnormal cells through the Fallopian tubes with distension of the cavity with saline.

Examination under anaesthesia and a fractional curettage is commonly practiced as the preferred diagnostic procedure for endometrial cancer especially in low-resourced countries where hysteroscopic facilities are unavailable. It must be noted that curettage is carried out blindly and if due care is not taken in as much as 30% of cases an abnormal area may not be curetted. Hence, there is a possibility of a fairly high false negative rate.

A fractional curettage differs from a formal diagnostic curettage in that in the former, the endocervix is first curetted before the passage of the uterine sound as well as before cervical dilatation, and secondly, multiple endometrial samples are obtained from the fundus, the various walls of the cavity and the lower segment. The curettings are placed in separate labelled containers with formalin and sent to the histopathologist.

Vaginal ultrasonography is used to measure the endometrial thickness as well as to detect any adnexal masses. With an endometrial thickness of less than 3-5mm in the postmenopause, it is unlikely any significant pathology exists. However, if the patient continues to be symptomatic even with a thin endometrium further evaluation is warranted. If the endometrium is

thickened, then either hysteroscopy and biopsy or a fractional curettage should be considered urgently.

With endometrial hyperplasia, the endometrium is thickened and hyperechogenic, and the deeper endometrium is compacted but remains intact producing a subendometrial 'halo'. This pattern is absent with invasive disease.

Vaginal ultrasonography and endometrial sampling are safe and effective as first line measures.

Table 4. Prognostic factors for endometrial cancer

-
1. Tumour volume
 2. Histological grade
 3. Myometrial invasion and/or positive peritoneal cytology
 4. Cervical involvement
 5. Histological subtype
 6. Lymphovascular space spread
 7. Extra-uterine disease
 8. Nodal metastases
 9. DNA ploidy
 10. Patient' characteristics such as age and co-morbid conditions
-

Treatment

Surgery

The main modalities of treatment for endometrial carcinoma are surgery, irradiation, chemotherapy, hormonal treatment and palliative care. Surgery in the form of total abdominal hysterectomy and bilateral salpingo-oophorectomy remains the cornerstone of treatment. Whereas pelvic irradiation was common in the past, this is no longer so and rightly so.

Laparoscopic hysterectomy has the benefits of shorter hospital stay, reduced analgesic requirements, superior quality of life scores and earlier resumption of daily activities. In obese women, a sizeable number of laparoscopic operations are converted to open procedures. It has also been reported that it is not cost-effective to perform laparoscopic hysterectomy in women with high body mass index.

Vaginal hysterectomy and bilateral salpingo-oophorectomy performed under regional anaesthesia may be an alternative route for surgery in obese women not fit for general anaesthesia. However, this route does not allow direct visualization of the peritoneal surfaces and for pelvic and para-aortic lymphadenectomy.

Pre-operative pelvic irradiation alone or in combination carries disadvantages such as over-treatment of patients with early disease, inadequate staging, the specimen may not be suitable for proper histological evaluation, increased morbidity and delay in carrying out surgery.

Since most patients with endometrial cancer are diagnosed when the disease is at Stage 1A, then surgery alone is required. The role of pelvic and para-aortic lymphadenectomy is controversial. One must keep in mind that lymphadenectomy increases morbidity to many of these patients who are already high-risk for anaesthesia and surgery.

Patients with well-differentiated tumours without myometrial invasion and no extra-uterine spread have an almost 0% chance of nodal disease. Thus it appears inappropriate to consider lymphadenectomy in this group of patients.

On the other hand, when there is deep myometrial penetration and peritoneal disease, the chances of pelvic and para-aortic node involvement are about 40 and 20%, respectively. In this high risk group, lymphadenectomy may have a place at the time of the panhysterectomy.

Serum CA 125 levels are elevated in more than two-thirds of patients with extra-uterine disease whereas most women with early disease would have normal values. Vaginal ultrasound and pelvic MRI are accurate in assessing myometrial penetration in about 75% of patients. These investigations when done pre-operatively can assist in proper planning of the surgical procedure. Table 5 demonstrates the pre-operative work-up for endometrial carcinoma.

Table 5. Pre-operative evaluation

Complete blood count
Renal function test
Blood glucose
CA 125
Chest X-ray
ECG
Abdominal and pelvic ultrasound
MRI
Cross-matched blood or blood components

Radiotherapy

Radiation may be used as a single treatment in patients who refuse or who are unfit for surgery even with early disease. Although there are no direct comparisons of primary radiotherapy and surgery, early case series suggest that radiotherapy has inferior results. However, the main indication for radiation is postoperatively as an adjunct to surgery in patients where the risk of vault recurrence is high. These cases include deep myometrial spread, high grade, cervical involvement and lymphovascular space permeation. There may be some benefit in terms of survival when radiation (40 Grays) is added to surgery if the above prognostic factors are present.

Another indication for radiation is in the treatment of vault recurrences where a much higher dose (70 Grays) is used.

Advanced disease

Radiation, surgery, hormonal therapy, chemotherapy and palliative care are part of the armamentarium in battling a disease which carries a low 5-year survival. Even with advanced disease, a simple panhysterectomy may be carried to control vaginal bleeding.

Many cytotoxic agents such as doxorubicin, cisplatin and paclitaxel are active against advanced or recurrent disease especially when the drugs are combined. This of course increases adverse effects from these chemotherapeutic agents.

With metastatic endometrial cancer especially to the lungs, high doses of medroxyprogesterone acetate (50-100 mg orally 3 times daily) for 6 months seem reasonable in view of safety profile of this agent. However, many studies have not revealed any benefit from progestogen therapy with endometrial carcinoma.

For improved survival in endometrial cancer, each case should be individualized and treatment should be planned by a multi-disciplinary team. The patient's state of health and prognostic factors should be considered. There is an urgent need for more properly conducted research as we implement new and innovative treatment. Table 6 shows the 5-year survival rates.

Table 6. 5-year survival rates

Stage 1	80-90%
Stage 11	60-80%
Stage 111	30-40%
Stage 1V	10%
Overall	60%

Uterine Sarcoma

Sarcomas of the uterus are uncommon but they are highly malignant. They are to be more common amongst black women as well as women who have had prior pelvic irradiation.

Endometrial stromal sarcomas are subdivided into low-grade and high-grade based on the level of atypia. The treatment includes a total abdominal hysterectomy with bilateral salpingo-oophorectomy as well as possible adjuvant pelvic irradiation, progestational therapy and adriamycin.

Malignant mixed mullerian tumour also called mixed mesodermal tumour or carcinosarcoma arises from both the endometrial glands and the stroma. The mean age of the patient with aggressive tumour is 60 years. The patient usually presents with postmenopausal bleeding, abdominal pain or a mass. Apart from a panhysterectomy, pelvic and para-aortic nodal sampling is required. Chemotherapy such as cisplatin should also be considered.

Leiomyosarcoma

This tumour is the malignant variant of leiomyoma. It is the most common sarcoma of the uterus. Only about 5-10% of leiomyosarcomas arise from fibroids. In other words, the majority arises de novo. Those which appear as a result of malignant degeneration of fibroids have a better prognosis. One must always consider a leiomyosarcoma if a fibroid increases rapidly in size. Symptoms include abnormal uterine bleeding, abdominal mass, pain and constitutional symptoms such as weight loss. These tumours tend to spread within the abdomen and hence, a full surgical staging as with ovarian cancer is required.

Treatment includes panhysterectomy and pelvic and para-aortic lymphadenectomy. Adjuvant pelvic irradiation reduces local recurrences but not survival. Chemotherapy may be of value

with advanced or recurrent disease. Recurrences are most frequent when the number of mitotic figures exceeds 10 per 10 high power fields.

KEY POINTS

- 1. Endometrial cancer peaks in the sixth decade and the incidence varies with ethnicity as well as geographical areas.**
- 2. Although the prognosis for Stage 1 A disease is between 80-90%, the overall 5-year survival is about 60%.**
- 3. The majority of patients presents with postmenopausal bleeding.**
- 4. The staging is now surgical and pathological.**
- 5. Surgery is the main modality of treatment which may be combined with irradiation.**
- 6. Chemotherapy is usually reserved for advanced disease.**
- 7. Whether progestogen is beneficial is debatable.**
- 8. While adenocarcinoma has a good prognosis, the same is not so for papillary serous and clear cell carcinomas.**

CHAPTER TWENTY ONE

TROPHOBLASTIC TUMOURS (GESTATIONAL TROPHOBLASTIC NEOPLASIA)

Bharat Bassaw

The term gestational trophoblastic neoplasia (GTN) is used to describe a spectrum of disease of the trophoblast that includes hydatidiform mole, invasive mole and choriocarcinoma. Hydatidiform mole is a benign form of gestational trophoblastic neoplasia. Malignant gestational trophoblastic neoplasia includes invasive mole (chorioadenoma destruens) and choriocarcinoma.

HYDATIDIFORM MOLE

Hydatidiform mole is an abnormality of pregnancy in which the embryo dies and the entire villous system becomes transformed to a bulky mass of grapelike vesicles (Figure 1). It was first described by Hippocrates but it was not until 1895 when Marchand demonstrated that it originated from the trophoblast. Over the past 15 years, much advance has been made in the diagnosis, management and follow-up of patients with this condition.

Incidence

Hydatidiform mole, or molar pregnancy as it is sometimes called, is a relatively uncommon disease in Western countries. Table 1 shows that it occurs more commonly in the Far East and in Africa than in the Western world.

Table 1. Incidence of hydatiform mole(pregnancies)

United States	1 in 2000
Trinidad	1 in 1500
Jamaica	1 in 1000
Singapore	1 in 500
Taiwan	1 in 82
Japan	1 in 294
Iran	1 in 314
Nigeria	1 in 379

These geographical differences in the incidence of hydatidiform mole may be due to nutritional, environmental, socio-economic or genetic factors. However, Bagshawe has found no association with socio-economic status in the United Kingdom. In Singapore, there is no

difference in the incidence of this condition in Chinese, Indians and Malays. Of interest, however, is a study of multi-ethnic population of Hawaii, where higher incidences of hydatidiform mole were encountered among Japanese, Chinese and Filipino women than among Caucasians and Hawaiians.

Pathology

Hydatidiform mole is characterised by villous oedema, decreased vasculature, trophoblastic proliferation and usually, the absence of an intact fetus (Figure 1). It has been described as an aberrant placenta arising as a consequence of a defective ovum. The entire villous system becomes swollen and transformed to a disorderly aggregate of grapelike vesicles which are filled with fluid. The vesicles are at first minute and boat- shaped but later become large, round and bullous .

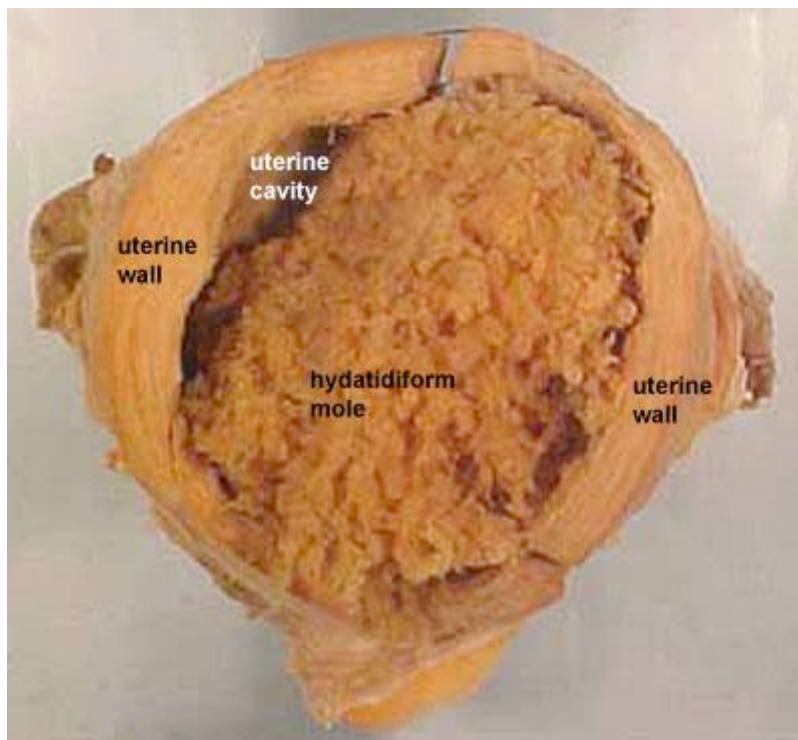


Figure 1. Molar mass

Hertig and Manswell believe that hydatidiform mole is a degenerative process but is capable of neoplastic change. It, therefore, represents a partly degenerative and a partly hyperplastic change affecting both layers of the trophoblast viz. syncytium and cytotrophoblast. The villi become cystic and the mesodermal core appears myxomatous and avascular.

On section of molar tissue, there is evidence of marked proliferation of the trophoblastic epithelium in some areas, while in others there is an almost complete absence of trophoblastic

activity. The former picture is seen in villi still attached to the uterine wall from which it derives its blood supply; the latter degenerative change is noticed in villi which have been expelled.

Novak and Woodruff have described the pathological changes as follows: the syncytium appears in the form of sheets of fenestrated amorphous masses with a large nucleus and scant cytoplasm: the cytotrophoblast (Langhan's layer) appears as well-differentiated mitotic cuboidal cells with large, hyperchromatic nuclei. It is believed that the cytotrophoblast actively produces human chorionic gonadotrophin (HCG) which is stored in the syncytium. The HCG in turn stimulates the ovaries to form multiple theca lutein cysts.

The stroma of the villi becomes oedematous, myxomatous and avascular. The waterlogging of the villus transforms it to a cyst which varies in size. In the majority of cases, the embryo dies and the molar mass fills the entire uterine cavity, although rarely, there may be a co-existent fetus.

While there is no tendency for a vesicular or hydatidiform mole to penetrate the uterine wall, it should always be regarded as a potentially malignant condition.

The above pathological description refers to that of a typical hydatidiform mole. This is called a complete or **classical mole**, and chromosomal analysis has demonstrated that they are **diploid**.

Occasionally, embryonic fragments or an embryo and normal chorionic villi are found in the midst of a typical mole. Such specimens have abnormal karyotypes, usually **triploid** or **trisomy**, with two of the three sets of chromosome being of paternal origin, and are called **partial** moles.

Aetiology

Attempts have been made to associate the aetiology of trophoblastic disease with some external factors such as viral, chemical, dietary and genetic but results are still rather inconclusive. The disease is therefore not preventable.

Clinical features

Molar pregnancy is usually associated with a spectrum of well-defined symptoms and signs (Table 2).

Table 2. Clinical features

Symptoms: 1. Vaginal bleeding
2. Hyperemesis gravidarum
3. Absent fetal movements

- Signs:
1. Uterine enlargement
 2. Inability to palpate fetal parts and absent fetal heart sounds
 3. Passage of vesicles per vaginam
 4. Pre-eclampsia
 5. Bilateral ovarian enlargement
 6. Anaemia
 7. Occasionally thyrotoxicosis
 8. Disseminated intravascular coagulation
-

Symptoms

1. Vaginal bleeding in virtually all patients is generally first observed at 6— 10 weeks amenorrhoea. The bleeding may be dark-brown in colour and is frequently described as a prune-juice discharge, because of liquefaction of the intrauterine clots. Sometimes, the bleeding is more profuse but this is a late manifestation.
2. Hyperemesis gravidarum is found in the majority of patients with hydatidiform mole. It is due to markedly elevated HCG levels. The patient may lose weight, and she appears dehydrated.
3. Absence of fetal movements in the latter part of the second trimester, in the presence of an enlarged uterus, should lead one to suspect a molar pregnancy.

Signs

1. A doughy uterus that is enlarged beyond the actual gestational age is seen in about 50 -75 per cent of cases. The uterus is enlarged both by the proliferating trophoblast and by clotted blood. Sometimes, the uterus may be “small for dates”.
2. Fetal parts are not felt and heart sounds are not detected, even with the use of Sonicaid or Cardiotocograph.
3. The vaginal discharge may contain “pearls” or “teardrops”; these are molar vesicles.
4. Signs of pre-eclampsia develop in 20 — 30 per cent of patients with molar pregnancy. Indeed, if pre-eclampsia occurs before 20 weeks of amenorrhoea, a diagnosis of hydatidiform mole must be entertained.
5. Vaginal examination reveals, apart for a large-for-dates uterus, bilateral ovarian enlargement. This finding is present in about 50 per cent of patients and is due to the formation of theca-lutein cysts. As stated above, these ovarian cysts are formed because of stimulation by HCG from the proliferating trophoblast.

6. Anaemia occurs because of intra-uterine bleeding.
7. In 10 per cent of cases, manifestations of thyrotoxicosis appear. The trophoblast cells produce molar thyrotrophin and there is an elevation of protein-bound iodine and serum thyroxine levels.
8. Disseminated intravascular coagulation is a complication of hydatidiform mole. Thromboplastins from molar tissue are released into the maternal circulation, leading to platelet consumption and fibrinogen depletion.

Diagnosis

Before the advent of ultrasound, invasive and non-invasive techniques were employed in order to make a pre-operative diagnosis of hydatidiform mole.

If ultrasound facilities are not available, a plain abdominal X-ray may be performed after 16 weeks of amenorrhoea. If a fetal skeleton is not seen, this is suggestive of a molar pregnancy. Difficulties arise (a) when this is actually less than the uterine size would suggest and (b) when there is a partial mole with a co-existent fetus.

Ultrasound

Ultrasound is the method of choice for the pre-operative diagnosis of hydatidiform mole. Its chief advantage is that a positive diagnosis can be made in the first trimester of pregnancy.

The characteristic sonogram of a molar pregnancy is produced by multiple echoes formed by the interface between the molar villi and the surrounding blood, decidua and trophoblastic tissue. When the vesicles are well-developed, a typical honeycomb pattern is seen. This snow-storm appearance is diagnostic (Figure 2).

Difficulties with ultrasound arise in patients with a blighted ovum, missed abortion, septic abortion, degenerating fibroids with pregnancy, and when hydatidiform changes are seen in a uterus that contains a fetus.

The ultrasound technologist should also search for the presence of theca-lutein cysts of the ovaries.

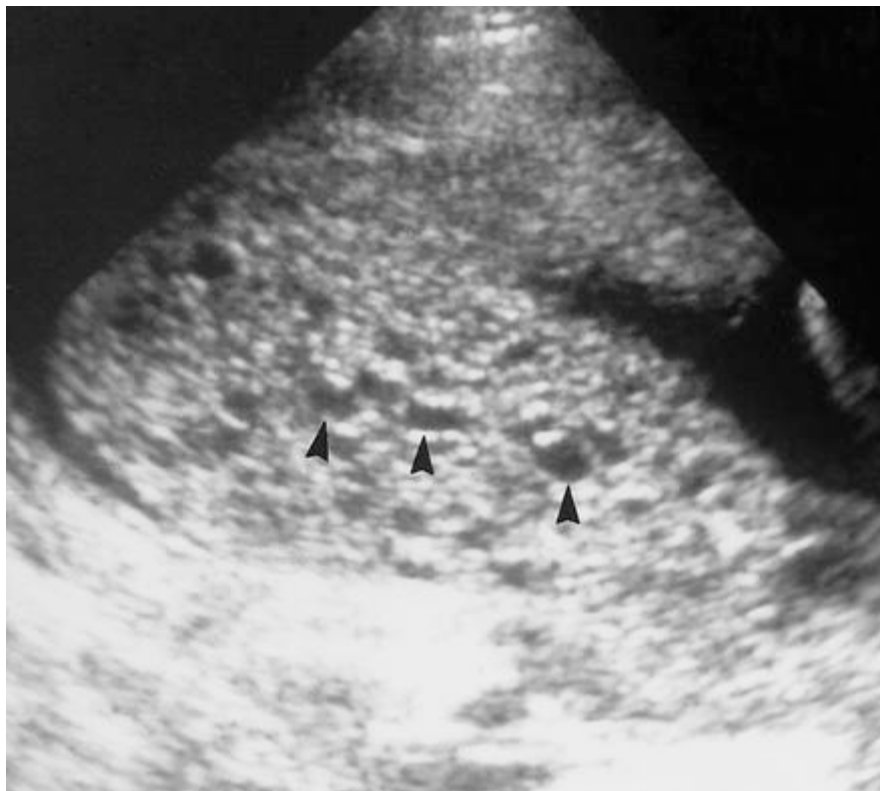


Figure 2. Molar pregnancy. US image demonstrates an echogenic mass in the uterine cavity with multiple small, hyperechoic areas (arrowheads).

Beta Human chorionic gonadotrophin (HCG)

Levels of Beta HCG are very elevated in patients with hydatidiform mole. Serial assays of HCG are used to monitor the response of treatment. By eight weeks after the evacuation of the uterus, the level of HCG should be undetectable. If it is still present, then a diagnosis of persistent trophoblastic disease should be entertained.

Differential diagnosis

In view of what has already been stated, it is clear that the following conditions may mimic molar pregnancy:

- a. Abortion
- b. Multiple pregnancy
- c. Polyhydramnios

Complications

Table 3 lists the complications of a hydatidiform mole.

Table 3. Complications

-
- | |
|---|
| 1. Haemorrhage |
| 2. Uterine perforation |
| 3. Trophoblastic embolization |
| 4. Disseminated intravascular coagulation |
| 5. Choriocarcinoma |
-

1. Haemorrhage

During and after evacuation of the uterus, profuse haemorrhage may occur, particularly in more advanced molar pregnancies.

2. Uterine perforation

Perforation of the uterus may occur during evacuation of a mole. The procedure calls for gentleness, especially if a sharp curette is used following suction curettage.

3. Trophoblastic embolization

This is a possible complication resulting from uterine contractions, and usually becomes manifest following evacuation of the uterine contents.

4. Disseminated intravascular coagulation (DIC)

Impairment of blood coagulation is seen from time to time in patients with hydatidiform mole. It is necessary therefore to determine platelet count, prothrombin time, fibrinogen

level and fibrin degradation products. If DIC manifests itself, correction of platelet and fibrinogen deficiency should be undertaken.

5. Choriocarcinoma

At least 50 per cent of choriocarcinomas follow hydatidiform mole. It is for this reason that after evacuation of a molar pregnancy, all patients should be monitored by serial serum HCG levels in order to ensure early diagnosis of trophoblastic malignancy.

Investigations

Table 4 summarises the investigations that should be carried out in patients with a molar pregnancy.

Table 4. Investigations

Haemoglobin, white cell count and platelet count
Prothrombin time and fibrinogen level
Group and cross-match 2 units of blood
Baseline Chest X-ray
Pelvic ultrasound
Serum HCG titre
Blood urea and electrolytes
Liver function tests
Thyroid function tests
Urinalysis

Management

The key to successful management of a molar pregnancy is prompt evacuation of the uterine contents. This can be achieved by:

1. Suction curettage
2. Hysterotomy
3. Hysterectomy

The patient should be referred to a regional centre where there is more experience in dealing with this problem. Needless to say, anaemia is corrected by packed red cell transfusion. If dehydration (from hyperemesis) is present, fluid replacement is necessary. At least two units of cross-matched blood should be reserved for the evacuation of the uterus.

1. Suction curettage

This is done under general anaesthesia. As soon as the suction curette is inserted, a syntocinon infusion of 40 IU in 1 litre Dextrose water is begun. After the bulk of the molar tissue is evacuated, rotatory and scraping motions of the curette are permitted. Following evacuation by suction, the uterine size shrinks and if less than 12 weeks, a sharp curette is employed for gentle scraping of any molar tissue.

Some workers have recommended a Syntocinon infusion or prostaglandin F_{2α} vaginal suppositories, in order to expel the bulk of molar tissue prior to curettage. As pointed out above, this carries the risk of trophoblastic embolization. Our policy is to begin a Syntocinon infusion during suction curettage and continue it for 24 hours postoperatively. If there is heavy bleeding during the evacuation, one ampoule of Syntometrine may be administered intravenously, provided that the blood pressure is normal.

2. Hysterotomy

This is no longer performed except in the rare instance of a co-existent fetus, either hysterotomy or Caesarean section may be indicated.

3. Hysterectomy

Hysterectomy with the mole in-situ is an option when no further childbearing is contemplated, for example, in a woman aged 40-years-old with a parity of 3 or more. However, hysterectomy is not commonly performed for molar pregnancy. This is so because hysterectomy is a major operation with significant morbidity and also with the advent of effective chemotherapy for choriocarcinoma. The choice of primary hysterectomy based on age and parity reduces the malignancy rate but does not in any way reduce the need for close follow-up and HCG monitoring. The theca-lutein cysts do not serve as an indication for oophorectomy because they regress spontaneously after a mole has been removed.

Postoperative management

All specimens must be sent to the laboratory for careful histological examination and cytogenetic analysis. If a villous or avillous choriocarcinoma is detected, immediate chemotherapy is instituted. When the mole is partial i.e. embryonic elements are present, an indirect Coombs' test should be performed on the patient if she is Rhesus negative. If unsensitized to the rhesus antigen, she should be given rhesus immune anti-D immunoglobulin (Rhogam).

Patients are followed up on a weekly basis, until the serum beta-HCG level has returned to normal. There is no contra-indication to a further pregnancy if the woman is desirous of having a baby, but she should be advised to avoid becoming pregnant for about six to twelve months after evacuation. The chief problem with an early pregnancy is the difficulty in distinguishing a rise in HCG titre due to trophoblastic neoplasia from that due to pregnancy.

If the serum HCG remains high after a thorough evacuation of a mole, or fails to return to normal after a period of 2 — 3 months, this may suggest progression to a malignant form of gestational trophoblastic neoplasia and chemotherapy should be instituted.

At each visit, the patient should be asked about irregular vaginal bleeding, vomiting and haemoptysis. An abdominal and a pelvic examination should be performed with special reference to subinvolution of the uterus, vaginal secondaries and the presence of theca lutein cysts. Serum HCG assays are also done each week until they return to normal non pregnant levels.

Contraception

The patient should be instructed to postpone pregnancy for 6 to 12 months because any rise in HCG levels would be difficult to interpret.

The safest contraceptive method initially is the barrier method but is unreliable. The intrauterine device should not be inserted because it may produce irregular uterine bleeding.

We recommend the use of the contraceptive pill after remission of serum HCG levels. If the "pill" is commenced too early, the HCG may remain elevated and hence, the patient may need chemotherapy. The reason for this is not clear.

Injectables should be avoided since they commonly result in abnormal bleeding which as mentioned above is a very important symptom of choriocarcinoma.

INVASIVE MOLE

When a hydatidiform mole penetrates the myometrium, it becomes known as an invasive mole. It possesses marked neoplastic propensities.

Pathology

Pathologically, invasive mole exhibits a persistence of the villous pattern but with penetration beyond the endometrium into the myometrium. Occasionally, there is metastasis to the lungs and vagina. The uterus is often enlarged and in some cases, nodules may be seen on its surface. In rare instances, protrusion of molar tissue is visible on the serosal coat of the uterus. A section of the uterus after hysterectomy will show molar tissue within the substance of the myometrium.

Clinical features

If the serum HCG levels continue to rise or if they plateau after a thorough evacuation of molar tissue, gestational trophoblastic neoplasia should be suspected.

Clinical examination may reveal an enlarged uterus which may or may not be tender. Other symptoms and signs of hydatidiform mole may also be present.

Occasionally, the patient may present with an “acute abdomen” if perforation of the uterus by the invasive mole has occurred.

Diagnosis

1. Endometrial curettage. Diagnosis by curettage is unreliable. If there is an intimate relationship between muscle and molar villi obtained by curettage, the degree of suspicion should be high.
2. Ultrasound examination may show a vesicular pattern within the myometrium.
3. Pelvic angiography reveals a localised area of increased vascularity within the myometrium.
4. Laparoscopic examination: some bluish or darkish nodules may be seen on the surface of the uterus.

Treatment

1. Hysterectomy is the treatment of choice in older women who have completed their childbearing.

2. Local uterine resection may sometimes be attempted in the case of solitary tumours.
3. Chemotherapy is advocated by most Gynaecologists. Patients who have had surgery as indicated above, should also be given chemotherapy postoperatively. The chemotherapy protocols are given below.

CHORIOCARCINOMA

Choriocarcinoma is a purely epithelial tumour composed of interlacing strands of syncytiotrophoblastic and cytotrophoblastic cells. It may commence during or following any type of pregnancy.

It is preceded by a hydatidiform mole in approximately 50% of patients, in 25% by a normal pregnancy, and in 25% by an abortion or ectopic pregnancy. Because of this, it is most important to consider choriocarcinoma in the differential diagnosis of secondary postpartum haemorrhage, and all curettings should be sent for histology.

Choriocarcinoma is one of the most malignant forms of neoplastic disease and is characterised by rapid dissemination.

Incidence

While choriocarcinoma is a rare tumour in the Western World, it is a fairly common tumour in the Far East and in Nigeria, where it is the third commonest malignancy among females, after cervical and breast carcinomas. In Indonesia, the incidence varies from 1 in 186 to 1 in 1035 pregnancies while in Nigeria, incidences of 1 in 1000 pregnancies have been reported.

Pathology

The growth usually resides in the uterine wall but may be found in the vagina, cervix, Fallopian tube or ovary. It is a raised, nodular, necrotic tumour of a purplish-red, granular, haemorrhagic appearance. It spreads rapidly by haematogenous means, the commonest sites of metastases being the vagina, vulva, lungs, liver and brain. Local extension through the uterus into the pelvis also occurs.

Histologically, there is a lack of villous pattern with areas of widespread haemorrhage within the stroma. Hence the term avillous choriocarcinoma is used.

The syncytiotrophoblast secretes large amounts of chorionic gonadotrophin (HCG) which can be quantitatively determined in the serum or urine. Even a small choriocarcinoma may secrete ten times or more HCG than the total secretion of a normal pregnancy.

Staging

A clinical staging for choriocarcinoma has been adopted by the International Federation of Gynecologists and Obstetricians (FIGO).

Stage I: The tumour is confined to the uterus.

Stage II: The tumour spreads to the pelvis and vagina.

Stage III: There are metastases to the lungs.

Stage IV: There are generalized metastases or distant metastases to the liver, brain, kidney and spleen.

Clinical features

The clinical diagnosis of choriocarcinoma may be extremely difficult. Irregular profuse uterine bleeding in a postpartum or post-abortion patient may be due to a choriocarcinoma. When the tumour becomes infected, an offensive vaginal discharge develops, and the patient becomes pyrexial and cachexic.

Most often, it presents only as a finding of distant metastatic disease, excision of which enables one to establish the diagnosis histologically. Pulmonary metastases present with haemothorax, dyspnoea or haemoptysis. Because of the frequency of a variety of sudden complications such as massive haemorrhage, and perforation of a viscus, the mortality rate from this neoplasm is very high.

Investigations

- a. Complete blood count
- b. Liver function tests, renal function tests and thyroid function test
- c. Frequent serum β -HCG levels
- b. Chest X-ray to demonstrate the typical cannon-ball appearance
- e. Electrocardiogram
- f. Intravenous pyelogram
- g. Computerized axial tomography of the brain (CAT scan)
- h. Ultrasound of the pelvis and the liver
- i. Pelvic arteriography to demonstrate a tumour focus

Treatment

Chemotherapy is the primary treatment of choice for choriocarcinoma. Hysterectomy may be indicated in addition to chemotherapy, if there is uncontrollable haemorrhage. Local excision of

vulval or vaginal secondaries may also be carried out to reduce sepsis and make the patient more comfortable.

A number of treatment regimes by way of chemotherapy have been adopted.

Regime A

This is the Methotrexate with Folinic acid (Citrovorum factor) rescue therapy. Methotrexate is a Folic acid antagonist.

I.M. Methotrexate 1.0 mg/kg body weight / 48 hours x 4 doses, followed 30 hours after each injection, by I.M. Folinic acid (Calcium leucovorin) 0.12 mg/kg body weight.

Regime B

Same as regime A plus I.V. Actinomycin-D, 2 mg single dose at the same time as the first I.M. dose of Methotrexate.

Regime C

Methotrexate 5 mg / 24 hours by slow I.V. infusion followed at 24 hours after beginning infusion, by I.M. or oral Folinic acid, 15 mg 6-hourly for 72 hours.

Regime D

I.V. Actinomycin-D 10 mg/kg daily x 5 doses. This regime is preferred if liver function tests are abnormal.

Regime E (the MAC regime)

Methotrexate 15 mg I.M.
Actinomycin-D 0.5 mg I.V.
Cyclophosphamide 3.0 mg/kg orally
or
Methotrexate 15 mg I.M.
Actinomycin-D 0.5 mg orally

The combination is given daily for 5 doses, every 2 — 3 weeks.

Regime F

5-Fluorouracil, 28 — 30 mg/ kg body weight/daily, dissolved in 500 ml 5% Dextrose- water, by slow I.V. infusion over 8 hours for 10 days, every 2 -3 weeks.
This regime is preferred if there is liver involvement.

**Another regime for high risk choriocarcinoma involves
Ethopside, Methotrexate, Actinomycin D, Cyclophosphamide and Vincristine (EMACO)
Choice of chemotherapy regime**

In order to decide on which regime to use, a simple guideline is as follows:

Invasive mole	Regime A or B
Low-risk choriocarcinoma	Regime C or D
High-risk choriocarcinoma	Regime E or F

To determine whether a tumour is low-risk or high-risk, factors such as age and parity, antecedent pregnancy, interval between antecedent pregnancy and treatment, blood group, presence of metastases, staging and HCG titre will have to be taken into consideration.

For example, a 38-year-old para 4 patient who develops the disease a year after a term pregnancy and with secondaries in the liver obviously falls into a high-risk category.

By contrast, a 21-year-old para 0 + I patient who develops the disease 3 months after expelling a hydatidiform mole and with no secondaries present, will fall into a low-risk category.

Monitoring chemotherapy

In developing countries, it is advisable that all patients should receive broad-spectrum antibiotics during chemotherapy, as prophylaxis against infection.

To prevent precipitation of Methotrexate in the renal tubules, the urine should be kept alkaline (ph 6.5 — 7.0) by giving Sodium bicarbonate during treatment and for 24 hours afterwards.

The following investigations should be done daily:

- (a) Hb and PCV
- (b) Platelet count
- (c) W.C.C. and differential count
- (d) Blood urea and creatinine
- (e) SGOT, SGPT and Alkaline phosphatase levels
- (f) Serum beta-HCG assays should be done weekly

Chemotherapy should be discontinued if:

- (a) Hb < 8g/dl
- (b) WCC < 3000/ml
- (c) Platelet count < $100 \times 10^9/L$
- (d) Pyrexia > 38°C

(e) Renal function tests are abnormal

After achieving maximum response which is recognized when the beta-HCG level approaches 10 mIU/ml, two courses of chemotherapy should be given again.

Drug toxicity

Toxicity may occur with the use of these chemotherapeutic drugs. Features of toxicity may include vomiting, stomatitis and gingivitis, diarrhoea, skin rash, alopecia, paronychia, anaemia, leucopenia and thrombocytopenia.

There are some drugs which may cause a potential for increased toxicity if used concurrently with Methotrexate. These drugs include Salicylates, Sulphonamides, Diphenyl hydantoin, Tetracycline, and Para-aminobenzoic acid and they should be avoided.

Prognosis

Choriocarcinoma carries a high mortality rate because of its rapid growth with early dissemination. This may be as high as 40-60 percent in the developing countries where patients are seen late and the interval between antecedent pregnancy and diagnosis or institution of treatment is much greater than four months. The prognosis is also worse if the disease follows term pregnancy as opposed to a hydatidiform mole. However, in the developed western countries, a remission rate of 85 — 90% has been reported.

KEY POINTS

1. This is an abnormal pregnancy where the embryo dies early and there is trophoblastic proliferation and hydropic degeneration.
2. The incidence of hydatidiform mole is about 1 — 2 per 1000 live births, and it is more common in certain ethnic groups and with advancing maternal age.
3. The main symptoms are vaginal bleeding (bright red, passage of clots, or prune-juice discharge) and amenorrhoea. There may also be hyperemesis gravidarum, pre-eclampsia before 20 weeks of gestation, features of hyperthyroidism and abdominal pain.
4. The uterus is usually “large-for-dates” and has a typical doughy consistency.
5. The patient may admit to the passage of vesicles or these may be seen with a speculum examination.
6. The HCG levels are markedly elevated, and a pelvic ultrasound will show the “snow-storm” pattern, absence of a fetus (in a complete mole) and possibly, bilateral theca lutein cysts.
7. Treatment comprises resuscitation and suction curettage. A sharp curettage may be performed following the suction and administration of Syntometrine.
8. Follow-up is critical to exclude the development of persistent trophoblastic disease.
9. Treatment of choriocarcinoma depends on the staging. Methotrexate with Folinic acid is commonly used for patients with low scores (early stage) while Ethopside, Methotrexate, Actinomycin D, Cyclophosphamide and Vincristine are used for advanced disease.
10. There is very little role for surgery with choriocarcinoma.

CHAPTER TWENTY TWO

TUMOURS OF THE OVARY

Joseph Hall, Hemant Persad

Benign tumours of the ovary are common, frequently asymptomatic and many may resolve spontaneously. Of major significance is the exclusion of a malignancy. Tumours of the ovary appear histologically similar to, or identical with, other neoplasms of the Mullerian system. Those that resemble the Fallopian tube are serous; those that resemble the endometrium are called endometrioid tumours, and those that appear similar to the endocervix are classified as mucinous.

Teratoma arises from primitive oocytes while serous and mucinous cystadenoma arise from the coelomic epithelium. Germ cell tumours include granulosa cell, theca cell, dysgerminoma and arrhenoblastoma. Fibromas are of mesenchymal origin (Table 1).

Table 1. Benign ovarian tumours

Physiological cysts

(a) Follicular cyst

(b) Luteal cyst

Benign germ cell tumours

Teratoma (Dermoid cyst)

Epithelial tumours

(a) Serous cystadenoma

(b) Mucinous cystadenoma

(c) Endometrioid cystadenoma

Benign sex cord stromal tumours

(a) Granulosa cell tumour

(b) Theca cell tumour

(c) Fibroma

Others

(a) Polycystic ovaries

(b) Theca lutein

Physiological cysts are large versions of the follicles which develop during the menstrual cycle. They are commonly found in young women and they tend to be asymptomatic.

Follicular cyst

This is the commonest benign ovarian tumour. It is usually an incidental finding. On rare occasions, they may rupture and cause acute pelvic pain. The cyst contains a thin fluid which is rich in oestrogen and as a result, there may be abnormal uterine bleeding. In most cases, spontaneous resolution occurs but if after three months the cyst is still present or is increasing in size, a laparotomy or a laparoscopy should be done. In order to expedite the disappearance of the cyst, one can recommend the combined oral contraceptive pill for three cycles.

At surgery, the cyst should be excised and the ovary reconstituted.

Luteal cyst

This cyst arises from the corpus luteum and contains blood. It may be mistaken as an endometrioma. The patient may present with abdominal pain and a period of amenorrhoea, and thus this condition mimics an ectopic pregnancy. If there is haemorrhage into the corpus luteal cyst in pregnancy, signs of an acute abdomen are elicited. A laparotomy is indicated but the aim is to secure haemostasis only and should not be to remove the corpus luteum, otherwise, a miscarriage will occur.

Benign germ cell tumours

Germ cell tumours are among the commonest ovarian swellings in young women. A small number of these tumours are malignant.

Dermoid cyst (Benign Cystic Teratoma)

This is a relatively common form of ovarian tumour which is bilateral in 10 per cent of cases. It also recurs in about 10 per cent of times and may reach a size of 10 cm in diameter. The cyst is usually unilocular and the contents include hair which has a characteristic echogenicity with ultrasound, teeth which are appreciable with a plain X-ray of the abdomen and pelvis, sebum and derivatives of endoderm and mesoderm (Figure 1).



Figure 1. Teratoma showing a tuft of hair

This tumour can be quite heavy and unlike other ovarian cysts may prolapse both anterior and posterior to the uterus. The patient may complain of urinary symptoms, abdominal pain and a swelling.

The pelvic examination may reveal an adnexal mass which is cystic and usually non- tender. The uterus may be displaced forward if the cyst is in the pouch of Douglas.

The diagnosis can be confirmed with an elevated serum alpha feto-protein, ultrasound or an X-ray of the abdomen and the pelvis.

The treatment entails an ovarian cystectomy and bisection of the contralateral ovary.

Epithelial tumours

Serous cystadenoma

This is a smooth, unilocular silvery-blue cystic tumour. It is the most common benign epithelial tumour and is bilateral in 10 per cent of cases. The cystic fluid is thin and serous. The serous cystadenoma is usually much smaller than the mucinous counterpart. The treatment is an ovarian cystectomy in a young patient, and a total abdominal hysterectomy and bilateral salpingo-oophorectomy in the older patient who has completed her reproductive ambition.

Mucinous cystadenoma

These constitute 15-25 per cent of all ovarian tumours. They are typically large, multilocular cysts with a smooth inner surface. The cystic fluid is thick and gelatinous. It may be complicated by pseudomyxoma peritonei.

The papillary cystadenoma is characterized by the formation of intra-ocular papilliferous projections which may also grow outwards and give the cyst a warty appearance. There may be calcium deposits in the papillae which are referred to as psammoma bodies. The patient may present with abdominal pain, ascites and later on with weight loss, constipation, loss of appetite t

The treatment is an ovarian cystectomy or an oophorectomy.

Endometrioid cystadenoma

Most of these tumours are malignant.

Sex cord stromal tumours

This group of tumours arises from the ovarian stroma. They do have the potential to secrete oestrogen or androgen.

Granulosa-theca cell tumours

These are small tumours which are slow-growing. The cut surface is of grayish-yellowish colour with many cystic cavities and areas of haemorrhage. These tumours almost always become malignant.

They may occur at any age from before puberty to the postmenopause. Signs of excess oestrogen are predominant. The patient may present with precocious puberty, menstrual irregularities, or postmenopausal bleeding. The hyperoestrogenism in the postmenopausal woman may lead to endometrial carcinoma.

In the young patient, a unilateral salpingo-oophorectomy may be considered but in the elderly woman, a pan-hysterectomy is advisable. If the tumour is malignant, then chemotherapy is required.

Fibroma

This is a small uncommon tumour with a smooth capsule and its cut surface has a pinkish-white colour. On rare occasions when it becomes large, it often develops a pedicle and becomes liable to torsion. Meig's syndrome comprises a fibroma, ascites and a hydrothorax. The treatment is

oophorectomy.

Others

Theca lutein cyst

This cyst typically arises in patients with hydatidiform mole and is due to the high levels of HCG. The size is variable. Once the source of the HCG disappears, the cyst also disappears.

Polycystic ovaries

The ovaries are usually 2-3 times larger and the capsule is thickened. There are numerous (>12) sub-cortical cysts 2-9 mm in diameter. Each ovary appears pearly-white with an ovarian volume of >10ml. The patient may present with irregular cycles, anovulation and infertility. There is insulin resistance and thus acanthosis nigricans is seen. Other associated features include obesity, hirsutism, galactorrhoea, hypertension, acne and a predilection to diabetes. Late complications include metabolic syndrome and endometrial carcinoma.

The treatment depends on the presenting complaints. Metformin is advised to redress the hyperinsulinaemia. Weight-loss is also very important. Clomiphene citrate is used to induce ovulation. The combined oral contraceptive pill or cyclical Provera is prescribed to regulate the cycles. Anti-androgen therapy is indicated if there is hirsutism. If medical treatment fails, then laparoscopic electrocautery drilling of the ovaries is required. Caution must be taken with the electrocautery of the ovaries since the ovaries are susceptible to heat. The power of the electrocautery is reduced to 30 Watts and the needle is applied for about 30 seconds only in the cortex of the ovary at four points. Immediately after, the ovaries must be cooled with normal saline. Electrocautery of the ovaries seem to work for about 12 months. It is not advisable to repeat this procedure on the same patient, the fear being ovarian damage leading to premature menopause.

Clinical presentation

The symptoms and signs of a benign ovarian tumour may be as follows:

1. Asymptomatic
2. Abdominal pain
3. Mass
4. Pressure effects on the bladder and the bowel
5. Menstrual disturbances
6. Hormonal effects

Asymptomatic

Many benign ovarian cysts are found as an incidental finding during a routine bimanual examination; for example, while performing a cervical smear or at an antenatal visit. Sometimes a cyst may be detected by an abdominal or pelvic ultrasound.

Abdominal pain

A large ovarian cyst may cause lower abdominal discomfort or heaviness. With endometriosis, there may be an endometrioma and the patient may complain of chronic pelvic pain, dysmenorrhoea and dyspareunia. It is possible for pain to be radiated to the medial aspects of the ipsilateral thigh via the obturator nerve.

Acute or sub-acute abdominal symptoms may result from torsion, rupture, haemorrhage into the cyst and infection. There is the possibility of nausea and vomiting as well as the signs of an acute abdomen such as tenderness, guarding and rebound tenderness. It may be difficult to differentiate between an accident to an ovarian cyst from a ruptured ectopic gestation, bleeding corpus luteum or an appendicitis.

Abdominal mass

The patient may have noticed that her abdomen is becoming larger either from the swelling or from ascites. The cyst is usually central if it is large. The characteristics include a soft, mobile swelling which is dull to percussion, and which may or may not be tender. Being a cystic mass, it should transilluminate. On bimanual palpation, the cervix will not move when the mass is moved except in the case where the mass is attached to the uterus by adhesions.

Others

Gastrointestinal and urinary symptoms may result from compression by the cyst. Occasionally, the patient may complain of menstrual disturbances. With sex cord stromal tumours, precocious puberty, menorrhagia or postmenopausal bleeding may occur. Secretion of androgen may lead to acne, hirsutism or obvious virilism.

Differential diagnoses include

1. Full bladder
2. Uterine fibroid
3. Pelvic inflammatory disease
4. Pregnancy
5. Cyst of Morgagni
6. Ectopic pregnancy
7. Appendicitis

8. Inflammatory bowel disease
9. Pelvic kidney

Ovarian cyst in pregnancy

This may be an incidental finding or the patient may present with sudden onset of severe abdominal pain and the other features of an acute abdomen. If the cyst is asymptomatic, it is prudent to defer surgical intervention until 14 weeks of gestation. This avoids the risk of removing a corpus luteal cyst. On the other hand, if the patient presents with acute pain from rupture, torsion or haemorrhage, then an emergency laparotomy is warranted regardless of the period of gestation. If the ovary has undergone ischaemia and gangrene, then an oophorectomy should be done.

A small ovarian cyst in the postmenopause

The finding of a simple ovarian cyst that is asymptomatic in a postmenopausal woman warrants a pelvic ultrasound to exclude any septa, papillary projections, solid areas or ascites, features consistent with a malignancy. If none of the above is present, then a repeat ultrasound should be done in one to three months. If the cyst is still present or enlarging, then other investigations for example, CA 125 levels and possibly a colour Doppler scan and/or a laparotomy should be considered. Many small simple ovarian cysts will disappear spontaneously and this is the rationale for a conservative approach in the management.

Ovarian carcinoma

Ovarian cancer remains the most common cause of death from gynaecological cancer in the United Kingdom with an estimated 4000 deaths each year in England and Wales. In Trinidad, cervical cancer is more common and accounts for more deaths than ovarian cancer. The lifetime risk of developing ovarian cancer is estimated to be 1 in 70 in the United Kingdom. The poor 5-year survival rate, at 30-35 per cent, has remained the same over the last few decades, mainly because ovarian cancer usually presents late when spread has already taken place to the abdominal cavity. It is estimated that the condition is diagnosed in 70 per cent of cases when the stage is advanced.

Most ovarian tumours are of epithelial origin. They are uncommon before 35 years of age but the incidence increases rapidly with age. The peak incidence is in the range of 50-70 year-old age group. Only 3 per cent of ovarian cancers are seen in women less than 35 years of age and most of these are of non epithelial origin. They are usually germ cell tumours.

Aetiology

The aetiology of ovarian carcinoma is unknown. They are associated with nulliparity, upper socio-economic group, Caucasians (especially Scandinavians), family history of breast,

endometrial or ovarian cancer. The use of the combined oral contraceptive pill protects against both ovarian and endometrial carcinoma. A family history of the BRCA 1 gene mutation increases the risk significantly. It is the current thinking that many ovarian cancers actually arise from the Fallopian tubes.

Classification

Ovarian tumours can be solid or cystic, but the malignant form is usually solid. Table 2 demonstrates a classification of ovarian carcinoma based on its histology.

Table 2. Histological classification of ovarian cancer

Epithelial tumours

Serous cystadenocarcinoma
Mucinous cystadenocarcinoma
Endometrioid tumour
Brenner tumour
Undifferentiated

Sex cord gonadal tumours

Granulosa-theca cell tumours
Sertoli-Leydig cell tumours

Germ cell tumours

Dysgerminoma
Yolk sac tumour
Teratocarcinoma

Secondary (metastatic) tumours

Examples: breast, endometrial, gastrointestinal

Serous cystadenocarcinoma

The majority of serous carcinomas show a mixture of solid and cystic elements. They tend to be bilateral, ranging from 50-90 per cent. The well-differentiated tumours have a papillary pattern with stromal invasion. This is the most common ovarian cancer.



Figure 2. Mucinous carcinoma

Mucinous carcinoma

This type of tumour accounts for 10 per cent of ovarian malignancies. They are typically multilocular, larger than the serous type, and the thin-walled cysts contain mucinous fluid (Figure 2).

Granulosa-theca cell tumour

The most common sex cord stromal tumours are the granulosa and theca cell tumours. They account for about 1.5 per cent of all ovarian tumours and 5-15 per cent of solid ovarian tumours. They often secrete oestrogen which leads to postmenopausal bleeding in older women and precocious puberty in very young girls.

Germ cell tumours

Dysgerminoma

This is uncommon ovarian tumour accounting for only 1 per cent of all primary ovarian malignancies. They tend to occur in young women with up to 90 per cent in women under 30 years old. They behave similarly to the seminoma in the male with spread mainly via the lymphatics to the para-aortic, mediastinal and supraclavicular nodes, and they are also very radiosensitive. Sometimes, they co-exist with choriocarcinoma, yolk sac and endodermal sinus tumours. These tumours are usually only solid.

Secondary tumours

Krukenberg tumour is an adenocarcinoma of the gastrointestinal tract especially the stomach with spread to the ovaries. Primary cancers from the endometrium, Fallopian tube and the breast may also spread to the ovaries.

Clinical features

Abdominal pain and discomfort are the common presenting complaints followed by the sensation of distension or a swelling in the abdomen. The patient may also complain of upper gastrointestinal symptoms such as dyspepsia, bloating, nausea and vomiting. Because of these symptoms and the common occurrence of ascites, it is not uncommon for the patient to be seen by many other Specialists before she comes to the Gynaecologist. Unfortunately, by the time the patient is seen, the disease has in most instances, already spread to the abdominal cavity (Stage 111).

Other complains include weight loss, anorexia, urinary frequency, irregular or postmenopausal bleeding. A hard mass arising from the pelvis is highly suggestive of ovarian carcinoma especially if ascites is also present. The patient may be cachexic. There may be palpable nodes in the neck. The abdominal distension is likely to be very evident. A fixed, hard, irregular pelvic mass may be felt with a combined vaginal and rectal examination.

The ascites can be so extensive that makes palpation of the abdominal mass impossible. Furthermore, she may complain of difficulty in breathing. This may, therefore, require a paracentesis. Cytological studies may reveal malignant cells. The ascitic fluid is typically sero-sanguinous with an ovarian carcinoma.

Investigations

Routine haematological investigations include a complete blood count, renal and liver functions. A chest X-ray is essential. If there are bowel complaints, a barium enema should be done. It is also advisable to request an intravenous pyelogram.

Table 3. FIGO stage for primary ovarian carcinoma

Stage I Growth limited to the ovaries
Stage Ia Growth limited to one ovary; no ascites; no tumour on external surface; capsule intact
Stage Ib Growth limited to both ovaries; no ascites; no tumour on external surface; capsule intact
Stage Ic Tumour either stage Ia or Ib but tumour on the surface, the capsule has ruptured, or with ascites
Stage II Growth involving one or both ovaries with pelvic extension
Stage II a Extension and/or metastases to the uterus or tubes
Stage II b Extension to other pelvic tissues
Stage II c Tumour either stage II a or II b but tumour on the surface of one or both ovaries; or with capsule ruptured; or with ascites present containing malignant cells; or with positive peritoneal washings
Stage III Growth involving one or both ovaries with peritoneal implants outside the pelvis or positive retroperitoneal or inguinal nodes.
Superficial liver metastasis equals stage III. Tumour is limited to the true pelvis but with histologically proven malignant extension to small bowel or omentum
Stage IIIa Tumour grossly limited to the true pelvis with negative nodes but with histologically confirmed microscopic seeding of abdominal peritoneal surfaces
Stage IIIb Tumour with histologically confirmed implants on abdominal peritoneal surfaces, none exceeding 2 cm in diameter; nodes are negative
Stage IIIc Abdominal implants greater than 2 cm in diameter or positive retroperitoneal or inguinal nodes
Stage IV Growth involving one or both ovaries with distant metastases
If pleural effusion is present there must be positive cytology to allot a case to Stage IV. Parenchymal liver metastasis equals Stage IV.

Tumour markers

Several antigens have been detected in association with ovarian carcinoma; among them CA 125 is the most popular. The CA 125 antigen is recognized by a murine monoclonal antibody, OC 125.

A radioimmunoassay to detect CA 125 is now available. CA 125 is elevated (above 35 units/ml) in 80 per cent of epithelial cancers. In premenopausal women, CA 125 may be elevated in other conditions for example, endometriosis, adenomyosis and ovarian hyperstimulation.

Serum CA 125 assays may detect ovarian cancers when they are still early. Hence, it is not surprising that CA 125 is used to screen for ovarian cancers (see Chapter 23). Monitoring of levels of CA 125 is also valuable in the assessment of the response of treatment. Rapidly falling CA 125 levels after chemotherapy indicate a better prognosis than when the level plateaus or remains elevated.

CA 125 is of greatest value in the follow-up of serous cystadenocarcinoma but it also elevated in other ovarian neoplasms such as mucinous cystadenocarcinoma and endometrioid tumours, but to lesser extent.

Some newer tumours have been identified and these have been combined with CA 125 to improve the predictive value of screening for ovarian cancer.

These tumour markers include macrophage colony-stimulating factor, tumour-associated glycoprotein-72, OVX1 and OVX2 monoclonal antibodies.

Ultrasound has been investigated as a screening method for ovarian cancer but the false positive rate remains high. Many of the tumours identified are benign and not malignant. Hopefully, with improvements in Doppler imaging the false positive rate will decrease and more ovarian cancers will be detected when they are still early.

Treatment of ovarian cancer

Surgery

Surgery is the mainstay of both the diagnosis and the treatment of ovarian cancer. A vertical incision is required for an adequate exploration of the upper abdomen. The entire abdomen must be inspected carefully with special attention being paid to the omentum, the sub-diaphragmatic areas, pelvic and para-aortic nodes and peritoneal fluid. One must keep in mind that up to 30 per cent of cases where it is felt that the disease is confined to the ovaries, there is involvement of the abdominal contents. A laparoscopy can prove to be of immense value in the accurate staging of the disease as well as to confirm or exclude nodal disease. However, tumour spillage and spread to the port sites are disadvantages of laparoscopy in the

management of ovarian cancer.

Aims of surgery are:

- i. to remove all of the malignant tissue, and if this is not possible, then to ensure that residual masses are less than 1.6 cm in diameter. This facilitates the efficacy of chemotherapy.
- ii. to perform omentectomy since the infra-colic omentum is a common site for micro-and macro- metastases.
- iii. to perform a pan-hysterectomy.
- iv. to remove pelvic and para-aortic nodes if involved.
- v. to obtain peritoneal washings or ascitic fluid for cytology.

The standard management for ovarian cancer is primary cytoreductive surgery followed by chemotherapy. The evidence for this is based on retrospective studies which have shown that prognosis is poorest when residual tumour volume is greatest.

Griffiths (1975) reported that the survival rate was poor if the diameter of the largest residual mass exceeded 1.6 cm, irrespective of the total tumour volume. The ideal aim of cytoreductive or debulking surgery should be to remove all intra-abdominal macroscopic disease. The rationale for attempting maximal cytoreductive surgery is to remove pharmacological sanctuaries of poorly perfused large tumour masses.

An exception to such radical surgery is the younger patient, in whom non-epithelial tumours such as germ cell tumours are more common. These are managed by conservative surgery, for example, unilateral salpingo-oophorectomy, bisection of the contralateral ovary to exclude bilateral ovarian involvement and where indicated, adjuvant chemotherapy; the aim is to maintain reproductive function.

In early stage disease, surgery alone affords an excellent outcome (90% or greater five- year survival).

In more advanced disease, the value of radical surgery is debatable. A randomised study to evaluate the role of major cytoreductive surgery has never been done. A large meta-analysis by Hunter et al (1992) involving nearly 7,000 patients revealed that the chemotherapeutic regimen rather than cytoreductive surgery was more important in influencing survival. One must also keep in mind that major surgery carries serious risks and sequelae. The Gynaecologist or Oncologist must balance the perceived benefits vs risks in terms of extensive debulking.

Interval debulking surgery has become a reasonable option in recent times in patients who have advanced disease where optimum debulking surgery is not feasible at the first operation. In these cases, three cycles of chemotherapy are first administered and this is followed by a repeat surgery. Finally, the patient receives another course of chemotherapy postoperatively.

In some patients who respond to chemotherapy, a further surgical intervention may improve survival. However, the overall value of interval debulking surgery remains unknown.

Second-look surgery

The rationale for either second-look laparotomy or second-look laparoscopy was an attempt to define when chemotherapy could be stopped or if the tumour is resistant to the current chemotherapeutic regime, an alternative may be introduced. The fact that second-look surgery can identify patients who are tumour-free could mean that chemotherapy may be discontinued and this would free the patients of continued risks and adverse effects of chemotherapy. A second benefit of second-look surgery is that resection of residual tumour may be accomplished. With the advent of reliable tumour markers, for example, CA 125, the role of second-look surgery has been questioned. Furthermore, a large prospective study (Luesley et al, 1988) revealed that second-look laparotomy failed to influence survival in epithelial ovarian cancer.

Palliative procedures

Surgery for palliation is primarily aimed to alleviate symptoms and not to prolong survival. One such procedure is surgery for bowel obstruction which is estimated to occur in 10— 40% of women with terminal ovarian cancer.

Chemotherapy

Chemotherapy plays a major role in the treatment of epithelial ovarian carcinoma. Surgery as the only modality of treatment is only indicated in Stage 1 disease with an intact capsule.

Chemotherapy is usually administered with all other cases.

One of the main chemotherapeutic agents employed in the treatment of ovarian carcinoma is Cisplatin which is a heavy metal compound. It is associated with vomiting, renal impairment and loss of hearing. The newer analogue, Carboplatin, which is as effective as Cisplatin, causes less nausea, vomiting and nephrotoxicity. These drugs are usually combined with Taxol (previously Cyclophosphamide) which is an alkylating agent in the treatment of epithelial ovarian cancers.

Vincristine, Actinomycin D and Cyclophosphamide are used in combination for dysgerminoma while the more toxic combination of Cisplatin, Vinblastine and Bleomycin are administered for poorly differentiated teratocarcinoma, endodermal sinus tumour and choriocarcinoma.

Chemotherapy of epithelial ovarian cancers with platinum agents, either alone or in combination, will prolong the patient's life but are unlikely to result in an improved long-term survival.

Radiotherapy

The role of radiotherapy is limited to Stage II and III ovarian cancers. Radiotherapy is administered to the entire abdomen and the pelvis. The efficacy of radiotherapy is enhanced if there is minimal disease following debulking surgery. Dysgerminomas are highly radiosensitive, and radiotherapy is effective for metastatic or recurrent disease.

General guidelines for chemotherapy:

- a. Before treatment, request
Hb, WBC, platelets
Serum electrolytes and blood urea nitrogen

Chest X-ray
Liver function test
- b. Withhold treatment if:

WBC is less than $3,000 \times 10^3/\text{L}$
Platelets are less than $150 \times 10^9/\text{L}$
Polymorphs are less than $1,200 \times 10^3/\text{L}$
- c. Treatment should be determined by oncologist
- d. Usually administer chemotherapy for two or three months
- e. Ensure adequate hydration and prescribe Zofran as an anti-emetic agent
- d. Discontinue the drugs if signs of toxicity appear.

KEY POINTS

1. **Ovarian cancer contributes a significant number of deaths from gynaecological malignancies.**
2. **In about 70 per cent of cases of ovarian carcinoma, the disease is diagnosed when it is already advanced.**
3. **The prognosis of ovarian cancer thus remains poor.**
4. **There is no reliable screening method for ovarian tumours.**
5. **A small proportion of women with ovarian cancer do have a family history of breast, ovarian, endometrial or colonic cancer (Lynch syndrome). Some women have BRCA I or II gene mutations.**
6. **Surgical excision of all tumours is an integral aspect of the management.**
7. **The response of chemotherapy depends on the diameter of residual tumour. The aim should be to ensure that no residual tumour should exceed 1.6 cm in diameter.**
8. **Chemotherapy involving Cisplatin prolongs the patient's life.**
9. **Radiotherapy has a limited role in the management of ovarian cancer.**

CHAPTER TWENTY THREE

SCREENING and DIAGNOSTIC PROCEDURES

Mary Singh-Bhola

INTRODUCTION

Screening is defined as a procedure to assist in identification in an organised way a specified disease or condition among asymptomatic individuals. Screening programmes may be applied to a general population or targeted at a specific group which is considered to be at a particular risk.

The aim is to detect disease early in order to facilitate effective treatment. Appropriate screening offers improved prognosis in some cases, less radical treatment for others and potential resource savings for society. However, screening also has its limitations such as false positive results which may cause undue anxiety and results in unnecessary further investigations, and false negative results which may lead to false reassurance and possible delay in diagnosis of serious illnesses.

Some of the desired criteria for an effective screening programme include:

1. The condition should be important in terms of morbidity and/or mortality.
2. An accepted treatment must be available for the condition.
3. The facilities for diagnosis and treatment must be available.
4. A sensitive and specific screening test must be available.
5. A latent or early symptomatic stage must exist.
6. The test must be acceptable to the population.
7. The natural history of the condition must be understood.
8. An agreed treatment policy must exist.
9. The cost must be reasonable.
10. Case finding must be a continuous process.

Sensitivity is the proportion of positives that are correctly identified by the test.

Specificity is the proportion of negatives that are correctly identified by the test.

Positive predictive value is the proportion of patients with positive test results who are correctly diagnosed.

Negative predictive value is the proportion of patients with negative test results who are correctly diagnosed.

Screening for Gynaecological Malignancies

Gynaecological cancers account for one-sixth of female cancers with an estimated 942,000 new cases worldwide per year. With increased life expectancy, early diagnosis and prevention of cancer is an increasingly important issue. Genetic testing is now available in Clinical Gynaecology to identify women at an increased risk of gynaecological cancer.

Ovarian cancer

Ovarian cancer is the fourth most common cause of female cancer with a lifetime risk of developing the disease of 1 in 70. The annual incidence of ovarian cancer in the general population increases with age to 40 per 100,000 in women over 45 years.

The 5-year survival for patients diagnosed with Stage 1 disease is more than 80% but this falls to 22% and 14% for Stages 111 and 1V, respectively. Despite the good prognosis for early stage disease, the overall 5-year survival rate is less than 45%, largely because most patients have disease which has spread outside the ovary by the time of clinical presentation.

The aim of screening is in detection of asymptomatic, early stage disease. Unlike cervical cancer, no precursor lesion has been identified so far and direct inspection is not possible without surgical intervention. Furthermore, the natural history of the condition is not well characterised at this time. The relatively low prevalence of the disease means that any screening test must have a high specificity to avoid unnecessary surgical interventions. Current screening strategies are based on serum tumour markers or ultrasound of the ovaries.

Tumour Markers

Serum CA 125 remains the most used tumour marker in ovarian cancer screening. It is a large glycoprotein that is expressed during normal fetal development and by more than 80% of epithelial ovarian cancers. Serum levels are elevated in approximately 1% of healthy women and in a number of benign conditions such as endometriosis, uterine leiomyoma and pelvic inflammatory disease as well as malignancies of the breast, liver, lung and pancreas.

In asymptomatic women, the sensitivity for preclinical lesions range from 70-80% with specificity in postmenopausal women of 98-99%. The low positive predictive value (3%) makes CA 125 measurements unsuitable as a single screening test for ovarian cancer. Improvements in sensitivity have been made using algorithms incorporating age, rate of change and absolute levels.

Although no other single tumour marker or analyte has been shown to have better predictive value for ovarian cancer than CA 125, a number of combination assays have been shown higher rates for the detection of the disease. Of these, the most effective to date appears to be an assay using a combination of serum proteins (leptin, prolactin, osteopontin and insulin-like growth factor) with CA 125 and macrophage inhibitory factor.

Ultrasound

Transvaginal ultrasound can be used as the sole screening method or as a secondary test after primary screening with serum CA 125. An ovarian volume over 20cm³ in premenopausal women and 10cm³ is considered abnormal. Unilocular cysts of less than 5cm in diameter are found in 3.3% of postmenopausal women but are associated with minimal risk of malignancy, whereas complex cysts are associated with significant risk. Other features suspicious of malignancy include solid areas and multiple septations. The use of colour Doppler assessment of tumour blood flow has not been shown to provide additional information.

Genetic testing

Over 80% of families with 2 or more ovarian and/or premenopausal breast cancer affected members are thought to have mutations in BRCA 1 gene on chromosome 17. Women having BRCA 1 or BRCA 2 gene mutations have a lifetime risk of developing breast cancer that is 10 times higher than that of the general population (80% vs 8%), and 25-60% risk of developing ovarian cancer compared with a background risk of 1.4%.

Screening at-risk populations

Prophylactic oophorectomy at the completion of childbearing is recommended for women with hereditary ovarian cancer syndromes and is associated with a significant reduction in subsequent disease compared with surveillance alone. For those women who have not yet completed their family or who do not wish to undergo surgery screening using a combination of CA 125 measurement and ultrasound is offered beginning between 30 and 35 years of age. There is however, no agreement as to the appropriate interval at which it should be undertaken or any evidence as yet that such a strategy would affect mortality.

Summary of recommendations for ovarian cancer screening

Screening of women with no family history of ovarian cancer is not recommended.

Women with a family history but without evidence of a high-risk pattern (e.g. a single affected first degree relative) should be counselled about their individual risk taking into account their age, parity and prior use of the contraceptive pill and the limitations of screening.

Women with suspected hereditary cancer syndrome should have testing for BRCA 1 and BRCA 2 mutations.

Women who have these mutations should be offered screening and prophylactic oophorectomy by the age of 35 or when they have completed their family.

For those women who desire screening current evidence would suggest that 6-monthly CA 125 with transvaginal scanning for those with levels above 35IU/L is the most appropriate option.

Cervical Cancer

Cervical carcinoma is the second most common cause of cancer-related death in women worldwide with nearly half a million new cases per year. In Trinidad and Tobago at least 500 new cases are recorded annually and about 250 women die from this malignancy per year. There is an 8-fold difference in mortality rates between developed and developing countries with 83% of new cases now occurring in the latter. Women who have never had sexual intercourse are at low risk for squamous cell carcinoma but adenocarcinoma can occur in women with no previous history of sexual activity. Although screening using the Pap smear has never been evaluated in a randomized controlled trial, there is strong evidence from observational studies of its effectiveness in reducing the incidence and mortality of cervical cancer.

The age at which screening is recommended is commenced and the interval between each test depends mainly on resources. In the UK screening is performed every 3 years commencing at age 25 until age 49, then every 5 years until age 64. A negative result implies low risk rather than no risk. Sensitivity for cervical intra-epithelial neoplasia grade 1 (CIN 1) or worse using conventional cytology is 30-87%. In conventional cytology, cells are sampled from the cervix (squamo-columnar junction) using a brush or the Ayre's spatula and plated directly onto a labelled slide and a fixative such as 95% alcohol is applied immediately.

Unsatisfactory smears may be as a result of: insufficient or unsuitable material; inadequate fixation; poor spreading; or a smear consisting mainly of blood or pus, inflammatory exudates or excessive cytolysis.

A newer method of processing samples is using liquid-based cytology (LBC). In this method, samples are collected as mentioned above using a brush but then transferred directly into the fixative solution rather than dispersed on a slide. The liquid is then passed through a filter and the cells are transferred on to a slide as a monolayer. The slide is then processed like a conventional Pap test.

LBC has significantly reduced the number of false negative results. Furthermore, it has also resulted in fewer unsatisfactory smears. The ability to interpret the slide is improved because the cells are more evenly distributed and there is less artefactual material such as blood and

mucus. An additional benefit of LBC is the ability to use the liquid buffer for testing for human papilloma virus (HPV).

The following are the indications for referral for colposcopy according to the National Health Service Cervical Screening Programme (NHSCSP):

1. Women with persistent (3) inadequate smears
2. Women with borderline smears with endocervical changes
3. Women with persistent (3) borderline squamous changes
4. Women with persistent (2) LGSIL
5. Women with HGSIL
6. Women previously treated for CIN 2 or worse and those with any glandular abnormalities where the cervix appears suspicious
7. Women who have 3 abnormal tests (of any grade) in a 10-year period.

HPV Testing

HPV DNA is found in 95% of the cervical carcinoma and precursor lesions worldwide. Studies have shown that 15-28% of HPV DNA positive women with normal cytology develop CIN within 2 years. The risk of progression is higher with the high-risk HPV subtypes especially 16 and 18. In women who are HPV DNA negative, only 1-3% would develop CIN.

Currently there is no consensus as to the optimum strategy for incorporating HPV testing in the cervical cancer screening programme. The most appropriate role at present seems to be as an adjunct to cytology by aiding the allocation of women with borderline or mild dyskaryosis smears to either immediate colposcopy or cytological surveillance.

Another possible role is for population screening in developing countries that are without a well-organized cervical cytology screening programme.

Endometrial Cancer

Endometrial cancer rates are increasing as one would expect because of the obesity epidemic among mature women. This cancer has a more favourable prognosis than ovarian and cervical cancers with a 5-year survival rate of about 80%. Measurement of endometrial thickness with transvaginal ultrasound, out-patient endometrial sampling, and cervical cytology have been proposed as potential screening tools.

Given that the majority of early endometrial cancers give rise to symptoms especially postmenopausal bleeding and the low pick-up for testing asymptomatic women mass screening is unlikely to be cost-effective. Screening by annual endometrial biopsy has been recommended by the American Cancer Society for women with hereditary non-polyposis colorectal carcinoma although there no data to show that this would reduce mortality. It is important to educate

women and their healthcare providers about the significance of postmenopausal bleeding as an early warning feature of possible endometrial cancer.

Screening for non-cancerous conditions

Screening for sexually transmitted infections like syphilis, gonorrhoea, Chlamydia and HIV are offered to asymptomatic women, women attending genito-urinary clinics and partners of infected males.

Chlamydia infection is the most common curable sexually transmitted infection in the Western world. Chlamydia infection tends to be largely asymptomatic. Numerous clinical conditions such as acute urethral syndrome, urethritis and mucopurulent cervicitis have been attributed to Chlamydia trachomatis. Up to 70% of infants delivered vaginally to women with Chlamydia acquire the infection. If left untreated, Chlamydia infection can have serious long-term consequences for women in the form of pelvic inflammatory disease, chronic pelvic pain, infertility, and ectopic pregnancy. Screening can be performed using urine testing (with DNA amplification such as PCR) with high sensitivity. An endocervical swab can also be used to test for Chlamydia. The less invasive method of using urine for testing is more acceptable to patients. Selective screening is more commonly performed as universal testing can be costly in many settings. The groups commonly targeted are those with symptoms of Chlamydia infection, those attending genito-urinary clinics or those seeking termination of pregnancy, opportunistic screening of women under the age of 25 years, women over 25 with a new sexual partner or more than two partners in the last year.

ENDOSCOPY

Laparoscopy

Laparoscopy may be carried out under general anaesthesia, preferably with muscular relaxation and intubation. However, local anaesthesia has been found to be adequate. The patient's bladder must be emptied and a vaginal examination is then performed. Under local anaesthesia, premedication with Diazepam with 10 mg and Pethidine 100 mg helps in allaying the patient's anxiety.

The cervix is exposed with a Sims' speculum and grasped with a volsellum. A uterine cannula is then introduced and locked in position, after which the speculum is removed. This arrangement allows the uterus to be manoeuvred easily, and facilitate the examination. The abdomen is then prepared with an antiseptic solution and the patient draped.

The skin around the lower border of the umbilicus is picked up with Allis forceps and a 1 cm transverse incision is made. The patient is placed in 45° Trendelenburg position and a special gas insufflations needle (e.g. Verres needle) is inserted into the peritoneal cavity. The abdomen is inflated with Carbon dioxide until the intra-abdominal pressure is 40 cm of water. Usually

about 3 litres of gas will be necessary. The needle used to introduce the gas is then withdrawn and the skin incision extended to 1.5 to 2.0 cm in length. The trocar and cannula are next introduced at an angle of 45° to the skin in a direction towards the pelvis. A hollow sound is heard as the abdominal cavity is entered, and as the trocar is removed, the escaping gas is heard. The laparoscope is then inserted and the examination is begun (Figure 1).

After a quick preliminary inspection of the pelvic organs, it may be desired to introduce the biopsy forceps, coagulation forceps or the Filshie clip applicator through the body of the laparoscope or through separate ports. At the end of the operation, the instruments are removed in reverse order of insertion. Before the laparoscope is withdrawn, the patient is returned to the horizontal position and the gas is allowed to escape with some assistance from abdominal compression. The wound is closed with a deep mattress suture to the skin.

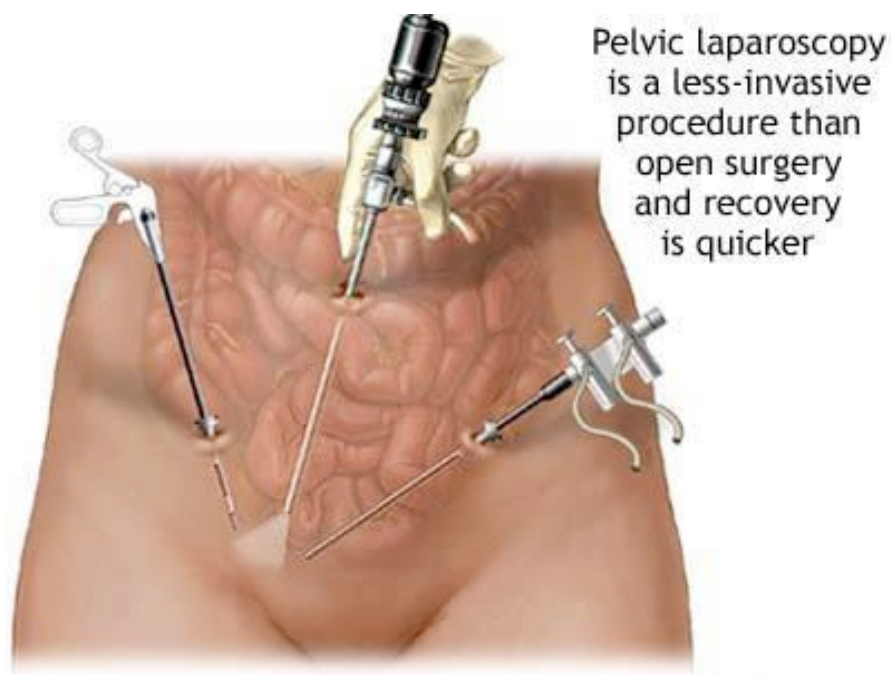


Figure 1. Laparoscopy

Application of laparoscopy in Gynaecology

1. Diagnostic indications

- a. Infertility
- b. Acute and chronic pelvic pain
- c. Suspected pelvic endometriosis
- d. Suspected tubal ectopic pregnancy
- e. Amenorrhoea associated with congenital anomalies
- f. Small pelvic masses
- g. Precocious puberty, ovarian dystrophies and intersex problems

- h. Certain gynaecological malignancies
- i. Genital tuberculosis

2. Surgical and therapeutic applications

- i. Tubal sterilization
- ii. Lysis of adhesions
- iii. Aspiration and puncture of ovarian cysts
- iv. Electrocoagulation or laser vaporization of endometriotic deposits
- v. Removal of foreign bodies such as the intrauterine device from the peritoneal cavity
- vi. Laparoscopic electrocautery drilling of the ovaries with PCOS
- vi. In oncology and in urogynaecology many procedures can be done with the aid of the laparoscope

Contra-indications to laparoscopy

- (a) Contra-indications to general anaesthesia (if this is the type of anaesthesia to be used)
- (b) Difficulty in establishing an adequate pneumoperitoneum e.g. extensive adhesions or morbid obesity
- (c) Presence of ileus, peritonitis or intestinal obstruction
- (d) Presence of extensive hernias especially of the diaphragm
- (e) Moderate to severe cardiorespiratory disease

Complications

- (a) Puncture of vessels such as inferior epigastric and iliac or rarely aorta
- (b) Perforation of a viscus such as bowel, or bladder
- (c) Effects of high pressure gas injection:
Cardiorespiratory embarrassment, gas embolism, mediastinal emphysema and pneumothorax, rupture of the diaphragm
- (d) Anaesthetic complications
- (e) Burns
- (f) Parietal emphysema
- (g) Death (3-8 in 100,000)

Hysteroscopy

Pantaleoni in 1865 was the first to examine the uterine cavity using a small tube inserted through the external os of the cervix with a kerosene lamp or a candle as the light source. During the last two or three decades, the invention of fibre-optic light sources and the Hopkins

lens system has made hysteroscopy a practical diagnostic outpatient procedure. It is also used for many therapeutic procedures as well.

Equipment

A simple hysteroscope consists of a telescope within a sheath. The Hopkins lens system consists of special glass rods placed at intervals along the axis of the telescope. This unique system provides high resolution and contrast and a wide viewing angle. The fibre-optic light source produces a bright image which allows still photography or closed circuit television. The sheath has a proximal valve through which gas e.g. Carbon dioxide or fluid e.g. Hyskon can be insufflated to distend the uterine cavity. If intra-uterine surgery is contemplated, a 4mm telescope should be enclosed in an operating sheath which contains a separate channel for instruments.

Indications

Hysteroscopy can be performed in all patients in whom a diagnostic curettage is indicated. Some of the main indications for hysteroscopy include abnormal uterine bleeding in the reproductive age, postmenopausal bleeding, intermenstrual bleeding, recurrent miscarriages to exclude congenital uterine abnormality for example, a septate uterus and suspected Asherman's syndrome. Recently, microcolpohysteroscopy has been introduced to visualise the entire transformation zone following an abnormal Pap smear in the postmenopausal woman.

Hysteroscopy is also invaluable in both the identification and retrieval of a lost intra-uterine contraceptive device. In puerperal bleeding, hysteroscopy may reveal a placental fragment. Operative hysteroscopy is indicated for sterilization and endometrial ablation with unexplained menorrhagia.

Pathological conditions seen with the hysteroscope include:

1. Benign polyps are frequently seen in women with abnormal uterine bleeding. They float in the cavity as the pressure of the distending medium varies. Polypectomy can be performed with the aid of the hysteroscope.
2. Submucosal fibroids are recognised by their pedicle, smooth surface and white appearance. If the fibroid is large, a course of GnRH analogue for example Goserelin (Zoladex) can be administered for a few months prior to the hysteroscopic myomectomy.
3. Lost intra-uterine contraceptive device can not be visualised but also can be retrieved under direct view.

4. Endometrial carcinoma: Hysteroscopy is beneficial for the following reasons with endometrial carcinoma: the tumour can be seen and the diagnosis confirmed by biopsy under direct vision; with a dilatation and curettage a small tumour may be missed, the size and extent of the growth can be recorded as well as spread to the cervix, this will help in accurate staging of the tumour, dual pathology such as associated fibroid can be excluded.

It is important to note that hysteroscopy is not useful in defining the depth of myometrial invasion, this can be determined with MRI.

5. Asherman's syndrome: Intra-uterine adhesions or synechiae may result from prior uterine surgery especially over-zealous curettage, septic abortion, Caesarean section and rarely, myomectomy. The patient may present with secondary amenorrhoea, oligomenorrhoea or infertility. It may also lead to recurrent first trimester miscarriages. Treatment is established with the aid of adhesiolysis with the hysteroscope followed by the insertion of an IUCD and the administration of the combined pill.
6. Congenital abnormalities of the uterus: the diagnosis of a septate or bicornuate uterus can be confirmed with hysteroscopic scrutiny of the uterine cavity. The mainstay of treatment of a septate uterus is now hysteroscopic metroplasty.
7. Atrophic endometritis is the leading cause of postmenopausal bleeding. The endometrium is atrophic, white and featherless except new fine capillaries are visible.

Contra-indications

Pregnancy and acute pelvic sepsis are the two main contra-indications. However, with unexplained postpartum bleeding, hysteroscopy may reveal retained placental tissue.

Complications

1. Distending medium: There is a small risk of anaphylactic reaction if Dextran is used to distend the uterus. Complications can also occur if the volume of CO₂ or if the pressure of the insufflations is too high. Close monitoring of the volume and pressure is mandatory during hysteroscopy since the uterine cavity is a potential space.
2. Perforation of the uterus: This is most likely if a retroverted uterus is not detected prior to hysteroscopic surgery or in pregnancy.

3. Sepsis: Pelvic inflammatory disease is a contra-indication for hysteroscopy because of the risk of peritonitis. The cervix must be thoroughly cleansed before the endoscope is inserted into the cervical canal.
4. Anaesthesia: The anaesthetic risks are uncommon since most diagnostic hysteroscopies are performed as outpatient procedure with no anaesthesia or just a paracervical block.

KEY POINTS

1. **Laparoscopy is a safe, endoscopic method for visualisation of the abdominal and pelvic cavities.**
2. **The most common indications for laparoscopy are in the evaluation of chronic pelvic pain, infertility and for tubal sterilization.**
3. **With endometriosis, laparoscopy is used to diagnose the condition, in staging and conservative surgery, as well as for second-look procedure to assess the efficacy of treatment.**
4. **Severe cardio-respiratory disease, extensive hernias and acute peritonitis are absolute contra-indications to laparoscopy.**
5. **Operative (therapeutic) laparoscopy allows for shorter hospital stay, less pain postoperatively and thus less need for analgesia, and earlier return to work. Of course, there is also the cosmetic benefit.**
6. **The initial cost for the equipment is significant which can be a limiting factor for the availability of laparoscopic services in some countries.**
7. **Most cases of sterilization in the UK are now performed with laparoscopic tubal occlusion with the Fishie clip.**
8. **Hysteroscopy with directed biopsies is now considered the gold-standard diagnostic procedure for postmenopausal bleeding.**
9. **The hysteroscope is of therapeutic benefit with submucosal fibroids, endometrial polyps, Asherman's syndrome and septate uterus.**